System 3 Manual

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Ship to:

TDT
11930 Research Circle
Alachua, FL 32615

* Custom hardware carries a one-year warranty on parts and labor. ES1 and EC1 carry a two year warranty.

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Part 1 RZ Z-Series Processors

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RZ2 BioAmp Processor

Overview

The RZ2 BioAmp Processor has been designed for high channel count neurophysiological recording and signal processing. The RZ2 features two (RZ2-2), four (RZ2-4), or eight (RZ2-8) Sharc digital signal processors networked on a multiprocessor architecture that features efficient onboard communication and memory access. The highly optimized multi-bus architecture realizes a device with up to nearly 20 gigaflops of processing power and four dedicated data buses to eliminate data flow bottlenecks— all transparent to the user. This architecture yields an extremely powerful system capable of sophisticated real-time processing and simultaneous acquisition on all 256 channels at sampling rates up to ~25 kHz and 128 channels at sampling rates up to ~50 kHz.

The RZ2 is typically used with a Z-Series Amplifier (such as the PZ2 or PZ3). High bandwidth data is streamed from the amplifier to the RZ2 over a lossless fast fiber optic connection.

The RZ2 also features 16 channels of analog I/O, 24 bits of digital I/O, two Legacy optical inputs for Medusa PreAmps, and an onboard LCD for system status display.

Power and Communication

The RZ2's Optibit optical interface ensures fast and reliable data transfer from the RZ2 to the PC and is integrated into the device. Connectors are provided on the back panel and are color coded for correct wiring. The RZ2's power supply is also integrated into the device and is shipped from the factory configured for the desired voltage setting (110 V or 220V). If you need to change the voltage setting, please contact TDT support at 386.462.9622 or email support@tdt.com.

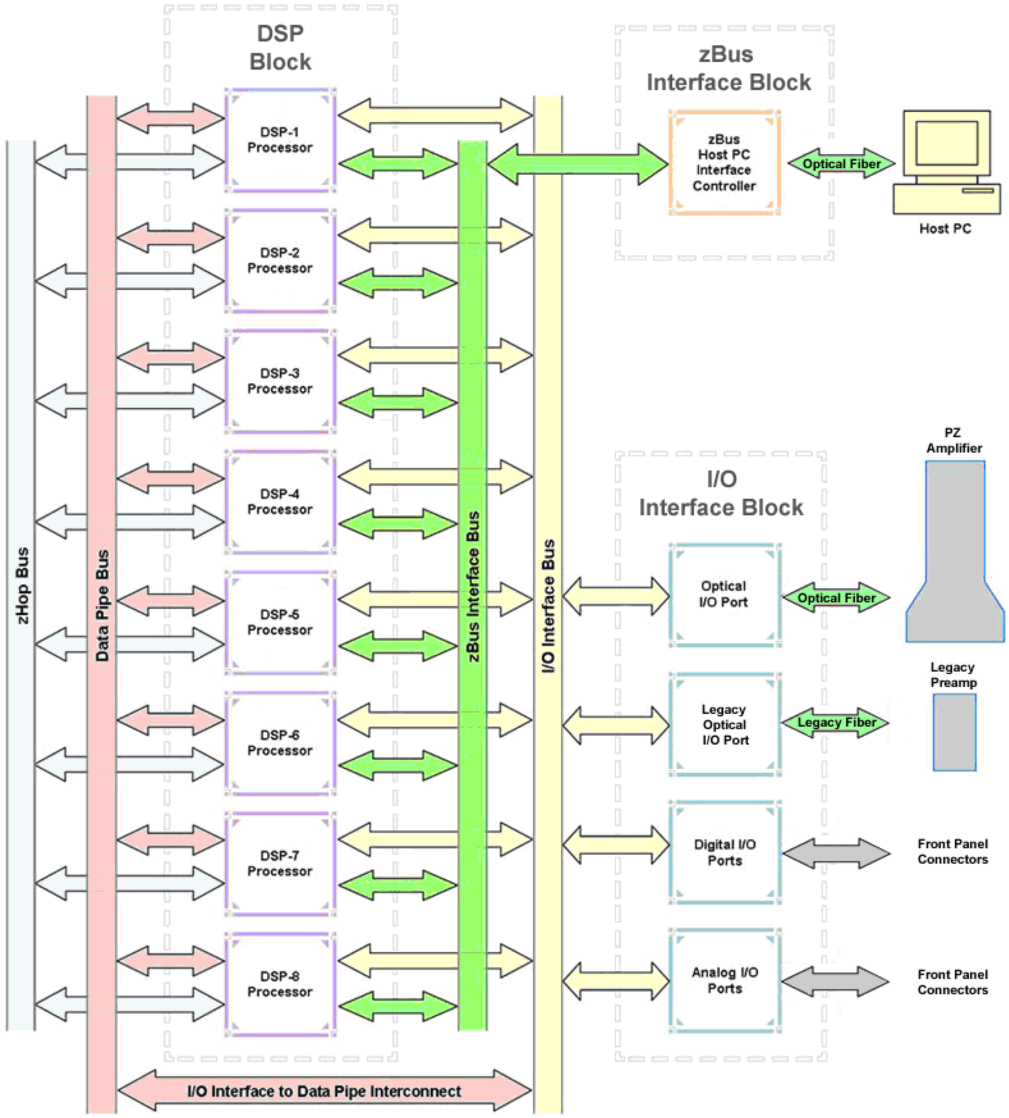
The RZ2 is UL compliant, see the *RZ2 Operations Manual* for power and safety information.

Software Control

Software control is implemented with circuit files developed using TDT's RP Visual Design Studio (RPvdsEx). Circuits are loaded to the processor through TDT run-time applications or custom applications. This manual includes device specific information needed during circuit design. For circuit design techniques and a complete reference of the RPvdsEx circuit components, see *MultiProcessor Circuit Design* and *Multi-Channel Circuit Design* in the *RPvdsEx Manual*.

RZ2 Architecture

The RZ2 processor utilizes a highly optimized multi-bus architecture and offers four dedicated, data buses for fast, efficient data handling. While the operation of the system architecture is largely transparent to the user, a general understanding is important when developing circuits in RPvdsEx.



As shown in the diagram above, the RZ2 architecture consists of three functional blocks:

The DSPs Each DSP in the DSP Block is connected to 64 MB SDRAM and a local interface to the four data buses: two buses that connect each DSP to the other functional blocks and two that handle data transfer between the DSPs (as described further in Distributing Data Across DSPs below). This architecture facilitates fast DSP-to-off-chip data handling.

Because each DSP has its own associated memory, access is very fast and efficient. However, large and complex circuits should be designed to balance memory needs (such as data buffers and filter coefficients) across processors. Memory use can be monitored on the RZ2 front panel display.

When designing circuits also note that the maximum number of components for each RZ2 DSP is 768.

The zBus Interface The zBus Interface provides a connection to the PC. Data and host PC control commands are transferred to and from the DSP Block through the **zBus Interface Bus**, allowing for large high-speed data reads and writes without interfering with other system processing.

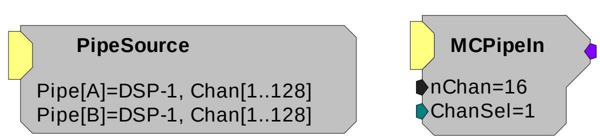
The I/O Interface The I/O Interface serves as a connection to outside signal sources or output devices. It is used primarily to input data from a PZ amplifier via the high speed optical port, but also serves the Legacy amplifier inputs and digital and analog channels. The **I/O Interface Bus** provides a direct connection to each DSP and the Data Pipe Bus.

Distributing Data Across DSPs

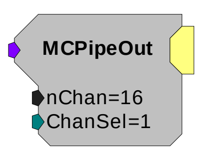
To reap the benefits of added power made possible by multi-DSP modules, processing tasks must be efficiently distributed across the available DSPs. That means transferring data across DSPs. The RZ2 architecture provides two data buses for this type of data handling.

The Data Pipe Bus The Data Pipe Bus is optimized for handling high count multi-channel data streams and efficiently transfers up to 256 channels of data between DSPs. The Data Pipe Bus also interconnects to the I/O Interface Bus allowing direct access to data from the PZ amplifiers.

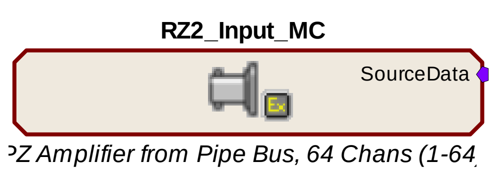
In RPvdsEx data can be transferred across the Data Pipe Bus using DataPipe components.



PipeSource and MCPipeIn components are used to select a data source (another DSP or the PZ amplifier) and feed data to a DSP circuit.

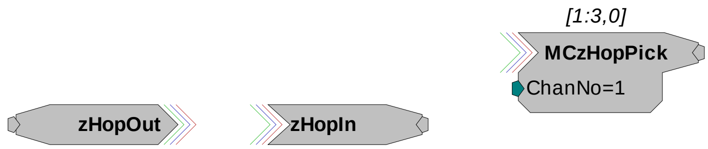


MCPipeOut feeds data off the DSP to the DataPipe Bus.



The RZ2_Input_MC macro also transfers inputs from the I/O interface to the PipeBus and DSPs.

The zHop Bus The zHop Bus is useful for transferring single or low channel count signals, such as timing and control signals.



In RPvdsEx data is transferred across the zHop Bus using paired zHop Components, including zHopIn, zHopOut, MCzHopIn, MCzHopOut, and MCzHopPick. Up to 126 pairs can be used in a single RPvdsEx circuit.

The zHopBus is less efficient than the Data Pipe Bus, so it is not recommended for multi-channel signals.

Bus Related Delays Standard delays are associated with the zHop and Data Pipe Bus. The zHop Bus introduces a single sample delay and the Data Pipe Bus adds a two sample delay. However, these delays are taken care of for the user in OpenEx when Timing and Data Saving macros are used.

50 kHz Sampling Rate Acquisition with the PZ Amplifier

The RZ2 and PZ amplifier support sample rates from ~6 kHz to ~50 kHz. When sampling at a rate of ~50 kHz, there are several important considerations:

- Only the first 128 PZ amplifier channels will be available.
- All DataPipes will have a max of 128 channels instead of 256.
- Both halves (A and B) of the PipeSource component must be selecting the desired source. For example, when acquiring data from a PZ amplifier, Pipe[A] and Pipe[B] both need to be set to Amp. Chan[1..128].

Data Transfer Rate

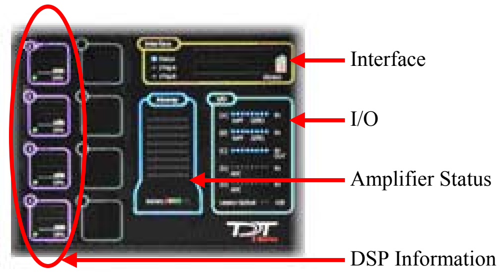
As with other devices, your expected sustained RZ-to-Host PC data rate should not exceed 1/2 to 2/3 of the rated data transfer speed. For the RZ2 device this is 160 Mbits/second (Mbps) so your designs should have a sustained data rate of no more than ~100 Mbps. When the RZ2 is processing, the current data transfer rate (Mbps) is displayed in the top right corner of the LCD Screen. This maximum rate may be further limited by your PC's ability to store the data to disk.

This equates to streaming a maximum of 160 channels at a sampling rate of ~25 kHz or 90 channels at a sampling rate of ~50 kHz. See *Calculating Data Transfer Rates* in the *OpenEx Manual* for more information.

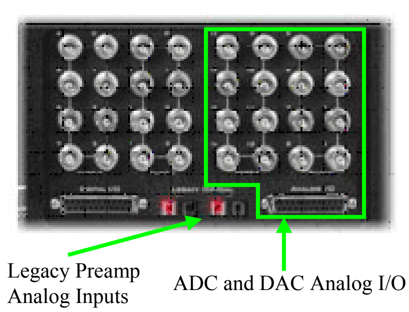
RZ2 Features

LCD Screen

The LCD screen shows information about each DSP, the optical PC interface, the PZ preamplifier and system I/O. A selection knob allows the user to highlight a section of the screen to display more detailed information. Rotate the selection knob to select a system component. Once the selection has been made, push the knob and expand the information view.



<table>
<tr><th>Selection</th><th>Available Information</th></tr>
<tr><td>DSPs</td><td>Component usage, memory usage and pipe source statistics for that processor
A stacked histogram shows cycle usage for each DSP with the bottom section (blue) showing the cycle usage taken up by circuit operation and the top section (pink) showing the cycle usage required for data transfer
If the cycle usage surpasses 100%, a bar is drawn above the 100% line in the cycle use histogram and will persist until the RZ2 is rebooted</td></tr>
<tr><td>Interface</td><td>Firmware version, MB data received/sent and transfer errors</td></tr>
<tr><td>Amp</td><td>Amp model, number of channels and firmware version of connected PZ series amplifier</td></tr>
<tr><td>I/O</td><td>Virtual indicator lights
<table>
<tr><td>[A], [B], and [C]:</td><td>Digital I/O
LED will light for an input bit or it will show the logic level for an output bit</td></tr>
<tr><td>[D] and [E]:</td><td>Analog I/O
16 lights will indicate the signal level, green when a signal is present and red to warn that the signal is approaching the maximum voltage (at which point clipping would occur)</td></tr>
<tr><td>Legacy Optical:</td><td>Amp Light For The Legacy Preamplifier Sync
Flash when no amp is connected and will be light light blue when the amplifier is correctly connected</td></tr>
</table></td></tr>
</table>



Amplifier and Onboard Analog I/O

The RZ2 is equipped with both optical port amplifier input and onboard analog I/O capabilities. The high speed fiber optic ports (located on the RZ2 back panel) and Legacy fiber optic ports (shown left) allow a direct connection to Z-Series or Medusa Preamplifiers. Physiological signals are digitized on the preamplifier and transferred across noiseless fiber optics.

The RZ2 also includes onboard D/A for stimulus generation and experiment control, and A/D for input of signals from a variety of other analog sources.

RZ2_Input_MC

Data Pipe, 16 Chans (1-16)

The RZ2_Input_MC macro provides a universal solution for analog input via the RZ2, automatically selecting the correct components, applying any scale factors or channel offsets, and performing data type conversion needed based on information the user provides about the input source.

The table below provides a quick overview of these I/O features and how they must be accessed during circuit design. When the RZ2_Input_MC macro is not used, reference the table and be sure to use the appropriate component, channel offset, scale factor and so forth. Further detail can be found below the table. Also, see the *RPvdsEx Manual* for more information.

Analog I/O	Description	Components	Channels	Notes
Port D	Analog Input	AdcIn	1-8	Standard Configuration (may vary) Accessed through Port D BNCs or Analog I/O labeled DB25
Port E	Analog Output	DacOut	9-16	Standard Configuration (may vary) Accessed through Port E BNCs or Analog I/O labeled DB25
High Speed Fiber Optic Port	Z-Series BioAmp Input (located on RZ back panel)	MCPipeIn PipeIn recommended	1-256	When the RZ2_Input_MC is NOT USED, use MCInt2Float or Int2Float with a scale factor of 1e-9
		MCAdcIn	1-256	No scale required.
Legacy Amp-A	Medusa PreAmp Input	AdcIn	17-32	When the RZ2_Input_MC is NOT USED, apply a scale factor of .000833
Legacy Amp-B	Medusa PreAmp Input	AdcIn	33-48	When the RZ2_Input_MC is NOT USED, apply a scale factor of .000833

Onboard Analog I/O

The RZ2 is equipped with eight channels of 16-bit PCM D/A and eight channels of 16-bit PCM A/D. All 16 channels can be accessed via front panel BNCs marked Port D and Port E or via a 25-pin analog I/O connector. See *RZ2 Technical Specifications*, page 1-10, for the DB25 pinout.

PZ Amplifier Fiber Optic Port

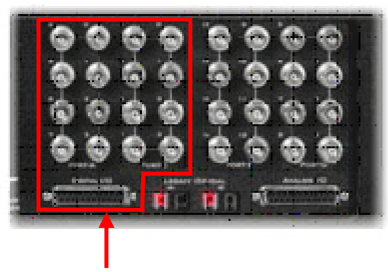
The RZ2's primary amplifier input, a high-speed fiber optic port is located on the back panel. The connectors on the fiber optic pair used for PZ amplifier communication are color coded for correct wiring. When designing circuits in RPvdsEx, the PZ Amplifier input channels are accessed using the Pipe components. When the DataPipe is used to feed signals from the Amplifier a MCInt2Float or Int2Float must be used with a scale factor of 1e-9. The Amplifier inputs can also be accessed using the RPvdsEx MCAdcIn component starting at channel 1; however, this access method is less efficient and not recommended for high channel count applications. Unlike the Legacy Port, this high speed port can input up to 256 channels at a maximum sampling rate of 25 kHz or 128 channels at a maximum sampling rate of 50 kHz.

Legacy Fiber Optic Ports

The base station can also acquire digitized signals from the Medusa preamplifier, RA8GA, or other legacy enabled device over a fiber optic cable using the Legacy ports. Two Legacy fiber optic ports labeled -A- and -B- are provided to support simultaneous acquisition from up to two Medusa preamplifiers. Each port can input up to 16 channels at a maximum sampling rate of 25 kHz. The Legacy fiber optic ports can be used with any of the Medusa preamplifiers including, the RA16PA and the RA4PA, or the RA8GA. The channel numbers for each port begin at a fixed offset regardless of the number of channels available on the connected device.

Digital I/O

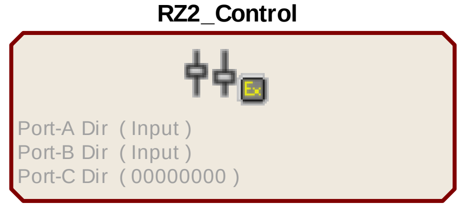
The digital I/O ports include 24 bits of programmable I/O. The digital I/O is divided into three ports (A, B, and C) as described in the chart below. All digital I/O lines are accessed via the 25-pin connector on the front of the RZ2 and ports A and C are available through BNC connectors on the front panel.



Digital I/O

See *RZ2 Technical Specifications*, page 1-10, for the DB25 pinout and BNC channel mapping.

See the Digital I/O Circuit Design section of the RPvdsEx Manual for more information on programming the digital I/O.

<table>
<tr><th>Digital I/O</th><th>Description</th><th>DB25</th><th>BNCs</th><th>Notes</th><th>Configuration</th></tr>
<tr><td>Port A</td><td>bits 0 - 7</td><td>Yes</td><td>Yes</td><td>byte addressable</td><td rowspan="3">RZ2_Control


Note: For more information on addressing and Digital I/O see the *RPvdsEx Manual*.</td></tr>
<tr><td>Port B</td><td>bits 0 - 7</td><td>Yes</td><td>No</td><td>byte addressable</td></tr>
<tr><td>Port C</td><td>bits 0 - 7</td><td>Yes</td><td>Yes</td><td>bit addressable</td></tr>
</table>

The data direction for the Digital I/O is configured using the RZ2_Control macro in RPvdsEx. Double-click the macro to access the settings on the Digital I/O tab. The RZ2_Control macro also offers a Direction Control Mode parameter that enables the macro inputs and allows the user to control data direction dynamically. For more information on using the RZ2_Control macro see the help provided in the macro's properties dialog box.

Technical Specifications

Specifications for the RZ2 Z-Series Base Station.

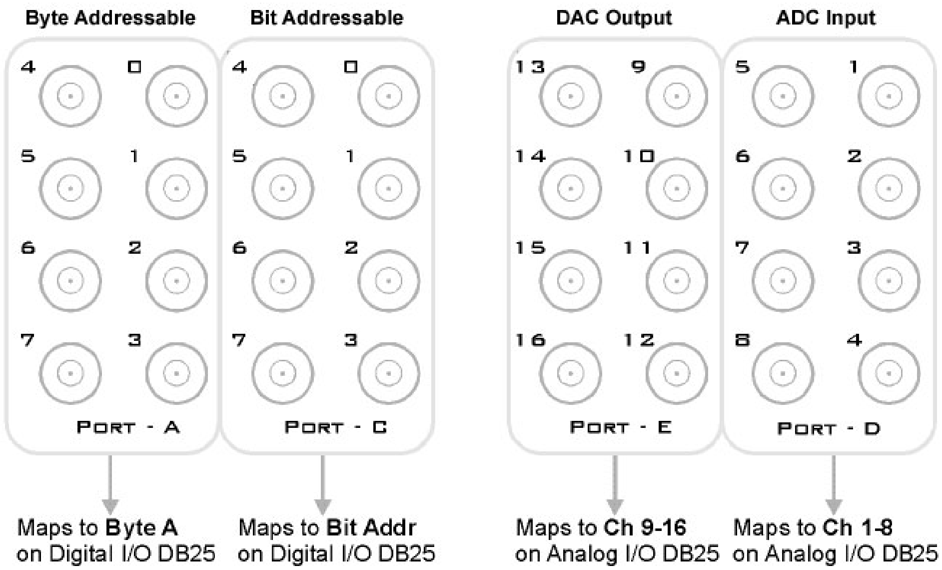
Note: Technical Specifications for amplifier A/D converters are found under the preamplifier's technical specifications.

DSP	400 MHz DSPs, 2.4 GFLOPS peak per DSP Two, Four, or Eight
Memory	64 MB SDRAM per DSP
D/A	8 channels, 16-bit PCM
Sample Rate	Up to 48828.125 Hz
Frequency Response	DC-Nyquist (~1/2 sample rate)
Voltage Out	+/- 10.0 Volts
S/N (typical)	82 dB (20 Hz - 20 kHz at 9.9 V)
A/D	8 channels, 16-bit PCM
Sample Rate	Up to 48828.125 Hz
Frequency Response	DC - 7.5 kHz (3 dB corner, 2nd order, 12 dB per octave)
Voltage In	+/- 10.0 Volts
S/N (typical)	82 dB (20 Hz - 20 kHz at 9.9 V)
Fiber Optic Ports	
Z-Series	One 256-channel input*
Legacy (Medusa)	Two 16-channel inputs
Digital I/O	24 bits programmable

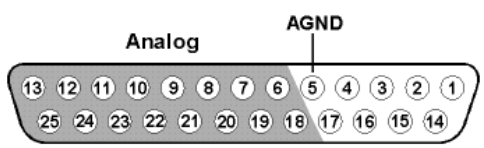
* The maximum sample rate is 48828.125 Hz when recording up to 128 channels or 24414.0625 Hz when recording 129 - 256 channels).

BNC Channel Mapping

Please note channel numbering begins at the top right block of BNCs for each port and is printed on the face of the device to minimize miswiring. The figure below represents the standard configuration and may vary depending on customer request.



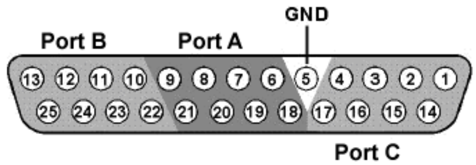
DB25 Analog I/O Pinout



Pin	Name	Description
1	NC	Not Used
2		
3		
4		
5	AGND	Analog Ground
6	A2	ADC Analog Input Channels (Port D)
7	A4	
8	A6	
9	A8	
10	A10	DAC Analog Output Channels (Port E)
11	A12	
12	A14	
13	A16	

Pin	Name	Description
14	NC	Not Used
15		
16		
17		
18	A1	ADC Analog Input Channels (Port D)
19	A3	
20	A5	
21	A7	
22	A9	DAC Analog Output Channels (Port E)
23	A11	
24	A13	
25	A15	

DB25 Digital I/O Pinout



Pin	Name	Description
1	C0	Port C Bit Addressable digital I/O Bits 0, 2, 4, and 6
2	C2	
3	C4	
4	C6	
5	GND	Digital I/O Ground
6	A1	Port A Word addressable digital I/O Bits 1, 3, 5, and 7
7	A3	
8	A5	
9	A7	
10	B1	Port B Word addressable digital I/O Bits 1, 3, 5, and 7
11	B3	
12	B5	
13	B7	

Pin	Name	Description
14	C1	Port C Bit Addressable digital I/O Bits 1, 3, 5, and 7
15	C3	
16	C5	
17	C7	
18	A0	Port A Word addressable digital I/O Bits 0, 2, 4, and 6
19	A2	
20	A4	
21	A6	
22	B0	Port B Word addressable digital I/O Bits 0, 2, 4, and 6
23	B2	
24	B4	
25	B6	

RZ5 BioAmp Processor

Overview

The RZ5 BioAmp Processor is available with either one or two 400 MHz Sharc digital signal processors networked on a multiprocessor architecture that features efficient onboard communication and memory access. The optimized multi-DSP architecture provides nearly five gigaflops of processing power, making the RZ5 a versatile solution for real-time processing and simultaneous acquisition.

The RZ5 acquires and processes up to 32 channels of neurophysiological signals in real-time. Data can be input from two Medusa preamplifiers at a sampling rate of ~25 kHz. The RZ5 also supports microstimulation applications. The RZ5 can be used with one of TDT's stimulus isolators (MS16 or MS4) and switching headstage (SH16) to comprise a complete microstimulation sytem. For more information see *MS4/MS16 Stimulus Isolator*, page 6-3.

The RZ5 also features eight channels of analog I/O, 24 bits of digital I/O and an oboard monitor speaker with volume control.

Power and Communication

The RZ5's Optibit optical interface ensures fast and reliable data transfer from the RZ5 to the PC and is integrated into the device. Connectors are provided on the back panel and are color coded for correct wiring. The RZ5’s power supply is also integrated into the device and is shipped from the factory configured for the desired voltage setting (110 V or 220V). If you need to change the voltage setting, please contact TDT support at 386.462.9622 or email support@tdt.com.

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Software control is implemented with circuit files developed using TDT's RP Visual Design Studio (RPvdsEx). Circuits are loaded to the processor through TDT run-time applications or custom applications. This manual includes device specific information needed during circuit design. For circuit design techniques and a complete reference of the RPvdsEx circuit components, see *MultiProcessor Circuit Design* and *Multi-Channel Circuit Design* in the *RPvdsEx Manual*.

RZ5 Architecture

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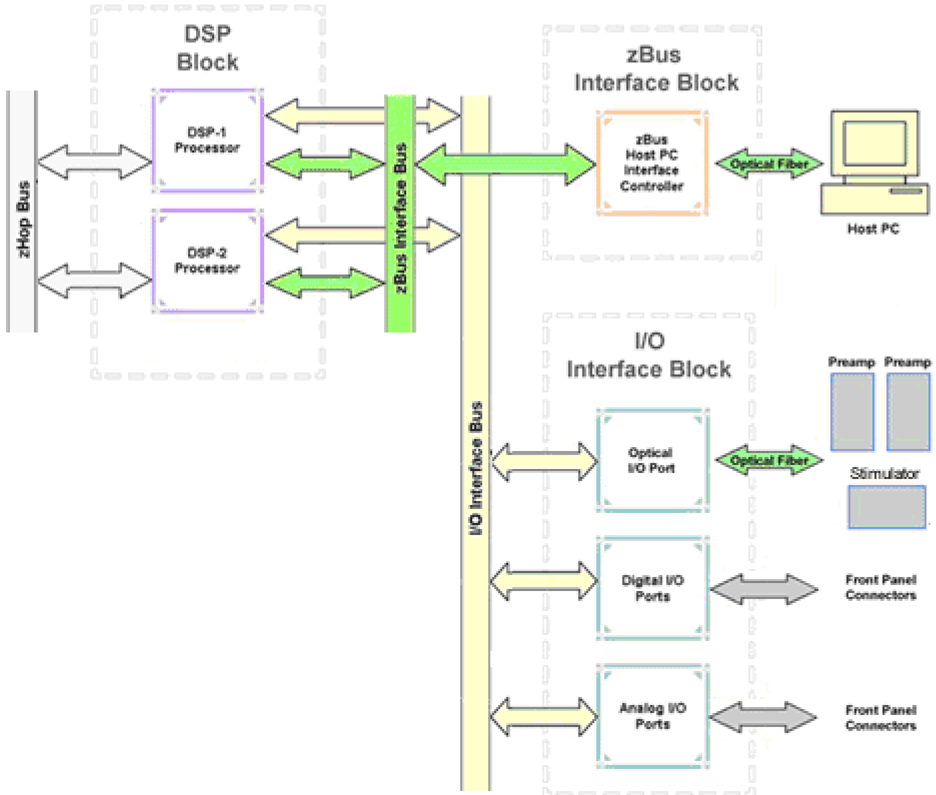


As shown in the diagram above, the RZ5 architecture consists of three functional blocks:

The DSPs Each DSP in the DSP Block is connected to 64 MB SDRAM and a local interface to the three data buses: two buses that connect each DSP to the other functional blocks and one that handles data transfer between the DSPs (as described further in Distributing Data Across DSPs below). This architecture facilitates fast DSP-to-off-chip data handling.

Because each DSP has its own associated memory, access is very fast and efficient. However, large and complex circuits should be designed to balance memory needs (such as data buffers and filter coefficients) across processors.

When designing circuits also note that the maximum number of components for each RZ5 DSP is 768.

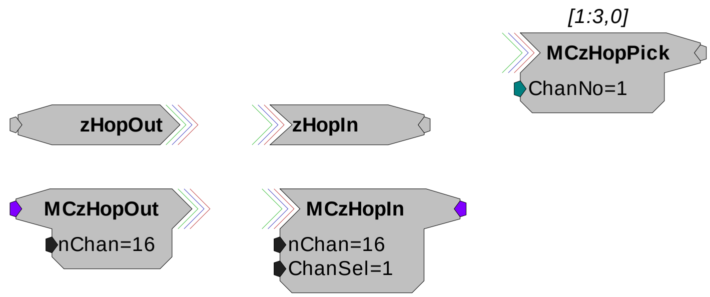
The zBus Interface The zBus Interface provides a connection to the PC. Data and host PC control commands are transferred to and from the DSP Block through the **zBus Interface Bus**, allowing for large high-speed data reads and writes without interfering with other system processing.

The I/O Interface The I/O Interface serves as a connection to outside signal sources or output devices. It is used to input data from the preamplifier inputs and digital and analog channels. The **I/O Interface Bus** provides a direct connection to each DSP.

Distributing Data Across DSPs

To reap the benefits of added power made possible by multi-DSP modules, processing tasks must be efficiently distributed across the available DSPs. That means transferring data across DSPs. The RZ5 architecture provides the zHop Bus for this type of data handling.

The zHop Bus The zHop Bus allows the transfer of single or multi-channel signals between each DSP in the RZ5.



In RPvdsEx data is transferred across the zHop Bus using paired zHop Components, including zHopIn, zHopOut, MCzHopIn, MCzHopOut, and MCzHopPick. Up to 126 pairs can be used in a single RPvdsEx circuit.

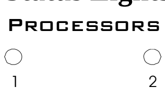
Bus Related Delays The zHop Bus introduces a single sample delay. However, this delay is taken care of for the user in OpenEx when Timing and Data Saving macros are used.

RZ5 Features

DSP Status Displays

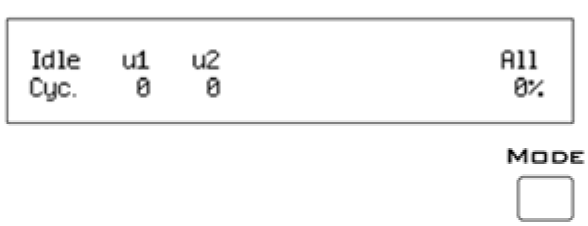
The RZ5 include status lights and a VFD (Vacuum Fluorescent Display) screen to report the status of the individual processors.

Status Lights



Two LEDs report the status of the multiprocessor's individual DSPs and will be lit solid green when the corresponding DSP is installed and running. The corresponding LED will be lit dim green if the cycle usage on a DSP is 0%. If the demands on a DSP exceed 99% of its capacity on any given cycle, the corresponding LED will flash red (~1 time per second).

Front Panel VFD Screen



The front panel VFD screen reports detailed information about the status of the system. The display includes two lines. The top line reports the system mode, Run!, Idle, or Reset, and displays heading labels for the second line. The second line reports the user’s choice of status indicators for each DSP followed by an aggregate value.

The user can cycle through the various status indicators using the Mode button to the bottom right of the display. Push and release the button to change the display or push and hold the button for one second then release to automatically cycle through each of the display options. The VFD screen may also report system status such as booting status (Reset).

Note: When burning new microcode or if the firmware on the RZ5 is blank, the VFD screen will report a cycle usage of 99% and the processor status lights will flash red.

Status Indicators

Cyc: cycle usage

Bus%: percentage of internal device's bus capacity used

I/O%: percentage of data transfer capacity used

Opt: Connection (sync) status of amplifiers A and B

Important Note!: The status lights flash when a DSP goes over the cycle usage limit, even if only for a particular cycle. This helps identify periodic overages caused by components in time slices.

Amplifier and Onboard Analog I/O

The RZ5 is equipped with both amplifier input and onboard analog I/O capabilities. The fiber optic ports allow a direct connection to Medusa Preamplifiers. Physiological signals are digitized on the preamplifier and transferred across noiseless fiber optics. The RZ5_AmpIn_MC and RZ5_AmpIn macros automatically apply the necessary scale factors and channel offsets for configuring the preamplifier fiber optic ports.

RZ5_AmpIn_MC

Amp -A-, 16 Chans (1-16)

RZ5_AmpIn

The following table provides a quick overview of the amplifier and analog I/O features and how they must be accessed during circuit design. When the RZ5_AmpIn_MC and RZ5_AmpIn macros are not used, reference the table and be sure to use the appropriate component, channel offset, scale factor and so forth. Also, see the *RPvdsEx Manual* for more information on circuit design.

Analog I/O	Description	Components	Channels	Notes
ADC Inputs	Analog Input	AdcIn	1 - 4	Accessed through ADC Input BNCs or Analog I/O labeled DB25
DAC Outputs	Analog Output	DacOut	9 - 12	Accessed through DAC Output BNCs or Analog I/O labeled DB25
Optical Amp-A	Medusa PreAmp Input	AdcIn	17 - 32	When the RZ5_AmpIn_MC or RZ5_AmpIn is NOT USED, apply a scale factor of .000833
Optical Amp-B	Medusa PreAmp Input	AdcIn	33 - 48	When the RZ5_AmpIn_MC or RZ5_AmpIn is NOT USED, apply a scale factor of .000833

Onboard Analog I/O

The RZ2 is equipped with four channels of 16-bit PCM D/A and four channels of 16-bit PCM A/D. All 8 channels can be accessed via front panel BNCs marked ADC and DAC or via a 25-pin analog I/O connector. See *RZ5 Technical Specifications*, page 1-10 for the DB25 pinout.

Fiber Optic Preamplifier Ports

The RZ5 acquires digitized signals from a Medusa preamplifier over a fiber optic cable. This provides loss-less signal acquisition between the amplifier(s) and the base station. Two fiber optic ports are provided to support simultaneous acquisition from up to two preamplifiers. Each port can input up to 16 channels at a maximum sampling rate of ~25 kHz.

The fiber optic ports can be used with any of the Medusa preamplifiers including the RA16PA, RA4PA, or RA8GA. The channel numbers for each port begin at a fixed offset regardless of the number of channels available on the connected device.

Channels are numbered as follows:

Amp-A	17 – 32
Amp-B	33 – 48

Note: When using the RZ5_AmpIn_MC and RZ5_AmpIn macros, the necessary scale factors and channel offsets for configuring the fiber optic ports are automatically applied.

Fiber Oversampling (acquisition only)

The fiber optic cable that carries the signals to the fiber optic input ports on the RZ5 has a transfer rate limitation of 6.25 Mbits/s. With 16 channels of data and 16 bits per sample, this limitation translates to a maximum sampling rate of ~25 kHz.

However, the need may arise to run a circuit at a higher sampling rate while still acquiring data via a fiber optic port. The two fiber optic ports on the RZ5 can oversample the digitized signals that have already been sampled up to 2X or ~50 kHz. This will allow the RZ5 to run a DSP chain at ~50 kHz and still sample data acquired through an optically connected preamplifier that digitized the incoming data stream at its maximum rate of ~25 kHz.

Oversampling is performed on the base station. The signals being acquired will still be sampled at ~25 kHz on the preamplifier. This means that, even with oversampling, signals acquired by an optically connected preamplifier are still governed by the bandwidth and frequency response of the preamplifier.

Fiber Optic Output (Stimulator) Port

The output port, labeled Stimulator, can be used to transfer microstimulation waveforms to the Stimulus Isolator and/or to control its digital output.

Important Note: This fiber optic port is disabled if the sampling rate of the system is set to a value greater than ~25 kHz.

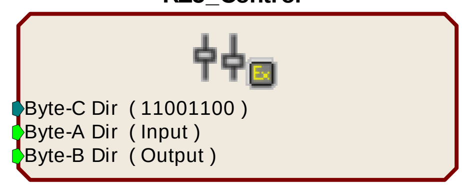
Monitor Speaker

The RZ5 is equipped with an onboard speaker. Maximum output is greater than 90 dB SPL at 10 cm. To use the speaker feed the desired signal to output channel 9 using a DacOut component. The speaker is provided primarily for audio monitoring of a single channel of electrophysiological potentials during recording.

Digital I/O

The digital I/O includes 24 bits of programmable I/O. The digital I/O is divided into three bytes (A, B, and C) as described in the chart below. All digital I/O lines are accessed via the 25-pin connector on the front of the RZ2 and bits 0 - 3 of byte C are available through BNC connectors on the front panel labeled Digital. See *RZ5 Technical Specifications*, page 1-10, for the DB25 pinout and BNC channel mapping.

See the Digital I/O Circuit Design section of the RPvdsEx Manual for more information on programming the digital I/O.

Digital I/O	Description	DB25	BNCs	Notes	Configuration
Byte A	bits 0 - 7	Yes	No	byte addressable	**RZ5_Control**  **Note:** For more information on addressing and Digital I/O see the *RPvdsEx Manual*.
Byte B	bits 0 - 7	Yes	No	byte addressable	
Byte C	bits 0 – 7	Yes	Yes*	bit addressable	
***Note:** Byte C Bits 0 - 3 are available via front panel BNCs					

The data direction for the digital I/O is configured using the RZ5_Control macro in RPvdsEx. Double-click the macro to access the settings on the Digital I/O tab. The RZ5_Control macro also offers a Direction Control Mode parameter that enables the macro inputs and allows the user to control data direction dynamically. For more information on using the RZ5_Control macro see the help provided in the macro's properties dialog box.

Note: By default, all digital I/O are configured as inputs.

LED Indicators

The RZ5 contains 16 LED indicators for the analog and digital I/O. These indicators are located directly below the VFD and DSP status LEDs and display information relative to the various analog and digital I/O contained on the RZ5. The following tables illustrate the possible display options and their associated descriptions.

Digital I/O - Byte C

8-bit, bit addressable byte C LED indicators are located to the bottom left of the RZ5 front panel.

Light Pattern	Description
Dim Green	Bit is configured for output and is currently a logical low (0)
Solid Green	Bit is configured for output and is currently a logical high (1)
Dim Red	Bit is configured for input and is currently a logical low (0)
Solid Red	Bit is configured for input and is currently a logical high (1)

Analog I/O - ADC Inputs and DAC Outputs

ADC and DAC LED indicators are labeled and located to the right of the byte C LED indicators.

Light Pattern	Description
Off	Analog I/O channel signal voltage is less than +/-100 mV
Dim Green	Analog I/O channel signal voltage is less than +/-5 V
Solid Green	Analog I/O channel signal voltage is between +/-5 V to +/-9 V
Solid Red	Analog I/O channel clip warning (voltage greater than +/-9 V)

Technical Specifications

Specifications for the RZ5 BioAmp Processor.

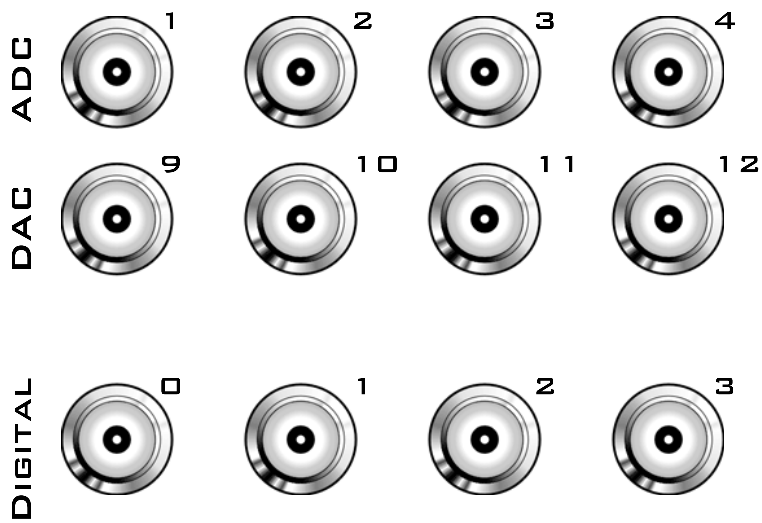
Note: Technical Specifications for amplifier A/D converters are found under the preamplifier's technical specifications.

DSP	400 MHz DSPs, 2.4 GFLOPS peak per DSP One or Two
Memory	64 MB SDRAM per DSP
D/A	4 channels, 16-bit PCM
Sample Rate	Up to 48828.125 Hz*
Frequency Response	DC-Nyquist (~1/2 sample rate)
Voltage Out	+/- 10.0 Volts
S/N (typical)	82 dB (20 Hz - 20 kHz at 9.9 V)
A/D	4 channels, 16-bit PCM
Sample Rate	Up to 48828.125 Hz *
Frequency Response	DC - 7.5 kHz (3 dB corner, 2nd order, 12 dB per octave)
Voltage In	+/- 10.0 Volts
S/N (typical)	82 dB (20 Hz - 20 kHz at 9.9 V)
Fiber Optic Ports	
Stimulator (MS16)	One output for MS16 Stimulus Isolator*
Preamplifier (Medusa)	Two 16-channel inputs
Digital I/O	24 bits programmable

* **Note:** When used with the Stimulus Isolator, the sampling rate is limited to 24.414 kHz.

BNC Channel Mapping

Please note channel numbering begins at the top left block of BNCs for both analog and digital I/O and is printed on the face of the device to minimize miswiring.

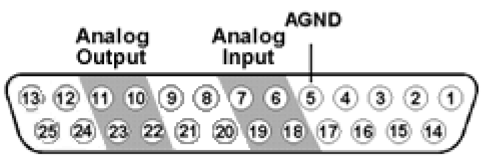


Maps to **Ch 1-4** on Analog I/O DB25

Maps to **Ch 9-12** on Analog I/O DB25

Maps to **Port C Bits 0-3** on Analog I/O DB25

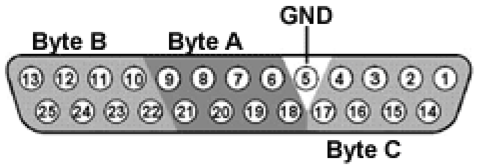
DB25 Analog I/O Pinout



Pin	Name	Description
1	NA	Not Used
2		
3		
4		
5	AGND	Analog Ground
6	A2	ADC Analog Input Channels (ADC Inputs)
7	A4	
8	NA	Not Used
9		
10	A10	DAC Analog Output Channels (DAC Outputs)
11	A12	
12	NA	Not Used
13		

Pin	Name	Description
14	NA	Not Used
15		
16		
17		
18	A1	ADC Analog Input Channels (ADC Inputs)
19	A3	
20	NA	Not Used
21		
22	A9	DAC Analog Output Channels (DAC Outputs)
23	A11	
24	NA	Not Used
25		

DB25 Digital I/O Pinout



Pin	Name	Description
1	C0	Byte C Bit Addressable digital I/O Bits 0, 2, 4, and 6
2	C2	
3	C4	
4	C6	
5	GND	Digital I/O Ground
6	A1	Byte A Word addressable digital I/O Bits 1, 3, 5, and 7
7	A3	
8	A5	
9	A7	
10	B1	Byte B Word addressable digital I/O Bits 1, 3, 5, and 7
11	B3	
12	B5	
13	B7	

Pin	Name	Description
14	C1	Byte C Bit Addressable digital I/O Bits 1, 3, 5, and 7
15	C3	
16	C5	
17	C7	
18	A0	Byte A Word addressable digital I/O Bits 0, 2, 4, and 6
19	A2	
20	A4	
21	A6	
22	B0	Byte B Word addressable digital I/O Bits 0, 2, 4, and 6
23	B2	
24	B4	
25	B6	

Part 2 RX Processors

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RX5 Pentusa Base Station

Overview

The RX5 Pentusa is a powerful multiple DSP device well suited for processing high channel count neurophysiology data in real-time. A streamlined hardware interface provides connections to up to 64 channels for neurophysiological data acquisition.

The RX5 is equipped with either two or five 100 MHz, 1600 MFLOPS Sharc DSPs and serves as a base station for up to four Medusa preamplifiers to form a powerful multi-channel amplifier system. The multiprocessor architecture provides simultaneous ~25 kHz sampling on every channel, 16-bit precision, fiber optic isolation, and the power of user-programmable real-time DSPs.

The RX5 also features front panel status indicators, 40 bits of configurable digital I/O, and four D/A converters for versatile experiment control and stimulus generation.

Power and Communication

The RX5 mounts in a System 3 zBus Powered Device Chassis (ZB1PS) and communicates with the PC using the Gigabit (PI5/FI5) or Optibit (PO5/FO5) PC interfaces. The ZB1PS is UL compliant, see the *ZB1PS Operations Manual* for power and safety information.

Software Control

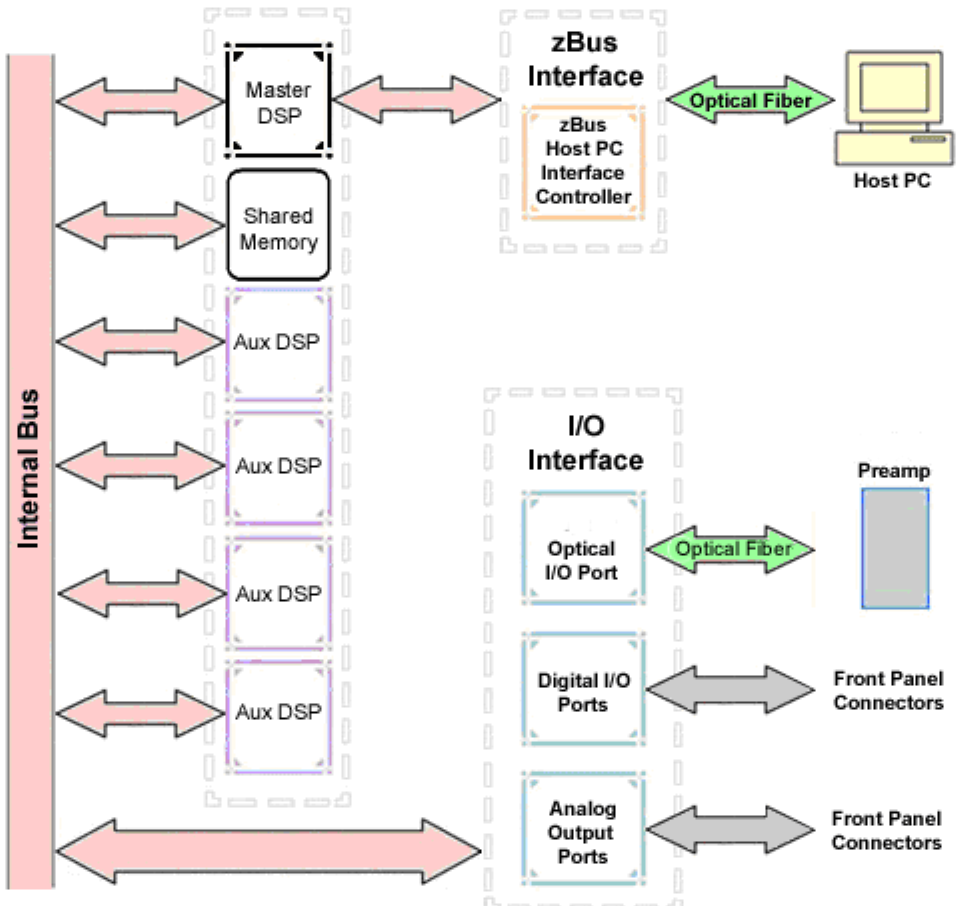
Software control is implemented with circuit files developed using TDT's RP Visual Design Studio (RPvdsEx). Circuits are loaded to the processor through TDT run-time applications or custom applications. This manual includes device specific information needed during circuit design. For circuit design techniques and a complete reference of the RPvdsEx circuit components, see the *RPvdsEx Manual*.

RX Architecture

Each RX multiprocessor device is equipped with either two or five digital signal processors (DSPs). The multi-DSP architecture allows processing tasks to be distributed across multiple processors and enables data to be transferred to the PC quickly and efficiently. The DSPs include one master and one or four auxiliary DSP(s). 128 MB SDRAM of system memory is shared by all DSPs. When designing circuits the maximum number of components for each RX DSP is 256.

Each DSP communicates with an internal bus to send and receive information from the I/O controller and the shared memory. The master DSP supervises overall system boot up and operation. The master DSP also acts as the main data interface between the zBus (host PC) and the multi-DSP environment.

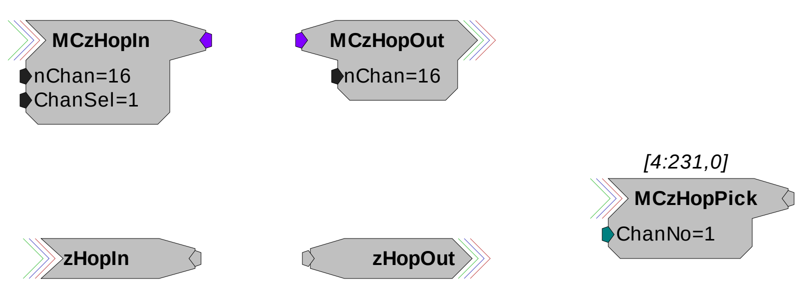
Because the zBus communicates only with the master processor, these devices operate most efficiently when the circuit related processing tasks assigned to the master DSP are minimized, allowing more processor power (cycles) for communication and overhead tasks.



The RX5 contains two DB25 connectors for interfacing with 40 bits of digital I/O and 4 channels of analog output. A BNC connector is provided for access to the first analog output channel. Four fiber optic Medusa preamp ports enable connections for up to 64 channels of analog input.

Distributing Data Across DSPs

In RPvdsEx data can be transferred between each of the auxiliary DSPs as well as the master DSP using zHop components.



Components such as MCzHopIn and MCzHopOut can be used for multi-channel signals while components such as zHopIn, zHopOut, and MCzHopPick are used with single-channel signals. Up to 126 pairs can be used in a single RPvdsEx circuit.

Bus Related Delays The zHop Bus introduces a single sample delay. However, this delay is taken care of for the user in OpenEx when Timing and Data Saving macros are used.

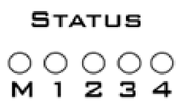
See *MultiProcessor Circuit Design* in the *RPvdsEx Manual* for these and other multiprocessor circuit design techniques.

RX5 Features

DSP Status Displays

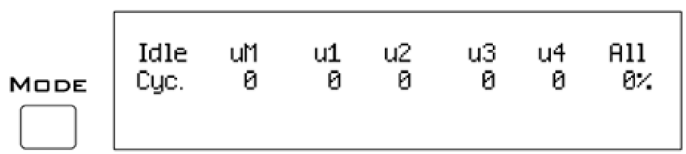
All high performance RX multiprocessors include status lights and a VFD (Vacuum Fluorescent Display) screen to report the status of the individual processors.

Status Lights



Up to five LEDs report the status of the multiprocessor's individual DSPs. When the device is turned on, they will glow steadily. If the demands on a DSP exceed 99% of its capacity on any given cycle, the corresponding LED will flash rapidly (~3 times per second).

Front Panel VFD Screen



The front panel VFD screen reports detailed information about the status of the system. The display includes two lines. The top line reports the system mode, Run! or Idle, and displays heading labels for the second line. The second line reports the user’s choice of status indicators for each DSP followed by an aggregate value.

The user can cycle through the various status indicators using the Mode button to the left of the display. Push and release the button to change the display or push and hold the button for one second then release to automatically cycle through each of the display options. The VFD screen may also report system status such as booting status (Booting DSP) or alert the user when the device's microcode needs to be reprogrammed (Firmware Blank).

Status Indicators

Cyc: cycle usage

Ovr: processor cycle overages

Bus%: percentage of internal device's bus capacity used

I/O%: percentage of data transfer capacity used

Important Note!: The status lights will flash (~3 times a second) to alert the user when a device goes over the cycle usage limit, even if only for a particular cycle. This helps to identify periodic overages caused by components in time slices.

Fiber Optic Ports

The RX5 base station acquires digitized signals from a Medusa preamplifier over a fiber optic cable. This provides loss-less signal acquisition between the amplifier and the base station. Two or four fiber optic ports are provided to support simultaneous acquisition from up to four preamplifiers. Each port can input up to 16 channels at a maximum sampling rate of ~25 kHz. The first two ports provide oversampling. See *Fiber Oversampling*, below for more information.

The fiber optic ports can be used with any of the Medusa preamplifiers including the RA16PA, RA4PA, or RA8GA. The channel numbers for each port begin at a fixed offset regardless of the number of channels available on the connected device.

Channels are numbered as follows:

Amp-A	1 – 16
Amp-B	17 – 32
Amp-C	33 – 48
Amp-D	49 - 64

Fiber Oversampling

The fiber optic cable that carries the signals to the fiber optic input ports has a transfer rate limitation of 6.25 Mbits/s. With 16 channels of data and 16 bits per sample, this limitation translates to a maximum sample rate of 24.414 kHz.

However, the need may arise to run a circuit at a higher sample rate while still acquiring data via a fiber optic port. The first two fiber optic ports can oversample the digitized signals that have already been sampled up to 4X or ~100 kHz. This will allow an RX5 to run a DSP chain at ~50 kHz or ~100 kHz, and still sample data acquired through an optically connected preamplifier that digitized the incoming data stream at a maximum rate of ~25 kHz.

Oversampling is performed on the base station. The signals being acquired will still be sampled at ~25 kHz on the preamplifier. This means that, even with oversampling, signals acquired by an optically connected preamplifier are still governed by the bandwidth and frequency response of the preamplifier.

When acquiring up to 16 channels of data on the first fiber optic input port of the RX5, the signals will be oversampled 4X to 100 kHz. If data is being acquired only on the first two fiber optic ports, the signals will be oversampled 2X to ~50 kHz.

Amp Status and Clip Warning Lights

Amp lights are located to the right of each fiber optic port. These lights are used to indicate the power status or provide a clip warning for the connected amplifiers.

When an amplifier is not connected the Amp light will flash in a slow steady pattern. The light is lit when the amplifier is connected and begins to flash quickly when the voltage on the battery for the corresponding amplifier is low. When any channel on the connected amplifier produces a voltage approaching the maximum input of the amplifier, the corresponding light will flash rapidly to warn that clipping may occur if the signal exceeds the maximum input voltage. See the corresponding preamplifier section for more information on input range and clip warnings.

Important Note!: The Li-ion batteries voltage decreases rapidly once the battery low light is on. Data acquisition will suffer if the battery is not charged soon after the light goes on.

Amplifier Status Patterns

Light Pattern	Amplifier Status
Solid	Connected
Very slow flash (~1 every two seconds)	Not connected
Slow flash (~1 per second)	Connected and charging
Rapid flash	Battery low
Very rapid flash	Clip Warning

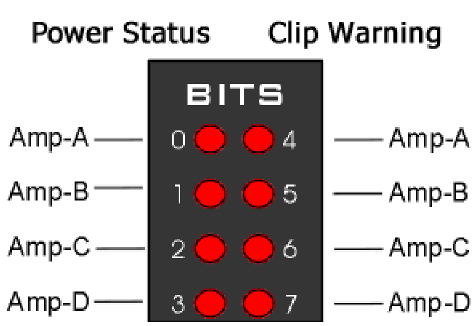
Note: If the amplifier appears to be connected and the amplifier status light is flashing slowly, check to ensure that the device is connected properly.

Bits Lights

The RX5's eight Bits lights are user configurable. By default the Bits lights indicate the logic level (light when high) for the eight bit-addressable digital I/O lines. The Bits lights can also be configured to provide information about amplifier status or act as logic level lights for any of the other four bytes of digital I/O.

Using the Bits Lights to Display Amplifier Status

Note: Because clip warning and amplifier status are always displayed using the Amp lights (located directly to the right of each fiber optic port), **TDT recommends using the Bits lights for other applications**. See *Amp Status and Clip Warning Lights* for more information.



When the Bits lights are configured to display the amplifier status, the left column of lights indicates the power status and the right column indicates a clip warning for the corresponding amplifier.

The table on page 2-7 shows the light pattern and corresponding amplifier status for the power status lights (0 - 3). Clip lights flash very rapidly when any channel on the connected amplifier produces a voltage approaching the maximum input of the amplifier.

Analog Output

The RX5 is equipped with four channels of 16-bit PCM D/A. The sampling rate is user selectable up to a maximum of ~100 kHz. The D/A is DC coupled and has a built-in upsampler for improved audio playback. The upsampler is controlled through one of the RX5's programmable bits and can be turned off to allow the D/A to drive external devices such as a stimulator. Channel one analog output can be accessed via a front Panel BNC (DAC-1). All four analog channels can be accessed via the DB25 Multi I/O connector (pins 14 – 17).

Digital I/O

The RX5 processor has 40 digital I/O lines. Eight bits are bit-addressable. The remaining 32 bits are four word-addressable bytes. Digital I/O lines are accessed via the two 25-pin connectors on the front of the RX5. *See the Digital I/O Circuit Design section of the RPvdsEx Manual* for more information on programming the digital I/O.

CAUTION!: The first eight bits of bit-addressable digital I/O on RX devices are unbuffered. When used as inputs, overvoltages on these lines can cause severe damage to the system. TDT recommends when sending digital signals into the device, (1) never send a signal with amplitude greater than seven volts into any digital input and (2) always use the byte-addressable digital I/O lines.

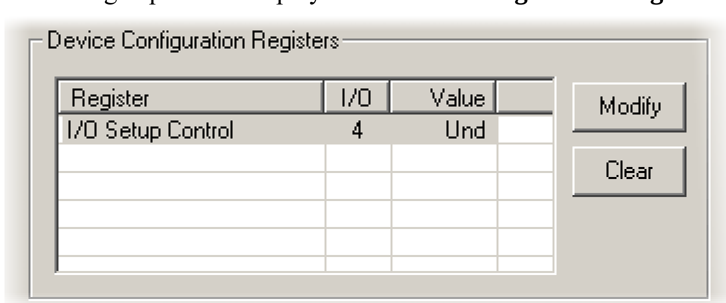
Configuring the Programmable I/O Lines

Each of the eight bit-addressable lines can be independently configured as inputs or outputs. The digital I/O lines can be configured as inputs or outputs in groups of eight bits – that is as byte A, byte B, byte C, and byte D. Note, however, that the bytes must be addressed as if part of a word, not as individual bytes. See *Addressing Digital Bits In A Word* in the *RPvdsEx Manual* for more information.

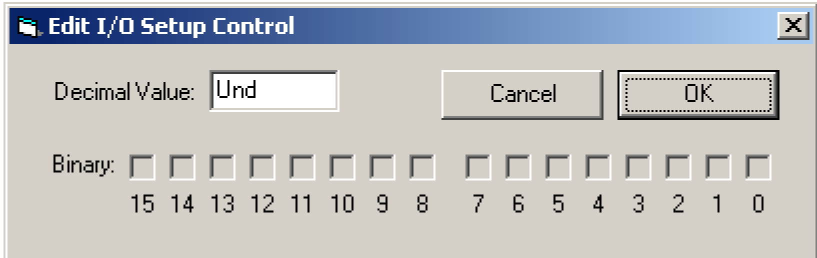
By default, all bits are configured as inputs. This default setting is intended to prevent damage to equipment that might be connected to the digital I/O lines. The user can configure the bits in the RPvdsEx configuration register. The configuration register is also used to determine what the eight front panel Bits lights represent.

To access the bit configuration register in RPvdsEx:

1. Click the **Device Setup** command on the **Implement** menu.
2. In the **Set Hardware Parameters** dialog box, click the **Device Type** box and select the **RX5 Pentusa** from the list.
3. The dialog expands to display the **Device Configuration Register**.



4. Click **Modify** to display the **Edit I/O Setup Control** dialog box.

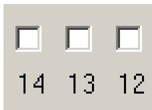


5. In this dialog box, a series of check boxes are used to create a bitmask that is used to program all bits.
6. To enable the check boxes, delete **Und** from the **Decimal Value** box.
7. To determine the desired value, select or clear the check boxes according to the table below. By default, all check boxes are cleared (value = 0). Selecting a check box sets the corresponding bit in the bitmask to one.
8. When the configuration is complete, click **OK** to return to the **Set Hardware Parameters** dialog box.

Bit #	**Description**
0-7	Each of these bits controls the configuration of one of the eight addressable bits as inputs or outputs. Setting the bit to one will configure that bit as an output.
8-11	Each of these bits controls the configuration of one of the four addressable bytes as inputs or outputs. Setting the bit to one will configure that byte as an output. bit 8 - byte A, bit 9 - byte B, bit 10 - byte C, and bit 11 - byte D
12-14	Create a bit code that determines how the front panel Bits lights are used, see table below.
15	Setting the bit to one will disable the D/A upsampler.

Bit Codes for Controlling the Bit Lights (Boxes 12-14)

By default, check boxes 12 –14 in the **Edit I/O Setup Control** dialog box (previous diagram) are cleared to create the bit code 000. This configures the eight front panel Bits lights to act as activity lights (lit when high) for the eight bit addressable digital I/O lines. The Bits lights can also be configured to provide information about amplifier status or act as activity lights for any of the other four bytes of digital I/O.

Bit Flags	Bits set to 1	Bit Lights Used For …
000 	None	Logical level lights for bit-addressable I/O lines
010 	13	Amplifier Clip Warning/Power Status display
100 	14	Enable logical level lights for byte A
101 	12, 14	Enable logical level lights for byte B
110 	13, 14	Enable logical level lights for byte C
111 	12, 13, 14	Enable logical level lights for byte D

XLink

The XLink is not supported at this time.

Pentusa Base Station Technical Specifications

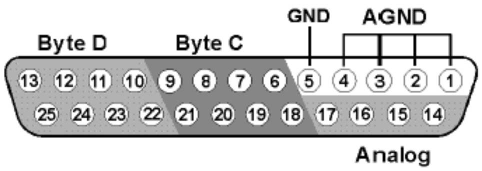
The RX5 has no onboard A/D converters. Technical Specifications for the A/D converters are found under the preamplifier's technical specifications.

DSP	100 MHz Sharc ADSP 21161, 600 MFLOPS Peak Two or Five
Memory	128 MB SDRAM (Shared)
D/A	4 channels, 16-bit PCM
Sample Rate	Up to 97.65625 kHz (8X upsampled to 200 kHz default operation)
Frequency Response	DC-Nyquist(~1/2 sample rate)
Voltage Out	+/- 10.0 Volts
Voltage Out Accuracy	+/- 10%
S/N (typical)	84 dB (20 Hz to 25 KHz) 82 dB with upsampling disabled
THD (typical)	-77 dB for 1 kHz output at 5 Vrms -74 dB with upsampling disabled
Output Impedance	10 Ohm
Fiber Optic Ports	Two or Four Inputs (Medusa)
Digital I/O	40 bits programmable (8 bits bit-addressable and a 32 bit word, addressable as 4 bytes)

DB25 Connector Pinouts

TDT recommends the PP24 patch panel for accessing the RX5 I/O.

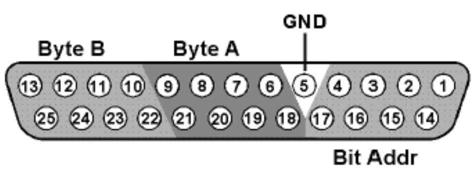
Multi I/O



Pin	Name	Description
1	AGND	Analog Ground
2		
3		
4		
5	GND	Digital I/O Ground
6	C1	Byte C Word addressable digital I/O Bits 1, 3, 5, and 7
7	C3	
8	C5	
9	C7	
10	D1	Byte D Word addressable digital I/O Bits 1, 3, 5, and 7
11	D3	
12	D5	
13	D7	

Pin	Name	Description
14	A1	Analog Output Channels
15	A2	
16	A3	
17	A4	
18	C0	Byte C Word addressable digital I/O Bits 0, 2, 4, and 6
19	C2	
20	C4	
21	C6	
22	D0	Byte C Word addressable digital I/O Bits 0, 2, 4, and 6
23	D2	
24	D4	
25	D6	

Digital I/O



Pin	Name	Description
1	BA0	Bit Addressable digital I/O Bits 0, 2, 4, and 6
2	BA2	
3	BA4	
4	BA6	
5	GND	Digital I/O Ground
6	A1	Byte A Word addressable digital I/O Bits 1, 3, 5, and 7
7	A3	
8	A5	
9	A7	
10	B1	Byte B Word addressable digital I/O Bits 1, 3, 5, and 7
11	B3	
12	B5	
13	B7	

Pin	Name	Description
14	BA1	Bit Addressable digital I/O Bits 1, 3, 5, and 7
15	BA2	
16	BA3	
17	BA4	
18	A0	Byte A Word addressable digital I/O Bits 0, 2, 4, and 6
19	A2	
20	A4	
21	A6	
22	B0	Byte B Word addressable digital I/O Bits 0, 2, 4, and 6
23	B2	
24	B4	
25	B6	

RX6 Piranha Multifunction Processor

Overview

The RX6 Piranha Multifunction Processor is a high performance multiple DSP device for researchers who need to acquire or produce high sample rate signals. The RX6 supports complex research, multimodal, and experimental paradigms on a single high-bandwidth device.

The RX6 equipped with either two or five 100 MHz, 1600 MFLOPS Sharc DSPs, combines a powerful multiprocessor architecture and high-speed data transfer with two channels of 24-bit sigma-delta D/A converters and two channels of 24-bit sigma-delta A/D converters to provide superior high frequency signal generation and acquisition. Optionally, the RX6 can be equipped with a fiber optic input, allowing it to serve as a base station for a Medusa preamplifier.

Power and Communication

The RX6 mounts in a System 3 zBus Powered Device Chassis (ZB1PS) and communicates with the PC using the Gigabit (PI5/FI5) or Optibit (PO5/FO5) PC interfaces. The ZB1PS is UL compliant, see the *ZB1PS Operations Manual* for power and safety information.

Software Control

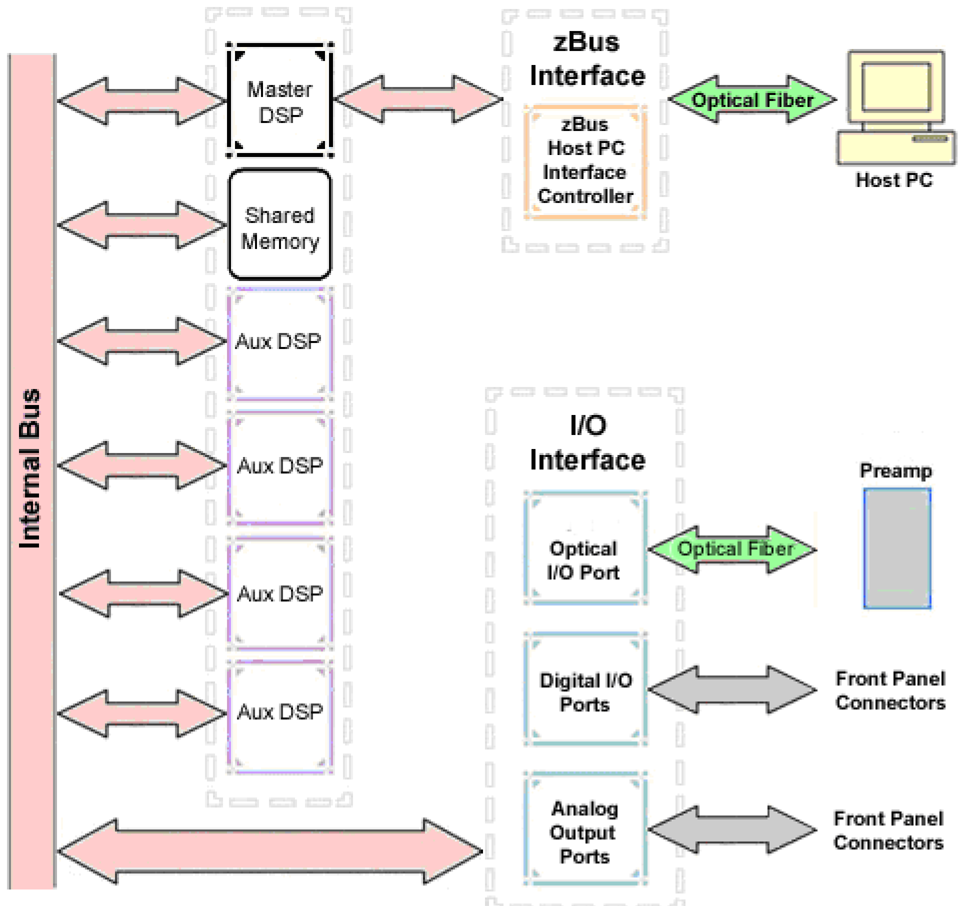
Software control is implemented with circuit files developed using TDT's RP Visual Design Studio (RPvdsEx). Circuits are loaded to the processor through TDT run-time applications or custom applications. This manual includes device specific information needed during circuit design. For circuit design techniques and a complete reference of the RPvdsEx circuit components, see the *RPvdsEx Manual*.

RX Architecture

Each RX multiprocessor device is equipped with either two or five digital signal processors (DSPs). The multi-DSP architecture allows processing tasks to be distributed across multiple processors and enables data to be transferred to the PC quickly and efficiently. The DSPs include one master and one or four auxiliary DSP(s). 128 MB SDRAM of system memory is shared by all DSPs. When designing circuits the maximum number of components for each RX DSP is 256.

Each DSP communicates with an internal bus to send and receive information from the I/O controller and the shared memory. The master DSP supervises overall system boot up and operation. The master DSP also acts as the main data interface between the zBus (host PC) and the multi-DSP environment.

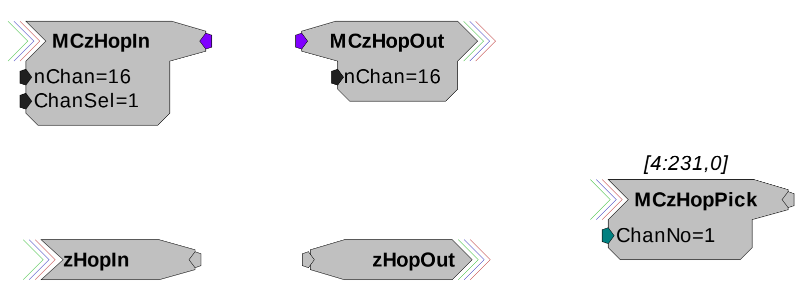
Because the zBus communicates only with the master processor, these devices operate most efficiently when the circuit related processing tasks assigned to the master DSP are minimized, allowing more processor power (cycles) for communication and overhead tasks.



The RX6 contains a DB25 connector for interfacing with 24 bits of digital I/O and four BNC connectors for interfacing with four channels of analog I/O. An optional fiber optic Medusa preamp port enables connections for up to 16 channels of analog input.

Distributing Data Across DSPs

In RPvdsEx data can be transferred between each of the auxiliary DSPs as well as the master DSP using zHop components.



Components such as MCzHopIn and MCzHopOut can be used for multi-channel signals while components such as zHopIn, zHopOut, and MCzHopPick are used with single-channel signals. Up to 126 pairs can be used in a single RPvdsEx circuit.

Bus Related Delays The zHop Bus introduces a single sample delay. However, this delay is taken care of for the user in OpenEx when Timing and Data Saving macros are used.

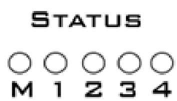
See *MultiProcessor Circuit Design* in the *RPvdsEx Manual* for these and other multiprocessor circuit design techniques.

RX6 Features

DSP Status Displays

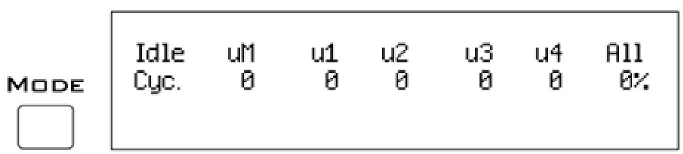
All high performance RX multiprocessors include status lights and a VFD (Vacuum Fluorescent Display) screen to report the status of the individual processors.

Status Lights



Up to five LEDs report the status of the multiprocessor's individual DSPs. When the device is turned on, they will glow steadily. If the demands on a DSP exceed 99% of its capacity on any given cycle, the corresponding LED will very flash rapidly (~3 times per second).

Front Panel VFD Screen



The front panel VFD screen reports detailed information about the status of the system. The display includes two lines. The top line reports the system mode, Run! or Idle, and displays heading labels for the second line. The second line reports the user’s choice of status indicators for each DSP followed by an aggregate value.

The user can cycle through the various status indicators using the Mode button to the left of the display. Push and release the button to change the display or push and hold the button for one second then release to automatically cycle through each of the display options. The VFD screen may also report system status such as booting status (Booting DSP) or alert the user when the device's microcode needs to be reprogrammed (Firmware Blank).

Status Indicators

Cyc: cycle usage

Ovr: processor cycle overages

Bus%: percentage of internal device's bus capacity used

I/O%: percentage of data transfer capacity used

Important Note!: The status lights will flash (~3 times a second) to alert the user when a device goes over the cycle usage limit, even if only for a particular cycle. This helps to identify periodic overages caused by components in time slices.

Fiber Optic Port - Optional

The RX6 can include a single fiber optic port most often used with the HTI3, but may also be used to acquire digitized signals from a Medusa preamplifier over a fiber optic cable. This provides loss-less signal acquisition between the amplifier and the base station. The port can input up to 16 channels at a maximum sampling rate of ~25 kHz. See *Fiber Oversampling*, below for more information. The fiber optic port can be used with any of the Medusa preamplifiers including the RA16PA, RA4PA, or RA8GA. The channel numbers for each port begin at a fixed offset regardless of the number of channels available on the connected device.

Channels are numbered as follows:

Amp-A 1 – 16

Fiber Oversampling

The fiber optic cable that carries the signals to the fiber optic input ports has a transfer rate limitation of 6.25 Mbits/s. With 16 channels of data and 16 bits per sample, this limitation translates to a maximum sample rate of 24.414 kHz.

However, the need may arise to run a circuit at a higher sample rate while still acquiring data via a fiber optic port. The fiber optic port on the RX6 can oversample the digitized signals that have already been sampled up to 4X or ~100 kHz. This will allow an RX6 to run a DSP chain at ~50 kHz or ~100 kHz, and still sample data acquired through an optically connected preamplifier that digitized the incoming data stream at a maximum rate of ~25 kHz.

Oversampling is performed on the base station. The signals being acquired will still be sampled at ~25 kHz on the preamplifier. This means that, even with oversampling, signals acquired by an optically connected preamplifier are still governed by the bandwidth and frequency response of the preamplifier.

Amp Status and Clip Warning Lights

If the RX6 includes a fiber optic port for a Medusa Preamplifier, an Amp light is located to the right of the fiber optic port. This light is used to indicate the power status or provide a clip warning for the connected amplifier.

When an amplifier is not connected the Amp light will flash in a slow steady pattern. The light is lit when the amplifier is connected and begins to flash quickly when the voltage on the battery for the corresponding amplifier is low. When any channel on the connected amplifier produces a voltage approaching the maximum input of the amplifier, the corresponding light will flash rapidly to warn that clipping may occur if the signal exceeds the maximum input voltage. See the preamplifier user guide for more information on input range and clip warnings.

Important Note!: The Li-ion batteries voltage decreases rapidly once the battery low light is on. Data acquisition will suffer if the battery is not charged soon after the light goes on.

Amplifier Status Patterns

Light Pattern	Amplifier Status
Solid	Connected
Very slow flash (~1 every two seconds)	Not connected
Slow flash (~1 per second)	Connected and charging
Rapid flash	Battery low
Very rapid flash	Clip Warning

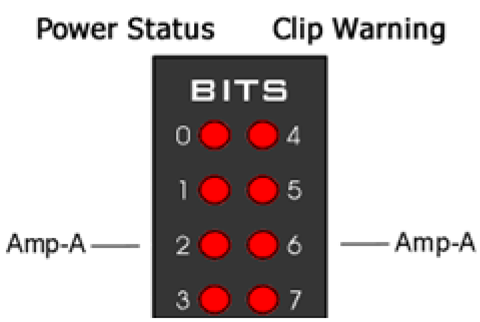
Note: If the amplifier appears to be connected and the amplifier status light is flashing slowly, check to ensure that the device is connected properly.

Bits Lights

The RX6's eight Bits lights are user configurable. By default the Bits lights indicate the logic level (light when high) for the eight bit-addressable digital I/O lines. The Bits lights can also be configured to provide information about amplifier status or act as logic level lights for any of the other two bytes of digital I/O.

Using the Bits Lights to Display Amplifier Status

Note: Because clip warning and amplifier status are always displayed using the Amp lights (located directly to the right of each fiber optic port), **TDT recommends using the Bits lights for other applications**. See *Amp Status and Clip Warning Lights* for more information.



When the Bits lights are configured to display the amplifier status, the left column of lights indicates the power status and the right column indicates a clip warning for the amplifier.

The table above shows the light pattern and corresponding amplifier status for the power status lights (0-3). Clip lights flash very rapidly when any channel on the connected amplifier produces a voltage approaching the maximum input of the amplifier.

Analog Input/Output

The RX6 has two channels of 24-bit, sigma-delta D/A and two channels of 24-bit, sigma-delta A/D, each accessible through BNC connectors. Sigma-delta converters provide superior conversion quality and extended useful bandwidths, at the cost of an inherent fixed group delay. **The RX6 DAC Delay is 43 samples and the RX6 ADC Delay is 70 samples.**

This device can sample at rates up to ~260 kHz for a realizable bandwidth of ~120 kHz. For specific information on the actual sampling rates see *Realizable Sampling Rates for the RX6*, page 2-21.

Important Note!: Because some RX6 models can acquire analog signals using a Medusa preamplifier via an optional fiber optic port, **the sigma-delta A/D inputs on all RX6 models are offset and accessed as ADC channels 128 and 129**.

Digital I/O

The RX6 processor includes 24 bits of programmable I/O in two eight bit word-addressable bytes and eight bits of bit-addressable I/O. Digital I/O lines are accessed via the 25-pin connector on the front panel and can be configured as inputs or outputs.

The first four bits of digital I/O (bits 0-3) can also be used for submicrosecond event timing. See *Nanosecond Event Timing*, page 3-12, for more information.

See the Digital I/O Circuit Design section of the RPvdsEx Manual for more information on programming the digital I/O.

CAUTION!: The first eight bits of bit-addressable digital I/O on RX devices are unbuffered. When used as inputs, overvoltages on these lines can cause severe damage to the system. TDT recommends when sending digital signals into the device, (1) never send a signal with amplitude greater than seven volts into any digital input and (2) always use the byte-addressable digital I/O lines.

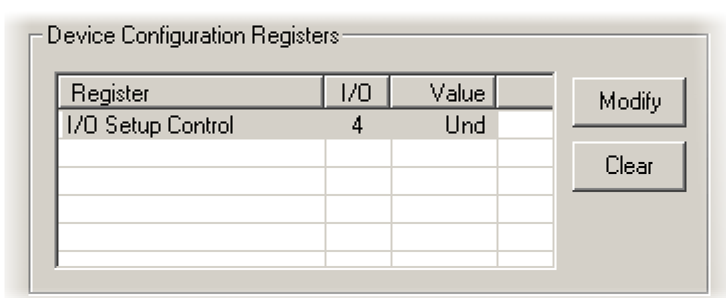
Configuring the Programmable I/O Lines

Each of the eight bit-addressable bits can be independently configured as inputs or outputs. The digital I/O lines can be configured as inputs or outputs in groups of eight bits – that is as byte A and byte B. Note, however, that the bytes must be addressed as if part of a word, not as individual bytes. See *Addressing Digital Bits In A Word* in the *RPvdsEx Manual* for more information.

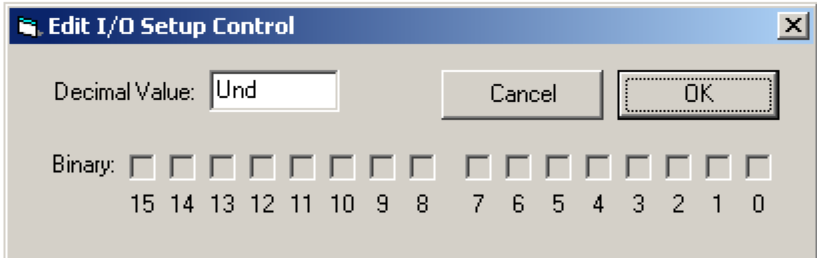
By default, all bits are configured as inputs. This default setting is intended to prevent damage to equipment that might be connected to the digital I/O lines. The user can configure the bits in the RPvdsEx configuration register. The configuration register is also used to determine what the eight front panel Bits lights represent.

To access the bit configuration register in RPvdsEx:

1. Click the **Device Setup** command on the **Implement** menu.
2. In the **Set Hardware Parameters** dialog box, click the **Device Type** box and select the **RX6 Multi-Function** from the list.
3. The dialog expands to display the **Device Configuration Register**.



4. Click **Modify** to display the **Edit I/O Setup Control** dialog box.

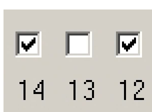


5. In this dialog box, a series of check boxes are used to create a bitmask that is used to program all bits.
6. To enable the check boxes, delete **Und** from the **Decimal Value** box.
7. To determine the desired value, select or clear the check boxes according to the table below. By default, all check boxes are cleared (value = 0). Selecting a check box sets the corresponding bit in the bitmask to one.
8. When the configuration is complete, click **OK** to return to the **Set Hardware Parameters** dialog box.

Bit #	Description
0-7	Each of these bits controls the configuration of one of the eight addressable bits as inputs or outputs. Setting the bit to one will configure that bit as an output.
8-11	Each of these bits controls the configuration of one of the four addressable bytes as inputs or outputs. Setting the bit to one will configure that byte as an output. bit 8 controls byte A, and bit 9 controls byte B. bits 10 – 11 are not used.
12-14	Create a bit code that determines how the front panel Bits lights are used, see table below.
15	Not used.

Bit Codes for Controlling the Bit Lights (Boxes 12-14)

By default, check boxes 12 –14 in the **Edit I/O Setup Control** dialog box (previous diagram) are cleared to create the bit code 000. This configures the eight front panel Bits lights to act as activity lights (glow when high) for the eight bit addressable digital I/O lines. The Bits lights can also be configured to provide information about amplifier status or act as activity lights for any of the other four bytes of digital I/O.

Bit Flags	Bits set to 1	Bit Lights Used For …
000 	None	Logical level lights for bit-addressable I/O lines
010 	13	Amplifier Clip Warning/Power Status display
100 	14	Enable logical level lights for byte A
101 	12, 14	Enable logical level lights for byte B

XLink

The XLink is not supported at this time.

Realizable Sampling Rates for the RX6

The following table shows the actual sampling rate values for the RX6. The X's on the table correspond to realizable frequencies for the ADC, DAC, Optical input, and Digital I/O. For example, the Digital I/O accepts a sampling rate up to 390625.0 Hz and the Audio ADC and DAC each accept a sampling rate up to 260416.67 Hz. The maximum realizable sampling rates are accepted as the maximum sampling rate without distortion. Each of the inputs and outputs will function above these sampling rates, but distortion will be present in the signal.

Standard Rate	Actual/Arbitrary Rate (Hz)	Audio ADC	Audio DAC	Optical/AMP Input	Digital I/O
6 kHz	6103.52	x	x	x	x
	6975.45	x	x		x
	8138.025	x	x		x
	9765.63	x	x		x
12 kHz	12207.03	x	x	x	x
	13950.89	x	x		x
	16276.04	x	x		x
	19531.25	x	x		x
25 kHz	24414.06	x	x	x	x
	27901.79	x	x		x
	32552.08	x	x		x
	39062.50	x	x		x
50 kHz	48828.13	x	x	x*	x
	55803.57	x	x		x
	65104.17	x	x		x
	78125.00	x	x		x
100 kHz	97656.25	x	x	x*	x
	111607.14	x	x		x
	130208.33	x	x		x
	156250.00	x	x		x
200 kHz	195312.50	x	x		x
	223214.29	x	x		x
	260416.67	x	x		x
	312500.00				x
400 kHz	390625.00				x

[x] = Fully functional [x*] = Sampling limited to 25KHz

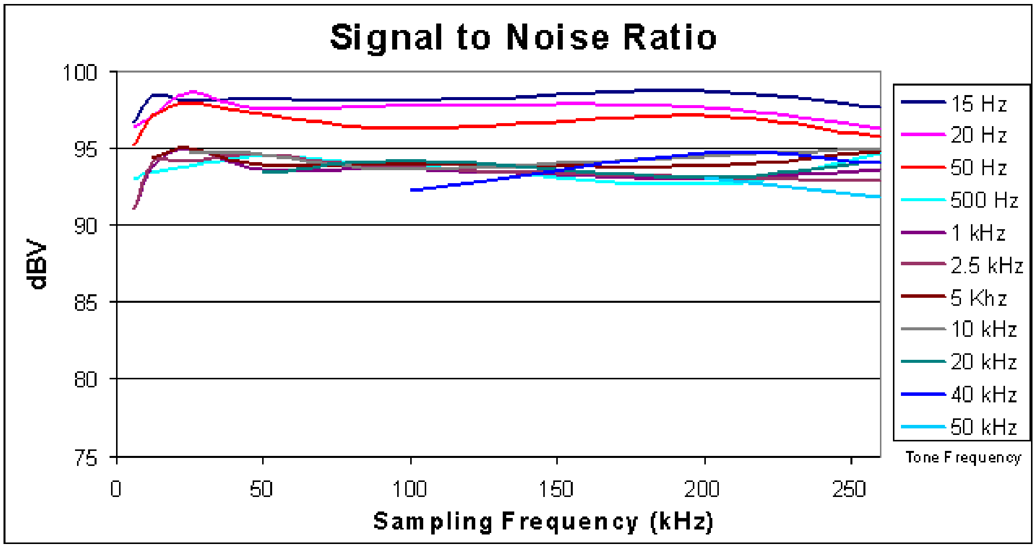
Piranha Technical Specifications

The RX6 can be equipped with a fiber optic input port which may be used with a Medusa or Adjustable Gain preamplifier. Specifications for the AD converters of those devices are found under the preamplifier's technical specifications.

DSP	100 MHz Sharc ADSP 21161, 600 MFLOPS Peak Two or Five
Memory	128 MB SDRAM
D/A	2 channels, 24-bit sigma-delta
Sample Rate	Up to 260.4166 kHz
Frequency Response	DC-Nyquist (~1/2 sample rate)
Voltage Out	+/- 10.0 Volts
S/N (typical)	105 dB (20 Hz - 20 kHz at 10 V)
THD (typical)	-92 dB (1 kHz output at 5 Vrms)
Sample Delay	43 samples
A/D	2 channels, 24-bit sigma-delta
Sample Rate	Up to 260.4166 kHz
Frequency Response	DC-Nyquist (~1/2 sample rate)
Voltage In	+/- 10.0 Volts
S/N (typical)	105 dB (20 Hz - 20 kHz at 10 V)
THD (typical)	-95 dB (1 kHz input at 5 Vrms)
Sample Delay	70 samples
Fiber Optic Ports	Optional Input (Medusa)
Digital I/O	24 bits programmable (8 bits addressable and a 16 bit word, addressable as 2 bytes)
Input Impedance	10 kOhms
Output Impedance	10 Ohms

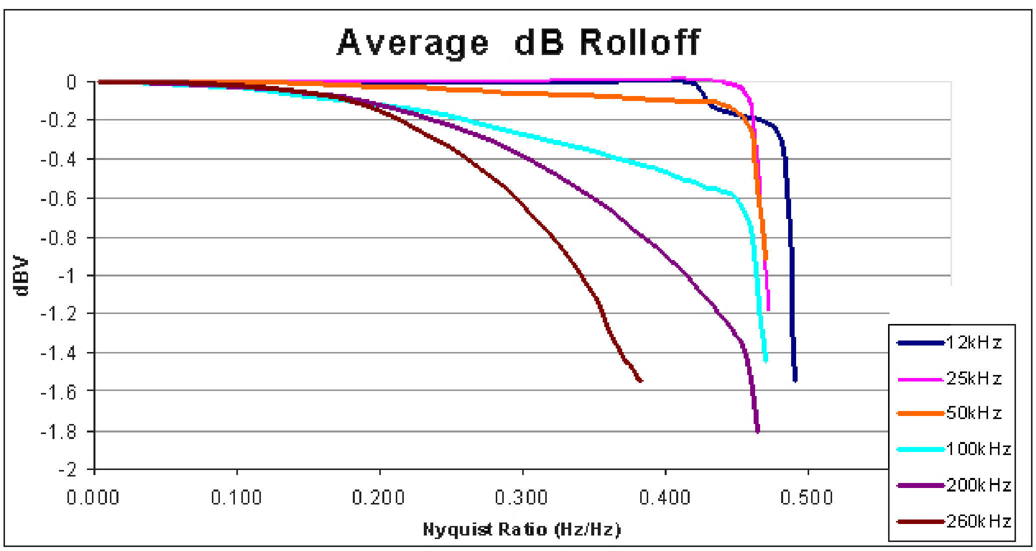
Signal-to-Noise Ratio Diagram

The following graph is of the signal to noise ratio with varying signal frequencies.



dB Rolloff Diagram

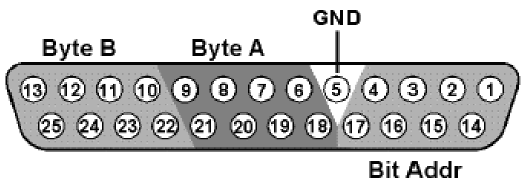
This graph shows the dB rolloff for the RX6 with varying sampling frequencies for both D/A and A/D. The sample delay remains relatively stable for varying frequencies.



DB25 Connector Pinout

TDT recommends the PP24 patch panel for accessing the RX6 I/O.

Digital I/O



Pin	Name	Description
1	BA0	Bit Addressable digital I/O Bits 0, 2, 4, and 6
2	BA2	
3	BA4	
4	BA6	
5	GND	Digital I/O Ground
6	A1	Byte A Word addressable digital I/O Bits 1, 3, 5, and 7
7	A3	
8	A5	
9	A7	
10	B1	Byte B Word addressable digital I/O Bits 1, 3, 5, and 7
11	B3	
12	B5	
13	B7	

Pin	Name	Description
14	BA1	Bit Addressable digital I/O Bits 1, 3, 5, and 7
15	BA3	
16	BA5	
17	BA7	
18	A0	Byte A Word addressable digital I/O Bits 0, 2, 4, and 6
19	A2	
20	A4	
21	A6	
22	B0	Byte B Word addressable digital I/O Bits 0, 2, 4, and 6
23	B2	
24	B4	
25	B6	

RX7 Stimulator Base Station

Overview

The RX7 base station is a high performance processor available with either two or five 100 MHz, 1600 MFLOPS Sharc DSPs. You can use the base station's onboard DSPs to design and generate complex arbitrary waveforms or complex patterns of biphasic pulses in real-time. The RX7 has been developed specifically for microstimulation applications. As part of TDT's RX7G MicroStimulator system, the RX7's primary role is to control the MS16 stimulus isolator, transferring hardware control and stimulation information across fiber optics. This proven digital communication system optically isolates the RX7 from the electrical stimulator, eliminating AC power surges and noise. For more information see *MS4/MS16 Stimulus Isolator*, page 6-3.

The RX7 includes 40 bits of digital I/O, analog output, and can include one or two fiber optic input ports for acquisition of digitized data from Medusa preamplifiers. Acquired signals can be filtered, rectified, or smoothed for stimulus output or dynamic real-time stimulus control based on analog control signals from virtually any signal source.

Power and Communication

The RX7 mounts in a System 3 zBus Powered Device Chassis (ZB1PS) and communicates with the PC using the Gigabit (PI5/FI5) or Optibit (PO5/FO5) PC interfaces. The ZB1PS is UL compliant, see the *ZB1PS Operations Manual* for power and safety information.

Software Control

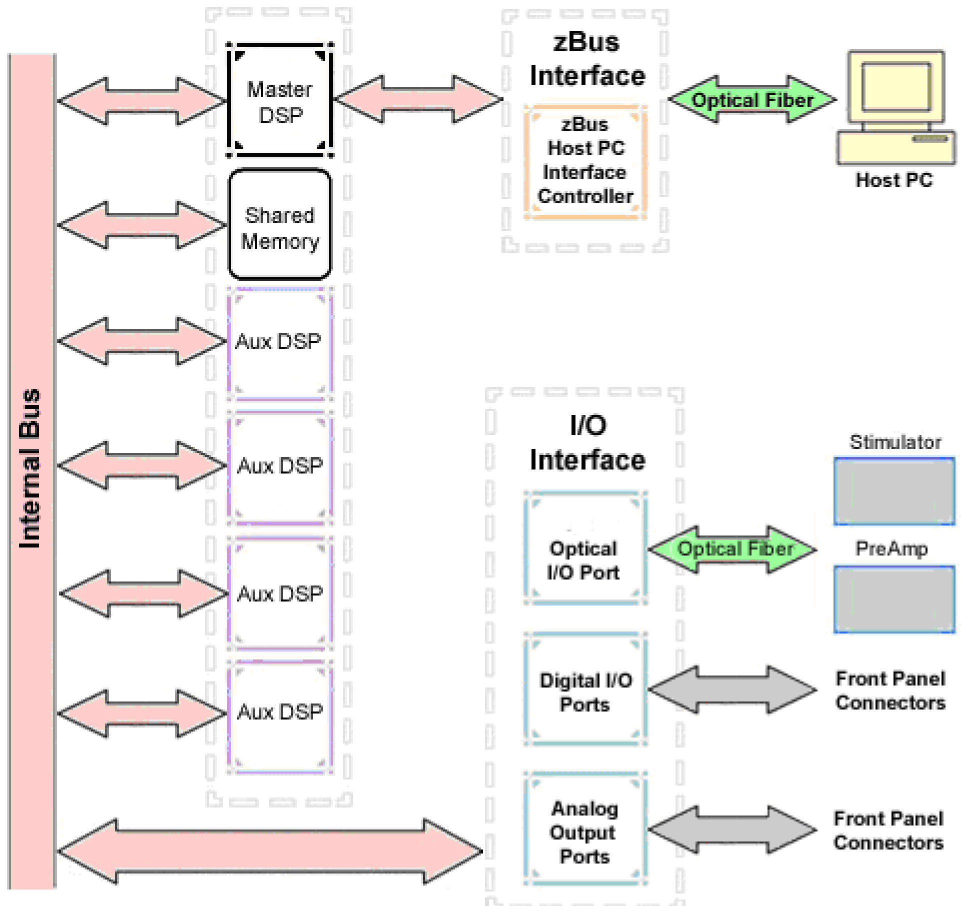
Software control is implemented with circuit files developed using TDT's RP Visual Design Studio (RPvdsEx). Circuits are loaded to the processor through TDT run-time applications or custom applications. This manual includes device specific information needed during circuit design. For circuit design techniques and a complete reference of the RPvdsEx circuit components, see the *RPvdsEx Manual*.

RX Architecture

Each RX multiprocessor device is equipped with either two or five digital signal processors (DSPs). The multi-DSP architecture allows processing tasks to be distributed across multiple processors and enables data to be transferred to the PC quickly and efficiently. The DSPs include one master and one or four auxiliary DSP(s). 128 MB SDRAM of system memory is shared by all DSPs. When designing circuits the maximum number of components for each RX DSP is 256.

Each DSP communicates with an internal bus to send and receive information from the I/O controller and the shared memory. The master DSP supervises overall system boot up and operation. The master DSP also acts as the main data interface between the zBus (host PC) and the multi-DSP environment.

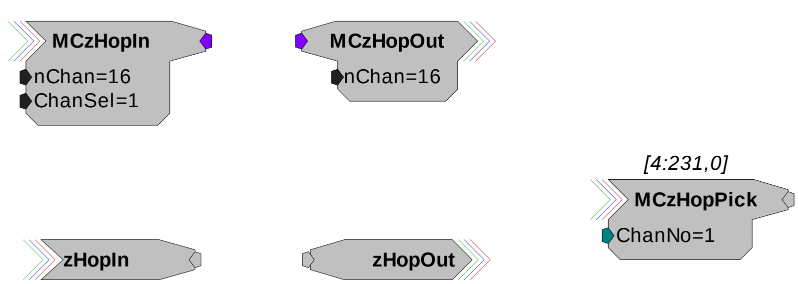
Because the zBus communicates only with the master processor, these devices operate most efficiently when the circuit related processing tasks assigned to the master DSP are minimized, allowing more processor power (cycles) for communication and overhead tasks.



The RX7 contains two DB25 connectors for interfacing with 40 bits of digital I/O and 4 channels of analog output. A BNC connector is provided for access to the first analog output channel. One or two fiber optic Medusa preamp ports enable connections for up to 32 channels of analog input.

Distributing Data Across DSPs

In RPvdsEx data can be transferred between each of the auxiliary DSPs as well as the master DSP using zHop components.



Components such as MCzHopIn and MCzHopOut can be used for multi-channel signals while components such as zHopIn, zHopOut, and MCzHopPick are used with single-channel signals. Up to 126 pairs can be used in a single RPvdsEx circuit.

Bus Related Delays The zHop Bus introduces a single sample delay. However, this delay is taken care of for the user in OpenEx when Timing and Data Saving macros are used.

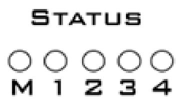
See *MultiProcessor Circuit Design* in the *RPvdsEx Manual* for these and other multiprocessor circuit design techniques.

RX7 Features

DSP Status Displays

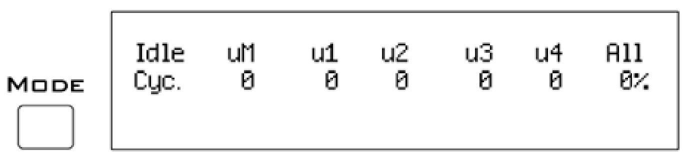
All high performance RX multiprocessors include status lights and a VFD (Vacuum Fluorescent Display) screen to report the status of the individual processors.

Status Lights



Up to five LEDs report the status of the multiprocessor's individual DSPs. When the device is turned on, they will glow steadily. If the demands on a DSP exceed 99% of its capacity on any given cycle, the corresponding LED will flash very rapidly (~3 times per second).

Front Panel VFD Screen



The front panel VFD screen reports detailed information about the status of the system. The display includes two lines. The top line reports the system mode, Run! or Idle, and displays heading labels for the second line. The second line reports the user’s choice of status indicators for each DSP followed by an aggregate value.

The user can cycle through the various status indicators using the Mode button to the left of the display. Push and release the button to change the display or push and hold the button for one second then release to automatically cycle through each of the display options. The VFD screen may also report system status such as booting status (Booting DSP) or alert the user when the device's microcode needs to be reprogrammed (Firmware Blank).

Status Indicators

Cyc: cycle usage

Ovr: processor cycle overages

Bus%: percentage of internal device's bus capacity used

I/O%: percentage of data transfer capacity used

Important Note!: The status lights will flash (~3 times a second) to alert the user when a device goes over the cycle usage limit, even if only for a particular cycle. This helps to identify periodic overages caused by components in time slices.

Fiber Optic Output Port (Stimulator)

The output port, labeled Stimulator, can be used to transfer microstimulation waveforms to the MS16/MS4 Stimulus Isolator and/or to control its digital output. See the *Stimulua Isolator* section, page 6-3, for more information.

Important Note: This fiber optic port is disabled if the sampling rate of the system is set to a value greater than ~25 kHz.

Fiber Optic Input Ports (Amp-A and Amp-B)

The RX7 base station can acquire digitized signals from a Medusa preamplifier over a fiber optic cable. This provides loss-less signal acquisition between the amplifier and the base station. Up to two fiber optic ports are provided to support simultaneous acquisition from up to two preamplifiers. Each port can input up to 16 channels at a maximum sampling rate of ~25 kHz. The fiber optic ports provide oversampling. See *Fiber Oversampling*, below for more information.

The fiber optic ports can be used with any of the Medusa preamplifiers including the RA16PA, RA4PA, or RA8GA. The channel numbers for each port begin at a fixed offset regardless of the number of channels available on the connected device.

Channels are numbered as follows:

Amp-A 1 – 16

Amp-B 17 – 32

Fiber Oversampling (acquisition only)

The fiber optic cable that carries the signals to the fiber optic input ports on the RX7 has a transfer rate limitation of 6.25 Mbits/s. With 16 channels of data and 16 bits per sample, this limitation translates to a maximum sampling rate of ~25 kHz.

However, the need may arise to run a circuit at a higher sampling rate while still acquiring data via a fiber optic port. The first two fiber optic ports on an RX device can oversample the digitized signals that have already been sampled up to 4X or ~100 kHz. This will allow an RX7 to run a DSP chain at ~50 kHz or ~100 kHz, and still sample data acquired through an optically connected preamplifier that digitized the incoming data stream at its maximum rate of ~25 kHz.

Oversampling is performed on the base station. The signals being acquired will still be sampled at ~25 kHz on the preamplifier. This means that, even with oversampling, signals acquired by an optically connected preamplifier are still governed by the bandwidth and frequency response of the preamplifier.

When acquiring up to 16 channels of data on the first fiber optic input port of the RX7, the signals will be oversampled 4X to ~100 kHz. If the RX7 is equipped with a second fiber optic input port and data is being acquired on both ports, the signals on second port will be oversampled 2X to ~50 kHz.

Amp Status and Clip Warning Lights

Amp lights are located to the right of each fiber optic port. These lights are used to indicate the power status or provide a clip warning for the connected amplifiers.

When an amplifier is not connected the Amp light will flash in a slow steady pattern. The light is lit when the amplifier is connected and begins to flash quickly when the voltage on the battery for the corresponding amplifier is low. When any channel on the connected amplifier produces a voltage approaching the maximum input of the amplifier, the corresponding light will flash rapidly to warn that clipping may occur if the signal exceeds the maximum input voltage. See the corresponding preamplifier section for more information on input range and clip warnings.

Important Note!: The Li-ion batteries voltage decreases rapidly once the battery low light is on. Data acquisition will suffer if the battery is not charged soon after the light goes on.

Amplifier Status Patterns

Light Pattern	Amplifier Status
Solid	Connected
Very slow flash (~1 every two seconds)	Not connected
Slow flash (~1 per second)	Connected and charging
Rapid flash	Battery low
Very rapid flash	Clip Warning

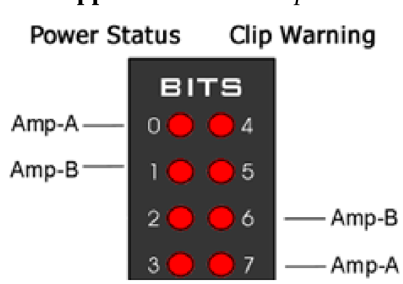
Note: If the amplifier appears to be connected and the amplifier status light is flashing slowly, check to ensure that the device is connected properly.

Bits Lights

The RX7's eight Bits lights are user configurable. By default the Bits lights indicate the logic level (light when high) for the eight bit-addressable digital I/O lines. The Bits lights can also be configured to provide information about amplifier status or act as logic level lights for any of the other four bytes of digital I/O.

Using the Bits Lights to Display Amplifier Status

Note: Because clip warning and amplifier status are always displayed using the Amp lights (located directly to the right of each fiber optic port), **TDT recommends using the Bits lights for other applications**. See *Amp Status and Clip Warning Lights* for more information.



When the Bits lights are configured to display the amplifier status, the left column of lights indicates the power status and the right column indicates a clip warning for the corresponding amplifier.

The table on page 2-29 shows the light pattern and corresponding amplifier status for the power status lights (0 - 3). Clip lights flash very rapidly when any channel on the connected amplifier produces a voltage approaching the maximum input of the amplifier.

Analog Output

The RX7 is equipped with four channels of 16-bit PCM D/A. The sampling rate is user selectable up to a maximum of ~100 kHz. The D/A is DC coupled and has a built-in upsampler for improved audio playback. The upsampler is controlled through one of the RX7's programmable bits and can be turned off to allow the D/A to drive external devices such as a stimulator. Channel one analog output can be accessed via a front Panel BNC (DAC-1). All four analog channels can be accessed via the DB25 Multi I/O connector (pins 14 – 17).

Important! When using the RX7 with the stimulus isolator, the sampling rate set for this device cannot exceed ~25 kHz—a limitation of the fiber optic connection between the RX7 and the stimulus isolator.

Digital I/O

The RX7 base station has 40 digital I/O lines. Eight bits are bit-addressable. The remaining 32 bits are four word-addressable bytes. Digital I/O lines are accessed via the two 25-pin connectors on the front of the RX7. *See the Digital I/O Circuit Design section of the RPvdsEx Manual* for more information on programming the digital I/O.

CAUTION!: The first eight bits of bit-addressable digital I/O on RX devices are unbuffered. When used as inputs, overvoltages on these lines can cause severe damage to the system. TDT recommends when sending digital signals into the device, (1) never send a signal with amplitude greater than seven volts into any digital input and (2) always use the byte-addressable digital I/O lines.

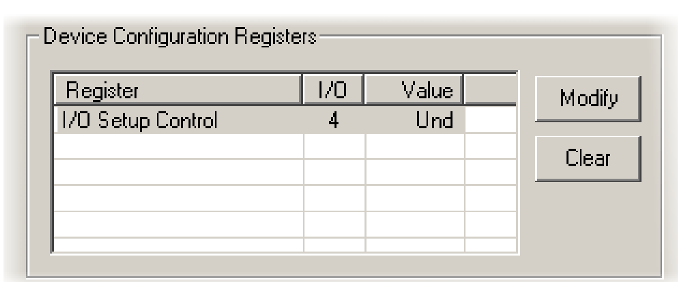
Configuring the Programmable I/O Lines

Each of the eight bit-addressable lines can be independently configured as inputs or outputs. The digital I/O lines can be configured as inputs or outputs in groups of eight bits – that is as byte A, byte B, byte C, and byte D. Note, however, that the bytes must be addressed as if part of a word, not as individual bytes. See *Addressing Digital Bits In A Word* in the *RPvdsEx Manual* for more information.

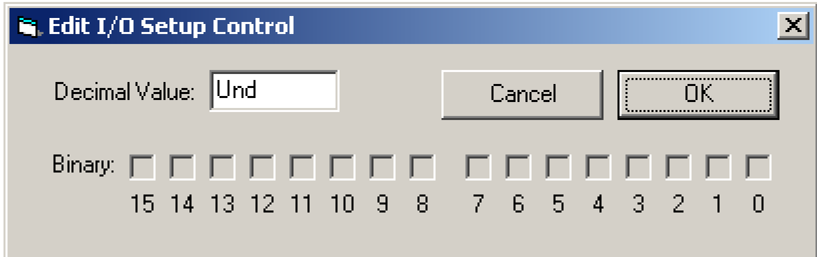
By default, all bits are configured as inputs. This default setting is intended to prevent damage to equipment that might be connected to the digital I/O lines. The user can configure the bits in the RPvdsEx configuration register. The configuration register is also used to determine what the eight front panel Bits lights represent.

To access the bit configuration register:

1. Click the **Device Setup** command on the **Implement** menu.
2. In the **Set Hardware Parameters** dialog box, click the **Device Type** box and select the **RX7 Elec-Stimulator** from the list.
3. The dialog expands to display the **Device Configuration Register**.



4. Click **Modify** to display the **Edit I/O Setup Control** dialog box.



5. In this dialog box, a series of check boxes are used to create a bitmask that is used to program all bits.
6. To enable the check boxes, delete **Und** from the **Decimal Value** box.
7. To determine the desired value, select or clear the check boxes according to the table below. By default, all check boxes are cleared (value = 0). Selecting a check box sets the corresponding bit in the bitmask to one.
8. When the configuration is complete, click **OK** to return to the **Set Hardware Parameters** dialog box.

Bit #	**Description**
0-7	Each of these bits controls the configuration of one of the eight addressable bits as inputs or outputs. Setting the bit to one will configure that bit as an output.
8-11	Each of these bits controls the configuration of one of the four addressable bytes as inputs or outputs. Setting the bit to one will configure that byte as an output. bit 8 - byte A, bit 9 - byte B, bit 10 - byte C, and bit 11 - byte D
12-14	Create a bit code that determines how the front panel Bits lights are used, see table below.
15	Setting the bit to one will disable the D/A upsampler.

Bit Codes for Controlling the Bit Lights (Boxes 12-14)

By default, check boxes 12 –14 in the **Edit I/O Setup Control** dialog box (previous diagram) are cleared to create the bit code 000. This configures the eight front panel Bits lights to act as activity lights (glow when high) for the eight bit addressable digital I/O lines. The Bits lights can also be configured to provide information about amplifier status or act as activity lights for any of the other four bytes of digital I/O.

Bit Flags	Bits set to 1	Bit Lights Used For …
000 	None	Logical level lights for bit-addressable I/O lines
010 	13	Amplifier Clip Warning/Power Status display
100 	14	Enable logical level lights for byte A
101 	12, 14	Enable logical level lights for byte B
110 	13, 14	Enable logical level lights for byte C
111 	12, 13, 14	Enable logical level lights for byte D

XLink

The XLink is not supported at this time.

Stimulator Base Station Technical Specifications

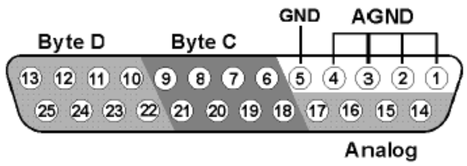
The RX7 is designed for use with the stimulus isolator. *The specifications for the stimulus isolator are found under that device's technical specifications.* The RX7 is also equipped with a fiber optic input port for use with Medusa of Adjustable Gain preamplifiers. *Specifications for the A/D converters of the preamplifiers are found in the corresponding technical specifications.*

DSP	100 MHz Sharc ADSP 21161, 600 MFLOPS Peak Two or Five
Memory	128 MB SDRAM (Shared)
D/A	4 channels, 16-bit PCM
Sample Rate	Up to 97.65625 kHz (8X upsampled to 200 kHz default operation)*
Frequency Response	DC-Nyquist(~1/2 sample rate)
Voltage Out	+/- 10.0 Volts
Voltage Out Accuracy	+/- 10%
S/N (typical)	84 dB (20 Hz to 25 KHz) 82 dB with upsampling disabled
THD (typical)	-77 dB for 1 kHz output at 5 Vrms -74 dB with upsampling disabled
Output Impedance	10 Ohm
Fiber Optic Ports	One or Two Inputs, Output for Stimulator *
Digital I/O	40 bits programmable (8 bits bit-addressable and a 32 bit word, addressable as 4 bytes)

* **Note:** When used with the microstimulator, the sampling rate is limited to 24.414 kHz by the Stimulator Fiber Optic Port.

DB25 Connector Pinouts

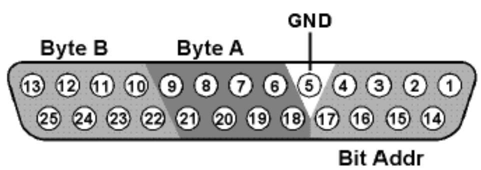
Multi I/O



Pin	Name	Description
1	AGND	Analog Ground
2		
3		
4		
5	GND	Digital I/O Ground
6	C1	Byte C Word addressable digital I/O Bits 1, 3, 5, and 7
7	C3	
8	C5	
9	C7	
10	D1	Byte D Word addressable digital I/O Bits 1, 3, 5, and 7
11	D3	
12	D5	
13	D7	

Pin	Name	Description
14	A1	Analog Output Channels
15	A2	
16	A3	
17	A4	
18	C0	Byte C Word addressable digital I/O Bits 0, 2, 4, and 6
19	C2	
20	C4	
21	C6	
22	D0	Byte C Word addressable digital I/O Bits 0, 2, 4, and 6
23	D2	
24	D4	
25	D6	

Digital I/O



Pin	Name	Description
1	BA0	Bit Addressable digital I/O Bits 0, 2, 4, and 6
2	BA2	
3	BA4	
4	BA6	
5	GND	Digital I/O Ground
6	A1	Byte A Word addressable digital I/O Bits 1, 3, 5, and 7
7	A3	
8	A5	
9	A7	
10	B1	Byte B Word addressable digital I/O Bits 1, 3, 5, and 7
11	B3	
12	B5	
13	B7	

Pin	Name	Description
14	BA1	Bit Addressable digital I/O Bits 1, 3, 5, and 7
15	BA2	
16	BA3	
17	BA4	
18	A0	Byte A Word addressable digital I/O Bits 0, 2, 4, and 6
19	A2	
20	A4	
21	A6	
22	B0	Byte B Word addressable digital I/O Bits 0, 2, 4, and 6
23	B2	
24	B4	
25	B6	

RX8 Multi I/O

Overview

The RX8 is a high channel count, high sample rate analog I/O system which provides a maximum of 24 channels of analog I/O and generates a maximum sampling rate of 100 kHz per channel. Each bank of four or eight channels of I/O is user configurable with either PCM or sigma-delta converters. The 24-bit sigma-delta converters are ideal for audio applications. The 16-bit PCM analog converters have an excellent dynamic range and almost no group delay. These converters are excellent for acquiring signal information and controlling external devices, such as motors.

The RX8 is equipped with either two or five 100 MHz, 1600 MFLOPS Sharc DSPs and can control audio feedback systems or motor controls in real-time. Built in digital filters, waveform generators, and logic control components give end users the ability to design and control virtually any presentation system.

Power and Communication

The RX8 mounts in a System 3 zBus Powered Device Chassis (ZB1PS) and communicates with the PC using the Gigabit (PI5/FI5) or Optibit (PO5/FO5) PC interfaces. The ZB1PS is UL compliant, see the *ZB1PS Operations Manual* for power and safety information.

Software Control

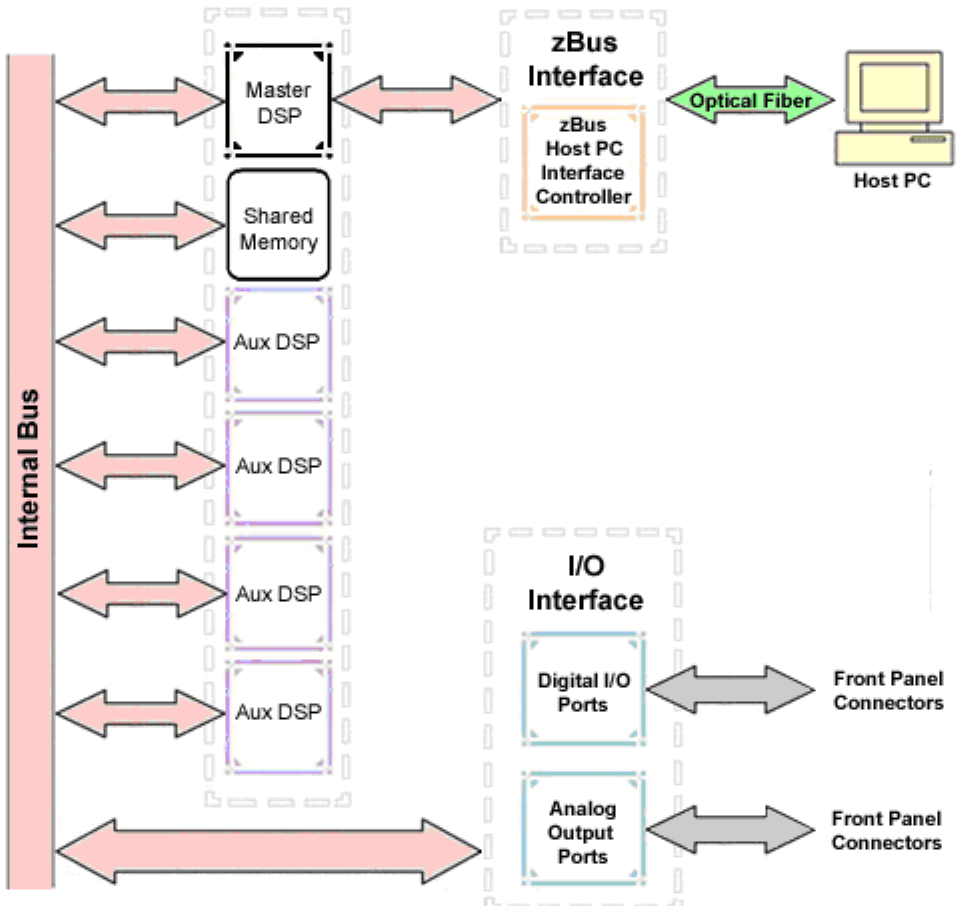
Software control is implemented with circuit files developed using TDT's RP Visual Design Studio (RPvdsEx). Circuits are loaded to the processor through TDT run-time applications or custom applications. This manual includes device specific information needed during circuit design. For circuit design techniques and a complete reference of the RPvdsEx circuit components, see the *RPvdsEx Manual*.

RX Architecture

Each RX multiprocessor device is equipped with either two or five digital signal processors (DSPs). The multi-DSP architecture allows processing tasks to be distributed across multiple processors and enables data to be transferred to the PC quickly and efficiently. The DSPs include one master and one or four auxiliary DSP(s). 128 MB SDRAM of system memory is shared by all DSPs. When designing circuits the maximum number of components for each RX DSP is 256.

Each DSP communicates with an internal bus to send and receive information from the I/O controller and the shared memory. The master DSP supervises overall system boot up and operation. The master DSP also acts as the main data interface between the zBus (host PC) and the multi-DSP environment.

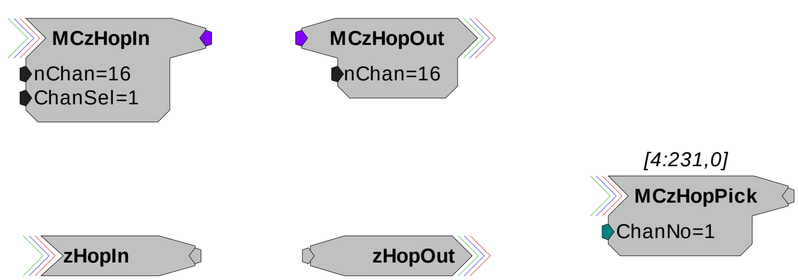
Because the zBus communicates only with the master processor, these devices operate most efficiently when the circuit related processing tasks assigned to the master DSP are minimized, allowing more processor power (cycles) for communication and overhead tasks.



The RX8 contains two DB25 connectors for interfacing with 24 bits of digital I/O and 24 channels of analog I/O.

Distributing Data Across DSPs

In RPvdsEx data can be transferred between each of the auxiliary DSPs as well as the master DSP using zHop components.



Components such as MCzHopIn and MCzHopOut can be used for multi-channel signals while components such as zHopIn, zHopOut, and MCzHopPick are used with single-channel signals. Up to 126 pairs can be used in a single RPvdsEx circuit.

Bus Related Delays The zHop Bus introduces a single sample delay. However, this delay is taken care of for the user in OpenEx when Timing and Data Saving macros are used.

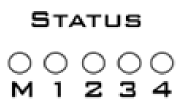
See *MultiProcessor Circuit Design* in the *RPvdsEx Manual* for these and other multiprocessor circuit design techniques.

RX8 Features

DSP Status Displays

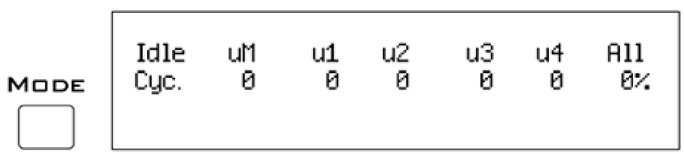
All high performance RX multiprocessors include status lights and a VFD (Vacuum Fluorescent Display) screen to report the status of the individual processors.

Status Lights



Up to five LEDs report the status of the multiprocessor's individual DSPs. When the device is turned on, they will glow steadily. If the demands on a DSP exceed 99% of its capacity on any given cycle, the corresponding LED will flash very rapidly (~3 times per second).

Front Panel VFD Screen



The front panel VFD screen reports detailed information about the status of the system. The display includes two lines. The top line reports the system mode, Run! or Idle, and displays heading labels for the second line. The second line reports the user’s choice of status indicators for each DSP followed by an aggregate value.

The user can cycle through the various status indicators using the Mode button to the left of the display. Push and release the button to change the display or push and hold the button for one second then release to automatically cycle through each of the display options. The VFD screen may also report system status such as booting status (Booting DSP) or alert the user when the device's microcode needs to be reprogrammed (Firmware Blank).

Status Indicators

Cyc: cycle usage

Ovr: processor cycle overages

Bus%: percentage of internal device's bus capacity used

I/O%: percentage of data transfer capacity used

Important Note!: The status lights will flash (~3 times a second) to alert the user when a device goes over the cycle usage limit, even if only for a particular cycle. This helps to identify periodic overages caused by components in time slices.

Bits Lights

The RX8's eight Bits lights are user configurable. By default the Bits lights indicate the logic level (light when high) for the eight bit-addressable digital I/O lines. The Bits lights can also act as logic level lights for any of the other two bytes of digital I/O.

Analog Input/Output

The RX8 can have a maximum of 24 channels of analog I/O accessed via the 25-pin connector on the front panel. Each bank of up to eight channels of I/O is user configurable with either PCM or sigma-delta converters.

Sigma-delta converters provide superior conversion quality and extended useful bandwidths, at the cost of an inherent fixed group delay. **When equipped with sigma-delta, the RX8 DAC Delay is 23 samples and the RX8 ADC Delay is 47 samples.**

Note: Because of device timing constraints at higher sampling rates, only the first 23 channels of analog I/O are processed when operating the RX8 at ~100 kHz.

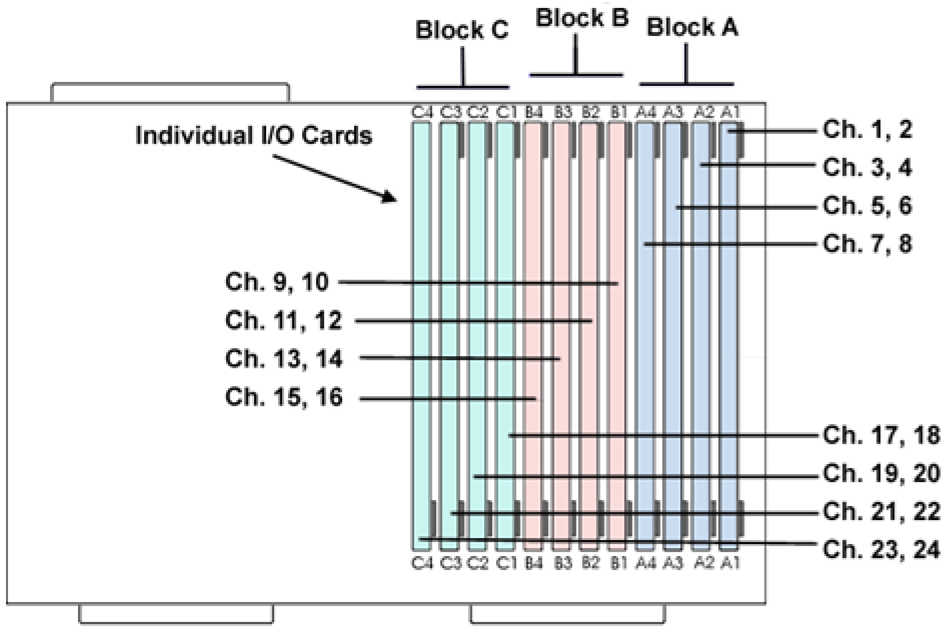
The analog I/O of each device is custom configured at the factory. Problems will arise if end users do not carefully note the configuration of their RX8 device. This topic provides information about configurations and channel numbering. The RX8's analog I/O channels are accessed via a 25-pin connector on the front panel. If you know what channel numbers your device uses, See the *RX8 Technical Specifications*, page 2-42, for the Analog I/O pinout diagram.

Organization of Analog I/O Blocks

The RX8 has three blocks of I/O ports. Each block can house up to eight channels for a total of 24 channels of analog I/O. Blocks can only be filled by analog I/O modules of the same type.

For example:

A block can be configured with all D/A's or all A/D's, but not a mixture of D/A's and A/D's. In addition, the D/A's and A/D's must be of the same type (either PCM or sigma-delta).



Note: Block C can only be configured with outputs.

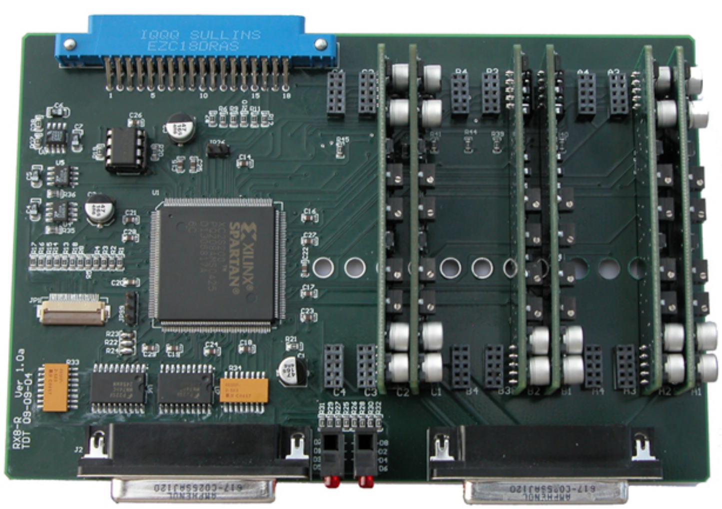
Channel Numbers

Starting with block A and ending with block C, channels are numbered sequentially from 1 to 24. The channel numbering is independent of whether the analog I/O board is an input or output.

For example:

The analog I/O of an RX8 that has four A/D's in the first two slots of Block A and four D/A's in the first two slots of Bank C, would be accessed with the A/D's as channels 1-4 and the D/A's as channels 17-20.

The photo below shows one possible configuration of the RX8's I/O boards. This configuration uses channels 1-4, 9-12, and 17-20.



Digital I/O

The RX8 processor includes 24 bits of programmable I/O in two eight bit word-addressable bytes and eight bits of bit-addressable I/O. Digital I/O lines are accessed via the 25-pin connector on the front panel and can be configured as inputs or outputs. *See the Digital I/O Circuit Design section of the RPvdsEx Manual* for more information on programming the digital I/O.

CAUTION!: The first eight bits of bit-addressable digital I/O on RX devices are unbuffered. When used as inputs, overvoltages on these lines can cause severe damage to the system. TDT recommends when sending digital signals into the device, (1) never send a signal with amplitude greater than seven volts into any digital input and (2) always use the byte-addressable digital I/O lines.

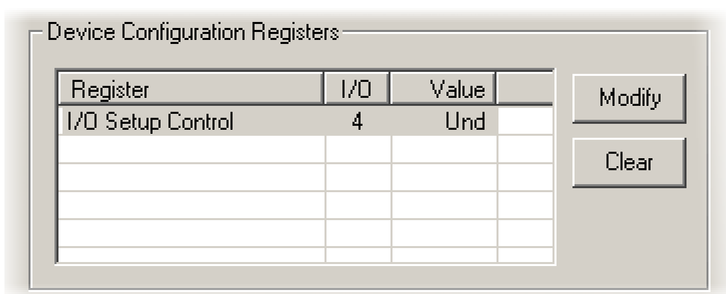
Configuring the Programmable I/O Lines

Each of the eight bit-addressable bits can be independently configured as inputs or outputs. The digital I/O lines can be configured as inputs or outputs in groups of eight bits – that is as byte A and byte B. Note, however, that the bytes must be addressed as if part of a word, not as individual bytes. See *Addressing Digital Bits In A Word* in the *RPvdsEx Manual* for more information.

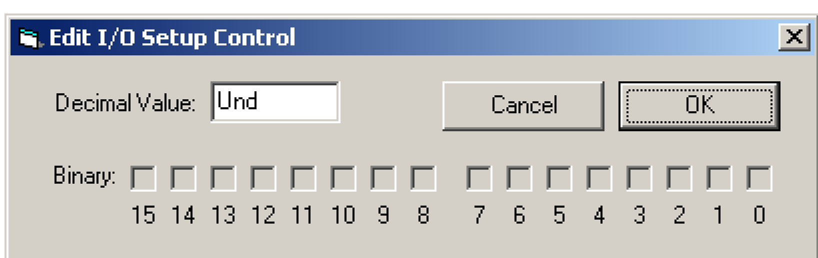
By default, all bits are configured as inputs. This default setting is intended to prevent damage to equipment that might be connected to the digital I/O lines. The user can configure the bits in the RPvdsEx configuration register. The configuration register is also used to determine what the eight front panel Bits lights represent.

To access the bit configuration register:

1. Click the **Device Setup** command on the **Implement** menu.
2. In the **Set Hardware Parameters** dialog box, click the **Device Type** box and select **RX8 Multi-Chan I/O** from the list.
3. The dialog expands to display the **Device Configuration Register**.



4. Click **Modify** to display the **Edit I/O Setup Control** dialog box.



5. In this dialog box, a series of check boxes are used to create a bitmask that is used to program all bits.

6. To enable the check boxes, delete **Und** from the **Decimal Value** box.
7. To determine the desired value, select or clear the check boxes according to the table below. By default, all check boxes are cleared (value = 0). Selecting a check box sets the corresponding bit in the bitmask to one.
8. When the configuration is complete, click **OK** to return to the **Set Hardware Parameters** dialog box.

Bit #	Description
0-7	Each of these bits controls the configuration of one of the eight addressable bits as inputs or outputs. Setting the bit to one will configure that bit as an output.
8-11	Each of these bits controls the configuration of one of the four addressable bytes as inputs or outputs. Setting the bit to one will configure that byte as an output. bit 8 controls byte A, and bit 9 controls byte B. bits 10 – 11 are not used.
12-14	Create a bit code that determines how the front panel Bits lights are used, see table below.
15	Not used.

Bit Codes for Controlling the Bit Lights (Boxes 12-14)

By default, check boxes 12 –14 in the **Edit I/O Setup Control** dialog box (previous diagram) are cleared to create the bit code 000. This configures the eight front panel Bits lights to act as activity lights (glow when high) for the eight bit addressable digital I/O lines. The Bits lights can also be configured to provide information about amplifier status or act as activity lights for any of the other four bytes of digital I/O.

Bit Flags	Bits set to 1	Bit Lights Used For …
000 	None	Logical level lights for bit-addressable I/O lines
100 	14	Logical level lights for byte A
101 	12, 14	Logical level lights for byte B

XLink

The XLink is not supported at this time.

Multi I/O Technical Specifications

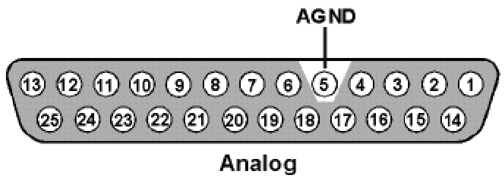
DSP	100 MHz Sharc ADSP 21161, 600 MFLOPS Peak Two or Five
Memory	128 MB SDRAM
D/A	up to 24 channels, 16-bit PCM or 24-bit sigma-delta
Sample Rate	Up to 97.65625 kHz*
Frequency Response	Sigma-delta or PCM: DC-Nyquist (~1/2 sample rate)
Voltage Out	+/- 10.0 Volts
S/N (typical)	Sigma-delta: 97 dB (20 Hz - 20 kHz at 10 V) PCM: 80 dB (20 Hz - 20 kHz at 10 V)
THD (typical)	Sigma-delta: -84 dB (1 kHz output at 5 Vrms) PCM: -70 dB (1 kHz output at 5 Vrms)
Sample Delay	Sigma-delta: 23 samples or PCM: 4 samples
A/D	up to 16 channels, 16-bit PCM or 24-bit sigma-delta
Sample Rate	Up to 97.65625 kHz*
Frequency Response	Sigma-delta: DC-Nyquist (~1/2 sample rate) PCM: DC - 7.5 kHz (3 dB corner, 2nd order, 12 dB per octave)
Voltage In	+/- 10.0 Volts
S/N (typical)	Sigma-delta: 97 dB (20 Hz - 20 kHz at 10 V) PCM: 80 dB (20 Hz - 20 kHz at 10 V)
THD (typical)	Sigma-delta: -84 dB (1 kHz output at 5 Vrms) PCM: -65 dB (1 kHz output at 5 Vrms)
Sample Delay	Sigma-delta: 47 samples or PCM: 4 samples
Digital I/O	24 bits programmable (8 bits addressable and a 16 bit word, addressable as 2 bytes)
Input Impedance	10 kOhms
Output Impedance	10 Ohms

***Note:** Because of device timing constraints at higher sampling rates, only the first 23 channels of analog I/O are processed when operating the RX8 at 100 kHz.

DB25 Connector Pinouts

TDT Reccomends accessing the RX8 I/O via a PP24 patch panel.

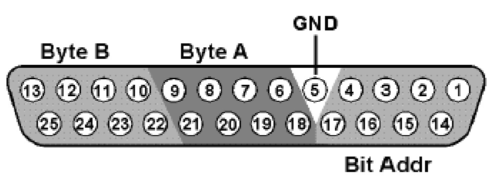
Analog I/O



Pin	Name	Description
1	A1	Analog I/O Channels Input or Output Depending on Custom Configuration
2	A3	
3	A5	
4	A7	
5	AGND	Analog Ground
6	A10	Analog I/O Channels Input or Output Depending on Custom Configuration
7	A12	
8	A14	
9	A16	
10	A18	Analog Outputs
11	A20	
12	A22	
13	A24	

Pin	Name	Description
14	A2	Analog I/O Channels Input or Output Depending on Custom Configuration
15	A4	
16	A6	
17	A8	
18	A9	
19	A11	
20	A13	
21	A15	
22	A17	Analog Outputs
23	A19	
24	A21	
25	A23	

Digital I/O



Pin	Name	Description
1	BA0	Bit Addressable digital I/O Bits 0, 2, 4, and 6
2	BA2	
3	BA4	
4	BA6	
5	GND	Digital I/O Ground
6	A1	Byte A Word addressable digital I/O Bits 1, 3, 5, and 7
7	A3	
8	A5	
9	A7	
10	B1	Byte B Word addressable digital I/O Bits 1, 3, 5, and 7
11	B3	
12	B5	
13	B7	

Pin	Name	Description
14	BA1	Bit Addressable digital I/O Bits 1, 3, 5, and 7
15	BA3	
16	BA5	
17	BA7	
18	A0	Byte A Word addressable digital I/O Bits 0, 2, 4, and 6
19	A2	
20	A4	
21	A6	
22	B0	Byte B Word addressable digital I/O Bits 0, 2, 4, and 6
23	B2	
24	B4	
25	B6	

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Part 3 RP Processors

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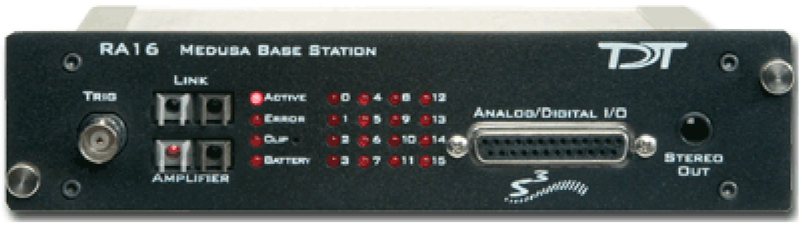
RA16 Medusa Base Station



Overview

Recommended for single or dual channel extracellular recordings and low channel count EEG's, EMG's and evoked potential recordings (such as ABRs), the Medusa Base Station is a versatile signal processor designed to acquire, filter, and process data digitized on one of our preamplifiers. The RA16 acquires digitized signals from a Medusa preamplifier over a fiber optic cable, providing loss-less signal acquisition between the amplifier and the base station.

PCM analog outputs can be used for a wide variety of signal production tasks, including control of motors, electrical stimulation, and monitoring analog signals during acquisition.

Power and Communication

The RA16 mounts in a System 3 zBus Powered Device Chassis (ZB1PS) and communicates with the PC using any of the zBus PC interfaces. The ZB1PS is UL compliant, see the *ZB1PS Operations Manual* for power and safety information.

Software Control

Software control is implemented with circuit files developed using TDT's RP Visual Design Studio (RPvdsEx). Circuits are loaded to the processor through TDT run-time applications or custom applications. This manual includes device specific information needed during circuit design. For circuit design techniques and a complete reference of the RPvdsEx circuit components, see the *RPvdsEx Manual*.

RA16 Features

Status Lights

The four lights on the left-hand side are status lights that relate to the amplifier.

Active - The active light blinks when there is no active connection between the base station and the amplifier. The active light is on when there is a connection to an amplifier and the amplifier is on.

Error - The error light blinks when there is a communication error between the base station and the amplifier.

Clip - The clip light is a warning light and flashes when any channel on the connected amplifier produces a voltage approaching the maximum input of the amplifier. The light will flash rapidly to warn that clipping may occur if the signal exceeds the maximum input voltage.

Battery - The battery light flashes when the battery voltage is low. The Li-Ion battery voltage decreases rapidly once this indicator light is on. Data acquisition will suffer if the battery is not charged soon after this warning.

Digital Out Lights

There is one digital out LED for each digital output bit. Each LED will light when a logical high (1) is sent out on the corresponding digital output bit. The digital out lights can be used to indicate clipping or spike detection on a channel.

Trigger

Allows input of an external digital trigger.

Link and Amplifier Ports

The Base Station has two sets of fiber optic ports. The Link port allows multiple base stations to be linked for complex or high channel count processing. The Amplifier port is used to connect the base station to a Medusa preamplifier.

Stereo Output

The stereo output samples from the first two channels of the digital-to-analog converters (DACs) so that users can monitor signal properties with headphones or speakers. The left speaker monitors channel one of the DAC and the right speaker monitors channel two.

Use the Ch (channel) parameter on the channel inputs to change which analog channels are being monitored.

Analog and Digital Outputs

Each base station comes with 16 digital output bits and eight analog output channels. See the technical specifications for DB25 pinout. Each DAC uses 18-bit sigma-delta parts for high quality signal conversion. Sigma-delta converters provide superior conversion quality and extended useful bandwidths, at the cost of an inherent fixed group delay. For the RA16BA the DAC Delay is 18 samples.

Sampling Rate Considerations

There are no onboard analog-to-digital converters (ADCs) on the Medusa base station. When acquiring data, a preamplifier does this conversion. Since the fiber optic connection from a preamplifier to the base station has a transfer rate limitation of ~25 kHz, circuits utilizing this data acquisition must use a sample rate of ~25 kHz or less. Otherwise (i.e. circuits with digital-to-analog conversion only), the maximum sample rate is ~50 kHz.

Force

Pushing a paper clip in to the pinhole next to the clip light deletes the microcode on the base station. Once the microcode is deleted the RA16 base station will need to be reprogrammed.

USB Transfer Rates

USB transfers are limited to 100,000 samples per second of 32-bit data. 16-channels of ~25 KHz data produces 400,000 samples of data per second. Data reduction techniques such as Compress to 16 and Shuffle to 16 will reduce the data size without significant loss of information. Selective channel analysis and filtering can further reduce the amount of data transferred.

Memory

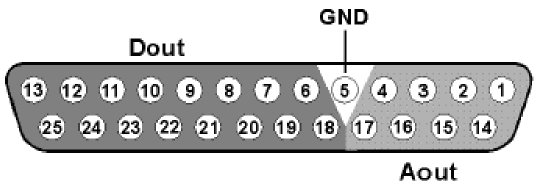
The RA16BA Medusa comes standard with 32MB of RAM. At 16-channels in 16-bit mode, 32MB would give around 40 seconds of continuous data acquisition. Each additional base station could add an additional 2.5 minutes of continuous data acquisition.

Medusa Base Station Technical Specifications

Note: The RA16BA has no onboard AD converters. Technical specifications for the AD converters are found under the preamplifier's technical specifications.

DSP	50 MHz Sharc 21065, 150 MFLOPS
Memory	16 MB SDRAM or 32 MB SDRAM
D/A	8 channels, 18-bit sigma-delta
Sample Rate	48.828 kHz maximum
Frequency Response	3 dB at 3 Hz - Nyquist (~1/2 sample rate)
Voltage Out	+/- 10.0 V (AC coupled)
S/N (typical)	90 dB (20 Hz to 25 KHz)
Distortion (typical)	-70 dB for 1 KHz output at 0.7 Vrms
Sample Delay	18 samples
Fiber Optic Ports	1 16-channel Input and 1 Link Port (24 kHz maximum sample rate)
Digital Inputs	1 bit
Digital Outputs	16 bits
Input Impedance	NA
Output Impedance	20 Ohm

DB25 Analog/Digital I/O Connector Pin Out



Pin	Name	Description
1	A1	Analog Output Channels
2	A3	
3	A5	
4	A7	
5	GND	Ground
6	D1	Digital Output Bits
7	D3	
8	D5	
9	D7	
10	D9	
11	D11	
12	D13	
13	D15	

Pin	Name	Description
14	A2	Analog Output Channels
15	A4	
16	A6	
17	A8	
18	D0	Digital Output Bits
19	D2	
20	D4	
21	D6	
22	D8	
23	D10	
24	D12	
25	D14	

Note: TDT recomends the PP16 patch panel for accessing the Digital I/O.

RP2.1 Real-Time Processor

Overview

The RP2 and RP2.1 real-time processors are flexible and powerful signal processing modules for TDT's System 3. The RP2 system consists of an Analog Devices Sharc floating point DSP with surrounding analog and digital interface circuits to yield a powerful programmable signal-processing device capable of handling a variety of tasks.

Power and Communication

The PR2.1 mounts in a System 3 zBus Powered Device Chassis (ZB1PS) and communicates with the PC using any of the zBus PC interfaces. The ZB1PS is UL compliant, see the *ZB1PS Operations Manual* for power and safety information.

Software Control

Software control is implemented with circuit files developed using TDT's RP Visual Design Studio (RPvdsEx). Circuits are loaded to the processor through TDT run-time applications or custom applications. This manual includes device specific information needed during circuit design. For circuit design techniques and a complete reference of the RPvdsEx circuit components, see the *RPvdsEx Manual*.

Features

Memory

The RP2 comes with 16MB of memory for data storage and retrieval. The RP2.1 has 32MB of memory for data storage and retrieval.

Digital Input/Output Bits

The digital I/O circuits include eight bits of digital input and eight bits of digital output that are accessed on the 25 pin connector on the front of the RP2. The eight bits of I/O can be used within the processing chain in a variety of ways including implementing triggers, timing trigger responses, and lighting LEDs. The first four bits of the digital inputs and digital outputs as well as the Trigger/Enable input are mapped to LED indicators on the front panel of the RP2. There is an additional TRIG input BNC on the front panel.

D/A and A/D

The RP2.1 is equipped with two channels of 24-bit, 200 kHz sigma-delta D/A and two channels of 24-bit, 200 kHz sigma-delta A/D.

Sigma-Delta converters provide superior conversion quality and extended useful bandwidths, at the cost of an inherent fixed group delay. See the technical specifications for the group dealy of each device.

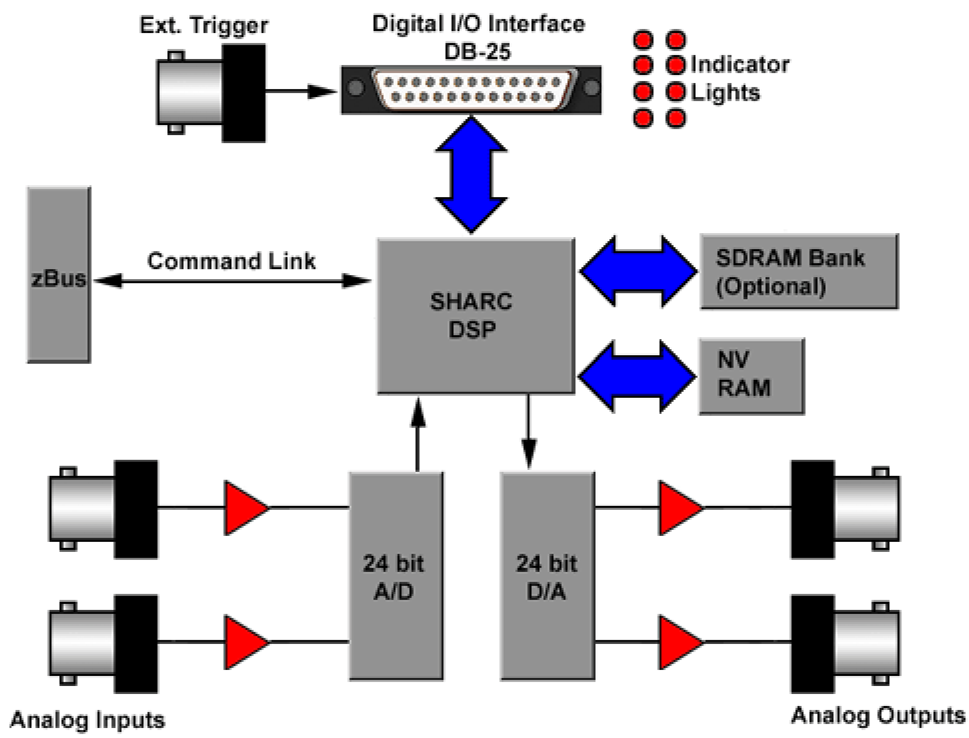
The original RP2 A/D's run at 100 kHz. An Optional RP2-5 (identifiable by its version number only) is equipped with 24-bit 50 kHz A/D and 50 kHz D/A. The RP2-5 device does not have SDRAM.

Hardware

Up to 32MB of SDRAM can be installed for storage of long waveforms and acquired data. An RP2 comes standard with 16MB of SDRAM while an RP2-5 has no SDRAM. All of the RPvdsEx buffer components, used to build circuits for the RP2, utilize the SDRAM memory and therefore will not work when used on an RP2-5 device.

The RP2 communicates with and is programmed through the zBus link.

The RP2 hardware also contains a powerful digital I/O sub-system, offering eight bits of digital input and eight bits of digital output as well as a dedicated trigger input connected to a BNC on the front panel. The first four bits of both input and output port and the trigger input have LED monitors for a quick indicator of bit state. The bits of these ports can be programmed individually or as a 'digital word' and used in a variety of ways within the RP2 processing circuit.

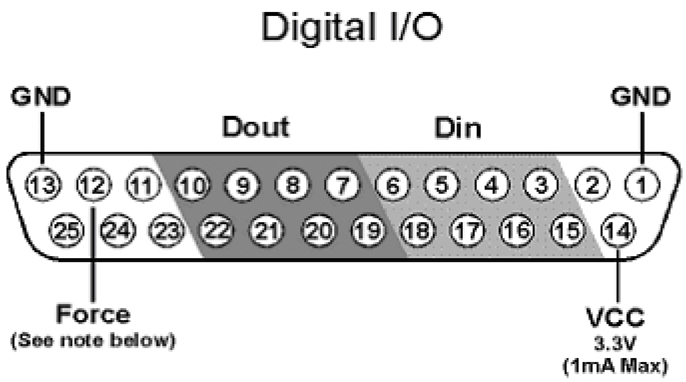


The RP2 is interfaced to the analog world via a two channel 24-bit analog to digital converter and a two channel 24-bit digital to analog converter. The RP2 system's I/O buffer handles +/- 10 Volt signals with excellent signal to noise performance. The RP2 contains a 100 kHz (50 kHz BW) A/D and a 200 kHz (100 kHz BW) D/A, while the RP2-5 has a 50 kHz (25 kHz BW) A/D and D/A. Both devices allow for user programmable sampling rates from the specified maximum down to 6.25 kHz. A special calibration program is used to calibrate the RP2's analog I/O offering very small gain and DC errors.

Real-Time Processor Technical Specifications

DSP	50 MHz Sharc 21065, 150 MFLOPS
Memory	RP2: 16 MB SDRAM RP2.1: 32 MB SDRAM RP2-5 has no SDRAM
A/D	2 channels, 24-bit sigma-delta
Frequency Response	DC-Nyquist (~1/2 sample rate)
S/N (typical)	105 dB (20 Hz to 20 KHz), 95 dB (20 Hz to 50 KHz)
Distortion (typical)	-95 dB for 1 KHz input at 5 Vrms
A/D Sample Rate	RP2.1: 195.312 kHz maximum RP2: 97.656 kHz maximum RP2-5: 48.828 kHz maximum
Sample Delay	RP2.1: 65 samples RP2: 41 samples
D/A	2 channels, 24-bit sigma-delta
Frequency Response	DC-Nyquist (~1/2 sample rate)
S/N (typical)	105 dB (20 Hz to 20 KHz), 95 dB (20 Hz to 50 KHz)
Distortion (typical)	-95 dB for 1 KHz output at 5 Vrms
D/A Sample Rate	RP2.1: 195.312 kHz maximum RP2: 97.656 kHz maximum RP2-5: 48.828 kHz maximum
Sample Delay	RP2.1: 30 samples RP2: 30 samples
Digital Inputs	8 bits + 1 TRIG input
Digital Outputs	8 bits
System Reset	**Force** input (see following section on how to reset)
Input Impedance	10 kOhm
Output Impedance	10 Ohm

DB25 Connector Pin Out



Pin	Name	Description
1	GND	Ground
2	NA	Not Used
3	DI1	Digital Input Bits
4	DI3	
5	DI5	
6	DI7	
7	DO1	Digital Output Bits
8	DO3	
9	DO5	
10	DO7	
11	NA	Not Used
12	Force	Used to reset the RP2.1

Pin	Name	Description
13	GND	Ground
14	VCC	3.3V (1A Max)
15	DI0	Digital Input Bits
16	DI2	
17	DI4	
18	DI6	
19	DO0	Digital Output Bits
20	DO2	
21	DO4	
22	DO6	
23	NA	Not Used
24		
25		

Note: TDT reccomends the PP16 Patch Panel for accessing digital I/O.

Important!:

Force is used to reset the RP2.1, including deleting the device's microcode. It has no function in data acquisition or manipulation.

To reset the device:

1. Connect a wire (or paper clip) from pin 12 to pin 13 on the Digital I/O port.
2. With pins 12 and 13 shorted, open the RPProg System 3 Device Programmer and select the device type (RP2) and interface in the *#1 Connection* group.
3. If necessary, select the desired device ID in the *#2 Erase* group. When the device is selected the device name in the #3 Program group will be similar to "*G21K_(1)*".
4. Next click the **Browse** button next to the *uCode File* field and select **RP21.dxe**.
5. Remove the short from pins 12 and 13, and click the **Program Device!** button.
6. Do not use your computer until the device reprogramming is complete (approximately five minutes).

RV8 Barracuda

Barracuda Overview

The Barracuda features include nanosecond accurate event-timing, fast DAC's for high frequency stimulus presentation and user control of sample frequencies. In addition the Barracuda gives users precise control over stimulus presentation. The system has 16-digital inputs and 8-digital outputs.

Power and Communication

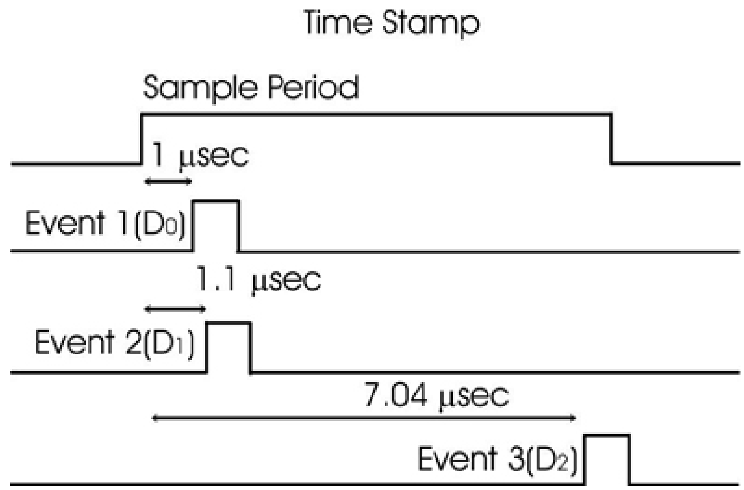
The RA16 mounts in a System 3 zBus Powered Device Chassis (ZB1PS) and communicates with the PC using any of the zBus PC interfaces. The ZB1PS is UL compliant, see the *ZB1PS Operations Manual* for power and safety information.

Software Control

Software control is implemented with circuit files developed using TDT's RP Visual Design Studio (RPvdsEx). Circuits are loaded to the processor through TDT run-time applications or custom applications. This manual includes device specific information needed during circuit design. For circuit design techniques and a complete reference of the RPvdsEx circuit components, see the *RPvdsEx Manual*.

Nanosecond Event-Timing

The Barracuda is a nanosecond accurate event timer. The TimeStamp component uses the high-speed clock on the system to record when a TTL event occurred during a sampling period. This means that event times are independent of sample rate. When an event occurs the TimeStamp sends out the time in microseconds from the start of that sample period. At the end of each sample period the event timer is reset to zero. In the figure below three events occurred during a sample period of ten microseconds. For each digital input a unique time stamp is recorded for that sample period.



Fast Digital-Analog Converters

The Barracuda ships with PCM DAC's with up to 500 kHz sample rate. The fast DAC's can be used for high frequency presentations. In addition the Barracuda's PCM DAC's give users precise control over voltage outputs for microelectrode stimulation.

Variable Sample Frequency

The Barracuda allows users to set the sample period in 40 nanosecond steps. Users can select sample frequency from 10 to 500,000 Hz.

User Control of System Devices

The Barracuda has two control modes: Free-run and Triggered. In Free-run mode the circuit runs continuously and gating functions are required to control the signal outputs and inputs. In Trigger mode the circuit only runs after it has been triggered. It then runs for a set number of samples and then stops. The system can be triggered once or multiple times. The circuit must be reset before it can trigger again. Gating functions are not required for turning on and off stimuli.

Additional Features

To simplify signal synchronization it is possible to send out the sample clock and the system clock (50 MHz) on the digital outputs. Users can also send out the sample clock period.

Barracuda Features

Trigger

Takes an external TTL pulse and triggers components (free run mode) or triggers the circuit (trigger mode).

Status Lights

The status lights indicate the state of the RV8. Armed, Running, DC (DoCount), and FreeRun. Combinations of the status light describe the state of the RV8.

Free Run Mode	Free Run Mode w/Circuit Running	Trigger Mode	Trigger Mode with System Armed	Trigger Mode with System Running:
				

Digital Input Lights

Lights are on when there is a TTL pulse on the digital input line. Pulse times may be too brief to see in many cases. Only channels 0-7 have indicator lights.

Digital Output Lights

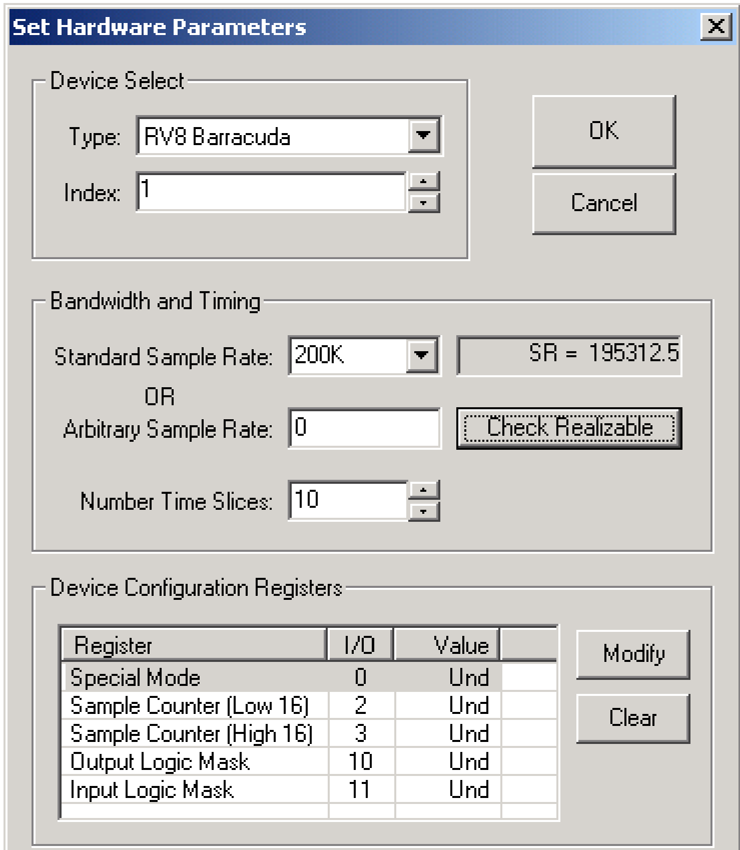
Lights are on when a TTL pulse is sent out of a digital output line. All eight channels (0-7) have a TTL indicator light.

25-pin Connector for Digital Inputs and Outputs

A 25-pin connector gives access to all 24 channels of digital I/O. The pin outs for the connector are shown in the technical specifications, page 3-16. TDT provides the PP16 with 24 connectors to give users easy access to all the digital output channels of the Barracuda.

Barracuda Device Setup

The Barracuda has several additional features not found in other RP devices. An expanded dialog box opens after selecting the RV8 option.



Bandwidth and Timing

Standard Sample Rates are in powers of two from 6 kHz to 400 kHz. The actual sample rate is given in the box to the right.

Arbitrary Sample Rate can be from 10 Hz to 500,000 Hz. In the Arbitrary Sample Rate box type a number between 10 Hz and 500,000 Hz. To reset to the Standard Sample Rates type 0 in the Arbitrary Sample Rate box. To determine the true sample rate click **Check Realizable**. The sample rate is based on the system clock (25 MHz) or a sample period of 40 nanoseconds ($40 * 10^{-09}$). To calculate the true sample rate, take the reciprocal of the required sample period in seconds.

Device Configuration Parameters

The device configuration parameters allow RPvdsEx access to unique features on the RV8. To access a particular parameter either double-click on the parameter name or click on the parameter and click the **Modify** button. To reset the parameter value to the default mode click **Clear**.

Special Mode

The Special Mode is a bit-masked value that determines which features of the Barracuda are activated. The default mode for the Special Mode is zero. This makes the system behave like other RP devices. There are seven modes that are accessed through the bit-mask shown below. Special Mode can be accessed with the ActiveX controls SetDevCfg and GetDevCfg.

Bit-number	Enabled Value	Name	Function
0	1	DoCount	Sets up system to run under trigger mode.
1	2	AutoClr	Clears the DAC out buffers after a trigger event.
2	4	TickOut	Sends a pulse at the beginning of each tick period on Digital Out 7. Pulse length is 40 nanoseconds.
3	8	ClkOut	Sends pulses at 1/2 the clock frequency (25 MHz).
4	16	UseZTRGA	Starts the Barracuda when a ZtrgA goes high. Only works in the trigger mode (must also have bit-number 1 enabled).
5	32	UseZTRGB	Starts the Barracuda when a ZtrgB goes high. Only works in the trigger mode (must also have bit-number 0 enabled).
6	64	UseEXTR	Starts the Barracuda using the external trigger. Only works in the trigger mode (must also have bit-number 0 enabled).
7	128	MTRIG	Enables multiple trigger mode. Users can repeatedly trigger the Barracuda without stopping and rerunning the circuit. 0=Very Large Number of Triggers

The Special Modes are set with a bit-masked pattern. For example, to set the trigger mode using a zTRGA the value for the Special Mode would be set to 1 + 16 or "17". To use the Mtrig function the value would be 1 (DoCount) + 16 (UseZTRGA) + 128 (MTRIG) or "145".

DoCount

Enable DoCount to use the trigger mode. If this is not enabled then the device is in free-run mode.

AutoClr

AutoClr works in trigger mode. AutoClr clears the output of the DAC's to zero after the last value is played. Otherwise the output of the DAC is set to the last value converted.

Trigger Mode

In trigger mode the circuit only runs after it has been triggered. After a trigger it runs for set number of samples and then stops. Using the trigger mode requires three steps:

1. Set the value of the Special Mode parameter.
2. This value is a bit-masked value. To calculate the value needed sum the individual bit-masks (see above). The bit-masks include DoCount (1) the trigger mode (16, 32 or 64 depending on what trigger option) and possibly enabling MTRIG (128).
3. Determine the number of samples that the circuit runs. The Barracuda can play out over 4 Gsamples ($4*10^9$ samples) on one trigger. Sample Counter (Low 16) sets the sample number between 0 and 65535 Sample Counter (High 16) sets it between 65536 and a large number. For example, to play out 80000 samples the Sample Counter (High 16) would be set to 1 (65,536) and Sample Counter (Low 16) to 14,464.
4. Load and trigger the circuit.

Sample Count Options

Sample count parameters set the number of samples the circuit will run. The Sample Counter (Low 16) values are between 0 and 65536 (lower 16-bits of data). Sample Counter (High 16) values are multiples of 65536. For example, a value of 2 in Sample Counter (High 16) will cause the circuit to run for 131,072 samples. If the system needed to run for 200,000 samples you would set Sample Counter (High 16) = 3 (196,608 samples) and Sample Counter (Low 16) = 3,392.

Sample count is only used when in trigger mode. At all other times the circuit is free running.

Sample Counter (Low 16) = the lower 16bits of the sample counter (0-65535)

Sample Counter (High 16) = the upper 16bits of the counter. A value of 1 in Sample Counter (High 16) = 65536.

Logic

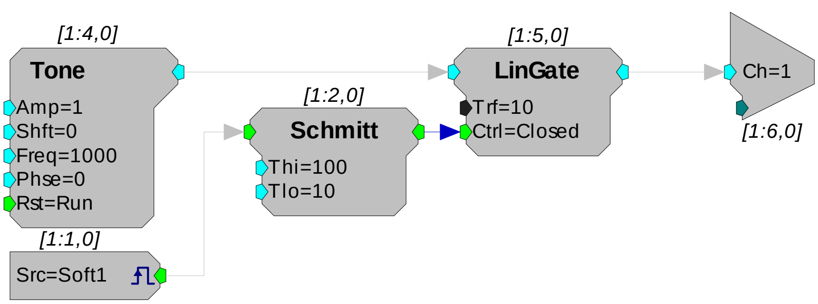
User selects whether a high voltage on a digital line is a logical 1 or logical 0 on the Barracuda.

The default state for a high voltage on a digital line is 1 (high true). Setting InLogic = 1 inverts the logic (low true) and makes a high input voltage produce a 0 and a low input voltage produce a 1. Similarly, when setting OutLogic = 1, a high voltage on a digital output line will produce a 0 and a low voltage will produce a 1.

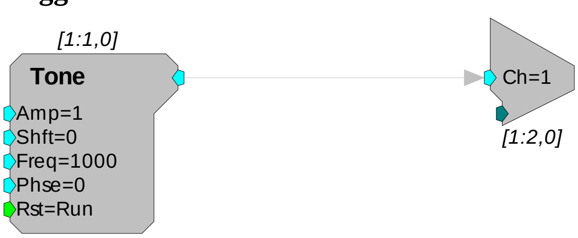
Software Control

The Barracuda has two modes: free-run and trigger. In free-run mode the circuit is always running and signals are constantly generated, acquired, and filtered. In the trigger mode the circuit runs for a set length each time it is triggered. The advantage of the trigger mode is that some circuit design is simplified. The example below shows two circuits that present a tone burst of 100 milliseconds. The first circuit works under the free-run mode and the second with trigger.

Free-Run Mode



Trigger Mode



The first circuit requires three additional components: LinGate gates the output on and off, Schmitt opens and closes the gate and Src (Soft1) starts the Schmitt trigger. The second circuit requires that the Barracuda be controlled from the trigger mode. Trigger mode is accessible within RPvdsEx or from the ActiveX controls.

TimeStamp

The TimeStamp component is unique to the Barracuda. The event-timer, with its submicrosecond accuracy, is independent of the sample period. This allows users to have separate control of both slow processes, such as button presses, and fast events, such as neural activity, all on one circuit with little or no loss of processing power.

PCM DAC Outs

The PCM DACs have a sample delay of only 2 samples. This makes them ideal for use with time critical presentation of signals. These DACs are excellent for neurophysiological stimulation for examining motor behavior.

Multiple Triggering

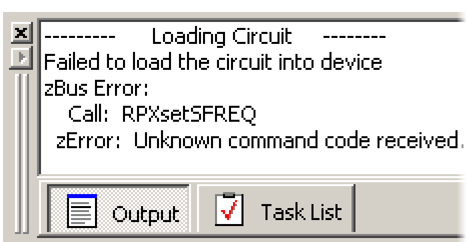
Multiple triggers allow users to repeatedly trigger the Barracuda without resetting (Halting and then Running the chain). To use multiple triggering with RPvdsEx add the bit-masked value of 128 to the Special Mode value. For example, to configure the Barracuda for multiple triggering from the zBUSTrigA, you would set the value to 1 (Trigger Enabled) + 16 (ZbusTRIGA) + 128 (multiple triggers). RPvdsEx has no way to control the number of presentations.

To generate an RPvdsEx circuit for multiple triggering, use the Setup Device command on the Implement menu to open the Set Hardware Parameters dialog box, then modify the Special Mode register. Use the bit-masked values for the Special Mode to make a circuit trigger off either the zBUS or external trigger. In general this will be 1(trigger mode enabled) + (trigger type) + 128 (mTrig enabled).

The multiple trigger does not require the addition of the trigger component. The circuit runs when the trigger pulses high. The RPvdsEx circuit will trigger for a near infinite number of times before stopping.

Arbitrary Sample Rates

The Barracuda is the only System 3 module that has arbitrary sample rates. To set the arbitrary sample, click **Device Setup** on the **Implement** menu, and then set the sample rate in the Arbitrary Sample Rate box. To check the true sample rate, click **Check Realizable**. This will display the true sample rate. Sample periods are in increments of 40 nanoseconds. To calculate the true sample rate determine the sample period in seconds that you require and then divide by 1/(sample period). These circuits work only with the Barracuda. If the circuit is run on a different RP module it will give the following error:

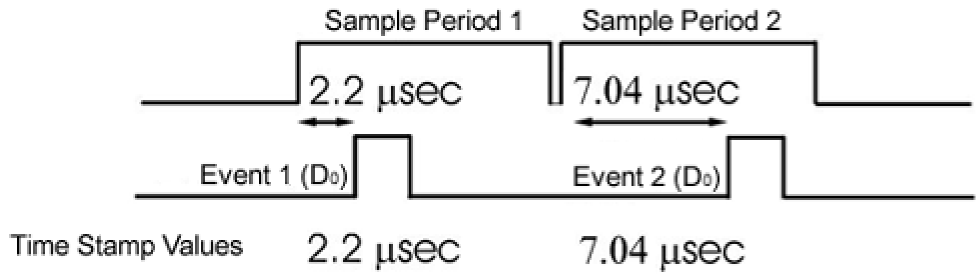


RP Control Object files (RCO) will produce similar problems. If you attempt to run an RCO file (compiled RPvdsEx files for use with ActiveX controls and turn-key software programs) that has an arbitrary sample rate on another RP device the same error will occur.

Using the TimeStamp Component

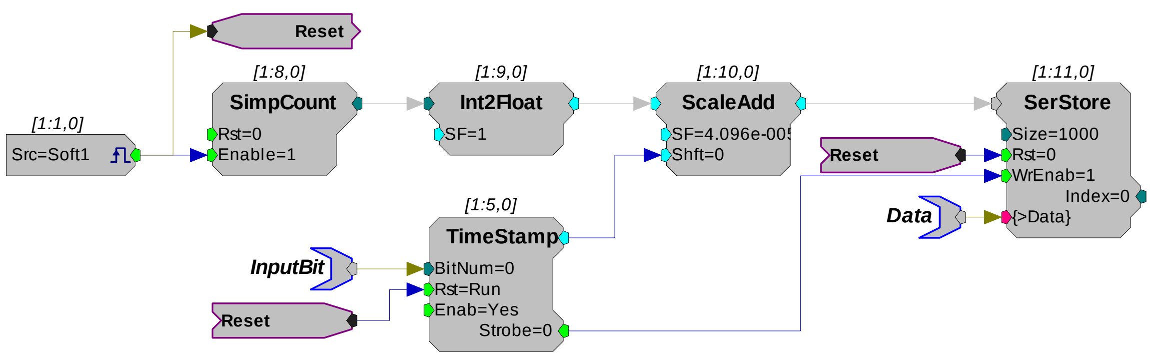
The TimeStamp component is an event timer with submicrosecond accuracy. With other RP systems the resolution of the TimeStamp is no better than the sample clock period. TimeStamp uses the system clock to determine when, within a sample period, the event occurred. After each sample period the TimeStamp component is reset.

The diagram below shows how TimeStamp works. The first event occurs 2.2 microseconds after the start of the first sample period so a value of 2.2 is generated. The second event occurs 7.04 microseconds after the start of the second sample period so a value of 7.04 is generated.



The circuit below saves the event time (in microseconds) to a SerStore buffer. The circuit has two parameter tags: *InputBit* and *data*. The *InputBit* tag sends the digital input channel number (to which the Event trigger will be sent) to the TimeStamp. This determines which of the Barracuda's digital input lines will be monitored for triggers. The *data* tag reads the stored event-time data to a PC buffer.

A software trigger resets the SimpCount, starting the clock, and will also reset the TimeStamp component and the SerStore buffer. The SimpCount increments the count value at every sample tick. The ScaleAdd divides the SimpCount output by the sample period (40.96 microseconds) to keep track of the time in milliseconds. When an event is detected, the TimeStamp output is added to the SimpCount output to get the event time in microseconds.



ActiveX

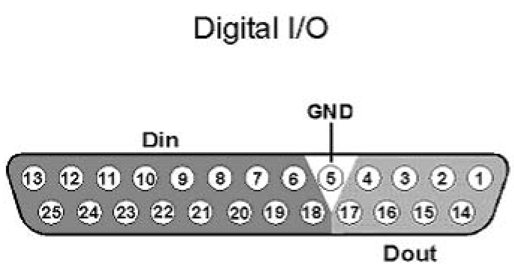
The Barracuda uses two additional ActiveX methods SetDevCfg and GetDevCfg. Detailed information about them is included in the ActiveX help.

Barracuda Technical Specifications

Specifications for the RV8 Barracuda Processor.

DSP	50 MHz Sharc 21065, 150 MFLOPS
Memory	32MB SDRAM
Digital Inputs	16 bits + 1 TRIG input
Digital Outputs	8 bits
Input Impedance	10 kOhm
Output Impedance	10 Ohm

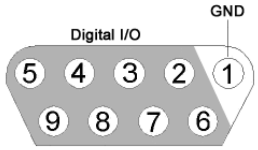
DB25 Connector Pin Out



Pin	Name	Description
1	Do0	Digital Output Channels
2	Do2	
3	Do4	
4	Do6	
5	GND	Ground
6	Di1	Digital Input Channels
7	Di3	
8	Di5	
9	Di7	
10	Di9	
11	Di11	
12	Di13	
13	Di15	

Pin	Name	Description
14	Do1	Digital Output Channels
15	Do3	
16	Do5	
17	Do7	
18	Di0	Digital Input Channels
19	Di2	
20	Di4	
21	Di6	
22	Di8	
23	Di10	
24	Di12	
25	Di14	

Option I/O DB9 Connector Pin Out



Pin	Name	Description
1	GND	Ground
2	A1	Analog Channels
3	A2	
4	A3	
5	A4	
6	A5	
7	A6	
8	A7	
9	A8	

Part 4 RM Mobile Processors

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RM Mobile Processors

The RM Family

The System 3 platform includes two self-contained real-time processors: the Mini Processor and the Mobile Processor. Designed as an affordable test-bed system for designing and debugging RPvdsEx circuits, each device includes stereo A/D and D/A, an adjustable onboard speaker, and can drive headphones at up to 100 dB SPL. The devices draw power from the USB interface of the computer and work well with laptop computers for maximum portability. These economical mobile systems can also be used for basic psychoacoustics.

For detailed information on each member of the RM family check the technical specifications of the module.

Power Requirements

Power is provided across the USB connection to a host PC. The RM draws approximately 300 mAmps from a 6 Volt input. The draw on a portable PC battery will depend on the power requirements of the portable PC and the properties of the battery. In many cases, the user may see less than 10% decrease of the battery life.

Users can attach an external power supply such as an AC adapter (supplied with the system) or an external pack such as a motorcycle battery (input range of 6-9 Volts).

Software Control

Software control is implemented with circuit files developed using TDT's RP Visual Design Studio (RPvdsEx). Circuits are loaded to the processor through TDT run-time applications or custom applications. This manual includes device specific information needed during circuit design. For circuit design techniques and a complete reference of the RPvdsEx circuit components, see the *RPvdsEx Manual*.

Mobile Processor Hardware

The RM1 Real-time Mini Processor and RM2 Mobile Processor combine a signal processor, a power supply, and a computer interface in one small form factor. The RM consists of an Analog Devices Sharc floating point DSP with surrounding analog and digital interface circuits and 32 MB of memory for data storage and retrieval. The RM2 also includes a fiber optic connection for the RA4/RA16PA Medusa amplifier.

D/A and A/D

The RM is equipped with stereo 24-bit sigma-delta A/D and D/A that can sample at rates up to 97.656 kHz. Sigma-delta converters provide superior conversion quality and extended useful bandwidths, at the cost of an inherent fixed group delay. For the RM1 and RM2, the DAC Delay is 17 samples and the ADC Delay is 16 samples.

Digital Input/Output Bits

The TTL I/O circuits include four bits of digital input and four bits of digital output that are accessed via the 9-pin connector on the back of the RM. BitO can also be accessed through a BNC connector on the front panel. The RM's digital I/O can be used to implement triggers, time trigger responses, and light LEDs.

Analog Output

The RM is equipped with an external speaker for use when previewing stimulus during the circuit design process. The RM's stereo analog output can drive a headphone at up to 100 dB SPL.

USB Input Port

An USB Input port allows multiple devices to be connected for increased processing power.

Mobile Processor Front Panel Features

Bit0

The BNC connector for Bit0 allows for a direct input or output to the first bit of the RM device. This allows for a more convenient connection for a typical trigger input. Access to the other digital inputs and outputs are from a 9-pin connector on the back panel.

Status Lights

The status lights indicate the state of the RM.

Power

The power light indicates that the device is connected to a power supply. The power may be supplied by an external power supply (included) or by a computer (powered on) via the USB interface.

Comm (Communication)

The communication light blinks when the device is sending or receiving information to or from the PC. (This requires the system to be connected to a PC.)

Err (Error) or Amp (RM2)

The error light indicates one of the following:

An error communicating with the host PC.

An error communicating with the RA4/RA16PA (RM2 Only)

Status

The status light blinks when a circuit is running. The rate at which the light blinks is a general indicator of cycle usage, with faster blinking indicating a higher cycle usage.

Bits Lights

Bit lights indicate when a bit input is set high. The LED(s) will light if the input signal is set high or if the output bit is set high. Voltage high is 3.3 volts and voltage low is nominal 0 Volts. Access to the digital I/O port is through a 9-pin connector on the back panel. The Bit In's are set logical high by default.

Analog I/O

The analog inputs and outputs use a 1/8" stereo plug and deliver or accept a +/- 1 Volt signal with a dynamic range of over 45 dB. The RM uses 24-bit Sigma-delta A/D and D/A converters.

In

The maximum analog input is +/- 1 Volt with a peak sample rate of 97.656 kHz. The input impedance is 10 kOhm.

Out

The maximum analog output is +/- 1 volt with a peak sample rate of 97.656 kHz. The low-level output impedance (10 Ohm) of the system allows users to drive earphones at up to 100 dB SPL. Because of the 0.16 Hz high pass filter on the D/A converter, the RM cannot play out DC or very low frequency (<1 Hz) signals.

Level

The RM has an internal speaker that is driven by channel 1 output. The Level knob controls the volume of the speaker and analog channels 1 and 2 when connected to the 1/8” audio jack labeled OUT. To achieve the full output level specified in your circuit on these two channels, set the Level knob to Max.

Mobile Processor Back Panel Features

USB In

The USB input on the RM acts as a USB hub. Multiple RM devices can be ganged together to increase signal processor power. A standard USB, A to B, cable is required for setup.

USB Out

The USB output connects either to another RM device, a UB4, or to the host computer's USB interface. The RM can be connected to PCs with either USB 1.1 or USB 2.0 hubs.

Digital I/O

The female DB-9 connector allows direct access to the digital inputs and outputs. Pinout information is provided on the label above the connector. Bits 0 - 3 (which map to pins 5, 9, 4, and 8 on the male DB-9 connector) are inputs and bits 4 - 7 (which map to pins 3, 7, 2, and 6 on the male DB-9 connector) are outputs. Ground is labeled G (which maps to pin 1 on the male DB-9 connector).

Note: The digital lines drive about 25 milliamps.

Amplifier (RM2 only)

A fiber optic connector is found on the RM2 for use with the Medusa RA4/RA16 preamplifier, the Loggerhead RA8GA, and the associated headstage assemblies.

Ext. Pow. (External Power)

An external power supply can be used as an alternative to drawing power from the USB connection. An adapter is supplied with the device allowing the device to be powered form an AC power source. A battery with an output range of 6-9 volts, such as a motorcycle battery, could also be used to power the device.

TDT recommends separate external power sources when using multiple RM devices.

Mobile Processors Digital Input/Output

The Mobile Processors are equipped with 8 bits of programmable digital input/output, accessed via the Digital I/O 9 pin connector on the back panel. See the *Mobile Processor Technical Specifications* for a pinout diagram.

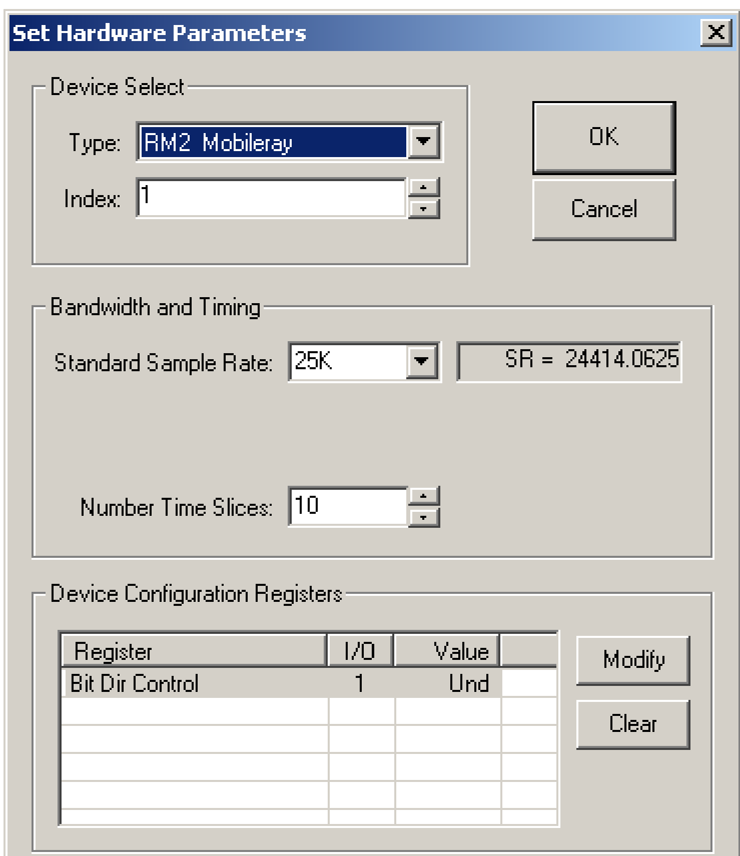
Note: The digital lines drive about 25 milliamps.

Configuring the Programmable I/O Lines

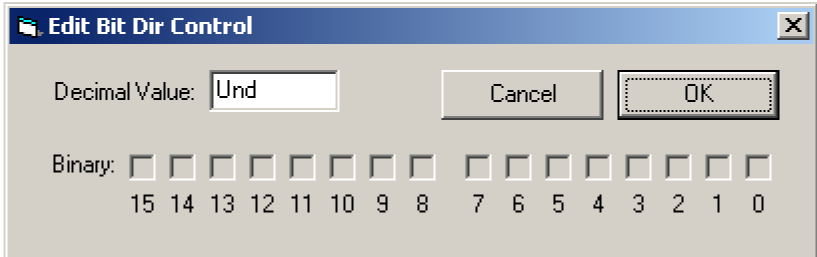
All 8 digital lines are independently configurable as inputs or outputs. By default, bits 0-3 are configured as inputs and bits 4-7 are configured as outputs. In RPvdsEx, bits 0-7 in the bit configuration register control the configuration of the eight addressable bits as inputs or outputs. Setting a bit to one will configure that bit as an output.

To access the bit configuration register:

1. Click the **Device Setup** command on the **Implement** menu.
2. In the **Set Hardware Parameters** dialog box, click the **Type** drop-down box and select **RM1** or **RM2** from the list.
3. The dialog expands to display the **Edit Bit Dir Control** dialog box.



4. Click **Modify** to display the **Edit Bit Dir Control** dialog box.



5. In this dialog box, a series of check boxes are used to create a bitmask that is used to program all bits.
6. To enable the check boxes, delete **Und** from the **Decimal Value** box.
7. To determine the desired value, select or clear the check boxes. By default, all check boxes are cleared (value = 0) and all bits are configured as inputs. Click the check boxes for desired bits (0 -7) to set the bit to one and configure that bit as an output.
8. When the configuration is complete, click **OK** to return to the **Set Hardware Parameters** dialog box.

Using the RM2 Fiber Optic Port

The RM2 Fiber Optic Port can be used with a Medusa or Loggerhead preamplifier; however, it is unlikely that a single RM2 device can acquire 16 channels of high frequency activity. Instead we recommend that the RM2 be used for low channel count (up to four channels) high sample rate acquisition or for high channel count low sample rate activity (e.g. 16 channels of slow EEG activity). Using the RM2 as part of a Medusa/Loggerhead system effectively provides two channels of high quality A/D inputs and up to 16 channels of signal input running at 25 kHz. The signal input lines accessed via the analog I/O and fiber optic port are mapped as described below to allow for simultaneous use of the high quality A/D and the amplifier input channels.

	RM2 Channel		**RM2 Channel**
Analog I/O Input Channel 1	Channel 1	**Amp Channel 8**	Channel 24
Analog I/O Input Channel 2	Channel 2	**Amp Channel 9**	Channel 25
Amp Channel 1	Channel 17	**Amp Channel 10**	Channel 26
Amp Channel 2	Channel 18	**Amp Channel 11**	Channel 27
Amp Channel 3	Channel 19	**Amp Channel 12**	Channel 28
Amp Channel 4	Channel 20	**Amp Channel 13**	Channel 29
Amp Channel 5	Channel 21	**Amp Channel 14**	Channel 30
Amp Channel 6	Channel 22	**Amp Channel 15**	Channel 31
Amp Channel 7	Channel 23	**Amp Channel 16**	Channel 32

For more information about the medusa, see the *RA16 Medusa Amplifier*, page 5-20.

Software Control for the Mobile Processor

In general, the RM processors can use any circuit that has been designed for the RP2.1. There are a few caveats that relate to the number of digital inputs and outputs, the positioning of the input channels from the fiber optics on the RM2, and the maximum signal voltage.

Digital I/O

The RM has only eight digital I/O channels. Circuits that use more than four TTL outs or four TTL ins will not work with the RM.

RM2 Acquisition Channel Input

The channels from the preamplifier to the RM2 are mapped so that the system can acquire from both the high quality analog inputs and the preamplifier. For acquisition channels across the fiber optic connection, channel numbers are offset by 16. Channel one from the preamp maps to channel 16 of the RM2, channel two maps to 17, and so forth. Users must modify existing circuit designs and OpenEx files by setting an offset value to match the channel organization of the RM2.

There is no fiber optic repeater to allow multiple RM2s to be linked for data acquisition from a single preamplifier. All acquisition from the preamplifier must take place on a single RM2.

Signal Voltage

The maximum signal voltage for acquisition and presentation is +/- 1 volt. Circuits that have components generating signals greater than +/- 1 volt will cause the device to clip either on input or output.

Mobile Processor Technical Specifications

Technical specifications for the RM1 and RM2 processors.

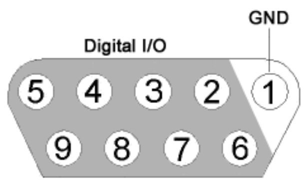
DSP	50 MHz Sharc 21065, 150 MFLOPS
Memory	32 MB
A/D	2 channels 24-bit sigma-delta A/D
S/N (typical)	85 dB (20 Hz to 20 kHz)
Distortion (typical)	80 dB for 1 kHz input at 630 mV rms
Sample Delay	16 samples
D/A	2 channels 24-bit sigma-delta D/A
S/N (typical)	85 dB (20 Hz to 20 kHz)
Distortion (typical)	80 dB for 1 kHz input at 630 mV rms
Sample Delay	17 samples
Highpass Filter	0.16 Hz

Digital I/O	8 user selectable
System Reset	Front panel next to ERR light
Input Impedance	10 kOhm
Output Impedance	10 Ohm

RM2 Fiber Optic Inputs

Input	up to 16 channels
Sampling Rate	24.414 kHz max

Digital I/O DB9 Female Connector Pin Out



Pin	Name	Description
1	GND	Ground
2	D6	Digital Input/Output Channels
3	D4	
4	D2	
5	D0	
6	D7	
7	D5	
8	D3	
9	D1	

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Part 5 Preamplifiers

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PZ2 Preamplifier

Overview

The PZ2 is a high channel count preamplifier suitable for extracellular recordings. The PZ2 preamplifier features a custom 18-bit hybrid A/D architecture that offers the advantages of Sigma-Delta converters at significantly lower power and a fast fiber optic connection capable of simultaneously transferring up to 256 channels. The extended bandwidth offered by this connection supports sampling rates up to ~50 kHz and improves signal fidelity, spike discrimination, sorting, and analysis. Used exclusively with Z-Series base stations, PZ2 preamplifiers are available in 32, 64, 128, or 256-channel models.

Note: When sampling at a rate of ~50 kHz only the first 128 amplifier channels will be available.

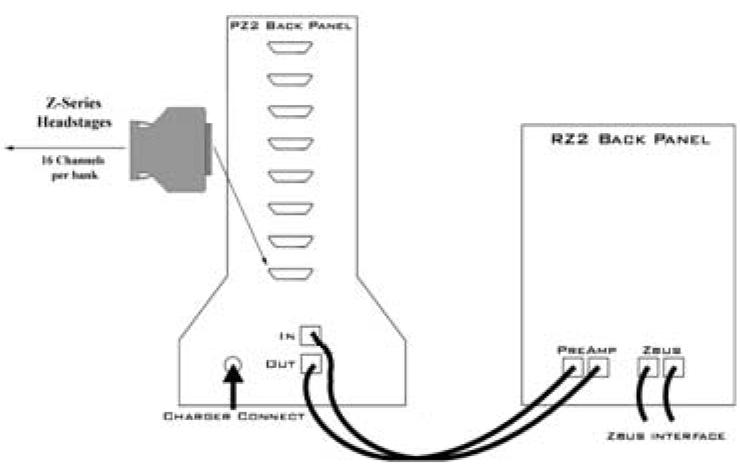
System Hardware

All PZ2 channels are organized into groups of 16 channel ***banks*** with each bank corresponding to a rear panel headstage connector and front panel LED display. Recorded signals are digitized, amplified, and transmitted to the RZ2 base station via a single fiber optic connection for further processing. In addition, configuration information is sent from the RZ2 to the PZ2 preamplifier across the fiber optic connection.

A standard configuration for neurophysiology recordings includes electrodes (chronic or acute), one or more Z-Series high impedance headstages, a PZ2 preamplifier, and an RZ2 base station.

Hardware Set-up

The diagram below illustrates the connections necessary for PZ2 preamplifier operation.



One or more Z-Series headstages can be connected to the input connectors on the PZ2 back panel.

A 5-meter paired fiber optic cable is included to connect the preamplifier to the base station. The connectors are color coded and keyed to ensure proper connections.

The PZ2 battery charger connects to the round female connector located on the back panel of the PZ2 preamplifier.

Important!: To avoid introducing EMF noise, DO NOT connect the charger to the PZ2 while collecting data.

Powering ON

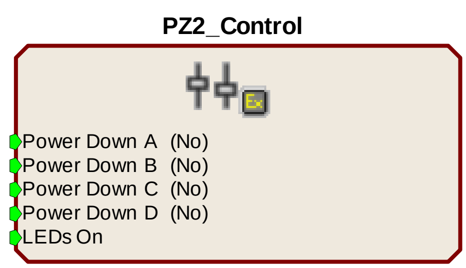
- To turn the preamplifier on, move the three position battery switch located on the front panel of the PZ2, to either the Bat-A or Bat-B position.

Powering OFF

- To turn the preamplifier off, move the three position battery switch located on the front panel of the PZ2, to the OFF position.

Important Note: Channels are grouped by 16-channel banks and each bank will only power up when a headstage is connected. This design helps to increase battery life.

PZ2 Software Control



The preamplifier's hardware operation (power options and indicator LEDs) can be configured using the ***PZ2_Control*** macro within the RPvdsEx control circuits running on the RZ2 base station.

Double-clicking the macro in RPvdsEx displays the macro properties and allows users to easily configure the macro. Additional information on using the macro is available in the macro properties dialog box.

This macro is not required for preamplifier operation but is recommended if the user requires more control over the amplifier power/up or power/down status or front panel LEDs. See the relevant sections below for more information about these features.

PZ2 Features

Clip Warnings and Activity Display

256 front panel LEDs can be used to indicate spike activity and/or clip warning depending on display mode and configuration. See *Display Button and Status LED* below for more information.

Recording Channel LEDs:

When enabled, LEDs for each channel may be lit green to indicate activity or red to indicate a clip warning.

Green: Activity | Red: Clip Warning

Clip Warning When the input to a channel is greater than -3dB from the preamplifier's maximum voltage input the LED for the corresponding channel is lit red indicating clipping may occur.

Activity Whenever a unit (spike) occurs (the sensitivity threshold can be configured with the PZ2_Control macro) the LED for the corresponding channel is lit green.

Note: The LED Indicators are also mirrored on the RZ2 LCD display.

Display Button:

The Display button located on the front panel of the PZ2 toggles the clip warning and activity display LEDs between software control and standard operation.

To toggle between display modes:

- Press the **Display** button.

Status LED:

When recording, the **status LED** located below the Display button indicates the current display mode of the LED Indicators.

Green Software Control of LEDs

Use the PZ2_Control macro to configure LED Indicators. LEDs are turned off until enabled through software control.

Orange LEDs enabled for standard operation

In this mode, LEDs are automatically enabled for default activity and clip warning display as described above.

Battery Overview

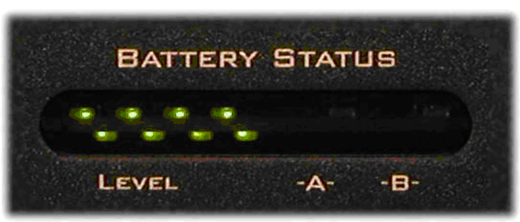
The PZ2 preamplifier features two Lithium ion batteries to allow for longer record times. A three-position switch selects the active battery between Bank-A, Bank-B, or both banks off.

Maximizing Battery Life

To increase battery life, individual banks of channels will only power up when a headstage is connected to the corresponding input.

The PZ2_Control macro can also be added to the circuit running on the RZ2 to further specify how PZ2 channel banks are powered. When a headstage is connected, banks may be powered on or off statically through the **Power Control** options within the macro or dynamically by using the **PZ2_Control macro** inputs. See the internal macro help for more information.

Battery Status LEDs



Battery Level: Eight LEDs indicate the voltage level of the selected battery. These LEDs can be found on the front of the PZ2 preamplifier by the heading Level. When the battery is fully charged, all eight LEDs will light green. When the battery voltage is low, only one green LED will be lit. If the voltage is allowed to drop further, the last LED will flash red. TDT recommends charging the battery before this flashing low-voltage indicator comes on. While charging, the Level LEDs will flash green.

Status	**Description**
8 Green	Fully Charged
1 Green, 7 Unlit	Low Voltage
1 Flashing Red	Low Voltage - Charge Immediately!
8 Green Flashing	Charging in Progress

Charging the Batteries

Operate the preamplifier with the charging cable disconnected. Connecting the PZ2 charger will simultaneously charge both batteries. TDT recommends putting the three-position switch in the **OFF** (middle) position while charging the PZ2.

Charging Indicators: When powered on the PZ2 battery status LEDs are also used for each battery to indicate which battery, if any, is charging. These LEDs are found next to the Level LEDs by the headings -A- and -B-. A green indicator denotes the battery bank is fully charged while a red indicator designates the battery is currently charging. When the device is in operation (charger is not connected) the -A- and -B- LEDs are not lit.

Status	**Description**
Red	Charging
Green	Fully Charged
Unlit	Operation Mode (charger not connected)

PZ2 Technical Specifications

Technical specifications for the PZ2 Z-Series Preamplifier.

A/D	Up to 256 channels, 18-bit hybrid
Maximum Voltage In	+/- 10 mV
Frequency Response	3 dB: 0.35 Hz – 7.5 kHz 6 dB: 0.2 Hz – 8.5 kHz
Anti-Aliasing Filter	4^{th} order Lowpass (24 dB per octave)
S/N (typical)	73 dB
Distortion (typical)	< 1%
A/D Sample Rate	Up to 48828.125 Hz*
Input Impedance	10^5 Ohms
Power Requirements	2 Lithium Ion cells at 10 AmpHours each
Battery	Eight hours to charge both batteries Battery life between charges per battery: 32 ch ~ 13 hrs 64 ch ~ 11 hrs 128 ch ~ 8 hrs 256 ch ~ 5 hrs
Charger	External 6VDC, 3A power supply
Indicator LEDs	Up to 256 status or clip warning, battery life, active battery bank
Input inferred noise	2μV rms typical 300- 7000Hz, 8μV peak typical
Fiber Optic Cable	5 meters standard, cable lengths up to 20 meters**

***Note:** When sampling at a rate of 48.828 kHz the PZ2 preamplifier is limited to a maximum of 128 channels.

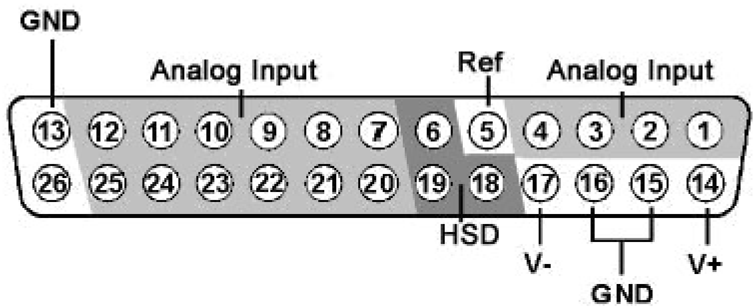
****Note:** If longer cable lengths are required, contact TDT.

Input Connectors

PZ2 Preamplifiers have up to 16, 26-pin headstage connectors on the back of the unit. A1 – A16 represent the 16 channels coming from each connected headstage. The PZ2 channels are marked next to the respective connector on the preamplifier. So, for the connector for channel 1 – 16, A1 is channel 1 while on the connector for channels 17 – 32, A1 is channel 17.

Important!: Each input connector uses its own unique ground and reference. When using multiple headstages, ground pins on all headstages should be connected together to form a single common ground. See the *Headstage Connection Guide*, page 5-33 for more information.

Pinout Diagram



Pin	Name	Description
1	A1	Analog Input Channels
2	A2	
3	A3	
4	A4	
5	Ref	Reference
6	HSD	Headstage Detect
7	A5	Analog Input Channels
8	A7	
9	A9	
10	A11	
11	A13	
12	A15	
13	GND	Ground

Pin	Name	Description
14	V+	Positive Voltage
15	GND	Ground
16	GND	
17	V-	Negative Voltage
18	HSD	Headstage Detect
19	HSD	
20	A6	Analog Input Channels
21	A8	
22	A10	
23	A12	
24	A14	
25	A16	
26	NA	Not Used

Note: Do not attempt to make any custom connections to pins 6, 18, or 19. These pins are intended for TDT use only.

PZ3 Low Impedance Amplifier

Overview

The PZ3 is a high channel count, low impedance amplifier well suited for ECOG, Evoked Potentials, EEGs, LFP's, EMGs, and other similar recording applications.

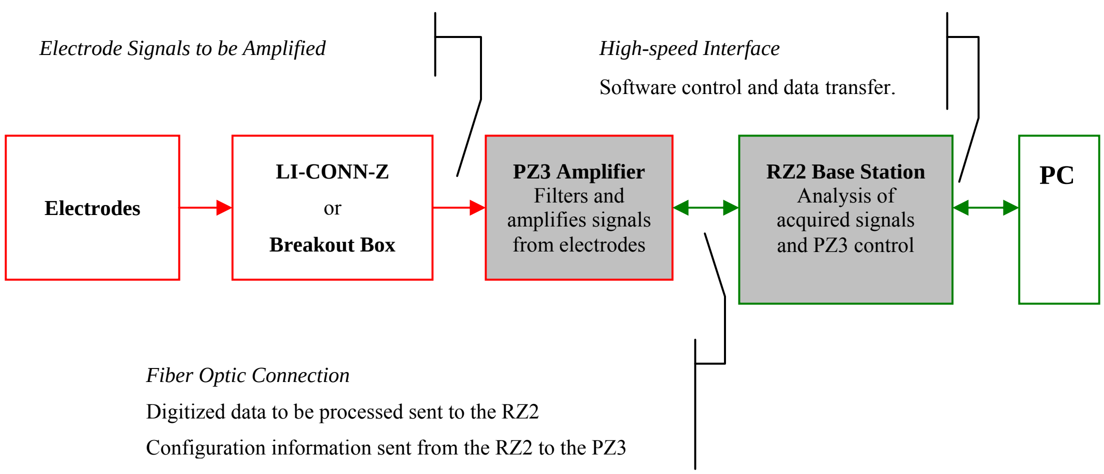
Available in 32, 64, and 128 channel models, the PZ3 amplifier offers shared or true differential operation, low input inferred noise, impedance checking, and an optional high input range mode.

System Hardware

A standard configuration for low sample rate, low impedance recordings includes 1.5 mm TouchProof connectors for electrodes, a PZ3 amplifier, and an RZ2 base station.

The battery powered PZ3 digitizes and amplifies signals recorded from each of the electrode channels. All digitized signals are sent via a single fiber optic connection to the RZ2 base station for further processing. The RZ2 also sends amplifier configuration information to the PZ3 across the fiber optics.

The diagram below illustrates this flow of data and control information through the system.

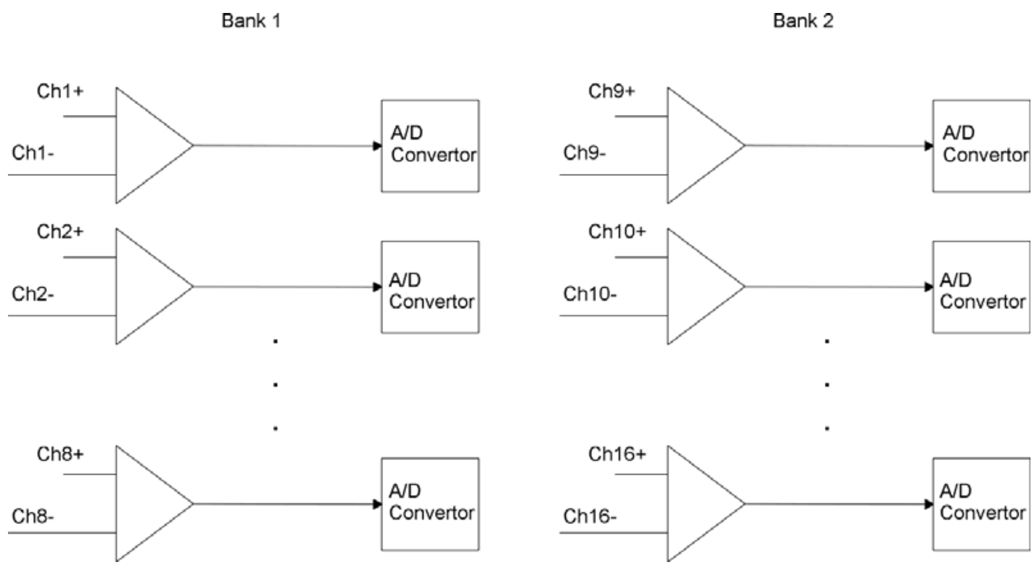


PZ3 Data and Control Flow Diagram

Recording Modes

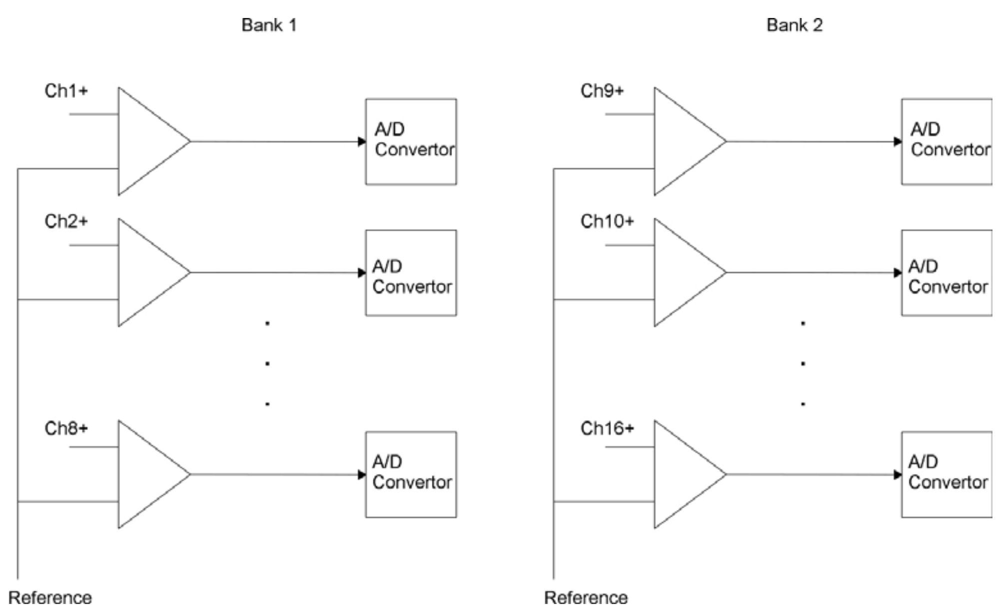
The PZ3 supports two recording modes: Individual Differential and Shared Differential.

For ***Individual Differential*** (true differential) operation, the amplifier inputs are grouped into banks of eight recording (+) channels, each with a paired alternate indifferent (-) channel *(inverting channel)*.



Individual (True) Differential, Bank 1 and 2 Functional Diagram

For ***Shared Differential*** operation, each bank of channels uses a separate shared reference.



Shared Differential, Bank 1 and 2 Functional Diagram

The PZ3's impedance checking and a high voltage range features can be used in both true and shared differential modes.

It is also important to note that in the various modes of operation, the RZ2 processor may use the alternate channels to report information such as impedance values or RMS. This occurs at the software level on the RZ2. For example, in Shared Differential mode the RZ2 maps RMS levels for each channel to the alternate channels. See the *PZ3-RZ2 Channel Data Chart* on page 5-17 for more information.

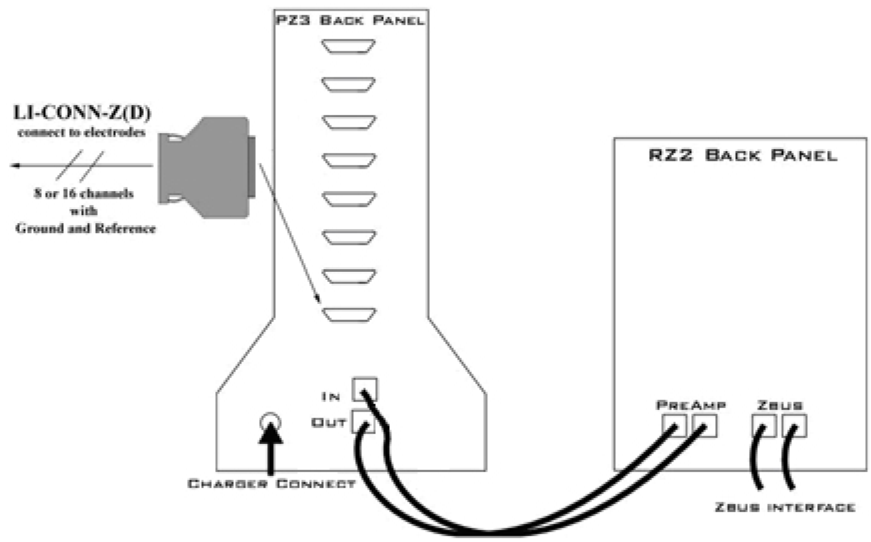
Electrode Connectors

The PZ3 is designed to record from low impedance electrodes and electrode caps with input impedances less than 20 kOhm. Signals are input via multiple DB26 connectors on the PZ3 back panel. A break out box or connector(s) are required for electrode connection.

TDT provides a version of our LI-CONN connector for the PZ3: the **LI-CONN-Z** for *Shared Differential* mode. It features standard 1.5 mm safety connectors and provides easy connections between electrodes and the amplifier.

Hardware Set-up

The diagram below illustrates the connections necessary for PZ3 amplifier operation.



One or more male connectors (such as the LI-CONN-Z) can be connected to the input connectors on the PZ3 back panel. Alternately, custom connectors and a breakout box can be used. If using custom connectors, see pinouts for the PZ3 connectors on page 5-18.

Note: In Shared Differential mode no connection should be made to the indifferrent (-) channels.

A 5 meter paired fiber optic cable is included to connect the preamp to the base station. The connectors are color coded and keyed to ensure proper connections.

The PZ3 battery charger connects to the round female connector located on the back panel of the PZ3 amplifier.

Important!: To avoid introducing EMF noise, DO NOT connect the charger to the PZ3 while collecting data.

PZ3 Software Control

The amplifier's mode of operation (shared or individual differential), other options, and channel mapping tasks are handled using PZ3 specific macros within the RPvdsEx control circuits running on the RZ2 Signal Processor.

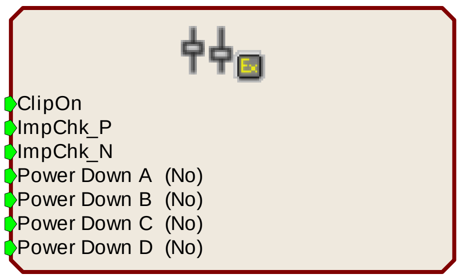
RPvdsEx includes two PZ3 specific macros:

- PZ3_Control macro
- PZ3_ChanMap macro

PZ3_Control Macro

The PZ3 Control macro should be added to your RPvdsEx circuit to configure all hardware features of the PZ3 amplifier.

PZ3_Control



Inputs are available on the macro for enabling/disabling the LED clip status lights, enabling Impedance mode for electrode (+) channels, enabling Impedance mode for alternate indifferent (-) channels, and dynamic power control for channel banks.

Macro Options

Double-clicking the macro in RPvdsEx, displays the macro properties dialog box and allows users to easily modify macro properties.

On the **Options** tab, in the properties dialog box:

- Setting the **Clip LEDs On** to **Yes** or **No** enables or disables the LED clip warning indicators.
- **Differential Mode** allows the user to select from **Shared (Shared Differential)** or **Individual (True-Differential)** modes.
- **Input Range** may be set to either **3mV** or **20mV** input ranges.
- The **Target Impedance** option allows the user to specify the impedance threshold for the status LEDs for each channel bank. Three inputs are available on the macro for enabling/disabling the LED clip status lights, enabling Impedance mode for electrode (+) channels, and enabling Impedance mode for indifferrent (-) channels.

Under the **Power Control** tab are additional options that specify how the PZ3 channel banks are powered.

Powering Down the Channel Banks

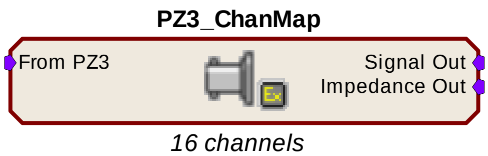
Channel banks may be powered down through the macro. As long as the **Power Control Mode** under the **Power Control** tab is set to **Static,** channel banks may only be powered up or down through the **Power Control Mode** options within the macro. **Dynamic** mode will allow channel banks to be powered on or off either through both the **Power Control Mode** options or by inputs on the macro through RpvdsEx components. Each of the letter indexed channel banks in the macro correspond to 32 channels of the PZ3. Selecting **No** will enable a bank of channels while selecting **Yes** will power down and disable that bank of channels.

For Example:

If you are using a PZ3 with 128 channels, powering down Bank A (Select **Yes**) would power down the first four blocks of 8 channels of the PZ3, disabling channels 1 – 32.

PZ3_ChanMap Macro

In the data stream on the RZ2, the odd numbered channels are the recording channels and the even numbered channels can report impedance measurements or RMS values. The PZ3_ChanMap should be added to your RPvdsEX circuit along with the RZ2_Input_MC macro to remap the data stream. The channel mapping macro selects the appropriate channels from the PZ3 input stream and builds two separate, sequential multichannel outputs containing either the amplified waveforms or alternate data (impedances or RMS values).



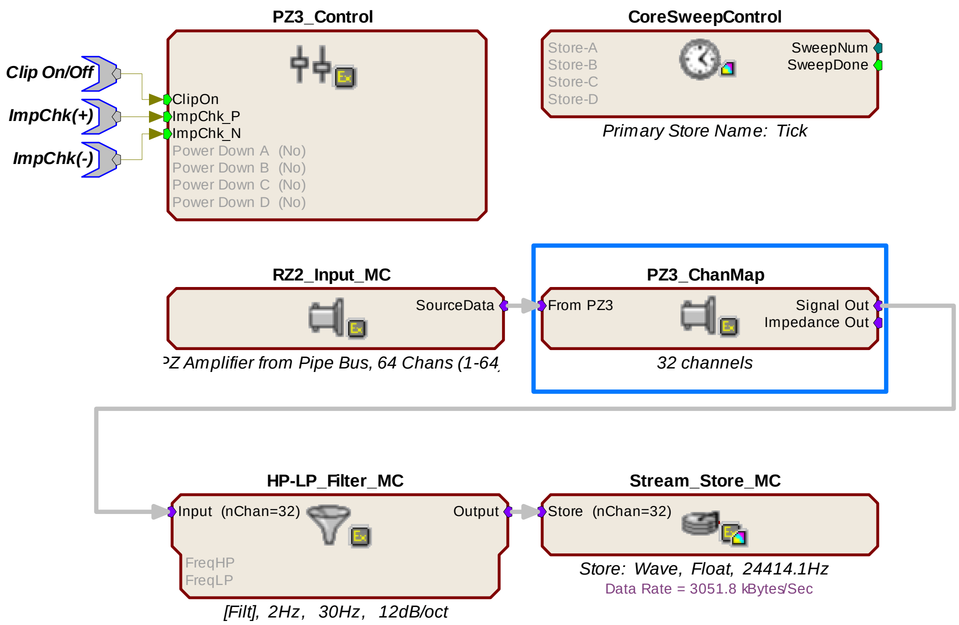
Macro Options

The user can set several different options under the **Options** tab.

- The designated number of channels to map and output.
- The ability to enable/disable the impedance measurement output.

PZ3 Circuit Example

The following illustration shows how macros can be used to create a simple OpenEx acquisition and control circuit for the PZ3.



The RZ2_Input_MC macro feeds the circuit with each digitally amplified signal acquired using the PZ3 amplifier. The data is fed first through the PZ3_ChanMap macro which separates the signals from their impedances (or RMS) values and builds the appropriate multi-channel data stream for further processing. In this case the signals are filtered and stored for post processing. A CoreSweepControl macro is included to handle the required timing functions used by programs such as OpenEx and a PZ3_Control macro configures the operation mode of the PZ3 as well as any additional options that may be necessary. Three parameter inputs allow toggling of clipping LEDs and toggling (+) or (-) channel impedance measurements.

PZ3 Operation

RCX control circuits running on the base station must include PZ3 specific macros to configure the amplifier's mode of operation; Shared Differential or Individual Differential and other configuration options such as input range and clip warning display. See *PZ3* on page 5-11 for more information. Impedance checking is also available from the front panel.

Powering ON

- To turn the amplifier on, move the three position battery switch to either the Bat-A or Bat-B position.

Powering OFF

- To turn the amplifier off, move the three position battery switch to the OFF position.

Operation Modes

Recorded signals are acquired in Shared or Individual differential mode.

Shared Differential

In shared differential mode a single shared reference and a ground are used for each bank of eight recording channels.

Note: In this mode no connection should be made to the alternate indifferent (-) channels. Use the LI-CONN-Z connector to ensure proper connections.

Enabling Shared Differential Operation

- To enable shared differential mode, use the PZ3 control macro and under the **Options** tab set the value of **Differential Mode** to **Shared**.

Individual Differential

When the PZ3 is operating in individual differential mode, each of the 8 (+) channels of an individual bank has a paired (-) differential reference.

Note: While operating in this mode no connections should be made to the Shared Reference (pin 5.)

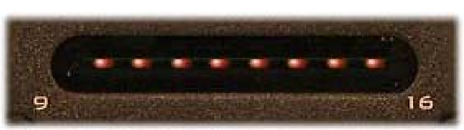
Enabling Individual Differential Mode

- To enable individual differential mode, use the PZ3 control macro and under the **Options** tab set the value of **Differential Mode** to **Individual**.

Clip Warnings

Analog clipping occurs when the input signal is too large. If analog clipping occurs, TDT recommends switching the PZ3 into high input range mode. For more information see *Modifying the Input Voltage Range on the PZ3*, page 5-15.

While the amplifier is recording, the front panel LEDs can act as clip warning indicators (according to configuration settings set using the PZ3_Control macro). If an analog signal approaches the PZ3s clipping range, the PZ3 LEDs for the corresponding channel are lit red.



Note: The LED Indicators are also mirrored on the RZ2 LCD display.

When recording, the **status LED** located below the Display Mode button indicates the status of the Clip Indicators. Solid green indicates that clip warning is disabled and orange indicates the clip warning is enabled.

- To enable clip warning, press the **Display Mode** button on the PZ3 front panel.

Alternatively the PZ3_Control macro can be used to enable or disable the clip warning indicators. For more information on the PZ3_Control macro see *PZ3 Macros,* page 5-11.

Modifying the Input Voltage Range on the PZ3

In the default mode, the PZ3 has an effective differential input range of +/- 3mV, which TDT recommends for EEG, LFP, and ECOG. If recordings demand a higher input range such as EMGs, the alternate High Input Range mode allows the input range to increase to +/- 20mV.

Note: The signal to noise performance is better while operating in the +/- 3mV input range.

Enabling the High Input Range Mode

The high input range mode can be enabled through the PZ3_Control macro.

- To enable the high range input mode, select **20 mV** from the **Input Range** option on the **Options** tab.

Testing your Electrode Impedance

Impedance measurement may be enabled programmatically or using the front panel **Display Mode** button.

Enabling Impedance Mode

- To enable impedance mode manually, push and hold down the **Display Mode** button on the PZ3 front panel.

During impedance checking all channels are tested in parallel using a ~375 Hz test signal and the impedance is measured relative to a target impedance (1kΩ – 15kΩ) specified by the user (set using the PZ3_Control macro). The LEDs on the PZ3 (and in the PZ3 display on the RZ2 LCD) will light green when the electrode impedance is greater than or equal to the target impedance or red when electrode impedance is less than the target impedance value.

Green: greater than or equal target impedance | Red: less than target impedance

Impedance Checking For True Differential Mode

Impedance values of either recording (+) or alternate indifferent (-) channels can be tested.

- To toggle between (+) and (-) channel impedance measurements, press the **Display Mode** button on the PZ3 front panel.

The **status LED** located below the **Display** button of the PZ3 will flash green while electrode (+) channel impedances are being tested or red while alternate indifferent (-) channel impedances are being tested.

Returning to Signal Acquisition Modes

- To leave Impedance mode, simply hold down the **Display Mode** button on the PZ3 front panel after enabling impedance mode.

Battery Overview

The PZ3 amplifier features two Lithium ion batteries to allow for longer record times. A three-position switch selects the active battery between Bank-A, Bank-B, or both banks off.

Battery Status LEDs

Battery Level: Eight LEDs indicate the voltage level of the selected battery bank. These LEDs can be found on the front of the PZ3 amplifier by the heading Level. When the battery is fully charged, all eight LEDs will be lit. When the battery voltage is low, only one green LED will be lit. If the voltage is allowed to drop further, the last LED will flash red. TDT recommends charging the battery before this flashing low-voltage indicator comes on. While charging, the Level LEDs will flash green.

Status	**Description**
8 Green	Fully Charged
1 Green, 7 Unlit	Low Voltage
1 Flashing Red	Low Voltage - Charge Immediately!
8 Green Flashing	Charging in Progress

Charging the Batteries

Operate the amplifier with the charging cable disconnected. Connecting the PZ3 charger will simultaneously charge both batteries. Ensure that the three-position switch is in the **OFF** (middle) position while charging the PZ3.

Charging Indicators: LEDs are also used for each bank to indicate which bank, if any, is charging. These LEDs are found next to the Level LEDs by the headings -A- and -B-. A green indicator denotes the battery bank is fully charged while a red indicator designates the bank is currently charging. When the device is in operation (charger is not connected) the A and B LEDs are not lit.

Status	**Description**
Red	Charging
Green	Fully Charged
Unlit	Operation Mode (charger not connected)

PZ3-RZ2 Channel Data Charts

The following charts show what data the user can expect to be available on the RZ2 for each channel depending on whether the amplifier is in a recording mode or in impedance checking mode. Please note that this does not necessarily reflect how the hardware channels are used on the PZ3. The RZ2 interprets input from the PZ3 then makes the data available as described below. To further simplify circuit design, the PZ3_ChanMap macro can be used to build separate multichannel data streams for waveform data and impedance values.

Unmapped Channel Index	**Recording Mode**	
	Shared Differential	**Individual Differential (True Differential)**
Channel 1	Analog Input Channel 1	Analog Input Channel 1(+)
Channel 2	RMS of Channel 1	Reference Channel 1(-)
. .	. .	. .
Channel 15	Analog Input Channel 8	Analog Input Channel 8(+)
Channel 16	RMS of Channel 8	Reference Channel 8(-)

Unmapped Channel Index	**Impedance Checking**	
	Shared Diferential	**Individual Differential (True Differential)**
Channel 1	NA	NA
Channel 2	Impedance of Channel 1	Impedance of Channel 1 (+) or (-)
. .	. .	. .
Channel 15	NA	NA
Channel 16	Impedance of Channel 8	Impedance of Channel 8 (+) or (-)

PZ3 Technical Specifications

Technical specifications for the PZ3 Low Impedance Amplifier.

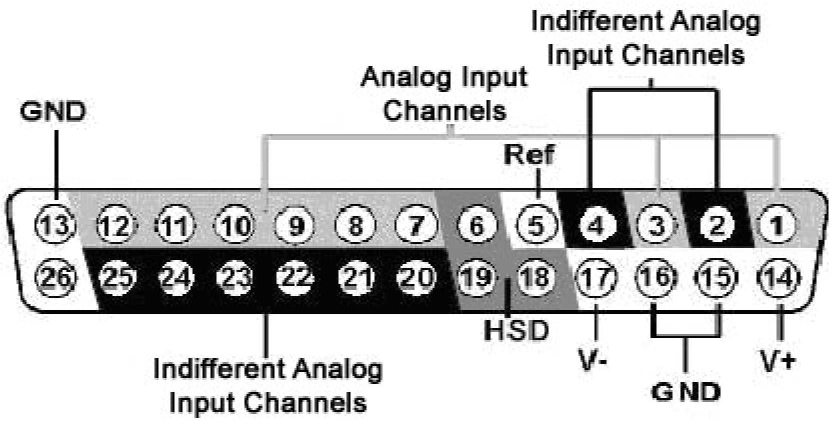
A/D	Up to 128 channels 18-bit hybrid
Maximum Voltage In	+/- 3mV - Default input range mode +/- 20 mV - High input range mode
Frequency Response	3 dB: 0.1 Hz – 5 kHz
S/N (typical)	71 dB - Default input range mode
Distortion (typical)	< 1%
A/D Sample Rate	Up to 48828.125 Hz
Input Impedance	10^6 Ohms
Power Requirements	2 Lithium Ion cells at 10 AmpHours each
Battery	Eight hours to charge Battery life between charges: 32 ch ~ 13 hrs 64 ch ~ 11 hrs 128 ch ~ 8 hrs
Charger	External 6VDC, 3A power supply
Indicator LEDs	Up to 128 status or clip warning, battery life, active battery bank
Input inferred noise	0.9µV rms typical 300- 5000Hz, 3mV input range 2.3µV rms typical 300- 5000Hz, 20mV input range
Fiber Optic Cable	5 meters standard, cable lengths up to 20 meters*

***Note:** If longer cable lengths are required, contact TDT.

Input Connectors

PZ3 amplifiers have up to 16 26-pin headstage connectors on the back of the unit. The PZ3 channels are marked next to the respective connector on the amplifier.

Pinout Diagram



Note: There are 8 (+) channels and 8 (-) channels per DB26 connector. Subsequent banks are indexed by an additional 8 channels.

Pin	Name	Description
1	A1(+)	Analog Input Channel
2	A1(-)	Indifferent Analog Input Channel
3	A2(+)	Analog Input Channel
4	A2(-)	Indifferent Analog Input Channel
5*	Ref*	Shared Reference*
6	HSD	Headstage Detect
7	A3(+)	Analog Input Channels
8	A4(+)	
9	A5(+)	
10	A6(+)	
11	A7(+)	
12	A8(+)	
13	GND	Ground

Pin	Name	Description
14	V+	Positive Voltage
15	GND	Ground
16	GND	
17	V-	Negative Voltage
18	HSD	Headstage Detect
19	HSD	
20	A3(-)	Indifferent Analog Input Channels
21	A4(-)	
22	A5(-)	
23	A6(-)	
24	A7(-)	
25	A8(-)	
26	NA	Not Used

***Note:** No connections should be made to pin 5 while operating in True Differential mode.

Medusa Preamplifiers

Overview

The Medusa Preamplifiers are low noise digital bioamplifiers and are available with either PCM or Sigma-Delta ADCs. The system amplifies and digitizes up to 16-channels of analog signal at a 24.414 kHz sampling rate. The amplified digital signal is sent to the base station via a noiseless fiber optic connector.

- Digitizes either four or 16 channels at acquisition rates of approximately 6, 12, or 25 kHz.
- Connects to the headstage via a 25-pin.
- Powered by a Lithium-ion battery that provides 20 hours of continuous data acquisition in 16-channel mode and 30 hours of operation in 4-channel mode.
- Clip warning lights indicate when any signal is -3db from the preamplifier's maximum voltage input.

Features

Analog Acquisition Channels

The **RA16PA** and **RA4PA** standard Medusa Preamplifiers acquire signals using 16-bit PCM ADCs, which provide quality acquisition with minimal delay. The **RA16SD** and **RA4SD** use Sigma-Delta ADCs, which have several characteristics that improve signal quality. Oversampling of the signal before conversion removes aliasing of high frequency RF signals.

RA16SD testing indicates that signals greater than 150% of the Nyquist frequency are removed from the signal. This allows users to acquire at lower sampling rates (6 kHz) without worry of significant aliasing. In addition, each converter also has a two pole anti-aliasing filter (12 dB per Octave) at 7.5 kHz. However, the sigma-delta ADC's have a fixed group delay of 20 samples (compared to four samples for the RA16PA). When using the RA16SD this group delay must be taken into account when the data is displayed or acquired (for example, adding a SampDelay to the RPvdsEx circuit).

Clip Warning Lights

When the input to a channel is greater than -3db from the preamplifier's maximum voltage input, a light on the top of the amplifier is illuminated. The first column of lights corresponds to channels 1-8 and the second column corresponds to channels 9-16. The clip warning light indicator can be turned off by flipping a switch on the end of the amplifier.

Power Light

The power light is in the top corner of the amplifier. It is illuminated when the device is on. It flashes quickly if the battery is low. It flashes slowly while the battery is charging.

Headstage Connector

The headstage connector is a 25-pin (16-channel) connector. Information on the pin inputs is provided with the technical specifications.

Base Station Connector - To Base

One end of the fiber optic cable connects to the amplifier and the other end connects to the amplifier input on the base station.

Power

A switch on the back powers up the amplifier. The fiber connector at the right will be illuminated when the amplifier is on.

LEDs

This switch turns the clip warning lights on top of the amplifier on or off.

Power Requirements

The Lithium-ion batteries charge in four hours. Keeping the battery charger connected to the amplifier does not affect the battery life. However, the charger will significantly increase the noise of the system if it is plugged in while an experiment is running. A 6 volt battery charger is included with the amplifier. The charger tip is center negative. If it is necessary to replace the charger make sure that the power supply has the correct polarity.

The Li-ion battery supplied with the system cannot be removed. If battery life longer than 30 hours is required, an external battery pack can be connected to the voltage inputs of the charger. TDT recommends a 6 (minimum) to 9 Volt (maximum) battery, such as lead acid batteries used for motorized wheel chairs. Contact TDT for more information.

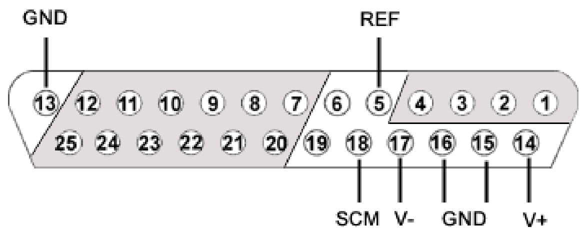
Medusa Preamplifier Technical Specifications

Technical Specifications for the RA4PA, RA16PA, and RA16SD Medusa Preamplifiers.

A/D	RA4PA: 4-channels 16-bit PCM RA16PA: 16-channels 16-bit PCM RA16SD: 16-channels 16-bit sigma-delta
Maximum Voltage In	RA4PA and RA16PA: +/- 4 millivolts RA16SD: +/- 5 millivolts
Frequency Response	3 dB 2.2 Hz - 7.5 kHz
Highpass Filter	2.2 Hz
Anti-Aliasing Filtering	RA4PA and RA16PA: 7.5 kHz (3 dB corner, 1st order, 6 dB per octave) RA16SD: 7.5 kHz (3 dB corner, 2nd order, 12 dB per octave)
S/N (typical)	RA4PA and RA16PA: 60dB
Input Inferred Noise	rms 3 microvolts bandwidth 300 - 3000 Hz 6 microvolts bandwidth 30 - 5000 Hz
Group Sample Delay	RA4PA and RA16PA: NA RA16SD: 20 Samples
A/D Sample Rate	6, 12, or 25 kHz
Input Impedance	AC (1 kHz): 10^5 Ohms
Power Requirements	500 mAmps while charging, 50 mAmps once charged
Battery	Li-ion Battery 1500 mAh, 20-30 hours between charges. 1000 cycles of charging, not removable by user
Charger	6-9 Volts DC, greater than 500 mAmps, center negative
Fiber Optic Cable	5 meters standard, maximum cable length 12 meters

Pin Diagrams

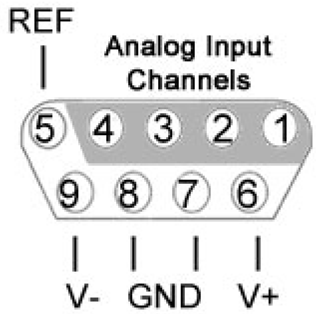
16/4-channel pin outs (all 16 channel models and 4 channel models built after 2002):



Pin	Name	Description
1	A1	Analog Input Channel Number
2	A2	
3	A3	
4	A4	
5	REF	Reference Pin
6	NA	TDT Use Only Pins 6, and 19 are for TDT use only and should not be used.
7	A5	Analog Input Channel Number
8	A7	
9	A9	
10	A11	
11	A13	
12	A15	
13	GND	Ground
14	V+	Positive Voltage Headstage Power Source (1.4 V as measured in reference to ground)
15	GND	Ground
16	GND	Ground
17	V-	Negative Voltage Headstage Power Source (1.4 V as measured in reference to ground)
18	SCM	Sixteen Channel Mode Indicator Pin The status of pin 18 determines whether the preamplifier is in four or 16-channel mode. To use the preamplifier in 16-channel mode with a custom headstage, connect pin 18 to pin 17.
19	NA	TDT Use Only Pins 6, and 19 are for TDT use only and should not be used.
20	A6	Analog Input Channel Number
21	A8	
22	A10	
23	A12	
24	A14	
25	A16	

Grounds (pins 13, 15, 16) are tied together.

4-channel pin outs (models shipped before January 2002):



Pin	Name	Description
1	A1	Analog Input Channel Number
2	A2	
3	A3	
4	A4	
5	REF	Reference Pin
6	V+	Positive Voltage Headstage Power Source
7	GND	Ground
8	GND	Ground
9	V-	Negative Voltage Headstage Power Source

Grounds (pins 7 & 8) are tied together.

Adjustable Gain Preamp

Overview

The RA8GA was designed to acquire and digitize multi-channel data from a variety of analog voltage sources such as eye-trackers, amplifiers (including grass, axon, and WPI amplifiers), PH meters, and temperature sensors. The RA8GA digitizes up to eight channels at acquisition rates of 6, 12, or 25 kHz. All channels have a variable group gain setting of 10 Volts, 1 Volt, or 100 millivolts. The system has a bandwidth to DC, which allows users to acquire low frequency DC signals. In addition a two-pole low pass filter (12 dB per Octave) is set at 7.5 kHz.

Power and Interface

The device is powered via the System 3 zBus (ZB1PS) and requires an interface to the PC. If the RA8GA is housed in one of several ZB1PS chassis in your system, ensure that it is connected in the interface loop according to the installation instructions: Gigabit, Optibit, or USB Interface.

Features

Max Input Lights

The Active light flashes once a second when the preamplifier is not connected to a base station. It glows steady when it is properly connected.

The 10V, 1V, and 0.1V lights indicate the current acceptable voltage range. If the signal input reaches -6db from the maximum input for the selected range, a clip warning light on the base station will be lit. On high performance processors, such as the RX5 or RX7 the LED located next to the fiber optic input port serves as the clip warning light.

Range Select Button

All channels use a group adjustable gain control i.e. all channels are either +/- 1 Volt, 10 Volts, or 0.1 Volt. A Range Selection button adjusts the gain setting among the following voltages: 0.1X gain = +/-10 Volts, 10X gain = +/- 100 milliVolts, 1.0 X gain = +/-1 volt. Press the button to scroll through the available voltage ranges. Max input lights located to the left of the button, indicate the current selection.

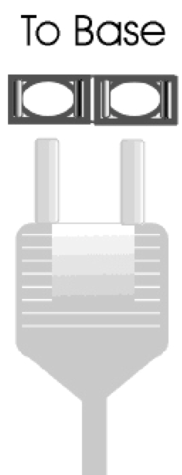
To Base

The To Base connector is used to connect the device to the base station (such as RA16BA, RX5, or RX7) using a fiber optic cable pair. One end of the fiber optic cable connects to the device using this connection pair and the other end connects to the input on the base station.

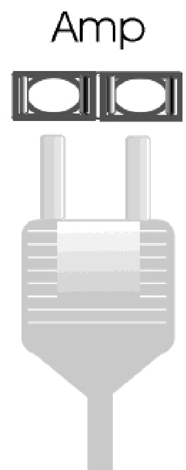
Connecting the Base Station to the Preamplifiers

To make the connection, plug one end of the cable into one of the fiber optic connectors as shown below and connect the other end of the cable to the fiber optic port on the base station. Both ends of the cable are the same but the two sides of the connector are different. See the diagram below to determine the correct way to make the connection for each device.

Preamplifier



Base Station



Analog Input

Each Preamp comes with eight channels of analog input. Each analog input uses 16-bit PCM parts for high quality signal conversion. See the technical specifications for a Pinout Diagram for the 25-pin Analog Input connector.

A PP16 patch panel can be used to simplify connection to the preamplifier's analog inputs. A ribbon cable can be connected from the RA8GA Analog I/O connector to the RA16 connector on the back of the PP16 allowing acquisition of signals via the first eight BNC connectors on the front of the PP16.

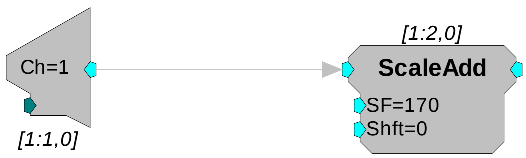
RA8GA Gain Settings

Gain	Voltage Range	RPvdsEx Scale Factor
0.1	+/-10 V	1700
1.0	+/- 1 V	170
10.0	+/- 0.1 V	17

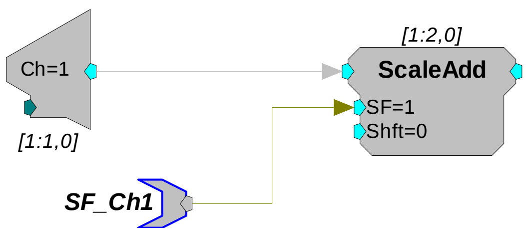
Accounting for Gain Settings in RPvdsEx

The output from a RA8GA generates a floating-point value of between +/- 6 mVolts (i.e. the voltage value of the RA16PA). A scale factor must be used in order for the acquired signal to display the correct voltage. The scale factor for each gain setting is listed in the table above. The scale factor should be added after the channel input (AdcIn).

The following example shows a circuit segment that could be used to add the scale factor for a +/- 1 Volt range:



A parameter tag may be used to allow the scale factor of the channel input to be modified at run-time.

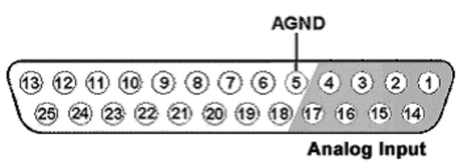


RA8GA Technical Specifications

Technical specifications for the RA8GA Adjustable Gain Preamplifier.

A/D	8-channels 16-bit PCM
Maximum Voltage In	Variable gain settings allow +/-10V, +/-1 V or +/- 100 mV
Frequency Response	DC - 7.5 kHz (2nd order 12 dB per octave)
S/N (typical)	70 dB (+/- 1 V 1000 kHz) at 1 V Gain Setting
THD (typical)	0.01%
A/D Sample Rate	6, 12, or 25 kHz
Cross Talk	< -70 dB (DC - Nyquist)
Input Impedance	10 kOhm
DC Offset	< 5 mV at +/- 10 V < 3 mV at +/- 1 V and +/- 100 mV

Analog Input Pinout Diagram



Pin	Name	Description
1	A1	Analog Input Channels
2	A3	
3	A5	
4	A7	
5	AGND	Ground
6	NA	Not Used
7		
8		
9		
10		
11		
12		
13		

Pin	Name	Description
14	A2	Analog Input Channels
15	A4	
16	A6	
17	A8	
18	NA	Not Used
19		
20		
21		
22		
23		
24		
25		

TB32 32-Channel Digitizer

Overview

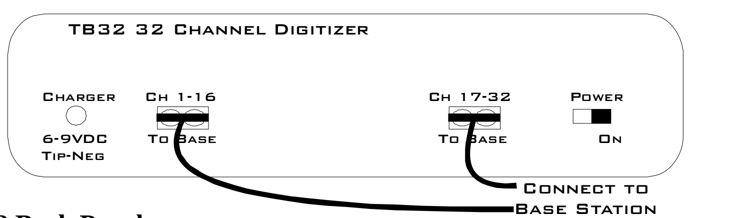
The TB32 32 channel digitizer interfaces directly with Triangle BioSystems, Inc. (TBSI) wireless headstage and receiver allowing up to 31-channels of recording from a free moving subject.

TBSI's wireless headstage captures the analog signals and wirelessly transmits them up to 3 meters from the subject to the TBSI receiver. The analog signals are then passed to the TB32 for digitization through a 37-pin connector. Signals are digitized at up to ~25 kHz on the digitizer and sent over two fiber optic links to a DSP device such as the Pentusa base station, where they are filtered and processed in real-time.

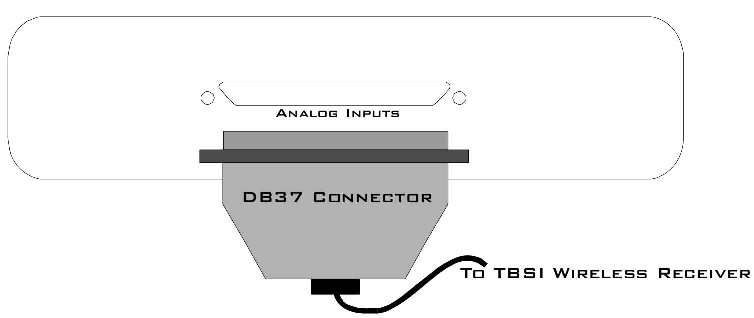
Hardware Setup

The diagram below shows the connections made to the front and back panels of the TB32 digitizer.

TB32 Front Panel



TB32 Back Panel



Features

Analog Acquisition Channels

The TB32 acquires signals using 16-bit sigma-delta ADCs, which provide superior conversion quality and extended useful bandwidths, at the cost of an inherent fixed group delay. Each converter has a two-pole anti-aliasing filter (12 dB per Octave) at 4.5 kHz.

Note: The TB32 16-bit sigma-delta A/D converters contain a 20 sample group delay.

Scale Factor

To determine the actual biopotential from the TB32, two scale factors should be applied in the DSP. The first scale factor is 400. This is used to convert the input from the TB32 into the standard voltage range expected by the DSP. The second scale factor is used to scale the signal according to the amplification of the TBSI headstage and receiver.

This can be simplified into a single conversion of 400/ G_{TBSI}

Where G_{TBSI} = Gain of TBSI wireless headstage and receiver

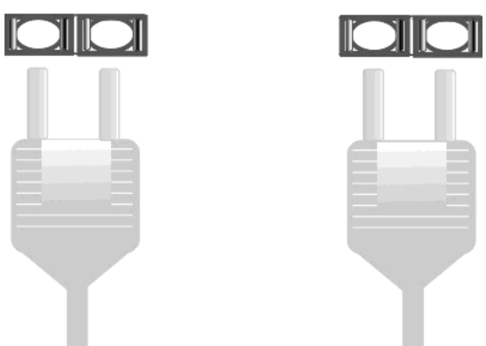
Headstage Connector

The headstage connector is a 37-pin (31-channel) female connector. Information on the pin inputs is provided with the technical specifications on page 5-31.

Base Station Connectors - To Base

One end of the fiber optic cable connects to the digitizer and the other end connects to the digitizer (amplifier) input on the base station. Two fiber optic base station connectors are provided. Connect each fiber optic cable as shown below.

Digitizer Output To Base Station **Base Station Connector For Digitizer Input**



Each connector on the TB32 is labeled and corresponds to the channels of the wireless headstage. Refer to the System 3 Manual for specific device channel configurations.

Power Switch

A switch on the front panel powers up the digitizer. The power light and fiber connectors at the left will be illuminated when the digitizer is on.

Power Light

The power light is illuminated when the device is on. It flashes quickly if the battery is low. It flashes slowly while the battery is charging.

Power Requirements

Onboard lithium-ion batteries charge in ten hours. Keeping the battery charger connected to the digitizer does not affect the battery life. However, the charger will significantly increase the noise of the system if it is plugged in while an experiment is running. A 6 Volt battery charger is included with the digitizer. The charger tip is center negative.

The Li-ion battery supplied with the system cannot be removed. If battery life longer than 20 hours is required, contact TDT for more information.

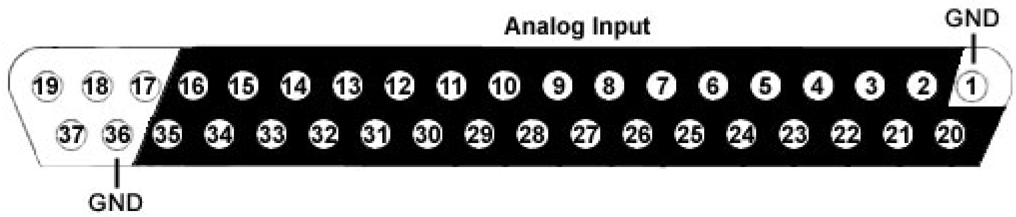
TB32 Digitizer Technical Specifications

A/D	31-channels: 16-bit sigma-delta
Maximum Voltage In	+/- 2 Volts
Frequency Response	3 dB 8 Hz - 4.5 kHz
Highpass Filter	8 Hz
Anti-Aliasing Filtering	4.5 kHz (3 dB corner, 2nd order, 12 dB per octave)
S/N (typical)	74 dB
Input Inferred Noise(Re 2V)	rms 400 microvolts bandwidth 300 - 3000 Hz* 1 millivolt bandwidth 30 - 5000 Hz*
Group Sample Delay	20 Samples
A/D Sample Rate	6, 12, or 25 kHz
Input Impedance	10^5 Ohms
Power Requirements	500 mAmps while charging, 50 mAmps once charged
Battery	Li-Ion Polymer Battery 5000 mAh, 20-30 hours between charges.
Charger	6-9 Volts DC, greater than 500 mAmps, center negative
Fiber Optic Cable	5 meters standard, maximum cable length 20 meters

***Note:** Given the standard gain on the TB32 these values are 1uV and 2.5uV respectively.

Pin Diagrams

31-channel pin out:



Pin	Name	Description
1	GND	Ground
2	A2	Analog input channels 2,4,6,8,10,12,14,16,18, 20,22,24,26,28,30
3	A4	
4	A6	
5	A8	
6	A10	
7	A12	
8	A14	
9	A16	
10	A18	
11	A20	
12	A22	
13	A24	
14	A26	
15	A28	
16	A30	
17	NA	Not Used
18	NA	
19	NA	

Pin	Name	Description
20	A1	Analog input channels 1,3,5,7,9,11,13,15,17,19, 21,23,25,27,29,31
21	A3	
22	A5	
23	A7	
24	A9	
25	A11	
26	A13	
27	A15	
28	A17	
29	A19	
30	A21	
31	A23	
32	A25	
33	A27	
34	A29	
35	A31	
36	GND	Ground
37	NA	Not Used

Note: No connections should be made to pins 17, 18, 19, and 37.

Headstage Connection Guide

Overview

Ground and Reference placement is important in all headstage configurations. They determine the operation of the headstage and can, if incorrectly wired, produce undesired results.

Important!: High channel count recordings (implemented either with PZ or multiple Medusa preamplifiers) may be implemented using multiple headstages. **When using multiple headstages, ground pins on all headstages should be connected together to form a single common ground.** This ensures that all headstage ground pins are at the same potential and eliminates additive noise from varying potentials across the subject's brain.

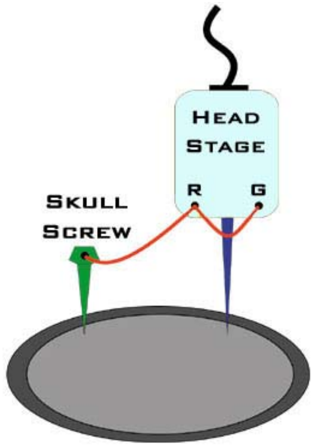
This section serves as a guide to headstage connection and will illustrate single and multiple headstage configurations. A common error example is provided for the final illustration.

Headstage Operation

Headstage operations can be categorized into three forms listed below. It is important that multiple headstage configurations use a common node for all grounds regardless of the operation of the headstage.

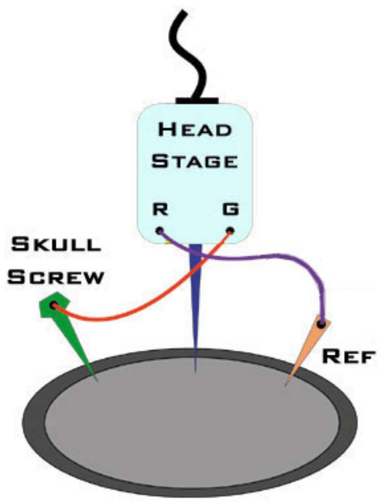
Headstage Operations	Description
Single-Ended	Ground and reference pins are tied together and the probe(s) reference all channels to ground.
Differential	Ground and reference pins are separate and the probes may use shared or multiple references.
Hybrid	A mixture of single-ended or differential operations when multiple headstages are used.

Single Headstage Configurations



Single headstage with a Shared Ground and Reference

When using a single headstage with a shared ground and reference, the ground and reference pins of the headstage should be tied together. A ground is used and attached to a skull screw. All recordings will reference this connection. This configuration is referred to as "Single-Ended".

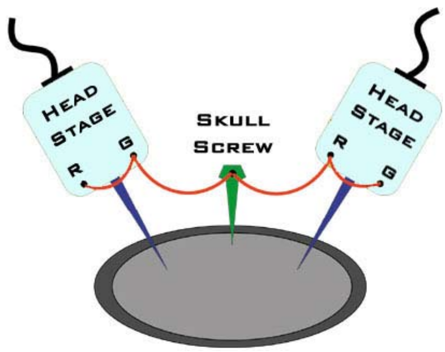


Single headstage with a Separate Ground and Reference

When using a single electrode with a separate ground and reference, it is important that the headstage itself is not single-ended, that is, its ground and reference pins are NOT tied together. This will allow the headstage to reference each channel to ground as well as an additional chosen site on the subject. This configuration is referred to as "Differential"

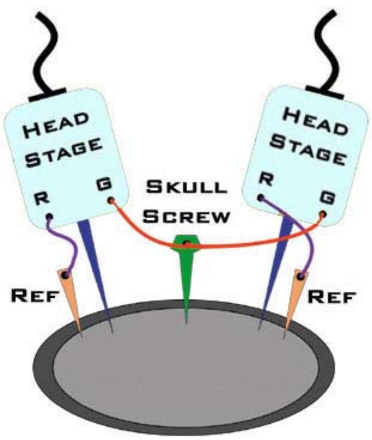
Multiple Headstage Configurations

Note: All headstages must use the same Ground wire. But not all headstages need to use the same Reference wire.



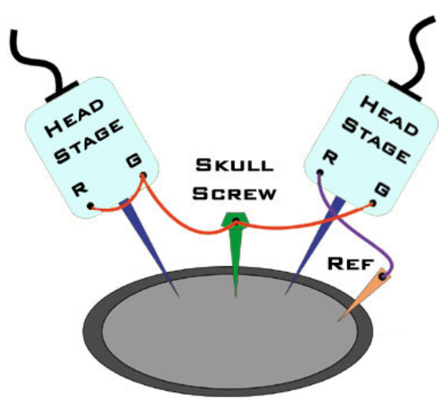
Multiple headstages with a Shared Ground or Reference

When using multiple headstages with a shared ground or reference, the ground and reference pins of each headstage should be tied together. A ground is used and attached to a skull screw. This ground is used by all headstages and ensures the headstages are referencing the same potential. This is a multiple single-ended configuration.



Multiple headstages with a Single Ground and Multiple References

This configuration uses multiple differential headstages each with their own separate references. Notice that all the headstages' ground pin are tied together. This is a multiple differential configuration.



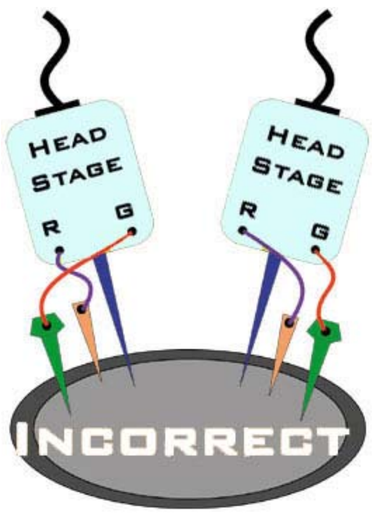
Multiple headstages with a Shared Ground and different Ground/Reference configurations

When using multiple electrodes with a shared ground and separate reference, all headstages' grounds are connected to the skull screw. A reference wire is present and connected to the desired headstage. This ensures all headstages have the same ground potential and provides a reference for the desired headstage. This is a hybrid configuration and uses a mixture of single-ended and differential headstages.

Alternatively, to use a single reference for all headstages you may tie all headstage reference pins to the site labeled "Ref".

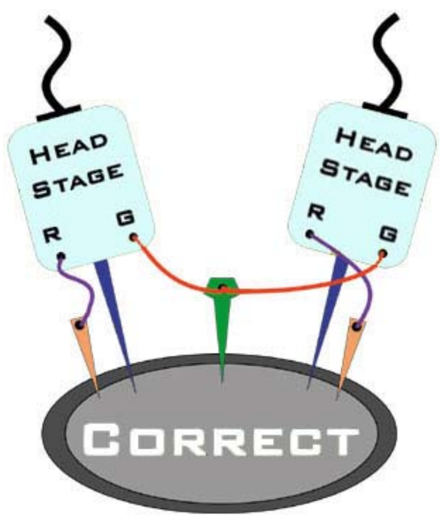
A Common Error to Avoid

When using multiple headstages a common error is to connect separate grounds for each headstage. This allows additional noise to corrupt signals increasing the number of artifacts present. To avoid this, ensure that all headstage ground pins are wired as a single ground.



Incorrect Configuration

Both headstages are connected to a unique node for ground. This will introduce additional noise artifacts into the recordings.



Correct Configuration

These headstages are correctly sharing a single node for ground. All headstages will be able to reference the same ground and will eliminate unnecessary noise artifacts from the recordings.

Part 6 Stimulus Isolator

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MS4/MS16 Stimulus Isolator

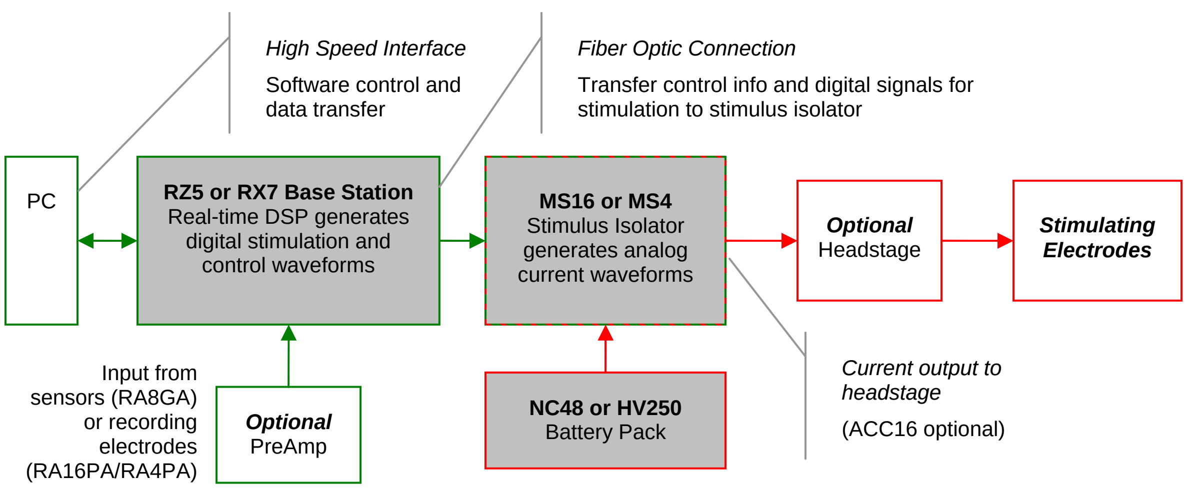
Overview

The MS4/MS16 Stimulus Isolator converts digital waveforms into analog current waveforms as part of a computer controlled neural microstimulator system that delivers user-defined current waveforms through multichannel electrodes.

The MicroStimulator System

A typical system consists of an RZ5 or RX7 processor base station (RX7 must be housed in a zBus Device Caddie with power supply and interface module), an MS4 or MS16 Stimulus Isolator, ACC16 AC Coupler (Optional) and NC48 or HV250 Battery Pack.

The block diagram below illustrates the functionality of the system.



Multichannel MicroStimulator System Diagram

As seen in the illustration above, stimulation control waveforms for each electrode channel are first defined on the base station and digitally transmitted over a fiber optic cable to the battery powered stimulus isolator. On the isolator, specialized circuitry for each electrode channel generates an analog current waveform as specified by the digital stimulation control waveform.

The final analog current output from the isolator is adjusted to match the stimulation control waveform by adjusting the isolator's driving voltage according to Ohm's law where: V=IR. That is, the driving voltage is adjusted for the stimulation control waveform level and the electrode impedance. In this way, the stimulation current specified by the user will be constant regardless of electrode impedance, within system limits.

The MicroStimulator System standard configuration is capable of delivering up to 100 µA of current simultaneously across up to 16 stimulating electrodes (impedances up to 1Mohm). *See Working with the MS16 MilliAmp Mode* on page 6-17, for information if your stimulus isolator has been configured for MilliAmp mode.

The Stimulus Isolator

The stimulus isolator features either four or 16 D/A converters that can deliver arbitrary waveforms of up to 10 kHz bandwidth. PCM D/As are used to ensure sample delays of only 4-5 samples and square edges on pulse stimulation waveforms.

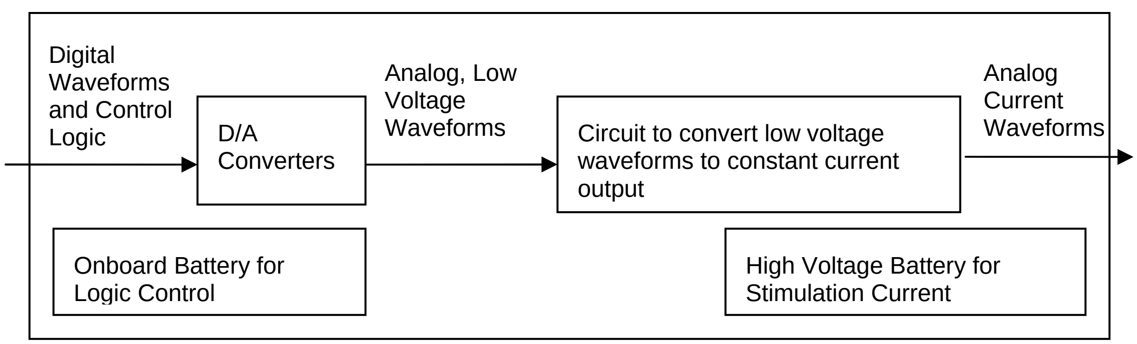
Each of the device's stimulation channels can be configured in one of three states:

Stimulate: Channels in stimulate mode pass current through the selected electrodes.

Reference: Channels in reference mode become part of the return path for the current. All channels in Reference mode use the same return path to analog ground on the stimulator. **Note:** Users can also use a dedicated global reference channel as a current return path. In this mode all channels can be used for stimulation.

Open: The Open mode is the default mode for all channels. In the open mode, the corresponding electrode channel is disconnected from output and internally grounded to eliminate noise and crosstalk. On multichannel electrodes, these electrodes might instead be connected to a recording preamp. In this mode a channel can be used to acquire neural signals.

The stimulus isolator utilizes an onboard, rechargeable Li-Ion battery for logic control and D/A converter operation. Special circuitry on the stimulus isolator draws on external high voltage battery packs to convert low voltage waveforms from the D/A converters to analog current waveforms as shown in the diagram below.



Stimulus Isolator Diagram

The ACC16 AC Coupler

The stimulus isolator may generate a DC bias current of up to 0.2% of full scale (up to 0.2 µA on 100 µA device) on any stimulation channel, even during a quiescent state. While this may not have significant short-term effects, over time, it may cause unintended tissue damage. This problem primarily affects researchers using electrodes with impedances of more than 100 kOhms. Users may connect the ACC16 AC coupler (supplied with all MS4/MS16s) directly to the Stim Output connector on the stimulus isolator to block any bias present on the Stim Output lines.

Note: Single-ended operation (G and Ref jumper pins tied together) is the only mode supported on the ACC16.

Each channel of the ACC16 coupler includes an RC circuit with a one µF capacitor in parallel with a one MOhm resistor. The coupler acts as a 1.6 Hz highpass filter, eliminating the DC bias current. It also acts as a voltage divider, decreasing the voltage and thus the current delivered through the electrode.

Note: When using the ACC16 you will NOT be able to deliver the MAXIMUM Rated current. See *Designing the Stimulus Signal,* page 6-9, for more information.

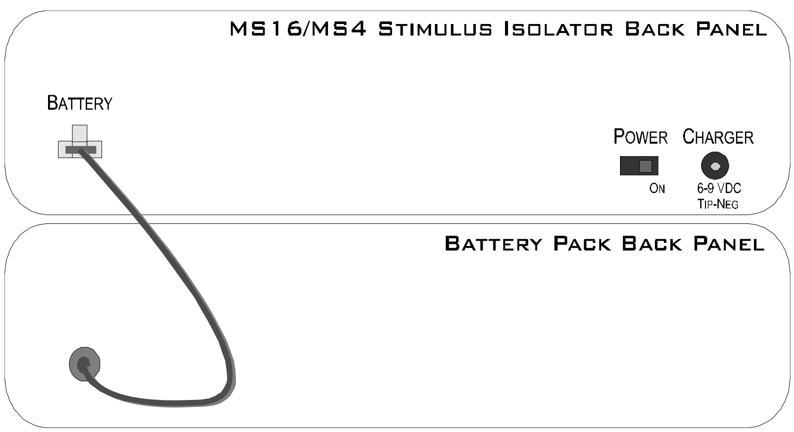
Stimulus Isolator Batteries

Power for stimulation is supplied by one of TDT's battery packs. Power requirements are determined by the amount of current needed for stimulation and the impedance of the electrode being used. When using a high impedance electrode (approximately 1 MOhm), the HV250 Battery Pack will most likely be required. With lower impedance electrodes (100 kOhms to 200 kOhms), the NC48 Battery Pack may be more suitable. Users should contact TDT for further information before attempting to use an external power supply. See *Battery Reference* page 6-20, for technical specifications and for more information.

Hardware Set-up

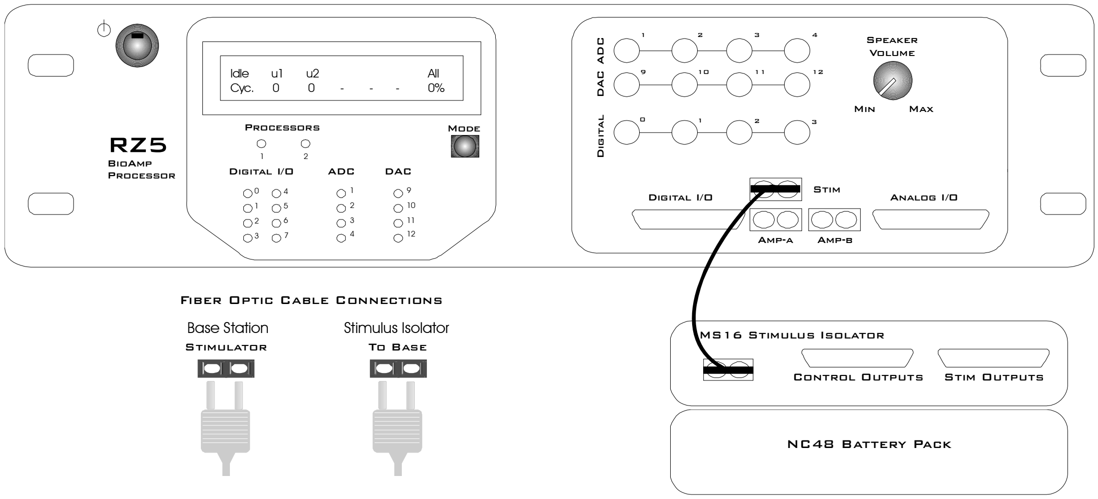
To connect the system hardware:

1. Ensure that the TDT drivers, PC interface, and device caddies are installed, setup, and configured according to the installation guide provided with your system.
2. Connect the battery pack to the back panel of the Stimulus Isolator via the connector labeled ***Battery***, as shown in the diagram below.



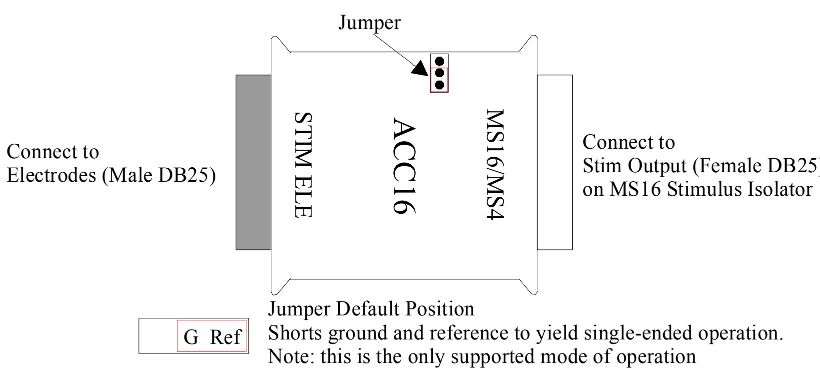
**Warning!:** The HV250 is a high voltage power source, capable of delivering up to 250 Volts DC at high currents. Shorting the battery connection pins can cause damage to the device and injury to the user. Always use caution when handling or connecting the devices.

3. Connect the Stimulus Isolator to the base station using the provided fiber optic cable.



Connect the fiber optic cable from the MS16 fiber optic port labeled **To Base** to the fiber optic port labeled **Stimulator** on either the RZ5 or the RX7 (not shown). Be sure to note the difference in the two sides if the fiber optic cable connectors and ensure they are inserted with the correct side up as shown under *Fiber Optic Cable Connections* above.

4. If desired, connect the ACC16 AC Coupler to the Stimulus Isolator's **STIM OUTPUT** port.



5. Connect the Stimulus Isolator's **STIM OUTPUT** or the ACC16's **STIM ELE** connector to the stimulating electrodes using your preferred method such as direct wiring, the SH16 switching headstage, or a custom pass through connector (available from TDT). See the *Stimulus Isolator Technical Specifications* section, page 6-18, for pinouts.
6. Power on the base station, then power on the stimulus isolator using the power switch on the isolator's back panel.

   Note: Ensure that the rechargeable batteries (onboard Li-Ion and NC48) are fully charged before starting your protocol.
7. The hardware is ready for use.

If using the system with other devices, such as a switching headstage or preamplifiers, see the documentation for those devices for hardware connection information.

Stimulus Isolator Features

Analog Outputs (Stim Outputs)

The Stimulus Isolator is equipped with four or 16 analog current output channels, arranged in four-channel banks that can be powered down when not in use. Channels can operate in three modes: Stimulate, Reference, or Open. Simultaneously setting any channel in a bank to both Stimulate and Reference mode turns off that entire bank of channels.

An ACC16 AC Coupler is supplied with all MS4/MS16 modules and may be connected directly to the Stim Output connector to block any DC current bias present on the Stim Ouput lines (this problem primarily affects researchers using electrodes with impedances of more than ~100 kOhms) when set in stimulate mode.

Note: When using the ACC16 you will NOT be able to deliver the MAXIMUM current.

Stim Lights

A Stim Light (one for each channel) indicates that a Stim Output channel is in use as a stimulus output. The Stim Lights are located above the Stim Output connector and are numbered 1 - 16, to indicate the active channel number. The LEDs will flash once every three seconds to indicate any bank of channels that has been powered off.

Ref Lights

A Ref Light (one for each channel) indicates that a Stim Output channel is in use as a reference. The Ref Lights are located above the Stim Output connector and are numbered 1 - 16, to indicate the active channel number.

Status Lights

Sync: Flashes once a second when the stimulator is not connected to a base station and glows steady when it is correctly connected.

Stim Ref: When lit, indicates that the stimulator has been configured to use a global reference.

Battery: When lit, indicates when the stimulator's onboard battery is low. The battery voltage decreases rapidly once the battery low light is on.

Fast: charging

Slow: low battery

High Voltage: When lit, indicates that the stimulator is correctly connected to the designated Battery Pack.

Solid - correct working voltage

Flashing - low voltage

Digital Output (Control Outputs)

The Control Output connector provides access to the stimulator’s 16 channels of Word addressable digital output. These outputs can control the relays on the SH16 switching headstage or other digital output device (maximum current 40 mA, maximum voltage 3.3 Volts).

Control Output Lights

A Control Output Light (one for each digital I/O) indicates that the digital output channel is set high (or active). The Control Output Lights are located above the Control Output connector and are numbered 1 - 16, to indicate the active digital output channel.

Fiber Optic Port (To Base)

The stimulus isolator's fiber optic input port (labeled To Base) provides an isolated connection to the base station (RZ5 or RX7). The fiber optic cable carries digital signals to D/A's on the stimulus isolator. It also carries control information and information about the state of the stimulation channels. One end of the fiber optic cable connects to the device using the To Base connection pair and the other end connects to the Stimulator input on the base station.

Keep in mind, because of the fiber optic cable data transfer rate, the corresponding Stimulator fiber optic output port on the base station (RZ5 or RX7) will be disabled if the system sampling rate is set to a value greater than 24.414 kHz.

High Voltage Input (Back Panel)

The stimulator uses either the NC48 or the HV250 High voltage Battery Pack for stimulation. The battery pack should be connected via the Battery connection on the back panel.

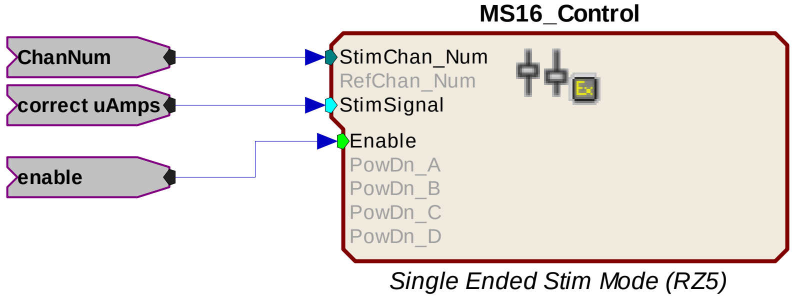
Warning! The HV250 battery packs are capable of delivering up to 250 Volts DC at high currents. Shorting the device can cause damage to the device and injury to the user. Always use caution when handling or connecting the devices.

Power Switch (Back Panel)

The Power switch turns the stimulus isolator power off or on. The fiber connector on the front panel will be illuminated when the stimulator is on.

Software Control

Operation of the MicroStimulator system is controlled via an RPvdsEx circuit loaded and run on the connected base station processor (RZ5 or RX7). TDT recommends using the MS16_Control Macro (pictured below) in your control circuits. This macro simplifies setup of stimulus and reference channels, stimulus signal output, and power conservation. The macro is also used to configure the correct scale factors and poke addresses for the RZ5 or RX7 processor. Select the correct device in the macro settings dialog.



When the MS16_Control macro is not sufficient for your task, a circuit can be designed using the Poke component to control the system. This component writes to special memory locations on System 3 devices and is intended primarily for TDT use. While both methods are described here, keep in mind that the Poke component should be used with caution.

Important Circuit Design Considerations

Sampling Rate

When using the RZ5 or RX7 with the stimulus isolator, the maximum sampling rate of the system is 24.414 kHz, a limitation of the fiber optic connection between the base station and the stimulus isolator.

Signal Resolution

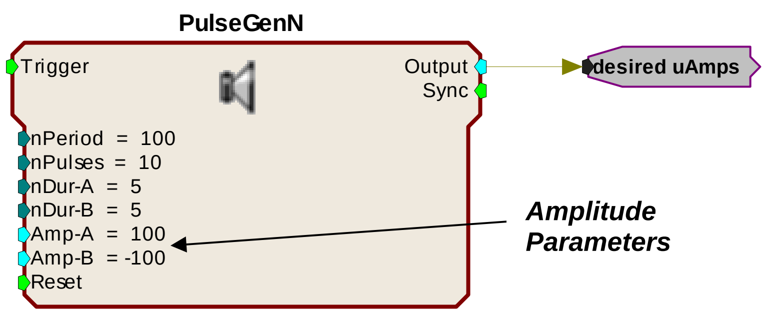
Signal resolution is dependant on the sampling rate used. The stimulus isolator's PCM D/A converters allow users to generate precise pulsed signals, including square waves with durations of only 1 sample. When using the maximum sampling rate of 24.414 kHz, the sample period is 40.96 microseconds. The stimulus isolator has an effective bandwidth of 10 kHz for continuous (non-pulsed) waveforms.

Designing the Stimulus Signal

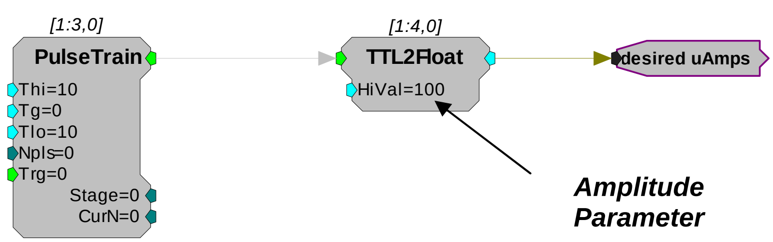
The MicroStimulator system offers flexible stimulus delivery capable of generating complex patterns of pulses or arbitrary waveforms. This allows you to make use of the full range of the waveform and pulse generators in the RPvdsEx component library, including the PulseGenN macro.

Desired Signal Range

When adding and configuring waveform components you must consider the output range of the system. The default configuration of the stimulus isolator can deliver stimuli in the range of +/-100 µA; be sure to set component amplitude parameters with this output range in mind. In the figure below, the amplitude of a biphasic pulse is defined in the Amp-A and Amp-B parameters.



When using components that output a logical signal, such as a PulseTrain, the output range can be defined when the output is converted to the desired data type. In the figure below the PulseTrain component sends out a standard TTL signal with a fixed duration. A TTL2Float component is then used to convert the signal to a user specified value between 0 and 100. This value indicates the desired stimulator output in microAmps.



If the ACC16 is not in use the desired uAmps in floating point format can be fed directly to the MS16_Control macro's Stim Signal input. If the ACC16 is being used a correction factor must be applied (see below).

ACC16 Correction Factor

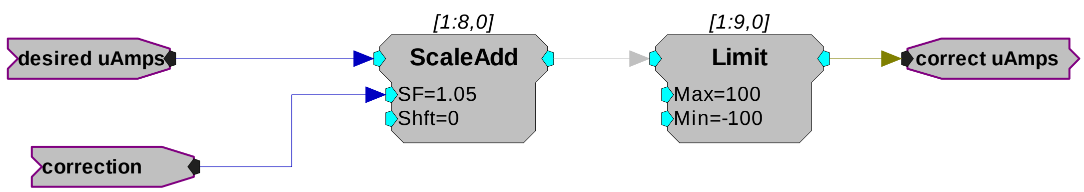
An ACC16 AC coupler can be used with the system in single-ended operation (global reference) to block any DC bias present on the Stim Output lines (a problem primarily affecting researchers using electrodes with impedances of more than 200 kOhms). When the ACC16 is in use, it acts as a voltage divider, decreasing the voltage and thus the current delivered through the electrode. The actual current delivered through the ACC16 depends on the ratio of the coupler impedance to the impedance of the electrode in use. For 50 kOhm electrodes the error is about 5%.

To calculate a correction factor for actual current delivered:

1. Determine the impedance of your stimulating electrode.
2. Calculate the following equation:

   Correction = 1/(1,000,000/(Electrode Imp+1,000,000))

   = (Electrode Imp +1,000,000)/1,000,000
3. In your circuit, scale the current output by this value.



In the example correction circuit above:

- The value for "**correction**" represents the results of the calculation above.
- The value for "**desired uAmps**" represents the desired amplitude of the stimulus signal.
- The values for the "**Limit**" component should be set based on the actual limits of your systems. The MS4/MS16 is available in 100 µA and 1 mA versions. **In either case, when using the ACC16 you will NOT be able to deliver the MAXIMUM current. The maximum current = 1/correction factor x 100. Calling for higher currents will deliver currents at the defined limit.**
- If using the recommended MS16_Control Macro, the ***correct uAmps*** value is fed to the macro's **Stim Signal** input.

Selecting Global or Local Reference Mode

The **MS16_Control** macro should be included in all circuits for stimulus isolator control. The **Stimulation Mode** setting on the **Setup** tab of the macro properties dialog box determines whether the stimulus isolator is configured to use a global reference (*Single ended*) or a local reference(s) (*Differential*).

Global Reference Mode

If a global reference is desired, set the **MS16_Control** macro's **Stimulation Mode** to *Single Ended* on the **Setup** tab of the macro properties dialog box. In this mode the **RefChan** input is disabled.

Local Reference Mode

If local reference is desired, set the **MS16_Control** macro's **Stimulation Mode** to *Differential* on the **Setup** tab of the macro properties dialog box. In this mode the **RefChan** input is enabled.

Note: In Local Reference (*Differential*) mode, writing a 0 to the **RefChan_Mask** macro input while the **Channel Select Method** is set to **With Chan Mask**, will disable all local reference channels and enable the global reference.

Configuring Reference and Stimulation Channels

The **MS16_Control** macro sets reference and stimulation channels. Feeding an integer value to the macro's **StimChan** and **RefChan** inputs will *turn on* channels for stimulation or reference, respectively. The **Channel Select Method** on the **Setup** tab of the macro properties dialog box determines whether the integer is read as a single channel number or as a mask value representing multiple channels.

Important Note! Configuring a channel, as both stimulus and reference will cause the unit to automatically turn off that bank of channels.

Setting a Single Channel for Stimulation or Local Reference

By default, the **Channel Select Method** on the **Setup** tab of the macro properties dialog box is set to **With Chan Number**. The **StimChan** and **RefChan** inputs accept an integer value of 0 through 16 and the macro will set the selected channel for stimulation or local reference.

Note: an integer value of 0 fed to StimChan disables all channels.

Setting Multiple Channels for Stimulation or Local Reference

To configure multiple reference channels, the **Channel Select Method** on the **Setup** tab of the macros properties box must be set to **With Chan Mask**. In this mode, **StimChan** and **RefChan** inputs accept an integer value channel mask representative of the desired channels (shown in the table below). The integer value is the sum of the channel masks for the channels.

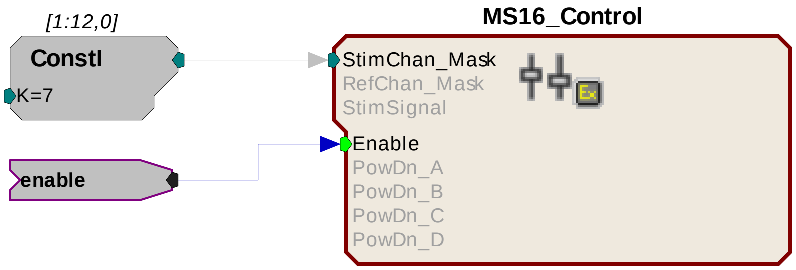
Channel Mask Table:

Channel #	Channel Mask
1	1
2	2
3	4
4	8
5	16
6	32
7	64
8	128

Channel #	Channel Mask
9	256
10	512
11	1024
12	2048
13	4096
14	8192
15	16384
16	32768

For example:

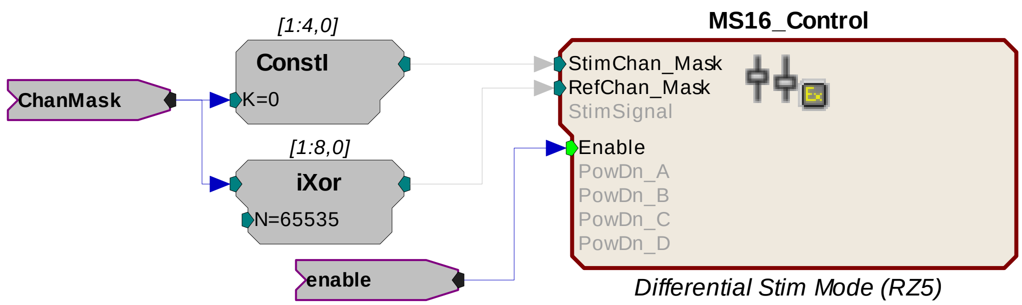
If you wish to simultaneously set channels 1 (channel mask 1), 2 (channel mask 2), and 3 (channel mask 4) to stimulation mode add their respective channel masks from the table above (1 + 2 + 4 = 7), and send that sum (7) to the StimChan_Mask input as shown in the figure below.



Single Ended Stim Mode (RZ5)

This example sets channels 1, 2, and 3 for stimulation. Unused banks of channels are powered down. The stimulus design and delivery are not included in this circuit segment.

The reference channels can be configured in the same way, using the integer values in the Channel Mask Table above. The iXor component can also be used to set all channels NOT set as stimulation to reference. In the figure below, an iXor is used to perform an ***exclusive*** bitwise OR function. The channel mask for stimulation is XORed with the integer mask value for all channels, resulting in a channel mask that sets all non-stimulus channels to reference channels.



Differential Stim Mode (RZ5)

Important!: Writing a 0 to the **RefChan_Mask** macro input while the **Channel Select Method** is set to **With Chan Mask**, will disable all local reference channels and enable the global reference.

Delivering the Stimulation

The stimulus delivery segment of the circuit can be handled within the MS16_Control macro or external to the macro using the Poke component. TDT recommends using the MS16_Control macro whenever possible.

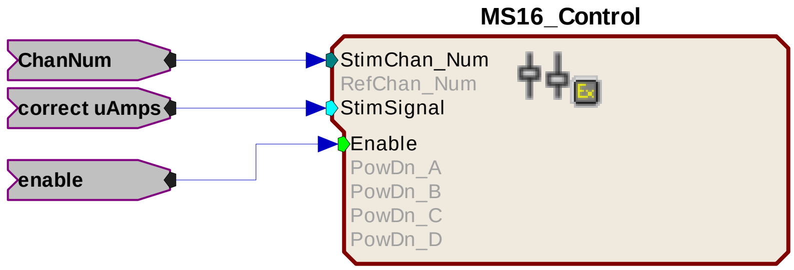
The Poke component should be used with caution; however, it is necessary for some tasks, including simultaneous stimulation on multiple channels.

Important!: The memory addresses used with the Poke component are different for the RZ5 and RX7. See the memory address table, page 6-14 for more information.

Single Channel Stimulation with Global Reference

When the global reference is used, the **MS16_Control** macro can be used for single channel stimulation. The **Stimulation Mode** on the **Setup** tab of the macro's properties box must be set to *Single Ended* and the **Channel Select Method** must be set to **With Chan Number** to enable the **StimSignal** input.

StimSignal accepts floating-point input, representative of the desired stimulus current waveform. The macro will send the stimulus signal to the channel set using the **StimChan_Num** input.



Single Ended Stim Mode (RZ5)

This example sends floating point values representing the amplitude of the waveform in microAmps to a user-specified channel of the stimulator as long as the enable is high. If using the ACC16 be sure to scale the signal by the necessary correction factor. See *ACC16 Correction Factor*, page 6-10 for more information.

Note: To conserve the life of the stimulus isolator's onboard and external batteries, remember to power down unused bank of channels on the **MS16_Control** macro's **Power Control** tab.

Simultaneous Stimulation on Multiple Channels and/or Local Reference Mode

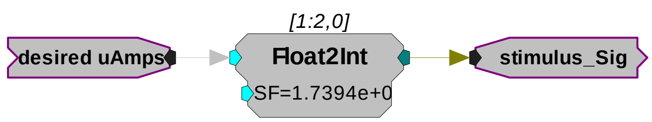
The MS16_Control macro’s StimSignal is disabled whenever the local reference mode is used or when a channel mask is used to set multiple stimulation channels. In these cases the macro should still be used to configure or turn on channels for stimulation (see *Configuring Reference and Stimulation Channels*, page 6-11), but stimulus delivery must be handled external to the macro.

Converting the Signal to an Integer Value

When designing the stimulus signal it is convenient to work with floating point values that represents the desired current in microAmps (See *Designing the Stimulus Signal*, page 6-9). However, when the macro is not used the stimulus signal must be converted to an integer value representing a voltage level in the proper range for the stimulus isolator. The scale factor required to scale the current in the desired range of +/-100 µA is dependent on the type of base station processor being used.

RZ5 When using the RZ5, use a scale factor of: **1.7394e+007**

RX7 When using the RX7, use a scale factor of: **265.41**

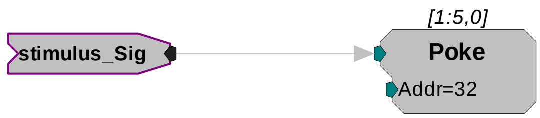


In this circuit segment, the desired floating point value in microAmps is fed to a Float2Int, which converts the data type and applies the scale factor.

Signal Output to Stimulus Channels

Once output waveforms are converted to an integer value they are poked (written) to memory locations on the MS4/MS16, using the Poke component. Memory addresses vary be processor as described here. *Reference tables are also provided below; page 6-14.*

RZ5 When using the RZ5, output to channels 1-16 must be written to memory addresses 32-47, respectively. To do so, offset the channel number by 31 and enter this value in the address parameter of the Poke component.

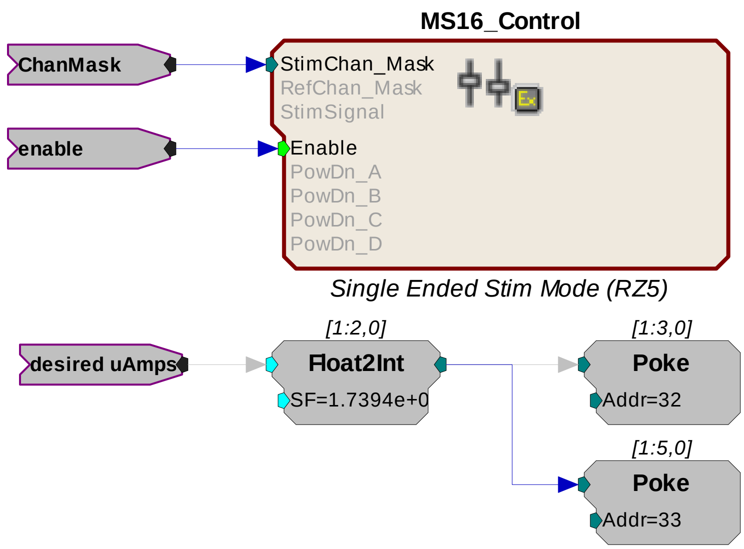


The circuit segment above sends out a stimulus signal to channel one of the stimulator.

RX7 When using the RX7, output to channels 1-16 must be written to memory addresses 20-35, respectively. To do so, offset the channel number by 19 and enter this value in the address parameter of the Poke component.

Summary: Simultaneous Stimulation on Multiple Channels

The example below shows a more complete picture, with the MS16_Control macro used to set or *turn on* multiple channels using the ChanMask hop, see page 6-11, and the Poke used to write the signal value to the MS4/MS16 memory location for channels one and two with the RZ5.



Circuit Design Using the Poke Component

Using the MS16_Control macro simplifies circuit design for the MicroStimulator System. If the macro **cannot** be used, you can use the RPvdsEx Poke component to control the stimulus isolator by writing information to memory addresses on the RZ5 or RX7.

Memory Address Reference for Using the Poke Component

The table below summarizes each stimulus isolator control function and its memory address.

Control	Value Description	Memory Address	
		RZ5	RX7
Stimulus Channels	Mask for channels between none and 16; integer value between 0 and 65535	48	7
Signal Output	Integer representing current level scaled for D/A (varies depending on device).	32-47	20-35
Global Reference	0 (off) or 1 (on)	50	9

Reference Channels	Mask for channels between none and 16; integer value between 0 and 65535	49	8
Digital Out	Mask for channels between none and 16; integer value between 0 and 65535	51	3

Signal Output to Stimulus Channels

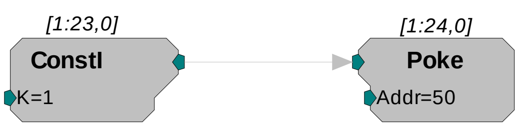
To generate signals on the stimulus isolator, the output waveforms are poked (written) to memory locations as integer values. See *Converting the Signal to an Integer Value*, page 6-13, for more information.

The table below maps the output channels of the RZ5 and RX7 to their poke address.

Isolator Output Channel	**Poke Waveform To Address**		**Isolator Output Channel**	**Poke Waveform To Address**	
	RZ5	**RX7**		**RZ5**	**RX7**
1	32	20	9	40	28
2	33	21	10	41	29
3	34	22	11	42	30
4	35	23	12	43	31
5	36	24	13	44	32
6	37	25	14	45	33
7	38	26	15	46	34
8	39	27	16	47	35

Global Reference Enable

Global reference uses the analog ground to complete the stimulation circuit. The global reference feature can be enabled by setting the value of a specific memory address to one. The StimRef indicator light on the front panel of the stimulus isolator is illuminated when the global reference has been set.



RZ5 To enable global reference when using an RZ5 **set the value of memory address 50 to one** as pictured above.

RX7 To enable global reference when using the RX7 **set the value of address 9 to one**.

Channel Masks

Memory addresses for stimulus, reference, or digital I/O channel setup expect an integer value between zero and 65535. Masked values for each channel are noted in the table below. Adding masked values together will set multiple channels.

The table below maps channel numbers to mask values.

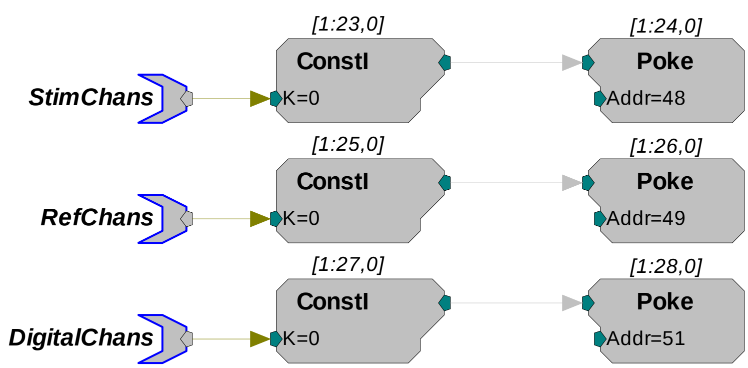
Channel #	Channel Mask	Channel #	Channel Mask
1	1	9	256
2	2	10	512
3	4	11	1024
4	8	12	2048
5	16	13	4096
6	32	14	8192
7	64	15	16384
8	128	16	32768

For example:

If channels 1 (channel mask 1), 2 (channel mask 2), and 3 (channel mask 4) are desired, use a channel mask of 7 (1 + 2 + 4 = 7).

Stimulus, Reference, or Control Channel Setup

To enable a given channel, an integer value is written to the appropriate memory address of the base station. The integer value is the sum of the channel masks (see table above for mask values) for all the stimulation channels that the user wishes to activate.



In the example circuit above, the StimChans parameter tag feeds a ConstI an integer value used to assign channels as stimulus channels, RefChans sets the reference channels, and DigitalChans sets the digital channels. This example above is configured for the RZ5.

Important!: The memory addresses for the RZ5 and RX7 are different. See the memory address table, page 6-14 for more information.

Note: When using the SH16 switching headstage, the digital I/O channels on the MS4/MS16 are used to control the switching headstage. These are accessed via a DB25 connector labeled ***Control***. For SH16 switching headstages (serial number 2000 and greater), channels 1-3 are used for communication and channels 4-8 are used to provide power to the SH16. When the SH16 is not being used, the MS4/MS16 digital I/O can be used for any type of digital control.

See *SH16 – 16 Channel Switchable Acute Headstage*, page 7-19, for more information about controlling the headstage.

Working with the MS16 MilliAmp Mode

The MS16 can be modified at the factory to deliver stimuli in the +/- 1 mA range. If your device has this modification, please note the following important differences in operation.

The HV250 battery pack **CANNOT** be used with milliAmp mode. This mode should only be used with the NC48 battery pack.

Circuit Design for the MS16 in MilliAmp Mode

MS16_Control Macro

When using the **MS16_Control** macro set **High Current Range** on the **Setup** tab of the macro's properties box to **Yes**. If High Current Range is set to Yes, all other circuit design considerations are handled automatically by the macro.

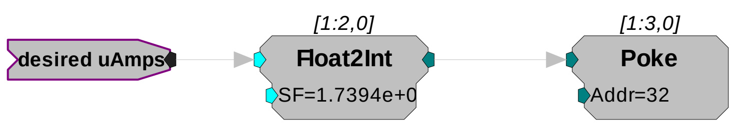
Scale Factor

When using the Poke component for stimulus delivery, use the appropriate scale factor for your processor to convert the signal in desired or corrected microAmps to the necessary voltage for A/Ds.

RZ5 When using RZ5, use a scale factor of **1.7394e+006.**

RX7 When using RX7, use a scale factor of **26.541.**

See *Converting the Signal to an Integer Value*, page 6-13, for more information.



In this circuit segment, the desired floating point value in microAmps is fed to a Float2Int, which converts the data type and applies the necessary scale factor for MilliAmp mode.

High Current Mode

When the MS16_Control is not used at all, the high current mode can be set by sending a specific value to the appropriate memory address for your processor. This memory address is the same address used to turn on or off the global reference. The value used to set the high current mode can be added to the global reference values 0 (off) and 1(on).

RZ5 When using the RZ5, the high current mode can be set by sending a **value of 54784 to memory address 50**.

Therefore, poking 54784 to the address turns on high current mode and turns off the global reference; while poking 54785 to the address turns on high current mode and turns on the global reference.

RX7 When using the RX7, the high current mode can be set by sending a **value of 214 to memory address 9**.

Therefore, poking 214 to address 9 turns on high current mode and turns off the global reference; while poking 215 to address 9 turns on high current mode and turns on the global reference.

Stimulus Isolator Technical Specifications

Technical specifications for the MS4/MS16 Stimulus Isolator.

Stimulus Output Channels	4 (MS4) or 16 (MS16)
Sampling rate	Up to 24.414 kHz
Stimulus Output Voltage	+/- 24 V with NC48 +/- 135 V with HV250
Stimulus Output Current	+/- 100 µA up to 1 MOhm load with HV250 +/- 100 µA up to 200 kOhms load with NC48*
DC Offset Current	Less than 0.2% of full range setting
Digital Output Max Current	40 mA
Digital Output Max Voltage	3.3 V
Selectable Reference	Local or Global
Power Control **Stimulation**	Onboard Rechargeable Li-Ion battery NC48 Rechargeable Battery with NiCad batteries* or HV250 Battery Pack with Carbon Zinc batteries

***Note:** the Stimulus Isolator may be modified at the factory for 1 MilliAmp Mode.

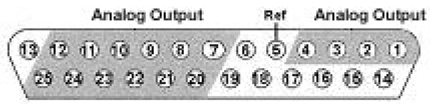
DB25 Connector Pinouts

STIM ELE Connector on the ACC16

The ACC16 AC Coupler is used to block DC bias and connects directly to this Stim Output Connector, passing signals through to its STIM ELE connector with the same pinout.

Stim Output Connector

The Stim Output connector provides access to the analog output channels. These channels are used primarily for stimulus output.



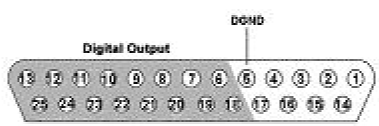
Pin	Name	Description
1	A1	Analog Channels Ch 1-4
2	A2	
3	A3	
4	A4	
5	Ref	Reference
6	NA	Not Used
7	A5	Analog Channels Ch 5, 7, 9, 11, 13, and 15
8	A7	
9	A9	
10	A11	
11	A13	
12	A15	
13	NA	Not Used

Pin	Name	Description
14	NA	Not Used
15		
16		
17		
18		
19		
20	A6	Analog Channels Ch 6, 8, 10, 12, and 14, 16
21	A8	
22	A10	
23	A12	
24	A14	
25	A16	

Note: Channels 5 - 16 not available on the MS4.

Control Output Connector

This connector provides access to control or relay output channels.



Pin	Name	Description
1	NA	Not Used
2		
3		
4		
5	DGND	Digital Ground
6	D1	Digital Output Bits 1, 3, 5, 7, 9, 11, 13, and 15
7	D3	
8	D5	
9	D7	
10	D9	
11	D11	
12	D13	
13	D15	

Pin	Name	Description
14	NA	Not Used
15		
16		
17		
18	D0	Digital Outputs Bits 0, 2, 4, 6, 8, 10, 12, and 14
19	D2	
20	D4	
21	D6	
22	D8	
23	D10	
24	D12	
25	D14	

Battery Reference

The stimulus isolator uses an onboard Lithium-Ion battery for general device operation. These batteries charge in four hours. A 6-9 Volt battery charger with 500 mA of current capacity is included with the stimulator and can be connected via the Charger connector on the stimulator's back panel. The charger tip is center negative. If it is necessary to replace the charger, ensure that the power supply has the correct polarity.

Issue	**HV250**	**NC48**	**Onboard Li-Ion**
Battery life	130 mA (up to 27 hours stimulation)	1000 mA (up to 240 hours of stimulation)	12-15 hours battery life between charges
Rechargeable	No	Yes	Yes
Compliance voltage	+/- 135 Volts	+/- 24 Volts	N/A
Maximum impedance for delivering a 100 microAmp current	1 MOhms	200 kOhms	N/A
Usable in MilliAmp Mode	No	Yes	Yes
Ambient temperature	Normal room temperatures	Normal room temperatures	Normal room temperatures

HV250 Battery Pack

The HV250 Battery Pack uses four Carbon Zinc batteries, each delivering 67 Volts. Because the HV250 Battery Back is non-rechargeable, it must be replaced periodically. The High Voltage LED on the front panel of the MS4/MS16 will flash to alert the user of a low voltage condition. To extend the life of the battery, we recommend enabling only the desired channels for stimulation.

WARNING The HV250 is a high-voltage power source, capable of delivering up to 250 Volts DC at high amperages. Shorting the device can cause damage to the device and injury to the user. Always use caution when handling or connecting the devices. Never attempt to charge the HV250.

NC48 Battery Pack

The NC48 Battery Pack uses 32 Nickel Cadmium (NiCad) batteries to supply a peak-to-peak voltage of 48 Volts with a range of +/- 24 Volts.

WARNING Just as with all batteries, shorting the NC48 Battery Pack can cause damage to the device and injury to the user. Always use caution when handling or connecting the devices.

WARNING Overcharging the NC48 battery pack can cause the cells to rupture.

The NC48 Battery Pack should be connected to its charger for a maximum of 16 hours. Overcharging shortens battery life and may burn out the battery in extreme cases. Although the batteries used in the NC48 are designed to provide the user with dozens of charge/discharge cycles, the performance of all rechargeable batteries deteriorates over time. The major sign that a battery is deteriorating is a shortened use cycle between charges.

Note! Used NiCad batteries must be recycled.

The NC48 Battery pack should be stored at normal room temperatures. Temperature extremes can affect the operation of the batteries. Battery packs stored for longer than two months should be tested prior to use.

MS4/MS16 Anomalies

If the stimulus isolator control bits and relay switching control bits do not work after power up, execute a hardware reset on the base station using zBusMon.

Serial numbers 4000 and above
Previous versions of the stimulator automatically switched banks of channels off when not in use. A recent change to the microcode eliminates this feature, giving users control over when channels are turned off. By default, all channels are on and must be turned off manually.

Serial numbers below 4008 (MS4) and 4015 (MS16)

When the NC48 is connected to the stimulus isolator, the High Voltage LED on the front panel of the MS4/MS16 will constantly flash even when the NC48 (+/-24 V) is at full charge, because the voltage monitoring circuitry was designed to detect a low voltage of the HV250 battery pack.

Serial numbers below 4000
The MS4/MS16 has undergone several design changes to improve performance and usability. TDT recommends that all users upgrade to the latest versions (serial numbers 4000 and above). Contact TDT for an RMA to upgrade your current module.

Serial numbers below 3000
Noise on outputs is high when the output is in 'Open" mode. The noise is especially evident during recording and stimulation events. Contact TDT for an RMA for upgrade of your current device.

Conservation of Power
The stimulus isolator's analog channels are arranged in four-channel banks. Each of these banks is powered up on reset of the device and will remain powered on. To conserve power, TDT recommends powering down unused banks of channels. The MS16_Control macro can be used to turn off unused banks of channels. When not using the macro, simultaneously setting any channel in a bank to both Stimulate and Reference mode turns off that four-channel bank.

Maximum Voltage Output
The stimulus output channels drive a current signal that ranges from 0-100 microAmps. The maximum voltage output from the MicroStimulator system using the TDT NC48 battery is the 24 volts and the maximum voltage output using the TDT HV250 battery is 125 Volts. The actual voltage output depends on the current waveform specified and the impedance of your electrodes, that is, V = ZI where V=Volts, Z = impedance and I = current.

Using the MicroStimulator with TDT's Switching Headstage
When using TDT's switching headstage, ensure that relays for channels used for stimulation have been switched to the correct position using the SH16_Control macro. Any stimulus channel for which the corresponding control channel has not also been set will fail to generate a signal. See *Switchable Headstage Operation*, page 7-21.

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Part 7 High Impedance Headstages

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ZIF-Clip® Headstage

Overview

The ZIF-Clip® headstage (Patent No. 7540752) features an innovative, hinged headstage design that ensures quick, easy headstage connection with almost no insertion force applied to the subject. ZIF-Clip® headstage contacts seat inside the probe array and snap in place, firmly locking the headstage and probe with very little applied pressure. These self aligning headstages provide long lasting low insertion performance for a variety of channel number and electrode configurations. An aluminum finish provides increased durability.

By default, ground and reference are separate on all ZIF Clip® headstages yielding a differential configuration. Reference and ground may be tied together on the headstage adapter or ZIF Clip® microwire array for single-ended configurations.

Note: ZIF-Clip® headstages are designed to connect directly to any Z-Series preamplifier but may be connected to a Medusa preamplifier with the use of an adapter.

Part Numbers (Patent No. 7540752):

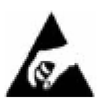
ZC16 - 16 Channel Aluminum ZIF-Clip® headstage

ZC32 - 32 Channel Aluminum ZIF-Clip® headstage

ZC64 - 64 Channel Aluminum ZIF-Clip® headstage

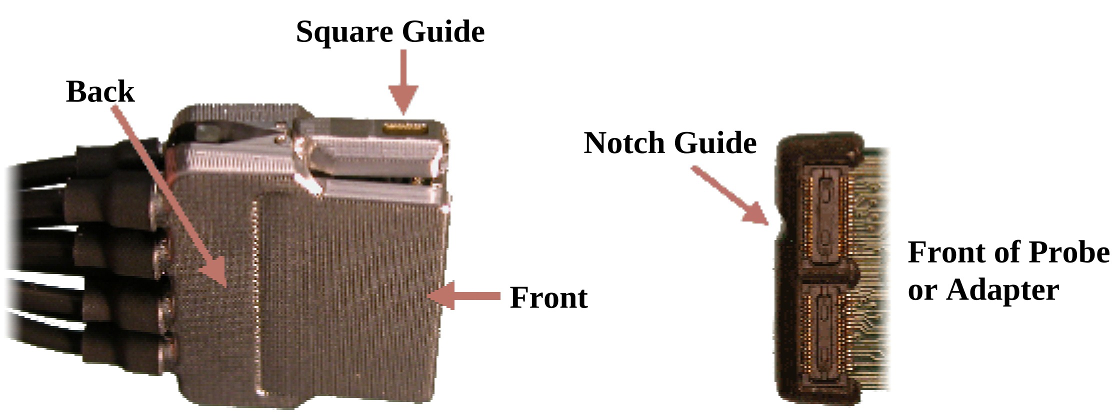
ZC96 - 96 Channel Aluminum ZIF-Clip® headstage

ZC128 - 128 Channel Aluminum ZIF-Clip® headstage

 The headstage has sensitive electronics. Always ground yourself before handling.

Adapter and Probe Connection

ZIF-Clip® headstages are designed to automatically position the high density connectors on the headstage and probe (or adapter) and are recommended for use with input impedances that range from 20 kOhm to 5 Mohm.



Connect the headstage as described below.

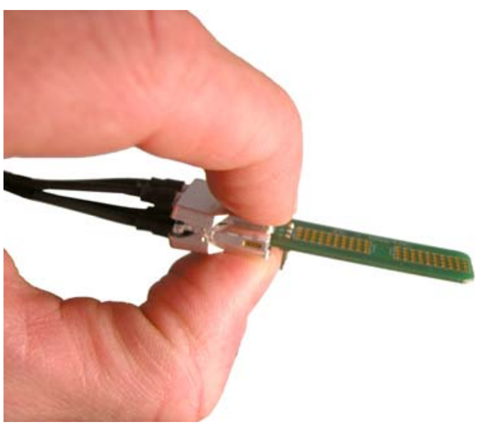
1. Firmly press and hold the **back** to open the headstage.

2. Align the **notch guide** of connector to the **black square guide** of the fully opened headstage then move headstage into position.

WARNING!

The ZIF-Clip® headstage **must be held** in the **fully open position** while being slid into position.

The headstage should only be closed when fully engaged. Sliding the headstage into position while applying pressure to the tip will **permanently damage** the ZIF-Clip® headstage and micro connectors.



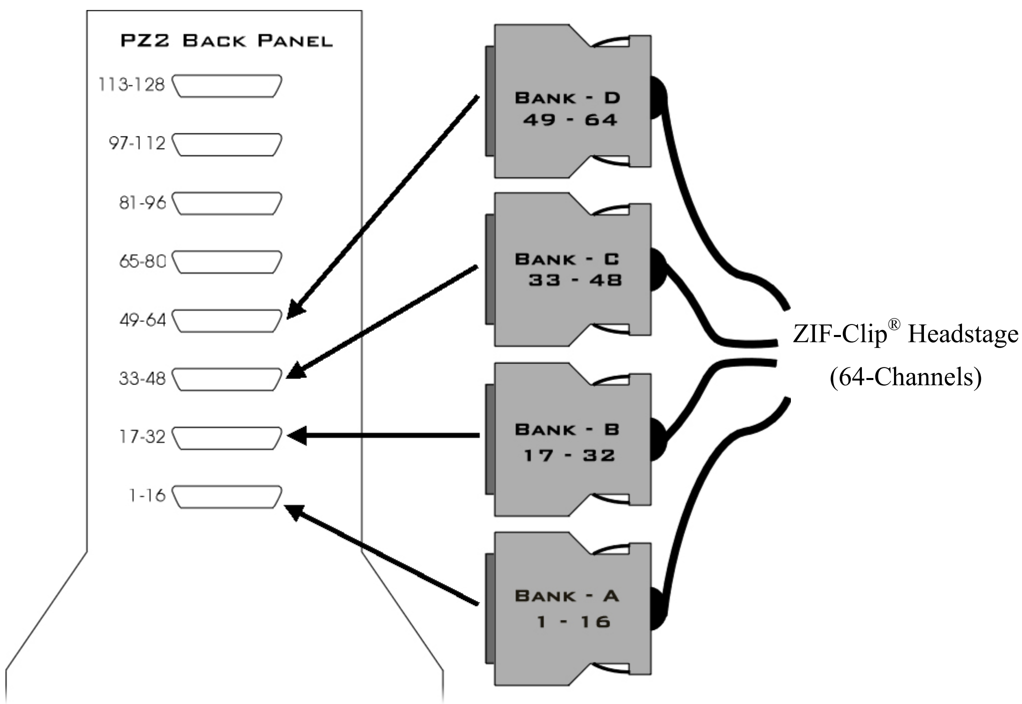
3. Press the **front** of the headstage together as shown to lock the connector in place.

Preamplifier Connection

One or more MiniDB26 connectors are used to connect the ZIF-Clip® headstage to a PZ2 preamplifier depending on the number of channels in the headstage. Each MiniDB26 connector is labeled with a bank letter that corresponds to its intended connecting bank on the preamplifier. For example the MiniDB26 connector labeled “Bank A” should connect to bank 1 on the PZ2 and will carry channels 1-16. Subsequently, banks B, C, D, etc, correspond to the next 16 channels of the headstage. Below is a table which shows the Bank labeled connectors as well as the necessary channels or banks they connect to on the PZ2.

ZIF-Clip® headstage	Bank Label on MiniDB26	Connect to PZ2 Bank
ZC16 (Connects Bank A)	Bank - A	1 (Channels 1 - 16)
ZC32 (Connects Banks A - B)	Bank - B	2 (Channels 17 - 32)
ZC64 (Connects Banks A - D)	Bank - C	3 (Channels 33 - 48)
ZC96 (Connects Banks A - F)	Bank - D	4 (Channels 49 - 64)
ZC128 (Connects Banks A - H)	Bank - E	5 (Channels 65 - 80)
	Bank - F	6 (Channels 81 - 96)
	Bank - G	7 (Channels 97 - 112)
	Bank - H	8 (Channels 113 - 128)

The diagram below illustrates the connection of a ZC64 ZIF-Clip® headstage to the PZ2 Preamplifier. Note that the bank channel numbering matches on both the preamplifier and headstage MiniDB26 connectors.



Headstage Voltage Range

When using a TDT preamplifier the voltage input range of the preamplifer is typically lower than the headstage and must be considered the effective range of the system. Check the specifications of your amplifier for voltage range. Also keep in mind that the range of the headstage varies depending on the power supply provided by the preamplifier. TDT preamplifiers supply +/- 1.5 VDC, but third party preamplifiers may vary. TDT recommends using preamplifiers which deliver +/- 2.5 VDC or less. Check the preamplifier voltage input and power supply specifications and headstage gain to determine the voltage range of the system.

The table below lists the input voltage ranges for the ZIF-Clip® headstage for either +/- 1.5 VDC or +/- 2.5 VDC power sources.

	Headstage input range when using +/- 1.5 VDC power source	Headstage input range when using +/- 2.5 VDC power source
ZIF-Clip® headstage	+/- 1.48 V	+/- 2.49 V

Technical Specifications

By default, ground and reference are separate on all ZIF Clip® headstages yielding a differential configuration. Reference and ground may be tied together on the headstage adapter or ZIF Clip® microwire array for single-ended configurations.

Important!: When using multiple headstages, ensure that a single ground is used for all headstages. This will avoid unnecessary noise contamination in recordings. See the headstage connection guide on page 5-33 for more information.

	ZIF-Clip® headstage			
S/N (typical)	rms 3 μ V bandwidth 300-3000 Hz rms 6 μ V bandwidth 30-8000 Hz			
Headstage Gain	Unity (1x)			
Input Impedance	10^{14} Ohms			
Dimensions	**Headstage**	**Length**	**Width**	**Height**
(Approx.)	ZC16/ZC32*	14.80 mm	10.60 mm	7.50 mm
	ZC64	17 mm	15 mm	7.50 mm
	ZC96	17.75 mm	18.60 mm	7.75 mm
	ZC128	18.70 mm	25 mm	7.75 mm

* Form factor for both the ZC16 and ZC32 is the same.

ZIF-Clip® Headstage Pinouts

TDT's ZIF-Clip® headstages are proprietary and are intended for use with ZIF-Clip® compatible electrodes or adapters. See page 10-4 for pinouts and connection information for ZIF-Clip® microwire arrays. If you are interested in using a third party electrode see page 9-6 for ZIF-Clip® adapter pinouts.

ZIF-Clip® Headstage Holder

Part Number: ZROD

The ZIF-Clip® headstage holder securely holds your ZIF-Clip® headstage during electrode insertion and can be used with most micromanipulators. The headstage holder is approximately 4.5” in length (the stabilizing rod is 3” in length and has a 3/32” diameter).

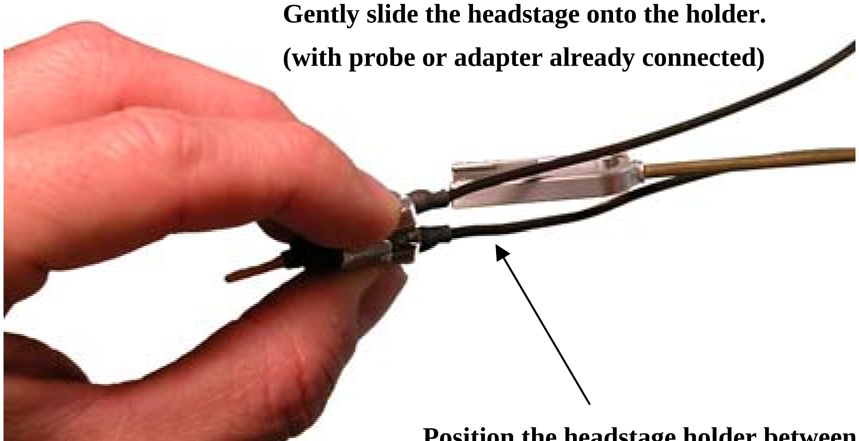
Each holder is designed for use with the selected ZIF-Clip® headstage.

Using the ZIF-Clip® Headstage Holder

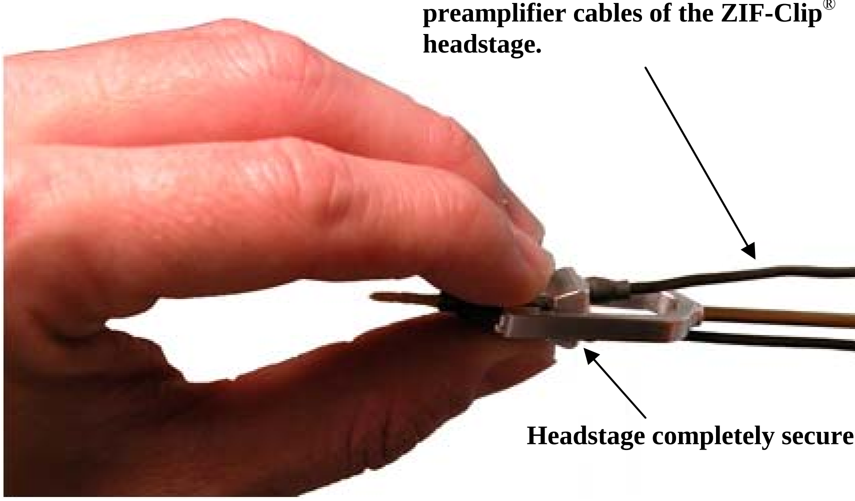
Connect the probe or adapter to your ZIF-Clip® headstage BEFORE putting the headstage in the holder (the square guide provided to ensure the probe or adapter is connected with the correct polarity is hidden from view when the headstage is in the holder). See the Adapter and Probe connection section on page 7-3 for more information. Gently slide the ZIF-Clip® headstage onto the holder until it is completely secure as shown in the images below.

Gently slide the headstage onto the holder.

(with probe or adapter already connected)



Position the headstage holder between the preamplifier cables of the ZIF-Clip® headstage.



Headstage completely secured in holder.

If you need to remove the headstage from the holder, grip the top and bottom of the ZIF-Clip® headstage and gently slide the holder back until it is no longer in contact with the headstage.

To remove, grip the top and bottom of the headstage and gently slide the holder off of the headstage.

Technical Specifications

Length	4.5” total length (stabilizing rod is 3” with a 3/32” diameter)
Weight	4.5g

RA16AC - 16 Channel Acute Headstage

Overview

The 16 Channel acute headstages is recommended for extracellular neurophysiology using silicon electrodes, metal microelectrodes or microwire arrays with recommended input impedances from 20 kOhm to 5 Mohm unless otherwise noted.

The 16 channel acute headstage has an 18-pin DIP connector that can be used with standard high impedance metal electrodes. The pinout of the RA16AC matches the wiring of NeuroNexus electrodes to allow for direct connection to the headstage. TDT recommends connecting electrodes to an 18-pin socket and then connecting the socket to the headstage to protect the headstage from unnecessary wear and tear. The RA16AC4 provides 4x gain and is used with electrodes with a recommended impedance range of 20 kOhm to 300 kOhm.

The headstage connects to a System 3 Medusa preamplifer (such as the RA16PA) via a DB25 connector or to a PZ series preamplifier via a mini 26-pin connector.

Part Numbers:

RA16AC – 16 Channel Acute Headstage for Medusa PreAmps, with unity (1x) gain

RA16AC4 - 16 Channel Acute Headstage for Medusa PreAmps, with 4x gain

RA16AC-Z - 16 Channel Acute Headstage for Z-Series (PZ) PreAmps, with unity (1x) gain

The headstage has sensitive electronics. Always ground yourself before handling.

Headstage Voltage Range

When using a TDT preamplifier the voltage input range of the preamplifer is typically lower than the headstage and must be considered the effective range of the system. Check the specifications of your amplifier for voltage range. Also keep in mind that the range of the headstage varies depending on the power supply provided by the preamplifier. TDT preamplifiers supply +/- 1.5 VDC, but third party preamplifiers may vary. TDT recommends using preamplifiers which deliver +/- 2.5 VDC or less. Check the preamplifier voltage input and power supply specifications and headstage gain to determine the voltage range of the system.

The table below lists the input voltage ranges for RA16AC headstages for either a +/- 1.5 VDC or +/- 2.5 VDC power source.

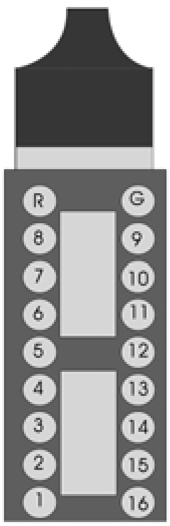
	Headstage input range when using +/- 1.5 VDC power source	**Headstage input range when using +/- 2.5 VDC power source**
RA16AC4	+/- 0.37 V	+/- 0.62 V
RA16AC	+/- 0.9 V	+/- 1.9 V

Technical Specifications

Warning!: When using multiple headstages ensure that all ground pins are connected to a single common node. See page 5-29 for more information.

S/N (typical)	rms 3 µ V bandwidth 300-3000 Hz rms 6 µ V bandwidth 30-8000 Hz
Headstage Gain	RA16AC - Unity (1x) RA16AC4 - 4x RA16AC-Z - Unity (1x)
Input Impedance	10^{14} Ohms

Pinout



(looking into connections)

The numbers in the diagram above show the channel connections to the amplifier. The electrode connector accepts 0.5 mm diameter male pins.

For pinouts for the preamplifier connector, see the corresponding preamplifier.

NN64AC - 64 Channel Acute Headstage

Overview

The 64 Channel Acute headstage is recommended for extracellular neurophysiology using silicon electrodes, metal microelectrodes or microwire arrays with input impedances from 20 kOhm to 5 Mohm.

The headstage features two 40-pin connectors designed for use with NeuroNexus Acute 64-channel probes. The headstage connects to a PZ series preamplifier via four mini 26-pin connectors or with System 3 Medusa preamplifers (such as four RA16PAs) via four DB25 connectors. In either case, each connector carries the signals for 16 channels, power and ground. Therefore, each connector can be connected independently. The connector labeled Bank-1 carries channels 1-16, Bank-2 carries 17-32, etc.

Part Numbers:

NN64AC – 64 Channel Acute Headstage for Medusa PreAmps

NN64AC-Z - 64 Channel Acute Headstage for Z-Series (PZ) PreAmps

 The headstage has sensitive electronics. Always ground yourself before handling.

Headstage Voltage Range

When using a TDT preamplifier the voltage input range of the preamplifer is typically lower than the headstage and must be considered the effective range of the system. Check the specifications of your amplifier for voltage range. Also keep in mind that the range of the headstage varies depending on the power supply provided by the preamplifier. TDT preamplifiers supply +/- 1.5 VDC, but third party preamplifiers may vary. TDT recommends using preamplifiers which deliver +/- 2.5 VDC or less. Check the preamplifier voltage input and power supply specifications and headstage gain to determine the voltage range of the system.

The table below lists the input voltage ranges for the NN64AC and NN64AC-Z headstages for either a +/- 1.5 VDC or +/- 2.5 VDC power source.

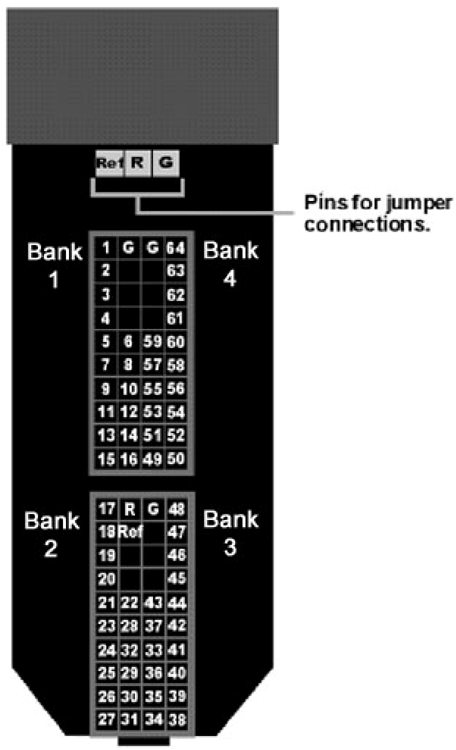
Headstage input range when using +/- 1.5 VDC power source	Headstage input range when using +/- 2.5 VDC power source
+/- 0.9 V	+/- 1.9 V

Technical Specifications

Warning!: When using multiple headstages ensure that all ground pins are connected to a single common node. See page 5-29 for more information.

S/N (typical)	rms 3 µ V bandwidth 300-3000 Hz rms 6 µ V bandwidth 30-8000 Hz
Headstage Gain	Unity (1x)
Input Impedance	10^{14} Ohms

Pinout



(looking into connections)

The numbers in the diagram above show the channel connections to the amplifier. The headstage also features jumper locations to short G, R and Ref (Ref refers to the built-in reference site on the NeuroNexus probe). The ground channel should either be tied to an external ground or to the reference for a single ended input.

See the table below, (NN32AC) for jumper configurations and associated requirements.

Important! When using the NN64AC with the NeuroNexus Acute 64-channel probe, keep in mind that there are several versions of the probe. Check the NeuroNexus Website for pin diagrams. Also, see *MCMap* for a description and examples on how to re-map channel numbers.

NN32AC - 32 Channel Acute Headstage

Overview

The 32 Channel Acute headstage is recommended for extracellular neurophysiology using silicon electrodes, metal microelectrodes or microwire arrays with input impedances from 20 kOhm to 5 Mohm. The headstage features a 40-pin connector designed for use with the NeuroNexus Acute 32-channel probe. The headstage connects to a PZ series preamplifier via two mini 26-pin connectors or to two RA16PA preamplifiers via two 25-pin connectors. For either headstage, Connector A carries the signals for channels 1-16, power and ground. This connector must be connected whether you are acquiring data from one of these channels or not.

Part Numbers:

NN32AC – 32 Channel Acute Headstage for Medusa PreAmps

NN32AC-Z - 32 Channel Acute Headstage for Z-Series (PZ) PreAmps


The headstage has sensitive electronics. Always ground yourself before handling.

Headstage Voltage Range

When using a TDT preamplifier the voltage input range of the preamplifer is typically lower than the headstage and must be considered the effective range of the system. Check the specifications of your amplifier for voltage range. Also keep in mind that the range of the headstage varies depending on the power supply provided by the preamplifier. TDT preamplifiers supply +/- 1.5 VDC, but third party preamplifiers may vary. TDT recommends using preamplifiers which deliver +/- 2.5 VDC or less. Check the preamplifier voltage input and power supply specifications and headstage gain to determine the voltage range of the system.

The table below lists the input voltage ranges for the NN32AC and NN32AC-Z for either a +/- 1.5 VDC or +/- 2.5 VDC power source.

Headstage input range when using +/- 1.5 VDC power source	**Headstage input range when using +/- 2.5 VDC power source**
+/- 0.9 V	+/- 1.9 V

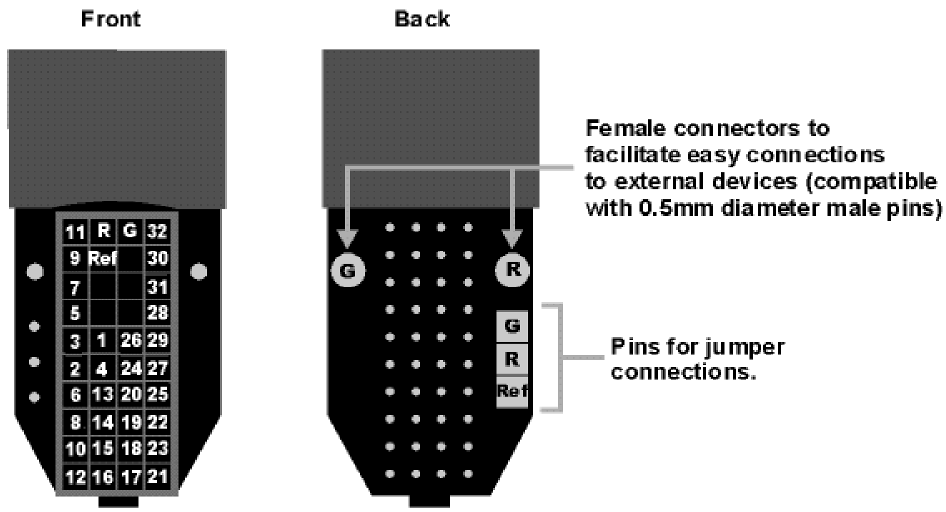
Technical Specifications

Warning!: When using multiple headstages ensure that all ground pins are connected to a single common node. See page 5-29 for more information.

S/N (typical)	rms 3 µ V bandwidth 300-3000 Hz rms 6 µ V bandwidth 30-8000 Hz
Headstage Gain	Unity (1x)

Input Impedance	10^{14} Ohms

Pinout



(looking into connections)

Important! When using the NN32AC with the NeuroNexus Acute 32-channel probe, keep in mind that there are several versions of the probe and the NN32AC was designed to correspond to the NeuroNexus rev 3 probe. Check the NeuroNexus Website for pin diagrams. Also, see *MCMap* in the *RPvdsEx User Guide*, for a description and examples on how to re-map channel numbers.

The numbers in the diagram above show the channel connections to the amplifier.

The surfaced connections on the back of the headstage include female connectors to simplify connections to external devices and jumper locations to short G, R and Ref (Ref refers to the built-in reference site on the NeuroNexus probe). The ground channel should either be tied to an external ground or to the reference for a single ended input.

Jumper Configuration

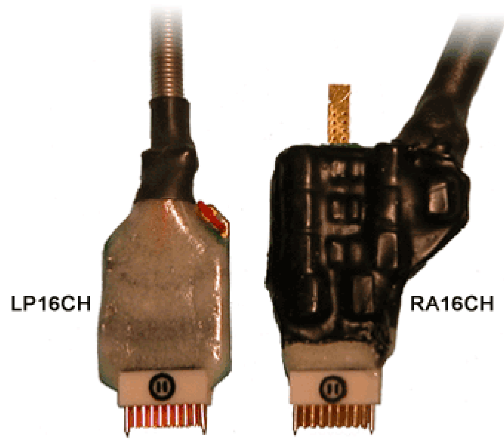
The following table describes the jumper configurations and associated requirements.

Jumper Connections	Operation	Requirements
[G R] Ref	Shorts headstage Ground and Reference inputs together, yielding single-ended amplification of signals relative to ground.	Connect common Ground/Reference wire to the headstage or electrode.
G [R Ref]	Shorts headstage Reference input to the pin labeled Ref (a low impedance site on the probe) yielding differential amplification of signals relative to the voltage of the Ref site.	Connect Ground wire to the headstage or electrode.
G R Ref	Headstage Ground and Reference separated and Ref pin is not used, yielding differential amplification of signals relative to the voltage of the Reference	Connect both a Ground wire and a Reference wire to the headstage or electrode.

RA16CH/LP16CH - 16 Channel Chronic Headstage

Overview

The 16 Channel Chronic headstages are recommended for extracellular neurophysiology using silicon electrodes, metal microelectrodes or microwire arrays with input impedances from 20 kOhm to 5 Mohm.



The 16-channel chronic headstages come in two configurations; RA16CH (standard profile) and LP16CH (low profile). The headstages provide the same performance with the smaller footprint of the LP16CH yielding better clearance in tight applications. The headstages use a low profile female Omnetics connector that is compatible with the NeuroNexus chronic electrodes. Users can also request the matching male Omentics connector (OMCON_ML_HB) from TDT for use in building electrode arrays.

Part Numbers:

LP16CH – 16 Channel Chronic Low Profile Headstage for Medusa PreAmps

LP16CH-Z – 16 Channel Chronic Low Profile Headstage for Z-Series (PZ) PreAmps

RA16CH – 16 Channel Chronic Headstage for Medusa PreAmps

RA16CH-Z – 16 Channel Chronic Headstage for Z-Series (PZ) PreAmps

 The headstage has sensitive electronics. Always ground yourself before handling.

Headstage Voltage Range

When using a TDT preamplifier the voltage input range of the preamplifer is typically lower than the headstage and must be considered the effective range of the system. Check the specifications of your amplifier for voltage range. Also keep in mind that the range of the headstage varies depending on the power supply provided by the preamplifier. TDT preamplifiers supply +/- 1.5 VDC, but third party preamplifiers may vary. TDT recommends using preamplifiers which deliver +/- 2.5 VDC or less. Check the preamplifier voltage input and power supply specifications and headstage gain to determine the voltage range of the system.

The table below lists the input voltage ranges for the 16 channel chronic headstages for either a +/- 1.5 VDC or +/- 2.5 VDC power source.

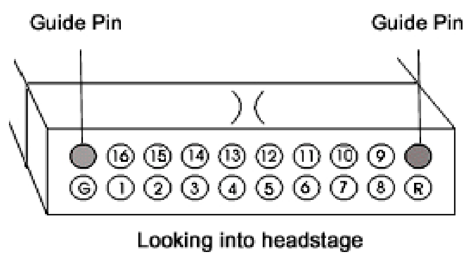
	Headstage input range when using +/- 1.5 VDC power source	**Headstage input range when using +/- 2.5 VDC power source**
LP16CH	+/- 1.48 V	+/- 2.49 V
RA16CH	+/- 0.9 V	+/- 1.9 V

Technical Specifications

Warning!: When using multiple headstages ensure that all ground pins are connected to a single common node. See page 5-29 for more information.

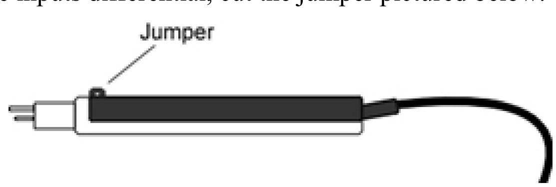
S/N (typical)	rms 3 µ V bandwidth 300-3000 Hz rms 6 µ V bandwidth 30-8000 Hz
Headstage Gain	Unity (1x)
Input Impedance	10^{14} Ohms

Pinout

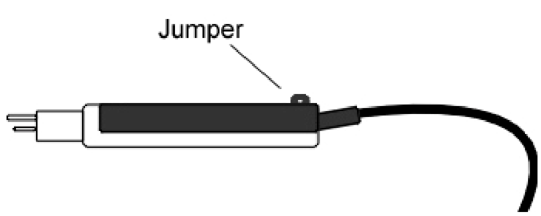


The numbers on the pinout diagram above show the channel connections to the amplifier. By default, the RA16CH/LP16CH inputs are single ended, with Ref and GND tied together. A jumper is provided to give the user the option of making the inputs differential.

To make the inputs differential, cut the jumper pictured below.



RA16CH:



LP16CH:

RA4AC - Four Channel Headstage

Overview

The 4 Channel Acute headstages are recommended for extracellular neurophysiology using silicon electrodes, metal microelectrodes, or microwire arrays with input impedances from 20 kOhm to 5 MOhm.

The RA4AC1 and RA4AC4 headstages have a low-profile 6-pin connector. The RA4AC1 provides unity gain (1x). The RA4AC4 provides 4x gain and is used with electrodes with a recommended impedance range of 20 kOhm to 300 kOhm. The 25-pin connector connects to the RA4PA 4-channel Medusa preamplifier.

Part Numbers:

RA4AC1 – 4 Channel Acute Headstage for Medusa PreAmps, with unity (1x) gain

RA4AC4 – 4 Channel Acute Headstage for Medusa PreAmps, with 4x gain

The headstage has sensitive electronics. Always ground yourself before handling.

Headstage Voltage Range

When using a TDT preamplifier the voltage input range of the preamplifer is typically lower than the headstage and must be considered the effective range of the system. Check the specifications of your amplifier for voltage range. Also keep in mind that the range of the headstage varies depending on the power supply provided by the preamplifier. TDT preamplifiers supply +/- 1.5 VDC, but third party preamplifiers may vary. TDT recommends using preamplifiers which deliver +/- 2.5 VDC or less. Check the preamplifier voltage input and power supply specifications and headstage gain to determine the voltage range of the system.

The table below lists the input voltage ranges for the RA4AC and RA4AC4 headstages for either a +/- 1.5 VDC or +/- 2.5 VDC power source.

	Headstage input range when using +/- 1.5 VDC power source	**Headstage input range when using +/- 2.5 VDC power source**
RA4AC4	+/- 0.37 V	+/- 0.62 V
RA4AC	+/- 0.9 V	+/- 1.9 V

Technical Specifications

Warning!: When using multiple headstages ensure that all ground pins are connected to a single common node. See page 5-29 for more information.

S/N (typical)	rms 3 μ V bandwidth 300-3000 Hz rms 6 μ V bandwidth 30-8000 Hz

Headstage Gain	RA4AC1 - Unity (1x) RA4AC4 - 4x
Input Impedance	10^{14} Ohms

Pinout

(looking into connections)

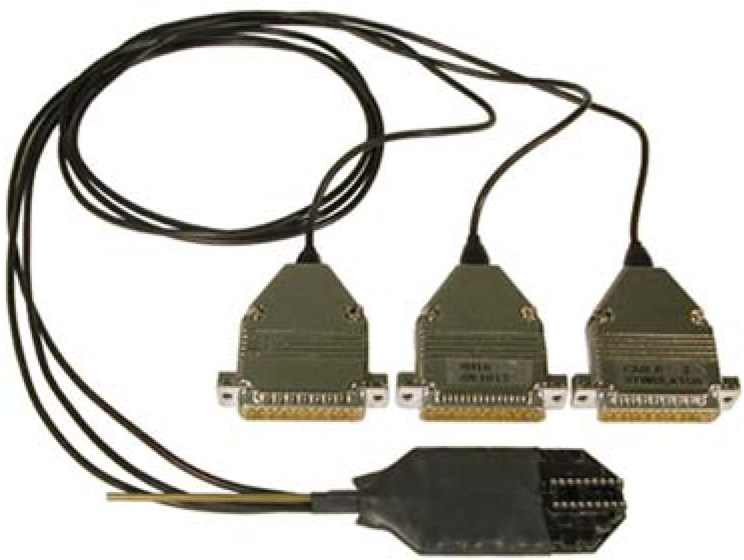
The numbers in the above diagram show the channel connections to the amplifier. The electrode connector accepts 0.76 mm diameter male pins.

The RA4AC1/RA4AC4 is also provided with a 6-pin male connector with flying leads. When connecting to the headstage, note that the silver dots marking channel 1 line up. The colors of the lead wires correspond to the headstage channels as follows:

Color	Channel
Black	1
Red	2
Orange	3
Yellow	4
Blue	Reference
Green	Ground

SH16 - 16 Channel Switchable Acute Headstage

Overview



The SH16 is a 16 channel acute headstage containing recording circuitry that can be bypassed for selected channels and connected to the stimulus isolator. It features high voltage, low leakage solid-state relays to allow for remote switching.

Note: The SH16 Switching headstage provides unity gain (1x) for its recording channels.

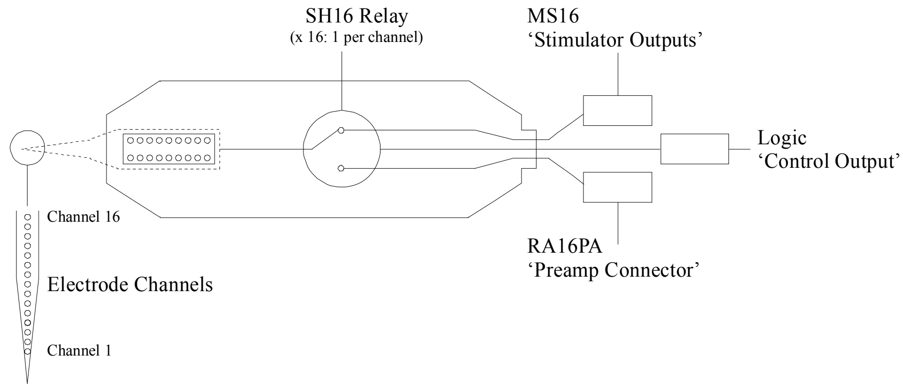
The minimum switching time for the SH16 is dependant on the length of time it takes to send the 24-bit serial control bit pattern (see *Creating the Serial Control Bit Pattern* for more information) that defines which channels are switched plus an inherent 2 ms delay associated with the solid state relay switches.

The minimum switching time can be calculated as follows:

[Number of bits in serial control pattern (24)] ÷ [Serial data transfer Rate (939 Hz Max)] + 2 ms

Serial Transfer Rate (Hz)	Minimum SH16 Switching Time (ms)
939	28
469	53

The diagram below illustrates how the relays are used to switch channels for recording (to RA16PA) or stimulation (from MS16).



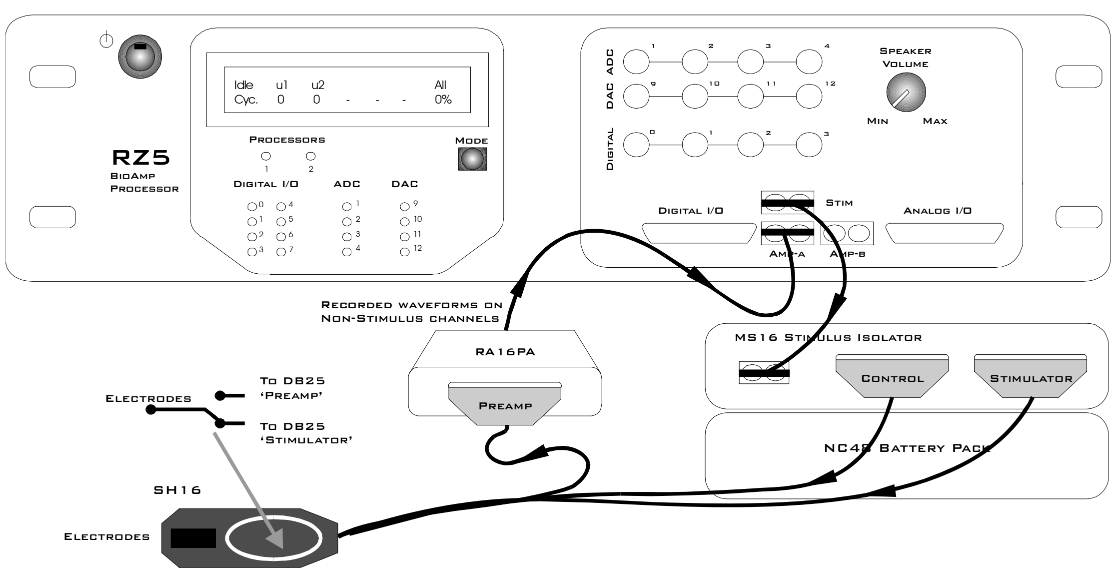
Switchable Headstage Diagram

The 16 channel switchable acute headstage has an 18-pin DIP connector that can be used with standard high impedance metal electrodes. The pinout of the SH16 matches the wiring of NeuroNexus electrodes, allowing direct connection to the headstage. TDT recommends connecting electrodes to an 18-pin DIP socket and then connecting the socket to the headstage to protect the headstage from unnecessary wear and tear.

Important! When using the headstage with the NeuroNexus probes, keep in mind that there may be several versions of the probe. Check the NeuroNexus Website for pin diagrams. Also, see *MCMap* for a description and examples on how to re-map channel numbers.

Connection Diagram

When using the SH16 with a microstimulator system, connect the system as shown. The diagram below shows a system configuration featuring the RZ5 BioAmp Processor, an MS16 Stimulus Isolator, and RA16PA Medusa PreAmp. Connections are much the same when using the RX7 in place of the RZ5.

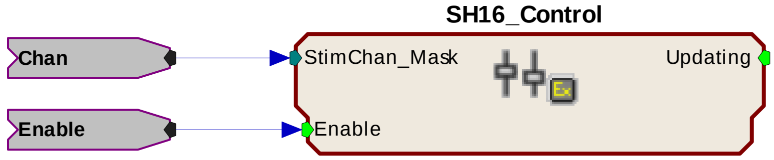


SH16 to MicroStimulator Connection Diagram

Switchable Headstage Operation

When using the SH16 switching headstage with an RZ5 or RX7 processor and an MS4/MS16 Stimulus Isolator, TDT recommends using the SH16_Control macro to set stimulation channels and mode of operation. Based on the macro settings, all necessary control signals are sent from the base station to the headstage via the MS4/MS16 Control output port.

Setup parameters determine which channels are used for stimulation and whether the headstage will be operated in single ended or differential mode.



Single Ended Stim Mode (RZ5)

See the Help text in the macro's properties dialog box for more information about this macro.

Note: The SH16 Headstage requires at least 10ms in order to initialize its control bits for use. If you are trying to trigger the enable input you must either use a trigger signal that is delayed 10ms from the point the circuit is run or use a manual trigger method to begin acquisition.

Operating the Switching Headstage without Using the Macro

The SH16_control macro (above) greatly simplifies control of the switching headstage. If the macro cannot be used, the SH16 can be controlled directly from RPvdsEx using the following information.

The SH16 is controlled using the digital I/O (digital control lines) on the MS4/MS16, which are in turn set by writing an integer value directly to memory (poke address values vary depending on the processor used). Channels 1 - 3 of the digital I/O (bits 0-2) are used to send a serial pattern that controls the state of all channels to the SH16.

Transmitting this data to the headstage from the MS4/MS16 is accomplished using the following 3 digital output lines.

Bit Number	Name (page 6-19)	Description	Pin # (Control DB25)
2	DO2	Serial Clock Line	19
1	DO1	Serial Data Line	6
0	DO0	Load/Latch Signal	18

DO0 (Bit 0) is the load/latch signal. This bit is pulsed for a minimum pulse width of 100 nanoseconds to latch the data to the relays on the headstage after the data has been transmitted.

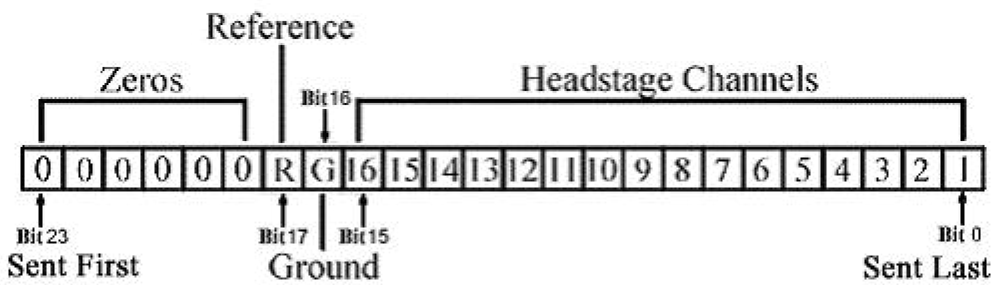
DO1 (Bit 1) is the serial data line. The 24-bit mask must be sent most significant bit (MSB) first. In other words, bit 23 is sent first, then bit 22, bit 21, etc.

DO2 (Bit 2) is the serial clock signal. When the SH16 is being controlled through a System 3 device such as the MS4/MS16, then the maximum rate for serial data transfer is 939 Hz.

Creating the Serial Control Bit Pattern

Channel setup and control are programmed by serially transmitting a 24-bit pattern to the headstage on the serial data line (DO1). The first bits in the pattern control the connection of a given channel to the Stimulus Isolator. Bit 16 controls the ground and bit 17 controls the record reference line. Bits 18-23 are not used and are always sent as zeros. By default, all channels are set in the record mode (disconnected from the stimulator). To connect a given electrode to the output of the stimulus isolator, send a binary ‘1’ on the appropriate bit of the pattern. Sending a binary ‘0’ on the appropriate bit will disconnect that electrode from the stimulus isolator and connect it to the recording preamp.

To disconnect the stimulator ground from the record ground during stimulation, a ‘1’ is sent in the mask at bit location 16. To disconnect the record reference line from the headstage and leave it floating during stimulation, a ‘1’ is sent at bit location 17.



SH16 Serial Control Bit Pattern

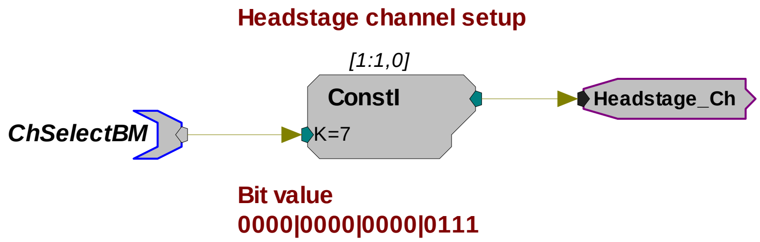
For example, to stimulate on channels 1 (1), 3 (4) and 4 (8), the following serial bit pattern with an integer value of 13 (1 + 4 + 8) should be sent to the headstage. Notice that bits 16 and 17 are not set (1), allowing non-stimulating channels to record using a preamplifier.

0000	0000	0000	0000	0000	1101

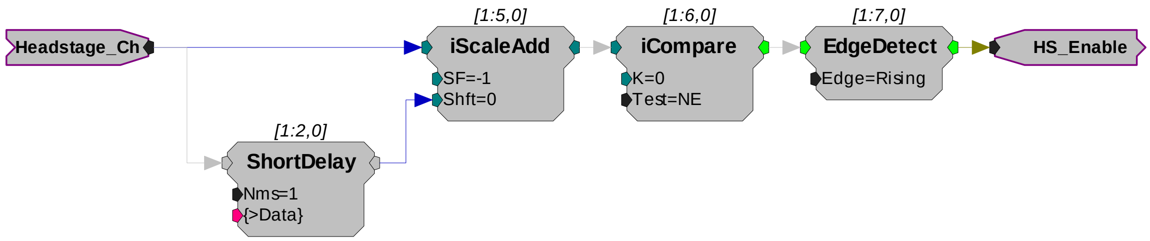
RPvdsEx Circuit

The following circuit illustrates the headstage channel setup and serial data load for the SH16 using an MS4/MS16 and RZ5 or RX7 processor.

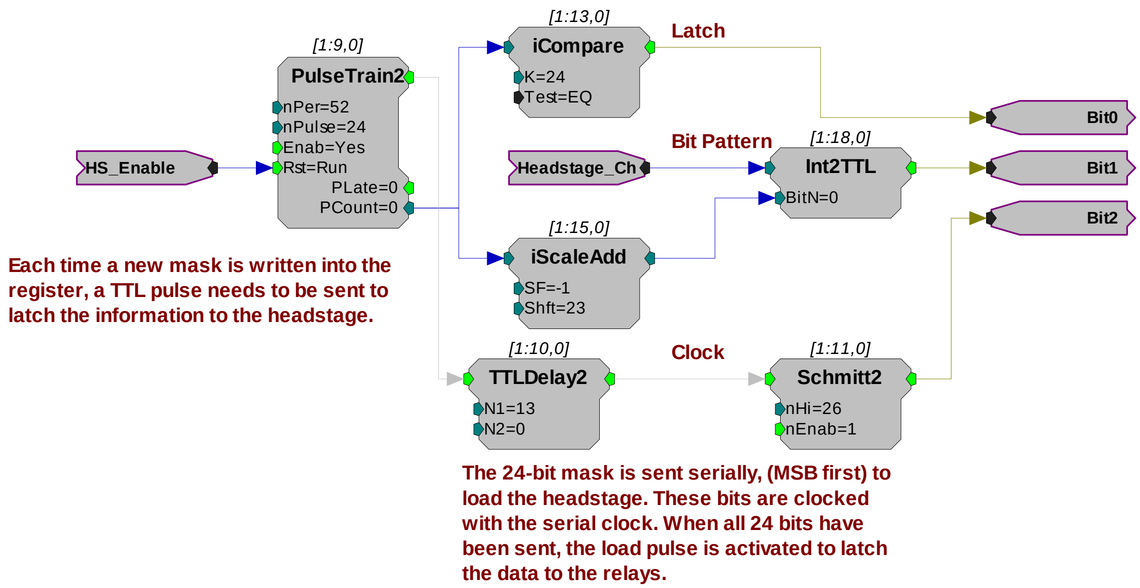
The first figure shows the headstage channel setup. The ChSelectBM parameter tag sets the value of the ConstI with an integer representing the serial control bit pattern discussed above.



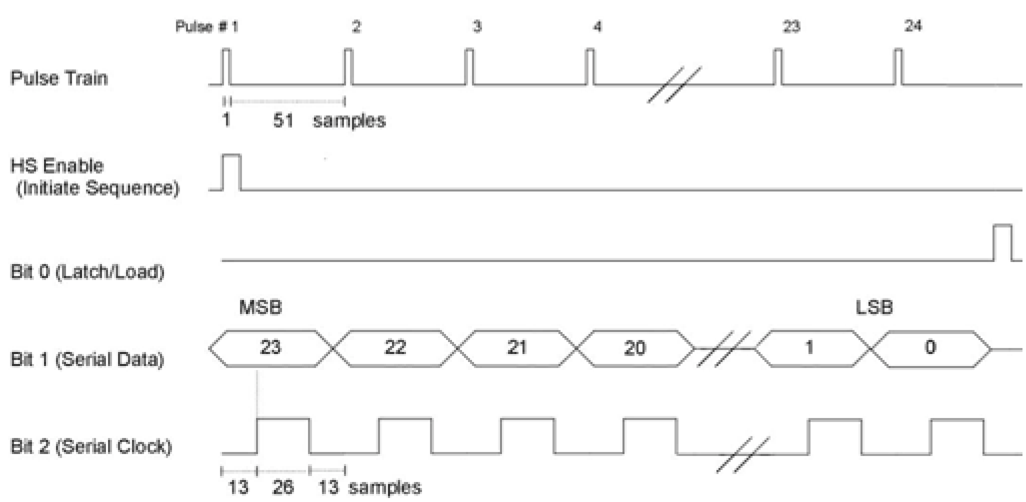
The next segment of the circuit detects a change to the headstage setup and generates a pulse that will reset the serial data transmission to send the new channel selection and control logic.



The third segment of the chain uses a pulse train to send the 24-bit pattern serially (MSB first) to the headstage. After all 24 bits have been sent; the data is latched to the relays.

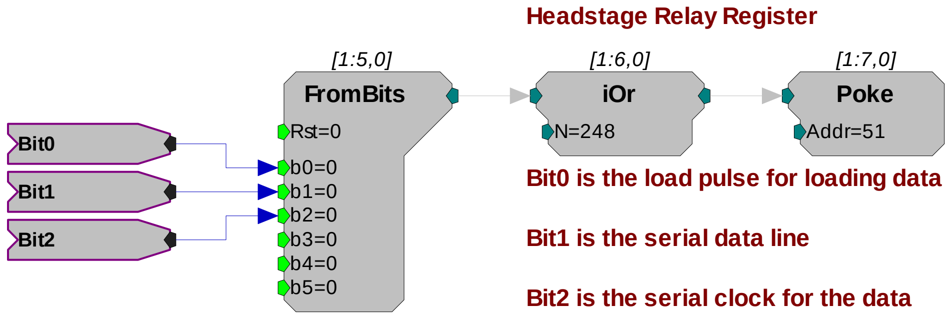


With the sampling rate set to 25 kHz in RPvdsEx and 'nPer' equal to 52 in the PulseTrain2 component, the serial clock (Bit 2) will run at 469 Hz. Setting 'nPer' equal to 26, will allow the clock to run at 939 Hz. The figure below (not to scale) shows the 25kHz pulse rate of 52 samples (1 sample high, 51 samples low) as well as the serial clock rate of 13 samples low, 26 samples high, and 13 samples low.



For headstages with serial numbers >2000, the headstage needs digital high voltages on the input lines of the control connector to power its circuits.

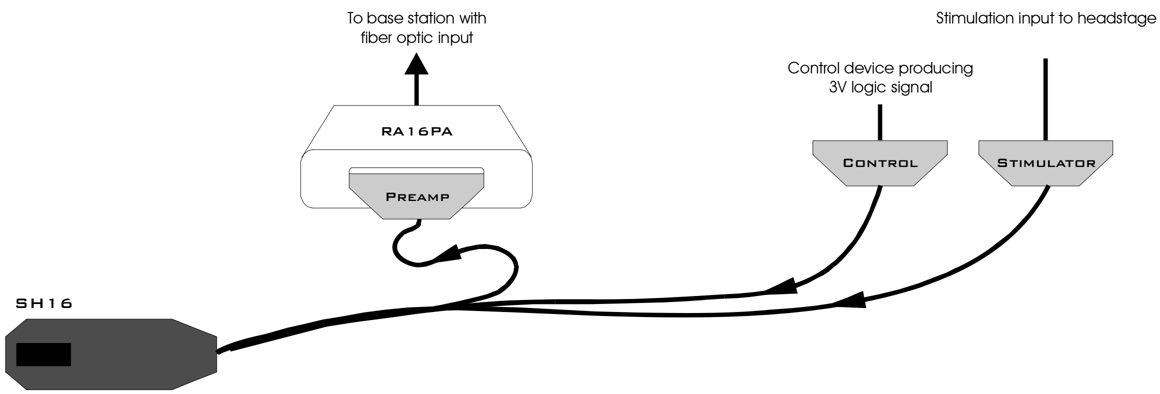
Power the headstage circuits by writing a logic '1' (high) to the MS16 control bits (bits 3-7). In the circuit segment below, the latch, data, and clock lines are fed directly to bits 0, 1, and 2 respectively on the FromBits component and bits 3-7 are set high by ORing the value from the FromBits component with the value 248 (binary: 0000 0000 1111 1000).



A poke component is used to send the resulting value to memory address 51 on the RZ5 processor or memory address 3 on the RX7. The Poke RPvdsEx component writes values to a specific device memory location and should be used with care.

Using the Switching Headstage with Other Devices

When using the SH16 with hardware other than a microstimulator System, connect as follows:



The Serial Control Bit Pattern that controls connection of a given channel to the Stimulus Isolator can be sent using any 3 digital logic lines that will produce a +3V logic signal. Circuit design is similar to the example above, designed for use with the RZ5 and RX7 processors, but must be modified by routing Bit 0, Bit 1, and Bit 2 to the appropriate digital outputs of the device (instead of using the Poke component).

Note: The serial clock (Bit 2) on the SH16 can be run at a maximum rate of 5 MHz for other devices.

Technical Specifications

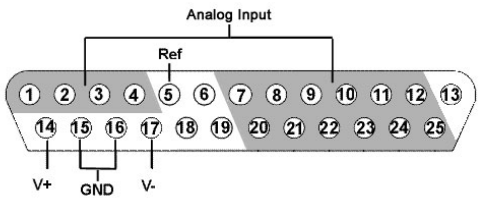
Headstage Gain	Unity (1x)
Input Impedance	10^{14} Ohms

SH16 Pinout Diagrams

PreAmp Connector

For SH16 headstages with serial numbers <2000, the DB25 connector labeled Preamp must be connected as it supplies power to the headstage. For headstages with serial numbers >2000, this connector does not need to be connected if the user is not recording on the non-stimulating channels.

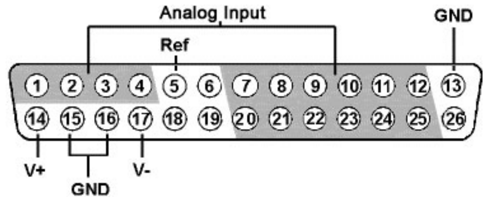
DB25 Pinout Connections for use with Medusa Preamps



Pin	Name	Description
1	A1	Analog Input Channel Number Ch 1-4
2	A2	
3	A3	
4	A4	
5	REF	Reference Pin
6	NA	Not Used
7	A5	Analog Input Channel Number Ch 5, 7, 9, 11, 13, and 15
8	A7	
9	A9	
10	A11	
11	A13	
12	A15	
13	NA	Not Used

Pin	Name	Description
14	V+	Positive Voltage
15	GND	Ground
16	GND	Ground
17	V-	Negative Voltage
18	NA	Not Used
19	NA	Not Used
20	A6	Analog Input Channel Number Ch 6, 8, 10, 12, 14, and 16
21	A8	
22	A10	
23	A12	
24	A14	
25	A16	

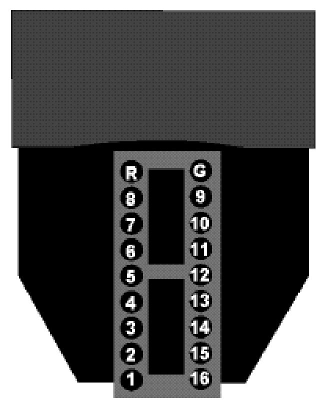
Mini DB26 Pinout Connections for use with PZ preamps



Pin	Name	Description
1	A1	Analog Input Channel Number Ch 1-4
2	A2	
3	A3	
4	A4	
5	REF	Reference Pin
6	NA	Not Used
7	A5	Analog Input Channel Number Ch 5, 7, 9, 11, 13, and 15
8	A7	
9	A9	
10	A11	
11	A13	
12	A15	
13	NA	Not Used

Pin	Name	Description
14	V+	Positive Voltage
15	GND	Ground
16	GND	Ground
17	V-	Negative Voltage
18	NA	Not Used
19	NA	Not Used
20	A6	Analog Input Channel Number Ch 6, 8, 10, 12, 14, and 16
21	A8	
22	A10	
23	A12	
24	A14	
25	A16	
26	NA	Not Used

Headstage Pinout



The numbers in the diagram to the right refer to the channel connections to the preamp connector or stimulator connector.

“G” on the diagram to the right is connected to the reference pin (Ref) on the stimulator connector and can also connect to the ground pin (GND) of the preamp connector through a switchable relay in the SH16.

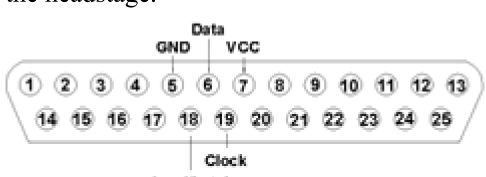
“R” on the diagram to the right is connected to a switchable relay that can connect to the “Ref” pin of the preamp connector.

The electrode connector accepts 0.5 mm diameter male pins.


The headstage has sensitive electronics. Always ground yourself before handling.

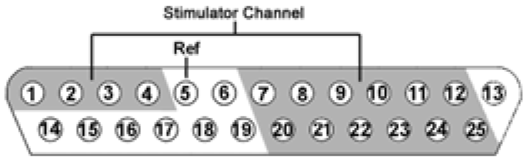
DB25 Control Connector

The Control DB25 can be connected to any control device that produces a 3V logic signal. For headstages with serial numbers >2000, this connector must be connected as it supplies power to the headstage.



Note: Pins that are not labeled are not connected.

DB25 Stimulator Connector



Note: The global reference (Ref) is connected to the SH16 ground pin (G of headstage pinout).

Pin	Name	Description
1	S1	Stimulator Channels Ch 1-4
2	S2	
3	S3	
4	S4	
5	Ref	Reference
6	NA	Not Used
7	S5	Stimulator Channels Ch 5, 7, 9, 11, 13, and 15
8	S7	
9	S9	
10	S11	
11	S13	
12	S15	
13	NA	Not Used

Pin	Name	Description
14	NA	Not Used
15		
16		
17		
18		
19		
20	S6	Stimulator Channels Ch 6, 8, 10, 12, 14, and 16
21	S8	
22	S10	
23	S12	
24	S14	
25	S16	

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Part 8 Low Impedance Headstages

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RA4LI - Four Channel Headstage

The RA4LI headstage is designed for low impedance electrodes with input impedances between <1 kOhm and 20 kOhm. Electrode connectors are standard 1.5 mm safety connectors making it easy to connect to standard needle and surface electrodes for recording evoked potentials and EEG's. The headstage connects directly to the RA4PA Medusa preamplifier's 25-pin connector. A built in impedance checker can be used to test each channel and the reference. Additional 20x gain on the headstage improves signal-to-noise of low voltage signals.

Impedance Checking with the Low-Impedance Headstage

The Impedance checker on the RA4LI provides a simple check of the channel impedance relative to ground. To check the impedance level, press the button next to the channel indicator. The highest-level light indicates the maximum impedance between the channel and the ground. If all impedance lights are illuminated it is likely that one of the channels is not properly connected. The (-) impedance button checks the impedance between the reference and the ground.

Headstage Voltage Range

When using a TDT preamplifier the voltage input range of the preamplifer is typically lower than the headstage and must be considered the effective range of the system. Check the specifications of your amplifier for voltage range. Also keep in mind that the range of the headstage varies depending on the power supply provided by the preamplifier. TDT preamplifiers supply +/- 1.5 VDC, but third party preamplifiers may vary. TDT recommends using preamplifiers which deliver +/- 2.5 VDC or less. Check the preamplifier voltage input and power supply specifications and headstage gain to determine the voltage range of the system.

The table below lists the input voltage ranges for the RA4LI headstage for either a +/- 1.5 VDC or +/- 2.5 VDC power source.

Headstage input range when using +/- 1.5 VDC power source	**Headstage input range when using +/- 2.5 VDC power source**
+/- 33 mV	+/- 80 mV

Headstage Technical Specifications

Warning!: When using multiple headstages ensure that all ground pins are connected to a single common node. See page 5-29 for more information.

S/N (typical)	rms 0.1 μ V bandwidth 300-3000 Hz 0.3 μ V bandwidth 2-8000 Hz
Headstage Gain	20x
Highpass Filter	2.2 Hz
Lowpass Filter	7.5 kHz
Input Impedance	10^6 Ohm

RA16LI - 16 Channel Headstage

The sixteen channel low impedance headstage (RA16LI) is a high quality, low-impedance headstage designed for recording high channel count EEG's.

The RA16LI headstage is designed for low impedance electrodes and electrode caps with input impedances between <1 kOhm and 20 kOhm. Either headstage unit connects to the Medusa preamplifier's 25-pin connector. The simple interface to the RA16PA preamplifier makes it easy to connect your electrodes to our system. An adapter is also available to connect a low impedance headstage to a PZ preamplifier. See *DBF-MiniDBM*, page 9-11 for more information. A built in impedance checker can be used to test each channel and the reference. Additional 20x gain on the headstage improves signal-to-noise of low voltage signals.

Impedance Checking with the Low-Impedance Headstage

The Impedance checker on the RA16LI provides a simple check of the channel impedance relative to ground. To check the impedance level, press the button next to the channel indicator. The highest-level light indicates the maximum impedance between the channel and the ground. If all impedance lights are illuminated it is likely that one of the channels is not properly connected. The (-) impedance button checks the impedance between the reference and the ground.

Headstage Voltage Range

When using a TDT preamplifier the voltage input range of the preamplifer is typically lower than the headstage and must be considered the effective range of the system. Check the specifications of your amplifier for voltage range. Also keep in mind that the range of the headstage varies depending on the power supply provided by the preamplifier. TDT preamplifiers supply +/- 1.5 VDC, but third party preamplifiers may vary. TDT recommends using preamplifiers which deliver +/- 2.5 VDC or less. Check the preamplifier voltage input and power supply specifications and headstage gain to determine the voltage range of the system.

The table below lists the input voltage ranges for the RA16LI headstage for either a +/- 1.5 VDC or +/- 2.5 VDC power source.

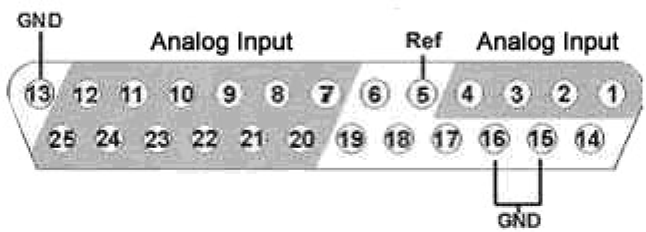
Headstage input range when using +/- 1.5 VDC power source	Headstage input range when using +/- 2.5 VDC power source
+/- 33 mV	+/- 80 mV

Headstage Technical Specifications

Warning!: When using multiple headstages ensure that all ground pins are connected to a single common node. See page 5-29 for more information.

S/N (typical)	rms 0.1 µ V bandwidth 300-3000 Hz 0.3 µ V bandwidth 2-8000 Hz
Headstage Gain	20x
Highpass Filter	2.2 Hz
Lowpass Filter	7.5 kHz
Input Impedance	10^6 Ohm

The electrode connector is a 25-pin connector. Information on the pin inputs is provided below.



Note: Pins 6, 14, 17, 18 and 19 are not connected.

Pin	Name	Description
1	A1	Analog Input Channels
2	A2	
3	A3	
4	A4	
5	Ref	Reference
6	NA	Not Used
7	A5	Analog Input Channels
8	A7	
9	A9	
10	A11	
11	A13	
12	A15	
13	GND	Ground

Pin	Name	Description
14	NA	Not Used
15	GND	Ground
16	GND	
17	NA	Not Used
18	NA	
19	NA	
20	A6	Analog Input Channels
21	A8	
22	A10	
23	A12	
24	A14	
25	A16	

RA16LI-D - 16 Channel Headstage with Differential

The RA16LI-D headstage is designed for fully differential recordings from low impedance electrodes and electrode caps with input impedances between <1 kOhm and 20 kOhm. It connects to the Medusa preamplifier's 25-pin connector. The simple interface to the RA16PA preamplifiers makes it easy to connect your electrodes to our system. An adapter is also available to connect a low impedance headstage to a PZ preamplifier. See *DBF-MiniDBM*, page 9-11 for more information.

The differential inputs allow for improved common mode rejection on all channels. Because of the increased complexity of the circuitry, the RA16LI-D does not have impedance checking. The headstage connector is a DB44. The pin out diagram is shown below.

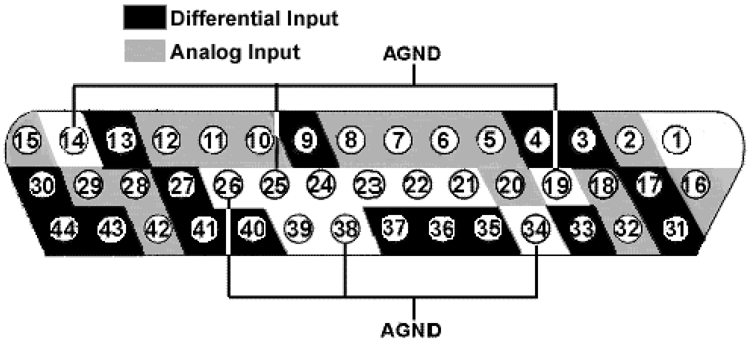
Headstage Voltage Range

When using a TDT preamplifier the voltage input range of the preamplifer is typically lower than the headstage and must be considered the effective range of the system. Check the specifications of your amplifier for voltage range.

Headstage Technical Specifications

Warning!: When using multiple headstages ensure that all ground pins are connected to a single common node. See page 5-29 for more information.

S/N (typical)	rms 0.1 µ V bandwidth 300-3000 Hz 0.3 µ V bandwidth 2-8000 Hz
Headstage Gain	20x
Highpass Filter	2.2 Hz
Lowpass Filter	7.5 kHz
Input Impedance	10^6 Ohm



Note: Pins 1, 21-24 and 39 are not connected.

Pin	Name	Description
1	NA	Not Used
2	A2	Analog Input
3	D3	Differential Input
4	D5	
5	A5	Analog Input
6	A7	
7	A8	
8	A9	
9	D9	Differential Input
10	A10	Analog Input
11	A11	
12	A12	
13	D13	Differential Input
14	AGND	Analog Ground
15	A16	Analog Input
16	A1	
17	D2	Differential Input
18	A4	Analog Input
19	AGND	Analog Ground
20	A6	Analog Input
21	NA	Not Used
22	NA	
23	NA	
24	NA	

Pin	Name	Description
25	AGND	Analog Ground
26	AGND	
27	D12	Differential Input
28	A14	Analog Input
29	A15	
30	D16	Differential Input
31	D1	
32	A3	Analog Input
33	D4	Differential Input
34	AGND	Analog Ground
35	D6	Differential Input
36	D7	
37	D8	
38	AGND	Analog Ground
39	NC	
40	D10	Differential Input
41	D11	
42	A13	Analog Input
43	D14	Differential Input
44	D15	

Part 9 Adapters and Connectors

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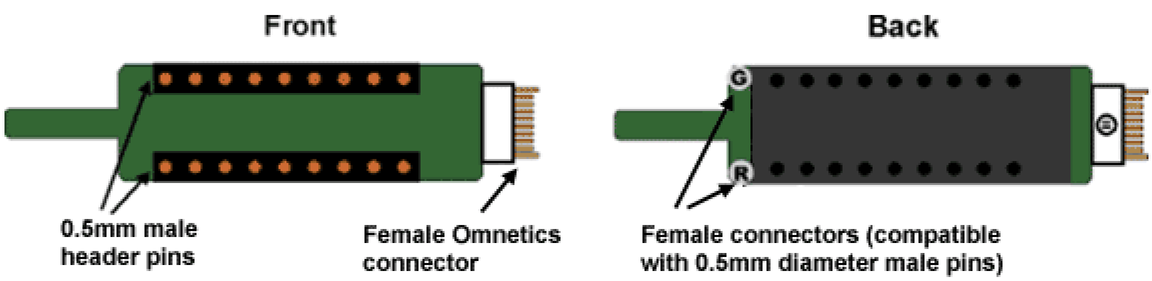
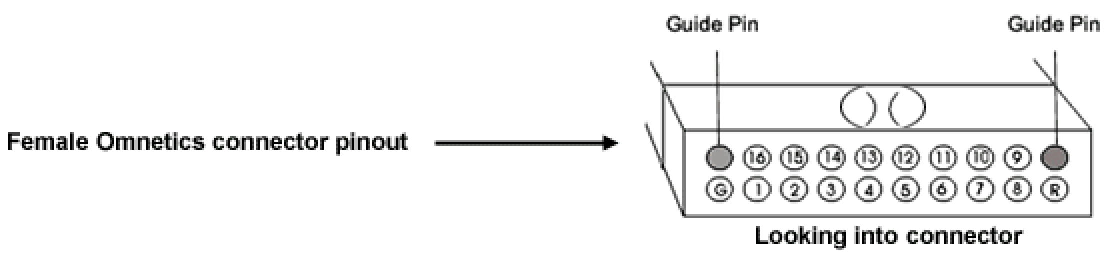
Probe Adapters

Each TDT headstage is designed for use with a particular style of probe. Probe adapters allow each headstage to be used with a wider variety of probes. When using adapters, keep in mind that standard operation (differential vs single ended) varies for acute and chronic preparations and headstages are designed accordingly. When adapting across preparations, carefully note and understand the use of the ground (G) and reference (R) connections provided on each adapter.

AC-CH Acute Headstage to Chronic Probe (16 Channels)

This adapter allows the user to connect a 16-channel chronic probe (such as a TDT 16 channel microwire array) to an acute TDT headstage (RA16AC/RA16AC4). Standard operation for chronic preparations is single ended with ground and reference shorted together in the chronic headstage. However, the acute headstage is designed for differential operation. When using the acute headstage with our microwire arrays, short G and R together on the adapter for single ended operation.

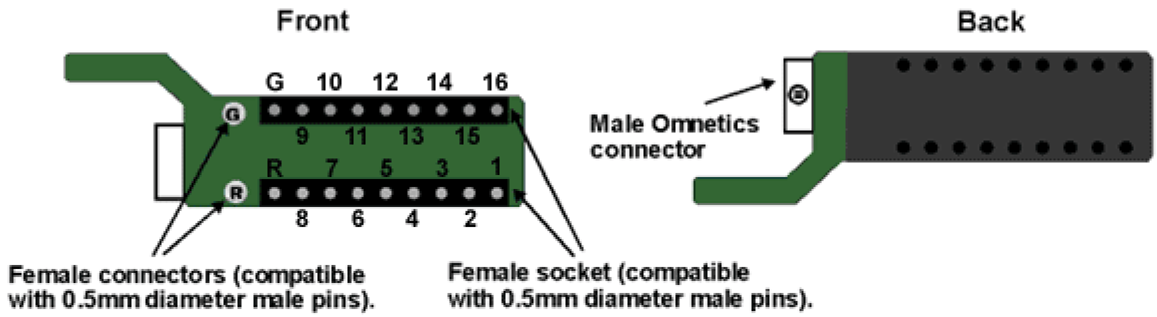




Pinouts are looking into the connector and reflect the preamplifier channels. TDT probe adapters are designed for specific TDT headstage to probe connections. If you are using a third party headstage, please contact TDT support for assistance.

CH-AC Chronic Headstage to Acute Probe (16 Channels)

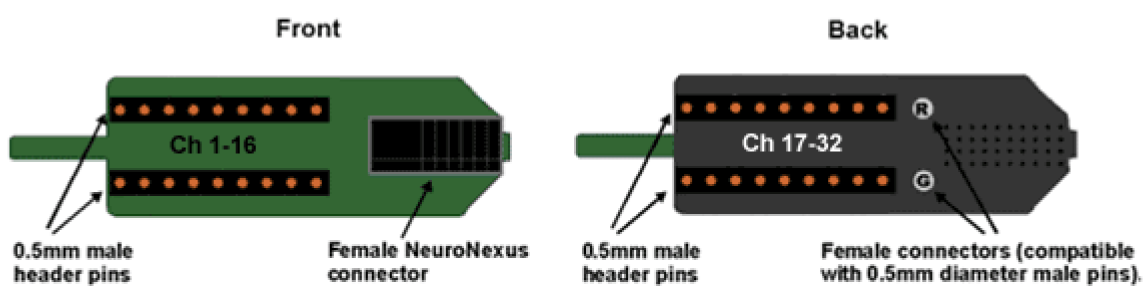
This adapter connects a 16-channel acute probe to a TDT chronic headstage (RA16CH). Reference and ground are tied together by default on the chronic headstage so in general only one pin connection is necessary. A jumper is provided on the RA16CH for differential operation. See *RA16CH*, page 7-15 for information.



Pinouts are looking into the connector and reflect the preamplifier channels. TDT probe adapters are designed for specific TDT headstage to probe connections. If you are using a third party headstage, please contact TDT support for assistance.

ACx2-NN 16 Channel Acute Headstage to 32 Channel Acute Probe

This adapter connects a 32-channel acute NeuroNexus probe to two 16-channel acute TDT headstages (RA16AC/RA16AC4). Standard operation with the NeuroNexus probe is differential. If you wish to use the Reference pad on the probe, do not tie G and R together.

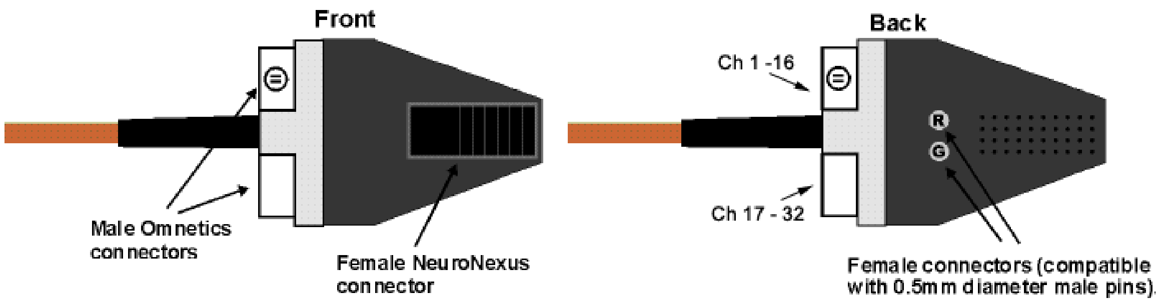


Pinouts are looking into the connector and reflect the preamplifier channels. TDT probe adapters are designed for specific TDT headstage to probe connections. If you are using a third party headstage, please contact TDT support for assistance.

Important!: When using these adapters with NeuroNexus probes, keep in mind that there are several versions of each of the probes. TDTs ACx2-NN is designed for use with Rev 2 of the 32-channel NeuroNexus acute probe. Check the NeuroNexus website for pin diagrams. Also, see *MCMap*, in the *RPvdsEx User Guide*, for a description and examples on how to re-map channel numbers.

CHx2-NN 16 Channel Chronic Headstage to 32 Channel Acute Probe

This adaptor connects a 32-channel acute NeuroNexus probe to two 16-channel chronic TDT headstages (RA16CH). Connect the first RA16CH headstage (channels 1-16) to the front of the adapter. Connect the second RA16CH (channels 17-32) to the back of the adapter. This adapter also features a holding rod for connection to a micromanipulator. As with the CH-AC adaptor, reference and ground are tied together by default on the chronic headstage so in general only one pin connection is necessary. If you wish to use the Reference pad on the probe, do not tie G and R together and cut the jumper on each headstage to make the inputs differential. See *RA16CH*, page 7-15 for more information.



Pinouts are looking into the connector and reflect the preamplifier channels. TDT probe adapters are designed for specific TDT headstage to probe connections. If you are using a third party headstage, please contact TDT support for assistance.

Important!: When using these adapters with NeuroNexus probes, keep in mind that there are several versions of each of the probes. TDTs CHx2-NN is designed for use with Rev 2 of the 32-channel NeuroNexus acute probe. Check the NeuroNexus website for pin diagrams. Also, see *MCMap*, in the *RPvdsEx User Guide*, for information on how to re-map channel numbers.

ZIF-Clip® Headstage Adapters

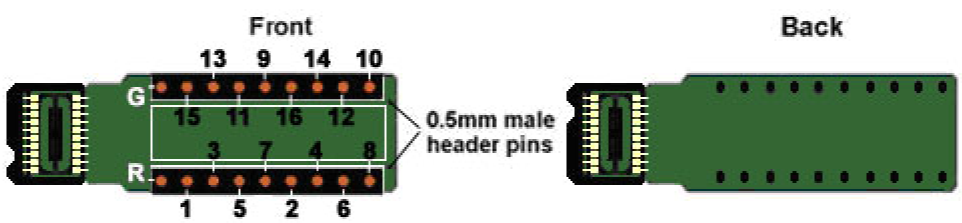
ZIF-Clip® headstage adapters are available for use with a variety of electrode styles. When using adapters, keep in mind that standard operation (differential vs single-ended) may vary for acute and chronic preparations. Carefully note and understand the use of the ground (G) and reference (R) connections provided on each adapter.

Standard operation for ZIF-Clip® headstages is differential. Headstage adapters can be configured for single-ended operation by tying ground (G) and reference (R) connections together on the adapter (if available). Refer to the electrode manufacturer's documentation for information on single-ended or differential configurations.

Note: When using these adapters with NeuroNexus, Gray Matter, or CyberKinetics probes, keep in mind that there may be updates to pin configurations. Check the suppliers' website for pin diagrams. Also, see *MCMap* for a description and examples on how to re-map channel numbers.

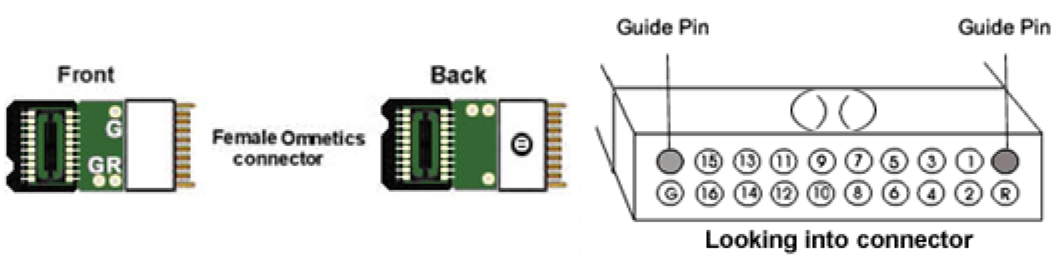
ZCA-DIP16 ZIF-Clip® Headstage to Acute Probe (16 Channels)

This adapter allows the user to connect a 16-channel acute probe (such as NeuroNexus) to a 16-channel ZIF-Clip® headstage. Ground and reference pins are located on the DIP connector and may be tied together for single-ended operation. **Pinouts are looking into the connector and reflect the preamplifier channels.**



ZCA-OMN16 ZIF-Clip® Headstage to Chronic Probe (16 Channels)

This adapter connects a 16-channel chronic Omnetics based probe to a 16-channel ZIF-Clip® headstage. Ground and reference pins may be tied together for single-ended operation. **Pinouts are looking into the connector and reflect the preamplifier channels.**



ZCA-NN32 ZIF-Clip® Headstage to 32 Channel Acute Probe)

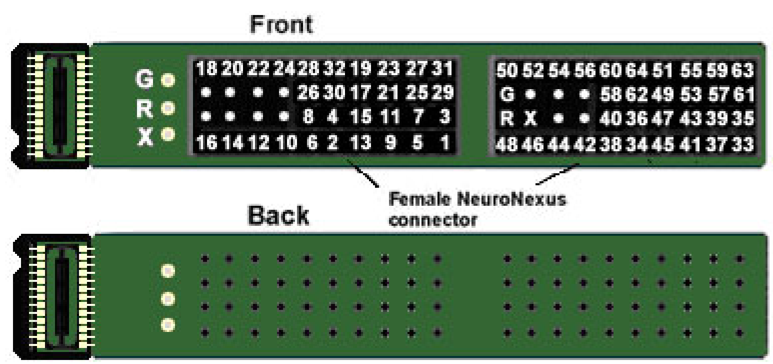
This adapter connects a 32-channel acute NeuroNexus probe to a 32-channel ZIF-Clip® headstage.

Note: X (Ref) is a reference pin that is connected from the adapter to the probe only. See the jumper configuration below for more information. **Pinouts are looking into the connector and reflect the preamplifier channels.**

ZCA-NN64 ZIF-Clip® Headstage to 64 Channel Acute Probe)

This adapter connects a 64-channel acute NeuroNexus probe to a 64-channel ZIF-Clip® headstage.

Note: X (Ref) is a reference pin that is connected from the adapter to the probe only. See the jumper configuration below for more information. **Pinouts are looking into the connector and reflect the preamplifier channels.**



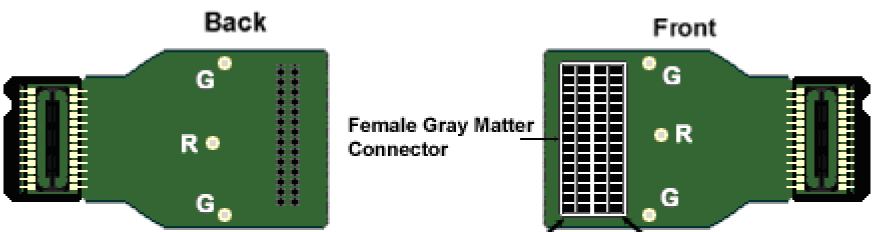
Jumper Configuration

The following table describes the jumper configurations for both the ZCA-NN32 and ZCA-NN64.

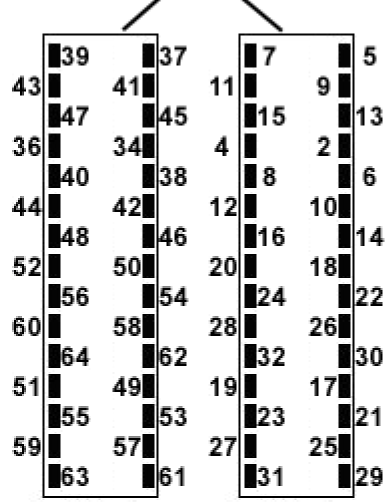
Jumper Connections	Operation
G R X (Ref)	Shorts headstage Ground and Reference inputs together, yielding single-ended amplification of signals relative to ground.
G R X (Ref)	Shorts headstage Reference input to the pin labeled X (a low impedance site on the probe) yielding differential amplification of signals relative to the voltage of the X (Ref) site.
G R X (Ref)	Headstage Ground and Reference separated and X (Ref) pin is not used, yielding differential amplification of signals relative to the voltage of the Reference

ZCA-GM60 ZIF-Clip® Headstage to 60-Channel Chronic Probe

This adapter connects a 60-channel chronic Gray Matter microdrive (SC60-1) to a 64-channel ZIF-Clip® headstage. Ground and reference pins are located on the adapter for access to single-ended and differential modes of operation. **Pinouts are looking into the connector and reflect the preamplifier channels.**

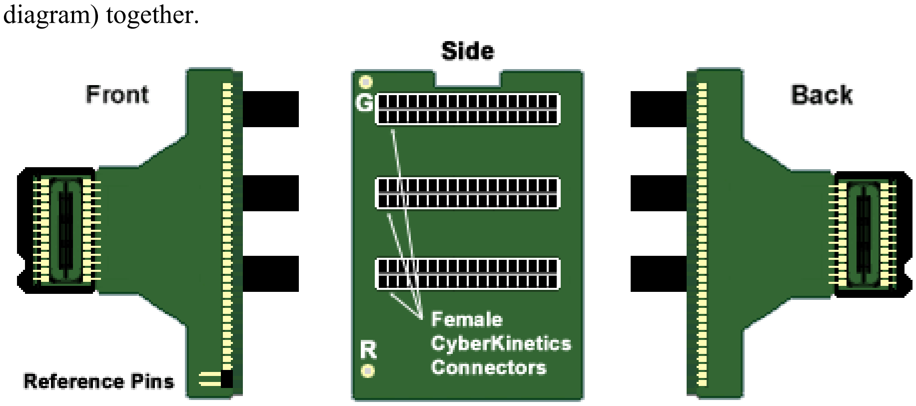


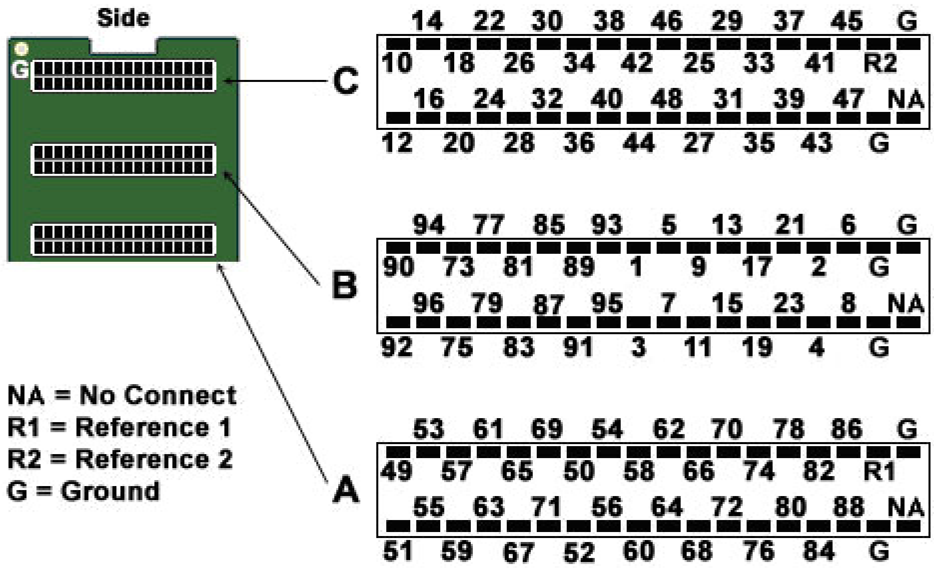
Note: Channels 1,3,33 and 35 are not connected.



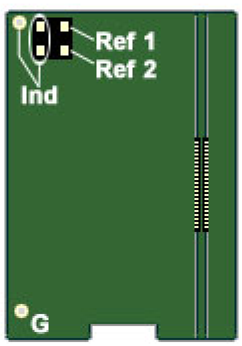
ZCA-CK96A ZIF-Clip® Headstage to 96-Channel Chronic Probe

This adapter connects a 96-channel chronic CyberKinetics CerePort connector to a 96-channel ZIF-Clip® headstage. For single-ended operation, tie the ground and reference pins (shown in diagram) together.





Pinouts are looking into the connector and reflect the preamplifier channels.



A four-pin header located on the backside of the adapter is provided for access to two probe reference pins. These pins are separate references and are connected internally to the adapter.

Connecting a jumper between the headstage reference pins (Ind) and either of the probe reference pins (Ref_1 or Ref_2) connects the headstage reference to the desired probe reference (see table below for more information).

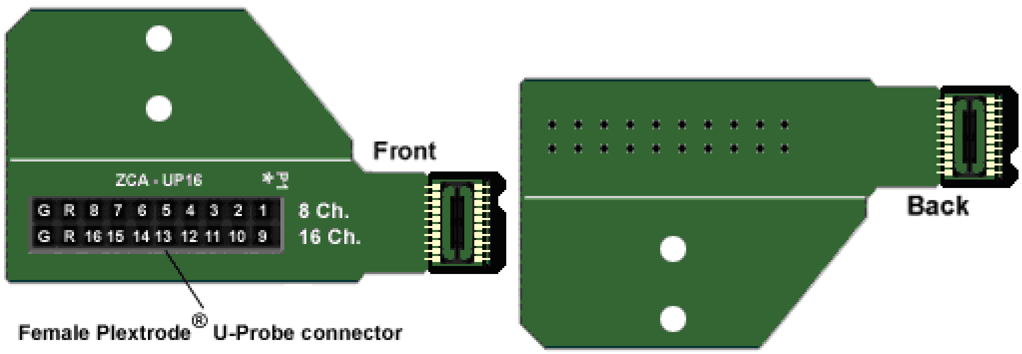
Jumper Configuration

The following table describes the jumper configurations for the ZCA-CK96A.

Jumper Connections	Operation
Ind Ref_1 Ind Ref_2	Headstage Ground and Reference separated and Ref_1, Ref_2 pins are not used, yielding differential amplification of signals relative to the voltage of the Reference (Ind). An external connection for the headstage reference (Ind) must be used for differential amplification.
$\boxed{\text{Ind } Ref_1}$ Ind Ref_2	Shorts headstage Reference input (Ind) to the pin labeled Ref_1 (a low impedance site on the probe) yielding differential amplification of signals relative to the voltage of the Ref_1 site.
Ind Ref_1 $\boxed{\text{Ind } Ref_2}$	Shorts headstage Reference input (Ind) to the pin labeled Ref_2 (a low impedance site on the probe) yielding differential amplification of signals relative to the voltage of the Ref_2 site.

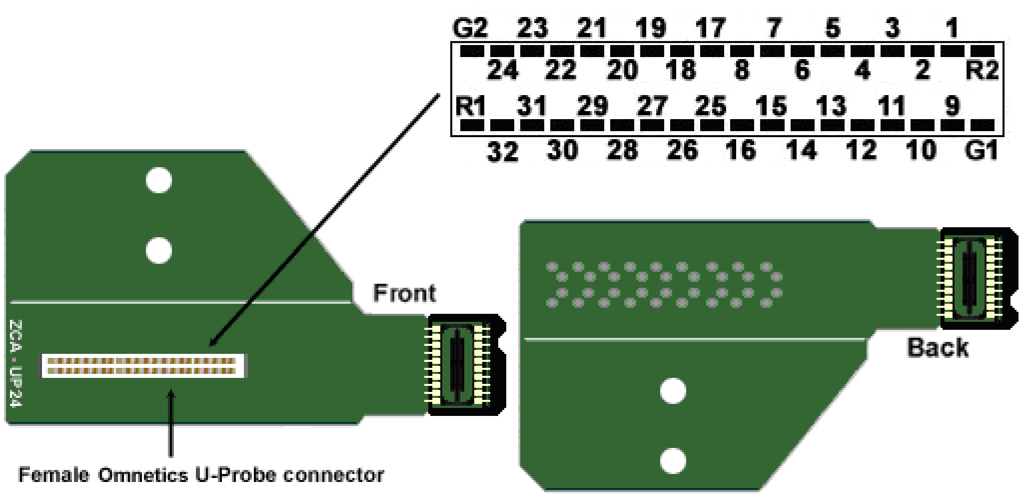
ZCA-UP16 16-Channel Plextrode® U-Probe to ZIF-Clip® headstage

This adapter connects an 8 or 16-channel acute Plextrode® U-Probe connector to a 16-channel ZIF-Clip® headstage. The adapter includes mounting holes for attachment to a micromanipulator. Configuration for single-ended or differential operation is provided on the electrode. Refer to the Plextrode® documentation for jumper configurations. **Pinouts are looking into the connector and reflect the preamplifier channels.**



ZCA-UP24 24-Channel Plextrode® U-Probe to ZIF-Clip® headstage

This adapter connects a 24-channel acute Plextrode® U-Probe connector to a 32-channel ZIF-Clip® headstage. The adapter includes mounting holes for attachment to a micromanipulator. Configuration for single-ended or differential operation is provided on the electrode. Refer to the Plextrode® documentation for jumper configurations. **Pinouts are looking into the connector and reflect the preamplifier channels.**

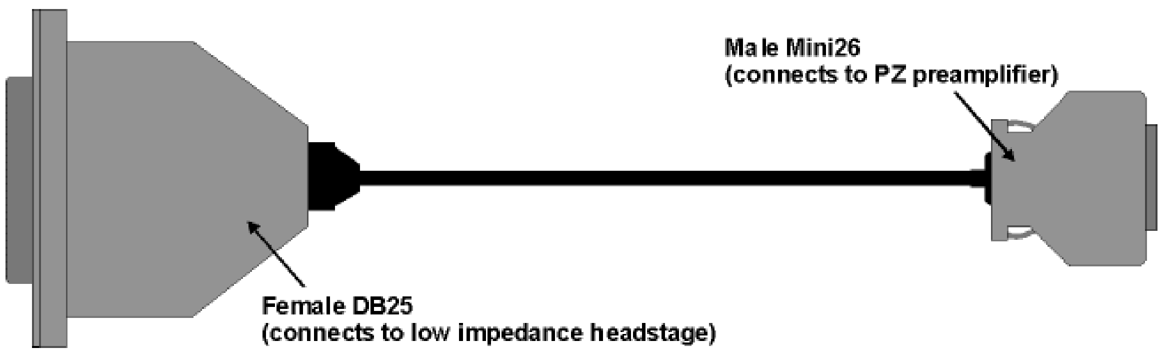


Preamplifier Adapters

Each TDT headstage is designed for use with either a Legacy or Z-Series preamplifier. Preamplifier adapters allow TDT headstages to be used with a variety of preamplifiers by converting the type of preamplifier connector.

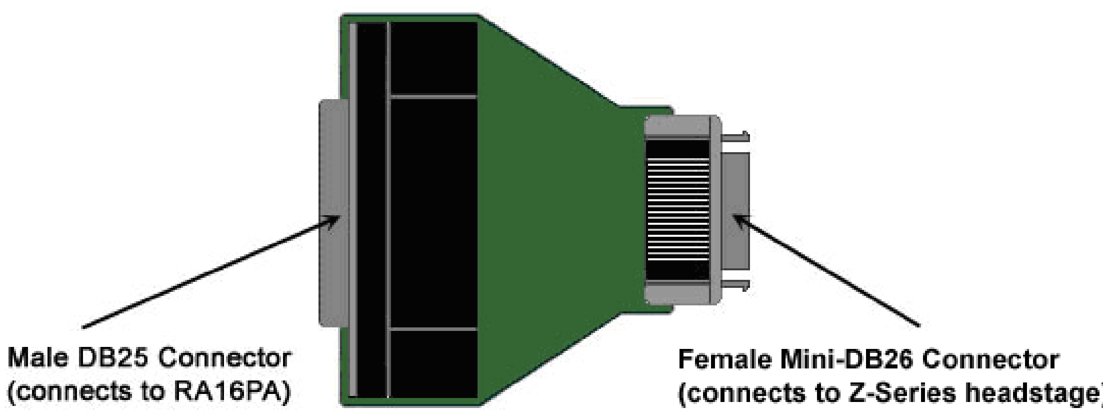
DBF-MiniDBM Low Impedance Headstage to PZ Preamplifier (16-channels)

This adapter connects a low impedance headstage (RA4LI or RA16LI) to a PZ preamplifier.



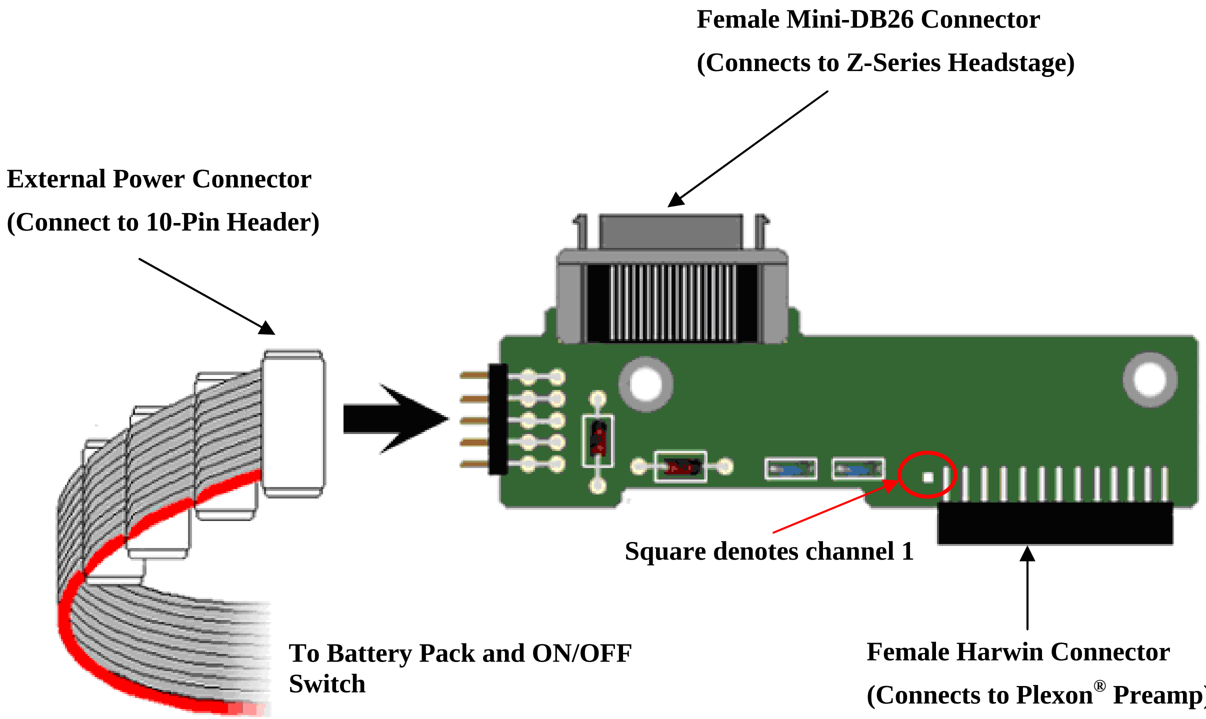
MiniDBF-DBM Z-Series Headstage Female Mini-DB26 to Male DB25 Cable Adapter

This adapter converts a Z-Series headstage Mini-D connector to a DB25 connector for use with a Medusa RA16PA preamplifier.



PLX-ZCA Z-Series Headstage to Plexon® Preamplifier

This adapter connects a Z-Series headstage to a Plexon® preamplifier. Each PLX-ZCA adapter board connects 16-channels. Multiple adapter boards can be stacked for a higher channel count and are fastened together using two screws on either side of the adapter board. An external power source is provided to power the headstage.



External Power Source Connector and a Single PLX-ZCA Adapter Board

External Power Source

In order to power TDT headstages when using this adapter, an external power source is required. Each external power source includes four connectors and can power up to four PLX-ZCA adapter boards. The external power source uses two 1.5 V D batteries and is enabled through a simple ON/OFF switch.

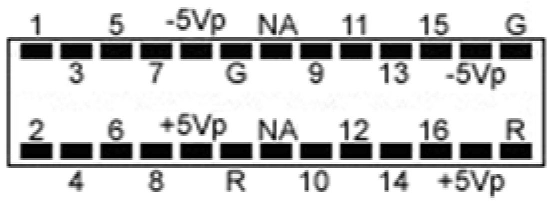
To power the PLX-ZCA adapter:

1. Align the red colored stripe to the Harwin connector side of the adapter (as shown in the diagram above).
2. Connect an external power connector to the 10-pin header located on the adapter.
3. Ensure that the batteries are correctly inserted in the battery pack then move the switch to the ON position.

Note: To power multiple PLX-ZCA adapters, simply connect each 10-pin header to one of the available external power connectors.

Plexon header pinout

Harwin Connector



10-Pin Header

(For external power connector)

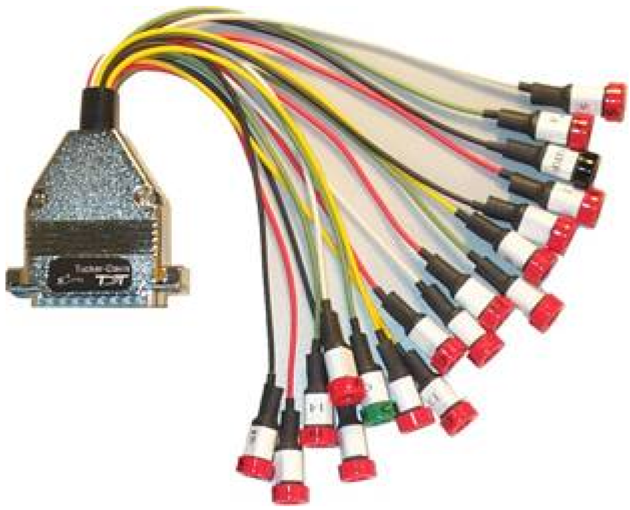
Pinouts are looking into the connector and reflect the preamplifier channels.

NA = Not Used, G = AGND, R = Reference

Connectors

LI-CONN - Low Impedance Connectors

A set of multi-channel low impedance connectors (LI-CONN) for the RA16LI is available for users who do not require a direct connection between the electrodes and the headstage. The LI-CONN uses standard 1.5 mm safety connectors to ensure proper connection between electrodes and the preamplifier.



LI-CONN-Z - Low Impedance Connector for the PZ3

The PZ3 is designed to record from low impedance electrodes and electrode caps with input impedances less than 20 kOhm. Signals are input via multiple DB26 connectors on the PZ3 back panel. A break out box or connector(s) is required for electrode connection.

The **LI-CONN-Z** for *Shared Differential* mode features standard 1.5 mm safety connectors and provides easy connections between electrodes and the amplifier.

Splitters

S-BOX - Amplifier Input Splitter

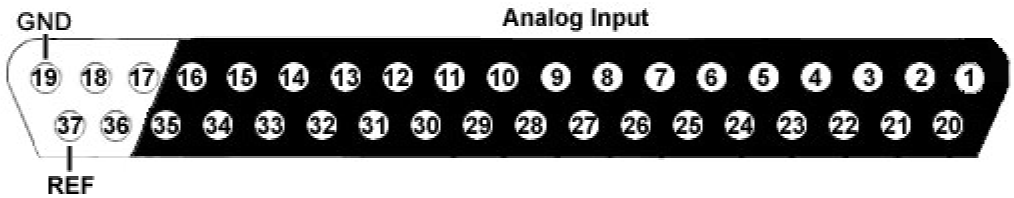
The S-BOX is a 32-channel passive signal splitter for use with the PZ3 Low Impedance Amplifier. The splitter provides a simple and effective means of routing low impedance biological signals to both a TDT acquisition system and a parallel recording system.

Four DB26 connectors provide direct connection to a PZ3 amplifier and a single DB37 provides a parallel output connection. Bank letters as well as channel number ranges are labeled on all the DB26 connectors (i.e. Bank A Channels 1-8).

Important! The S-BOX is NOT FDA approved and is intended for use with the PZ3 Amplifier in Shared Differential mode. It DOES NOT support Individual (True) Differential mode.

The S-BOX uses standard 1.5 mm safety connectors for input from electrodes. Front panel numbering of these inputs corresponds to TDT amplifier channels.

DB37 Pinout



Pin	Name	Description
1	A1	Analog input channels 1,3,5,7,9,11,13,15,17,19,21,23,25,27,29,31
2	A3	
3	A5	
4	A7	
5	A9	
6	A11	
7	A13	
8	A15	
9	A17	
10	A19	
11	A21	
12	A23	
13	A25	
14	A27	
15	A29	
16	A31	
17	NA	Not Used
18	NA	
19	GND	Ground

Pin	Name	Description
20	A2	Analog input channels 2,4,6,8,10,12,14,16,18,20,22,24,26,28,30,32
21	A4	
22	A6	
23	A8	
24	A10	
25	A12	
26	A14	
27	A16	
28	A18	
29	A20	
30	A22	
31	A24	
32	A26	
33	A28	
34	A30	
35	A32	
36	NA	Not Used
37	REF	Reference

Note: No connections should be made to pins 17, 18, and 36.

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Part 10 Microwire Arrays

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ZIF-Clip® Based Microwire Arrays

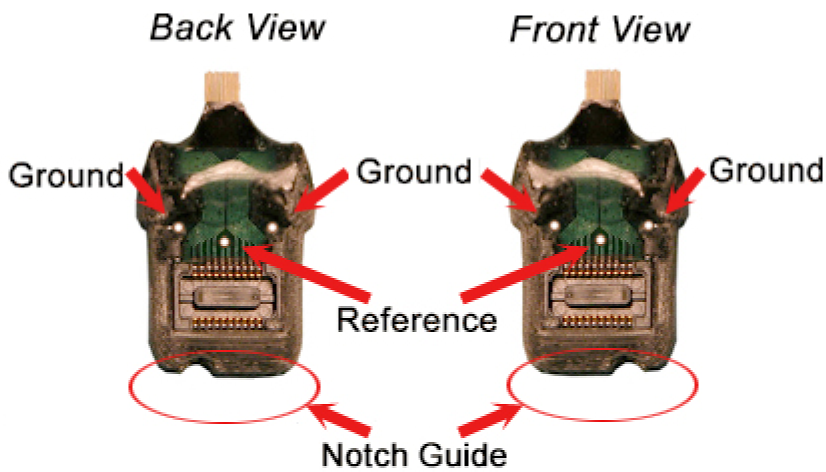
Part Number: ZIF2010, ZIF2011, ZIF2030

Standard 50 µm polyimide-insulated tungsten microwire gives the arrays excellent recording characteristics and the rigidity of tungsten facilitates insertion. The standard ZIF2010 array consists of sixteen channels configured in two rows of eight electrodes each and are accessed via our ZIF-Clip® headstage. A notch at the base of the connector facilitates correct insertion into the ZIF-Clip® headstage and also denotes the 1st row of electrodes as illustrated in the pinout diagram on page 10-4.

Grounding the Electrode

The following illustration shows the possible connections made for reference or ground wires. These wires are attached at TDT.

Important note! A notch guide provides easy connections to the ZIF-Clip® headstage. Ensure that the notch side is properly aligned with the arrow symbol on the headstage (as shown in the pinout diagram).



Caution! The microwire array can be damaged by extreme heat. Use caution when soldering.

Specifications might vary based on custom order:

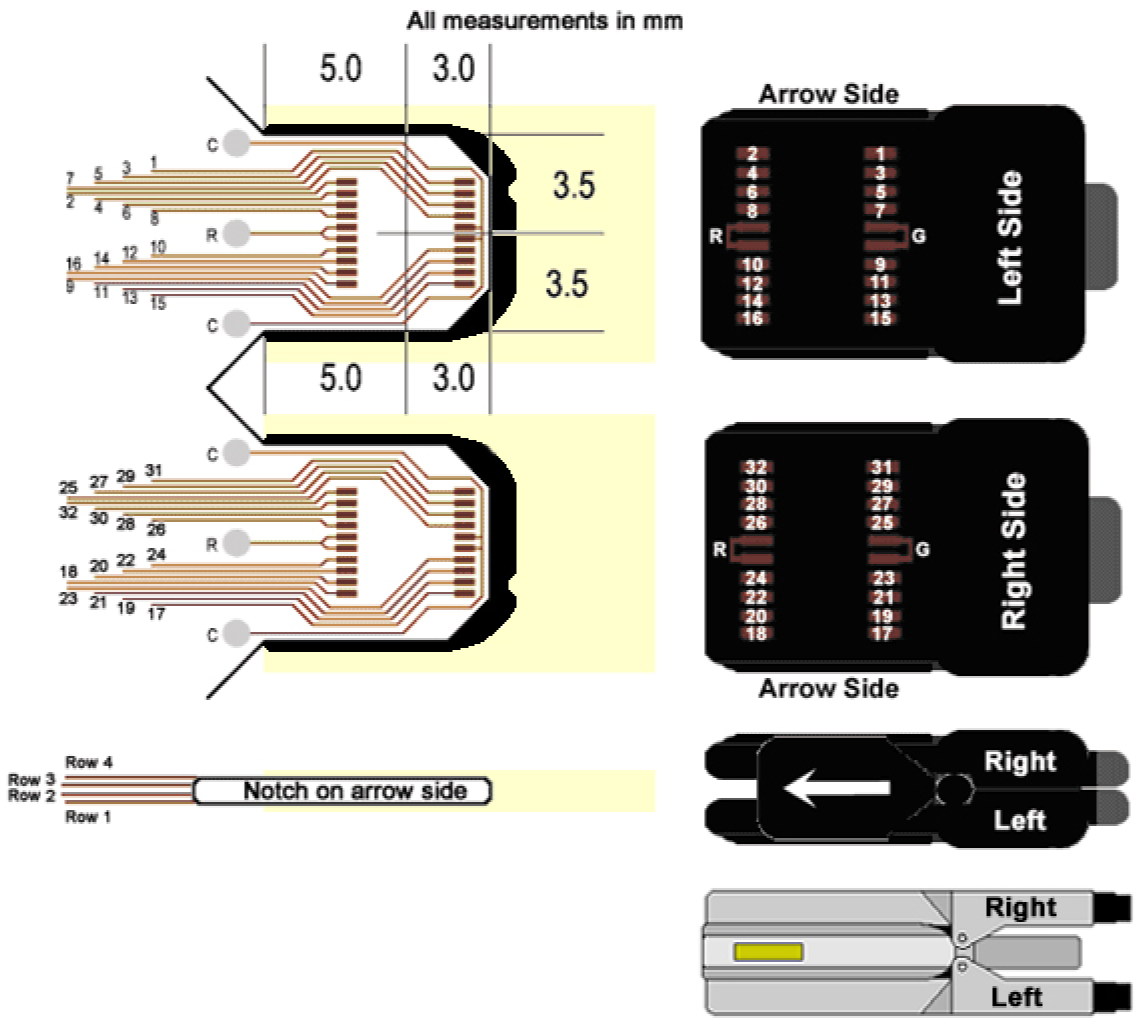
Specification	Default	Options
n Rows X n Electrodes	2X8	Max channels per connector = 64
Metal	Tungsten	
Wire Diameter	50 µm	33 µm
Insulation	Polyimide	
Electrode Type	Standard	Flex Ribbon
Flex Ribbon Site Specification	Attached	Separated

Electrode Spacing	250 µm	500 µm
Row Separation	375 µm	
Tip Angle	Blunt Cut (0 degrees)	30, 45, 60 degrees
Tip Length	2mm	0.5 - 10 mm
Ground and Reference Wires	Differential	Single-Ended

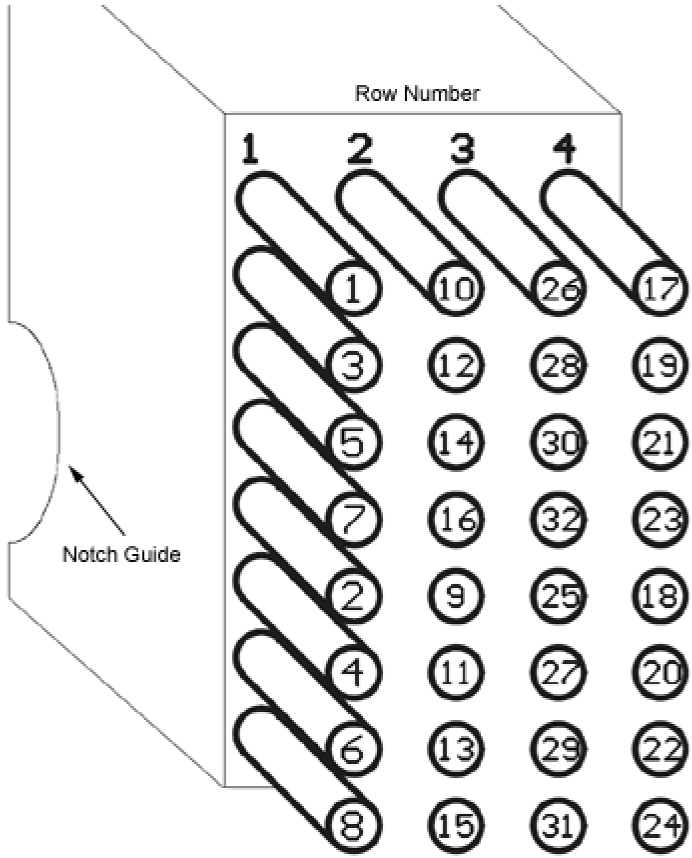
See the *Online Order Form* for more information on ordering specifications.

Pinout

The following pinout illustrates the numbering convention used for the ZIF-Clip® based microwire arrays. Shading denotes headstage area. Refer to the RPvdsEx Manual for information regarding channel remapping and use of the MCMap component.



Note: ZIF-Clip® headstages with an aluminum finish do not have an arrow side. It is instead replaced by a gold-colored square mark as shown. The orientation for the pinout remains the same as the image above.



The diagram to the left indicates the channel output to a TDT amplifier from the ZIF-Clip® based microwire array.

(Looking into the array)

ZCAP - Aluminum ZIF-Clip® Cap

Part Number: ZCAP

The Aluminum ZIF-Clip® Cap is designed to protect the ZIF-Clip® micro connector from potential damage in the absence of the ZIF-Clip® headstage. The caps are made of high quality aluminum and feature a rubber O-ring for easy handling and grip.

The ZCAP fits directly over all ZIF-Clip® compatible connectors protecting your ZIF-Clip® probe adapters and microwire arrays.

Using the ZCAP

Grip the ZCAP with two fingers and gently slide it onto the ZIF-Clip® micro connector. To remove, grasp both sides of the O-ring grip and gently pull away from the ZIF-Clip® micro connector until the ZCAP releases from the connector.

Omnetics Based Microwire Arrays

Part Numbers: OMN1010, OMN1005, OMN1020, OMN1030

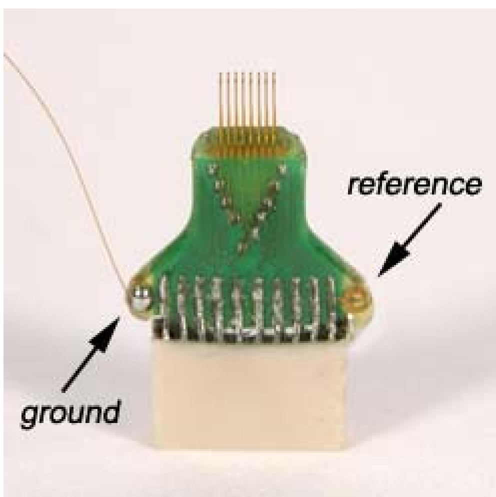
Standard 50 µm polyimide-insulated tungsten microwire gives the arrays excellent recording characteristics and the rigidity of tungsten facilitates insertion. The standard OMN1010 array consists of sixteen channels configured in two rows of eight electrodes each and are typically accessed via our RA16CH 16-channel headstages. OMN1005, OMN1020, and OMN1030 share this standard configuration with varying electrode separation specifications. Consult the documentation provided with your array for custom specifications.

Grounding the Electrode

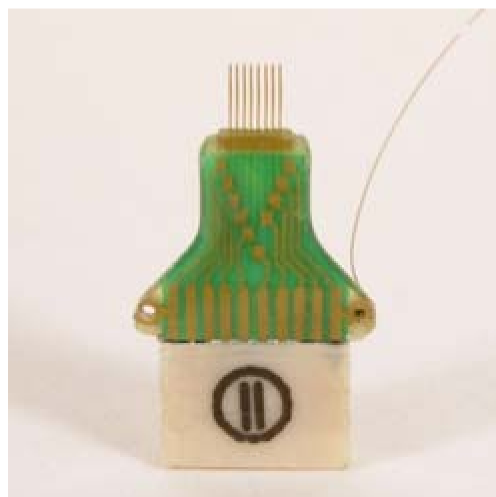
Our latest laser cut microwire arrays (OMN1010) have one location each to connect needed ground and reference wires. Because the reference and ground are shorted together in our RA16CH chronic headstages (unless the jumper is cut by the user) only one wire will be needed for most cases.

Important note! The solder pad is located on the backside of the microwire circuit board.

Back view



Front view



The illustrations above show a single wire connected to the ground pad located on the backside of the array.

Caution! The microwire array can be damaged by extreme heat. Use caution when soldering.

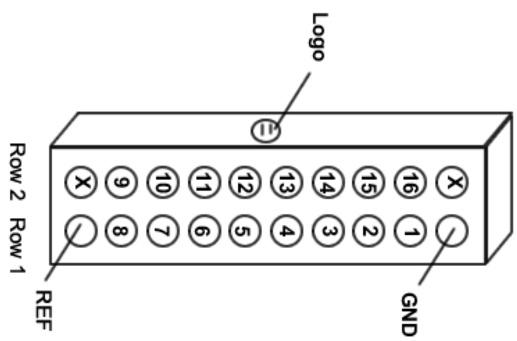
Specifications might vary based on custom order:

Specification	Default	Options
n Rows X n Electrodes	2X8	Max channels per connector = 16
Metal	Tungsten	
Wire Diameter	50 µm	33 µm
Insulation	Polyimide	
Electrode Spacing	250 µm	175 µm, 350 µm, 500 µm
Row Separation	500 µm	1000 µm, 1500 µm, 2000 µm
Tip Angle	Blunt Cut (0 degrees)	30, 45, 60 degrees
Tip Length	2mm	0.5 - 4 mm
Polymer Shroud	None	Shroud, Footed Shroud
Attached G/R Wires	None	Ground, Reference

See the *Online Order Form (PDF format)* for more information on ordering specifications.

Pinout

Omnetics dual row 18-pin nano connector(s) (0.025 mil pitch; <2x7x4mm)



(Looking into connector)

Suggestions for Microwire Insertion

I. General Procedures:

The following are general suggestions for insertion of TDT microwire arrays and may not comply with your animal care and use guidelines. Investigators should consult officials at their respective institutions to determine the regulations governing animal care and use in their laboratory.

We use aseptic techniques and avertin anesthesia for mouse, ketamine/xylazine anesthesia for rat.

We use the general procedures for rodent survival surgery described in: "Principles of Aseptic Rodent Survival Surgery: General Training in Rodent Survival Surgery - Part I" In: Laboratory Animal Medicine and Management, Reuter J.D. and Suckow M.A. (Eds.) International Veterinary Information Service, Ithaca NY (www.ivis.org), 2004; B2514.0604.

This can be downloaded from http://www.ivis.org/advances/Reuter/brown1/IVIS.pdf.

NIH offers instructional videos entitled: “Training in Basic Biometholodology for Laboratory Mice” and “Training in Survival Rodent Surgery” at their website: http://grants.nih.gov/grants/olaw/TrainingVideos.htm.

II. Stereotaxic Surgery:

We use procedures similar to those described in: "Stereotaxic Surgery In The Rat: A Photographic Series" by Richard K. Cooley and C.H. Vanderwolf. This reference is available from Amazon.com for $27.97 and is highly recommended.

III. Microwire Procedures:

General information, pictures, and available configurations for TDT microwire arrays can be found at:

http://www.tdt.com/products/MW16.htm and

http://www.tdt.com/products/OrderForm_Omn1010.pdf

A recent paper by Kralik et al. (2001) contains a very helpful description of microwire array insertion methods (Methods. 2001 Oct; 25(2): 121-50).

In rat and mouse, we recommend following the general and neurosurgical procedures as described in the references above.

We first prepare the subject and perform a craniotomy above the implantation site following the methods of Cooley and Vanderwolf (2004). Implant several skull screws as described in this reference to help bond the dental acrylic and array to the skull. A base coat of OptiBond FL (Kerr) applied to the skull works well to help bond the dental acrylic. Keep this out of the craniotomy.

For rat and mouse we recommend a durotomy, using the tip of a sterile syringe needle as a micro-scalpel to cut an "X" shaped incision through the dura. Reflect the flaps of dura aside, taking care not to disturb the pia or pial vasculature.

Advance the array to the pial surface using a stereotaxy and check that all electrodes are unobstructed by bone or dura. We have also used the stereotaxy to quickly advance the array through the pia and then to adjust the array to its final depth. This method has worked well for a number of our customers as well.

There have been two schools of thought on insertion speed. Fast insertion (e.g. Rousche PJ, Normann RA. Ann Biomed Eng. 1992;20(4):413-22) using an inserter device, and slow insertion (e.g. Nicolelis et al., Proc Natl Acad Sci U S A. 2003 Sep 16;100(19): 11041-6). A recent paper by Rennaker et al., 2004, (J Neurosci Methods. 2005 Mar 30;142(2):169-76) explores the relative merits of each method.

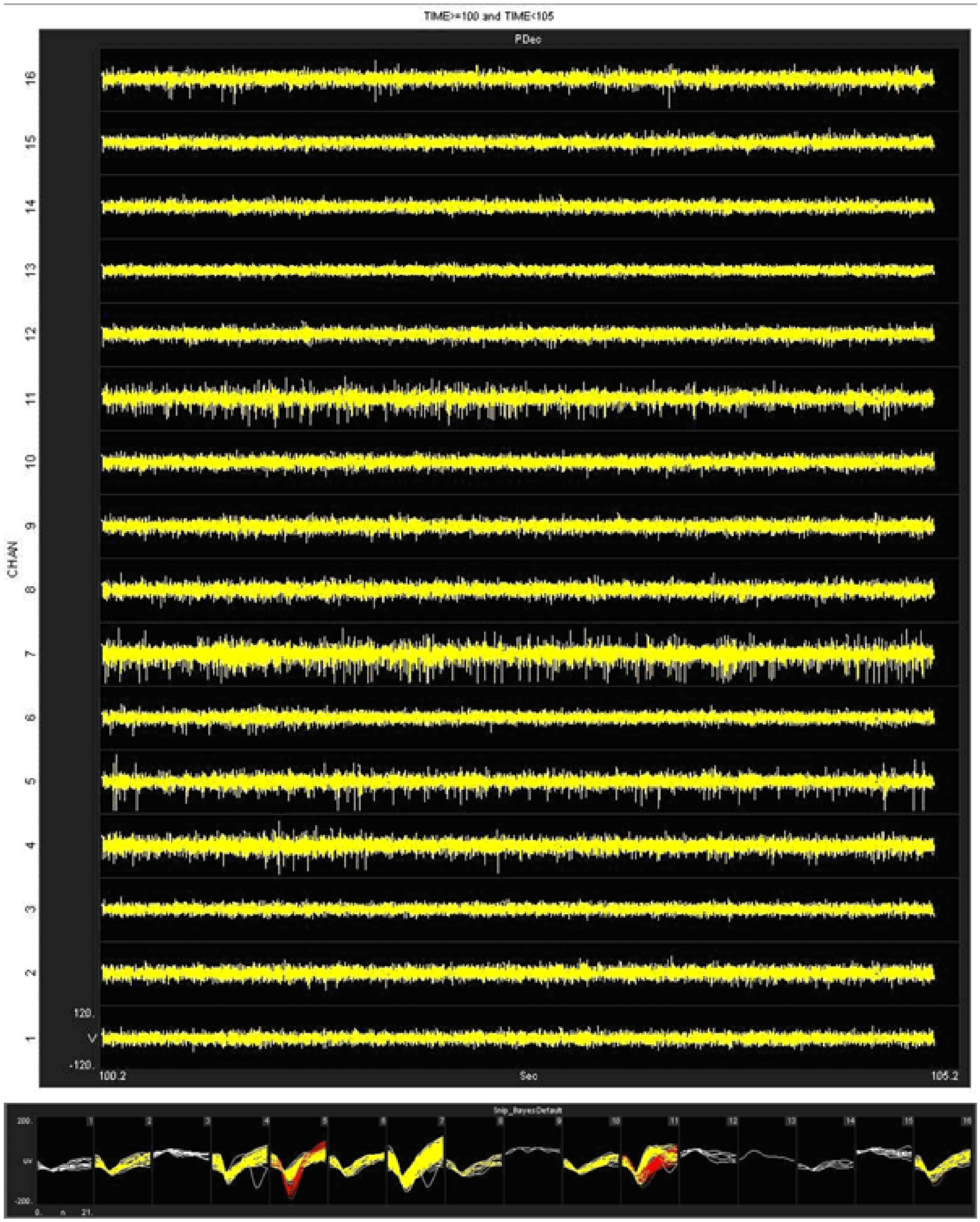
Regardless of which insertion method you choose, advance the array to its desired position, leaving it attached to the stereotaxy until it is fully bonded to the skull with dental acrylic. Prevent

CSF from weeping from the craniotomy by gently packing around the array with gelfoam. The CSF will eventually soak through and keep the acrylic around the craniotomy from curing, so perform this step quickly. Bone wax or Kwik-Cast would probably work better than the gelfoam, but we have not used these in our lab to date.

Attach the array to the skull using a thin layer of dental acrylic and the methods described by Cooley and Vanderwolf. Do not build up a large base of acrylic until the ground wire(s) of the array have been attached by wrapping them around the stainless skull screws. Make very sure that the ground wire(s) make good electrical contact to the screws. Pot the entire array/screw complex with dental acrylic using the methods described by Cooley and Vanderwolf.

In our hands, explanted arrays come out of the brain with roughly the same impedances they went in with. Here, recording duration seems to be more limited by surgical technique/capsule formation than by the arrays themselves. We recommend ethylene oxide gas sterilization of the arrays and good sterile surgical technique.

We have obtained good recordings in rat and mouse cortex for several weeks; using only alcohol sterilization of the arrays (we have no access to ethylene oxide). An example from rat with lots of active channels, ~150 µV spikes on ~20 µV background noise is below. We have seen up to ~300 µV spikes on the same noise floor. Our customers have reported recordings durations of several months in rat and monkey.


TIME>=100 and TIME<105
PDec
CHAN
100.2
Sec
105.2
Snip_BayesDefault

Part 11 Attenuator

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PA5 Programmable Attenuator

Overview

The PA5 Programmable Attenuator is a precision device for controlling signal levels over a wide dynamic range, providing 0 to 120 dB of attenuation for signals up to 100 kHz in frequency. The device is fully programmable; however, simple manual operation is also available using front panel controls.

When used **programmatically**, the module may be controlled via TDT's ActiveX Controls, as well as any programming environment that supports ActiveX or programs that allow scripts for implementing ActiveX controls, such as Microsoft Access and Excel. For information about how to control the module programmatically, see the *ActiveX Reference Manual*.

When used in manual operation, the attenuation level is adjusted in two modes of operation:

- The *Atten* mode permits the user to adjust the attenuation level of the signal from 0 to 120 dB in increments of 0.1 dB.
- The *UserAtt* mode permits the user to adjust the attenuation level of the signal using user-programmed parameters. Before using the UserAtt mode, attenuation parameters must be set up using the UserOps menu.

Power and Interface

The PA5 Programmable Attenuator is powered via the System 3 zBus (ZB1PS) and requires an interface to the PC (Gigabit, Optibit, or USB). Ensure that the ZB1PS chassis housing the PA5 is connected in the interface loop according to the installation instructions for the interface in use.

Important!: The chassis housing the PA5 must be powered and connected to a PC via the PC interface for BOTH manual and programmed operation.

Features

Display

Displays the current level of attenuation being applied to the signal or displays the manual operations menu. During manual operation it is used to set up user-defined attenuation parameters and to obtain descriptions for menu items. See *Display Icons*, page 11-11 for more information.

(ESC) Button

Exits the manual operations menu items without accepting changes.

SELECT (ENTER) Knob

During manual operation, allows the user to adjust the attenuation applied to the signal. In addition, it allows the user to scroll through the manual operation menus, set up user-defined attenuation parameters, and access descriptions of menu item.

Turn the Select knob to adjust attenuation or view menus. Press and release the knob to make a selection. The module must be in Attn or UserAtt mode to manually adjust attenuation. See *Manual Operation*, page 11-4 for more information.

INPUT BNC

Source signal input. The maximum input voltage is +/- 10V peak.

OUTPUT BNC

Attenuated signal output.

PA5 Manual Operation

Important!: The PA5 is powered via the zBus and must be connected to the PC via an interface module during manual operation.

In manual operation, the PA5 is operated using front panel controls. The menu options are viewed by turning the Select knob and entered by pressing and releasing the knob. The module must first be set to Attn or UserAtt mode to manually adjust attenuation.

To access a menu:

- Turn the knob until the name of the desired menu appears on the display, then press and release the knob. The module has two levels of menus.

  Top-level menu items are indicated by a single filled box in the upper left corner of the menu display, and sub-menus are indicated with an additional indicator box for each level. Only the UserOps menu item has sub-menu items. See *Display Icons*, page 11-11 for more information.

For a definition of each menu item:

- Turn the **Select** knob until the name of the menu appears on the display, then press and hold down the **Select** knob. A description of the menu function will scroll across the display.

To exit a menu without changing settings:

- Press and release the **ESC** button.

Operation in *Atten* Mode

In *Atten* mode, the user sets the desired level of attenuation with the Select knob. When the unit is powered on, it defaults to the Atten mode with 0.0 dB of attenuation.

To use Atten mode:

1. Turn the **Select** knob until *Atten* appears on the display, then press and release the **Select** knob.

   A small letter "A" appears in the upper left corner of the display, indicating the unit is in *Atten* mode, and a decibel reading appears on the right side of the display. See *Display Icons* for more information.

2. Turn the **Select** knob to adjust attenuation in 0.1 dB increments.

Operation in *UserAtt* Mode

In *UserAtt* mode, the user can adjust the attenuation level of the signal using user-programmed parameters available in the *UserOps* menu. Users can also save common parameter configurations in the PA5's nonvolatile memory. See *Using Preset Configurations* for more information.

To use UserAtt mode:

1. Turn the **Select** knob until *UserAtt* appears on the display, then press and release the **Select** knob.

   A small letter "U" appears in the upper left corner of the display, indicating the unit is in *UserAtten* mode, and a decibel reading appears on the right side of the display. See *Display Icons*, page 11-11 for more information.
2. Turn the **Select** knob to adjust attenuation according the current user programmable parameters (available in the *UserOps* menu). The default settings include a step size of 3.0 dB and dynamic update mode.

   Note: When the Update attenuation parameter is set to *Manual*, the intensity of the display will dim as the user turns the knob—this indicates that the changes have not been applied to the output signal. The user must press and release the Select knob to apply attenuation changes to the output signal.

To access the UserOps menu:

1. Turn the Select knob until UserOps appears on the display.
2. Press and release the Select knob.
3. Set the UserOps parameters as desired.
   - To set parameters such as step size (StpSize), update mode (Update), minimum attenuation (AbsMin), base attenuation (BaseAtt), and reference value (Refrnce); turn the Select knob to the desired value and then press and release to save changes.
   - To exit any menu without saving parameter changes, press and release the **ESC** button before the settings are saved.

About UserAtten Mode Parameters

In *UserAtten* Mode, the user may set parameters such as step size (*StpSize*), update mode (*Update*), and minimum attenuation (*AbsMin*). The scale can be adjusted using the base attenuation (*BaseAtt*) and reference value (*Refrnce*) parameters. Both base attenuation and reference can be used simultaneously, producing an actual attenuation equal to (*Refrnce*+*BaseAtt*-dial setting). See *Manual Operation Menus* for more information.

BaseAtt--Base Attenuation

Adds a fixed attenuation value, shifting the scale down and allowing attenuation to be displayed relative to this base level (useful for calibrating signals played over varying transducers). See *Setting Base Attenuation*, page 11-8 for more information.

StpSize--Step Size

Sets the increments in which attenuation is applied to the signal when using the Select knob.

Refrnce—Reference

Sets a reference value used to "flip" the scale of the display (useful for displaying actual signal level on the front panel of the PA5). May be used only when the intensity of the input signal is known. See *Setting a Reference Value*, page 11-9 for more information.

Update—Update

Determines whether attenuation changes dynamically as the selector knob is turned or only after pressing enter to select the current value.

AbsMin--Minimum Attenuation

Sets the minimum level of attenuation the user can apply to the signal (to avoid accidentally presenting excessively loud signals).

PA5 Manual Operation Menus

To access a menu:

- Turn the knob until the name of the desired menu appears on the display, then press and release the knob. The module has two levels of menus.

  Top-level menu items are indicated by a single filled box in the upper left corner of the menu display, and sub-menus are indicated with an additional indicator box for each level. Only the UserOps menu item has sub-menu items.

For a definition of each menu item:

Turn the Select knob until the name of the menu appears on the display, then press and hold down the Select knob. A description of the menu function will scroll across the display.

To exit a menu without changing settings:

Press and release the **ESC** button.

<table>
<tr><th colspan="2">PA5 Top Level Menu</th></tr>
<tr><td>Atten</td><td>Sets attenuation from 0.0 to 120.0 dB in 0.1 dB increments. The default setting is 0.0 dB. When Atten is in use, the letter "A" appears on the left side of the display, while the attenuation level appears on the right side of the display.</td></tr>
<tr><td>UserAtt</td><td>Sets attenuation based on UserOps settings. Before use, attenuation parameters must be set up via the UserOps sub-menus (see below). The default setting is 0.0 dB. When UserAtt is in use, the letter "U" appears on the left side of the display, while the attenuation level appears on the right side of the display.</td></tr>
<tr><td>UserOps</td><td>Access UserOps submenu
<table>
<tr><th colspan="2">UserOps Sub-menu</th></tr>
<tr><td>BaseAtt</td><td>Sets a fixed level of attenuation as a reference. The default setting is 0.0 dB and the range is 0 to 100.0 dB. When BaseAtt is set, a "+" symbol appears on the left side of the display.
When used, the attenuation level displayed is relative to BaseAtt. For example, with BaseAtt set to 60.0 dB, the attenuation level will be display from -60.0 dB to 60.0 dB.</td></tr>
<tr><td>StpSize</td><td>Sets the increments of attenuation. The default setting is 3.0 dB, and the range is 0.1 to 60.0dB.</td></tr>
<tr><td>Refrnce</td><td>Changes the display so it shows the output signal intensity rather than the attenuation level. This function may be used only when the input signal strength is known. When Refrnce is set, the letter "R" appears on the left side of the display. The default setting is 0.0, and the range is ± 300.0.
For example, when Refrnce is set to 136 and the attenuation level set to 0.0 dB, the display shows 136.0. When the attenuation level is adjusted to 30.0 the display shows 106.</td></tr>
<tr><td>Update</td><td>Determines when attenuation is applied to the signal. When set to Dynamic, attenuation is applied as the Select knob is turned. When set to Manual, attenuation is applied after the Select knob is pressed and released. The default setting is Dynamic.
Note that when Update is set to Manual, the attenuation level on the display changes as the Select knob is turned, but the attenuation is not applied to the signal until the Select knob is pressed and released. In this mode, the intensity of the display dims to indicate that the attenuation has not been applied to the signal.</td></tr>
<tr><td>MinAttn</td><td>Sets the minimum attenuation level for the UserAtt mode. This is used to avoid signals that are too loud for the subject or equipment. The default value is 0.0 dB and its range is 0.0 to 100.0 dB.
Note that setting this parameter limits the range of possible attenuation levels. For example, when it is set to 30.0 dB, the range of attenuation is 30 db to 120 dB.</td></tr>
</table></td></tr>
<tr><td>Load PS</td><td>Loads one of four preset UserAtt configurations from non-volatile memory. See Save PS (Below). The default is 1 and its range is 1 to 4.</td></tr>
<tr><td>Save PS</td><td>Saves the current UserAtt configuration in one of four non-volatile memory buffers. This permits the user to save commonly used UserAtt configurations. The default is 1 and its range is 1 to 4.
To save a configuration, first ensure that all UserAtt parameters are set as desired then turn the Select knob until the desired memory location is displayed, and press the Select knob. Saving appears on the display. The preset is ready of use.</td></tr>
<tr><td>Reset</td><td>Resets all menu items, including presets, to their default conditions.
<table>
<tr><td>Confirm</td><td>The user must confirm the reset by pressing and releasing the Select knob. While the module is resetting, Reseting appears on the display.
The user must confirm the reset by pressing and releasing the Select knob. While the module is resetting, Reseting appears on the display.
To exit without resetting, turn the Select knob until Cancel appears on the display and then press and release the Select knob, or press the Esc button.</td></tr>
<tr><td>Cancel</td><td>Cancels the reset.</td></tr>
</table></td></tr>
</table>

Setting Base Attenuation

When operating the PA5 manually in User Attenuation (UserAtt) mode, the Base Attenuation (BaseAtt) parameter can be used to apply a fixed attenuation level to the signal. Any additional attenuation to the signal is displayed relative to this base level within a range of 0 to 120 dB. For example: if the BaseAtt is set to 6 dB, when the user sets the attenuation to 3 dB the actual attenuation applied is 9 dB. This feature can be used to calibrate a number of different experimental setups, attenuating each by a different base attenuation so as to provide identical signal levels when each is set to 0.0 dB UserAtt.

When this feature is in use, a "+" symbol is displayed on the left side of the display. Note that the Base Attenuation and Reference parameters can be used simultaneously. When both of these features are in use, the letter "R" and a "+" symbol are displayed on the left side of the display. See *Display Icons* for more information.

To set the base attenuation:

1. Access the *UserAtt* mode, by turning the **Select** knob until *UserAtt* appears on the display, then pressing and releasing the knob.
2. Access the *UserOps* menu, and turn the **Select** knob until *BaseAtt* appears on the display.
3. Press and release the **Select** knob. *0.0 dB* appears on the display.
4. Turn the **Select** knob until the display shows the desired level of attenuation.
5. Press and release the **Select** knob. The level is saved and *BaseAtt* appears on the display.
6. To exit the *UserOps* menu, press and release the **ESC** button again.

Example 1: Adding Speaker Calibration Attenuation

A user wishes to equilibrate the level of stimuli applied to two different loudspeakers. Speaker #2 is 7.3 dB louder at the frequency of interest than speaker #1. This example requires the use of two PA5 Programmable Attenuators.

To more directly compare thresholds measured with both loudspeakers, set the BaseAtt parameter for speaker #1 to 0.0 dB and set the BaseAtt parameter for speaker #2 to 7.3 dB, so that the signal level delivered for a given UserAtt is the same for both loudspeakers. Actual attenuation versus displayed levels is shown in the following table.

Speaker 1: BaseAtt=0	
UserAtt Display Value	Actual Attenuation
0	0
120	120

Speaker 2: BaseAtt = 7.3	
UserAtt Display Value	Actual Attenuation
-7.3	0
0	7.3
112.7	120

Example 2: Multiple Signals of Varying Levels

The base attenuation feature is also useful when working with multiple signals of varying levels. BaseAtt can be configured so the intensity of each signal input is identical at 0.0 dB. When working with three signals 30, 34, and 36 dB SPL, the BaseAtt parameters are set and the actual versus displayed value of attenuation are shown in the table below.

This example requires three PA5 Programmable Attenuators.

Input Signal	BaseAtt	Displayed Value	Actual Attenuation
36 dB SPL	6.0 db	0	6
		4	10
		6	12
		8	14
34 dB SPL	4.0 dB	0	4
		4	8
		6	10
		8	12
30 dB SPL	0.0 dB	0	0
		4	4
		6	6
		8	8

Setting a Reference Value

The Reference parameter is used to display the intensity of the output signal. This parameter can be used only when the strength of the input signal is known. This serves to "flip" the scale, displaying larger numbers for smaller attenuation values.

When in use, a letter "R" is displayed on the left side of the display. Note that the Base Attenuation and Reference parameters can be used simultaneously. When both of these features are in use, the letter "R" and a "+" symbol are displayed on the left side of the display. See *Display Icons*, page 11-11 for more information.

To set the Reference parameter:

1. Access the *UserOps* menu, and turn the **Select** knob until *Refrnce* appears on the display.
2. Press and release the **Select** knob. *0.0 dB* appears on the display.
3. Turn the **Select** knob until the display shows the desired level.
4. Press and release the **Select** knob. The reference is saved.
5. To exit the *UserOps* menu, press and release the **ESC** button.

Example 1: Displaying Signal Level in SPL

A user wishes to use the PA5 to display the signal level in dB Sound Pressure Level (SPL) for the frequency of interest. Measurements with a sound level meter show a sound level of 96.4 dB SPL with 0.0 dB of attenuation in the PA5. The user sets the Refrnce parameter to 96.4. The actual attenuation versus the displayed value is as follows:

Display Value (in dB SPL)	Attenuation
0	96.4
50	46.4
96.4	0

Example 2: Combining Reference and Base Attenuation

When the Reference parameter is set to 110 dB and the Base Attenuation parameter is set to 6.0 dB, the actual attenuation versus displayed value is as follows:

Display Value (in dB SPL)	Attenuation
0	116
50	66
110	6

Using Preset Configurations

The PA5 Programmable Attenuator allows users to save four unique User Operation configurations that may be used in UserAttn mode. These configurations may include any of the *UserOps* parameters (such as step size, base attenuation, and minimum attenuation). Before a configuration can be loaded, it must be set up via the UserOps menu and saved via the SavePS menu.

Saving Preset Configurations

Warning: This procedure overwrites the contents of the selected preset location. Be certain that the existing configuration is not needed before continuing.

Before a configuration can be saved, it must be set up via the *UserOps* menu. Once the configuration is set up as desired, save the configuration by performing the following:

1. At any top-level menu, turn the **Select** knob until *SavePS* appears on the display.
2. Press and release the **Select** knob. *Preset-1* appears on the display.
3. Turn the **Select** knob until the desired preset location is displayed and then press and release the **Select** knob. *Saving* appears on the display and then *Atten* appears on the display. The configuration is saved.

Loading Preset Configurations

When a configuration has been set up via the *UserOps* menu and saved via the *SavePS* menu, load the configuration by performing the following:

1. Turn the **Select** knob until *LoadPS* appears on the display.
2. Press and release the **Select** knob. *Preset-1* appears on the display.
3. Turn the **Select** knob until the desired preset location is displayed, and then press and release the **Select** knob. First, *Loading* appears on the display and then *Attn* appears on the display. The configuration is loaded.

PA5 Display Icons

Menu Level Icons

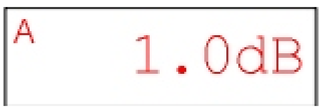
Display | Description

Single Box: indicates a top-level menu.

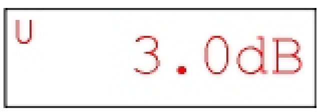
Double Box: indicates a second-level menu.

Attenuation Mode Icons

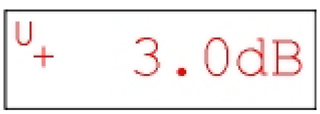
Display | Description



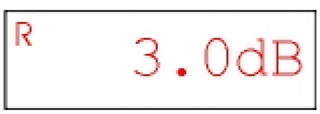
A: Normal Attenuation Mode



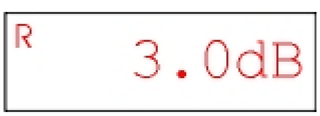
U: User Attenuation Mode



U+: User Attenuation Mode. Base attenuation value set.



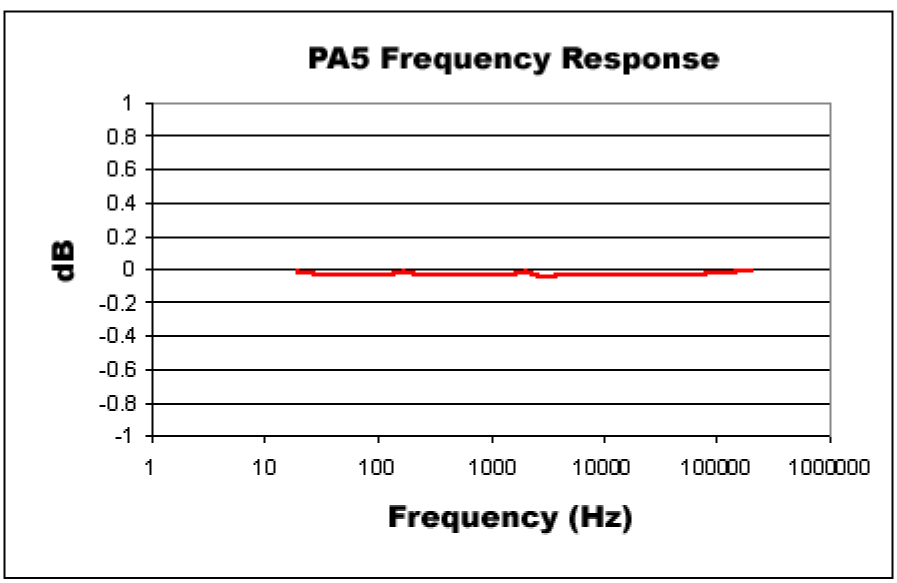
R: User Attenuation Mode. Reference level set.

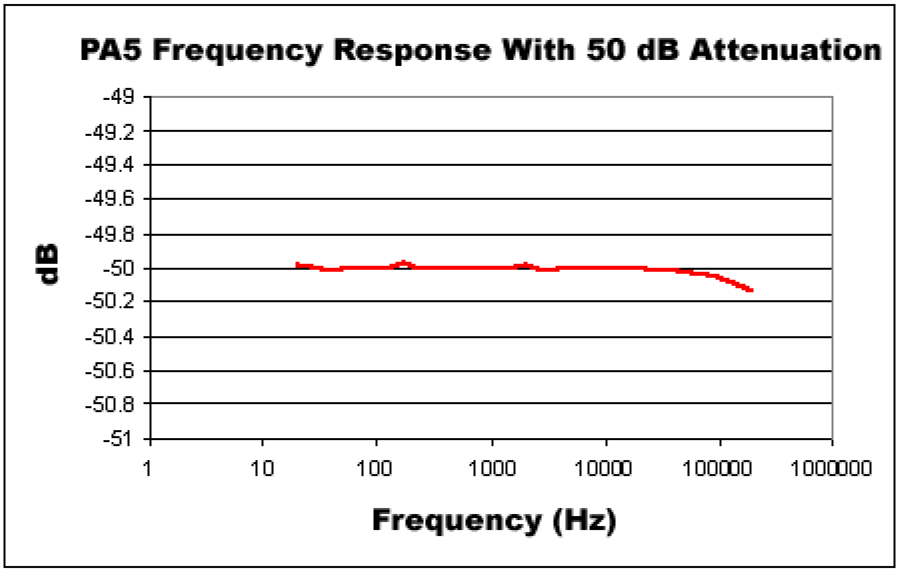


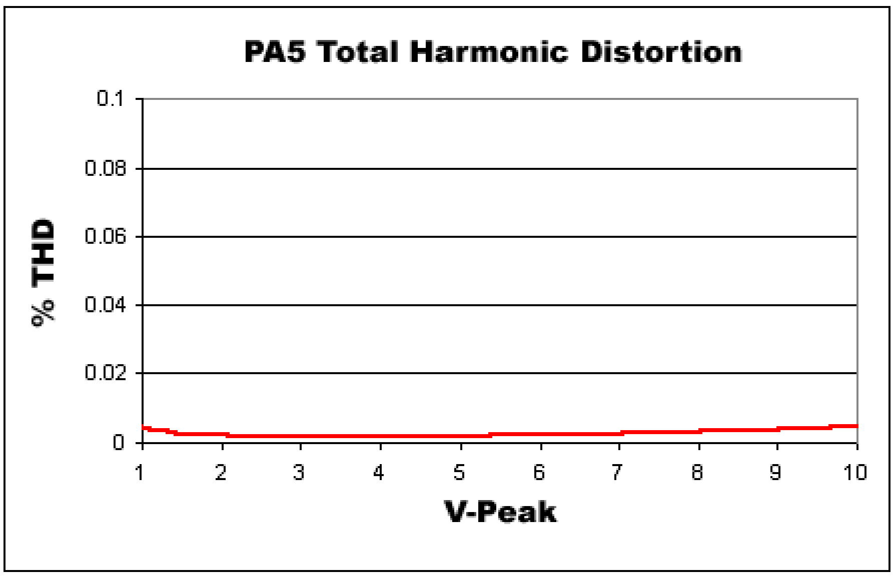
R+: User Attenuation Mode. Base attenuation value and reference level set.

PA5 Technical Specifications

Input Signal Range	±10V peak
Frequency Range	DC – 200 kHz
Attenuation Range	0.0 to 120.0 dB
Attenuation Resolution	0.1 dB
Attenuation Accuracy	0.05 dB
Spectral Variation	<0.04 dB (20Hz to 80 kHz)
DC Offset	< 10 mV
Signal/Noise	113 dB (20 Hz to 80 kHz at 9.9 V)
Noise Floor	16 µV rms (20 Hz to 80 kHz)
THD	<0.003 % (1kHz tone +/- 7V peak, 0 dB attenuation)
Attenuation Settling Time	5 ms
Switching Transient	< 8 mV (0 Hz to 80 kHz)
Input Impedance	10 kOhm
Output Impedance	10 Ohm




PA5 Frequency Response With 50 dB Attenuation
dB
-49
-49.2
-49.4
-49.6
-49.8
-50
-50.2
-50.4
-50.6
-50.8
-51
1
10
100
1000
10000
100000
1000000
Frequency (Hz)

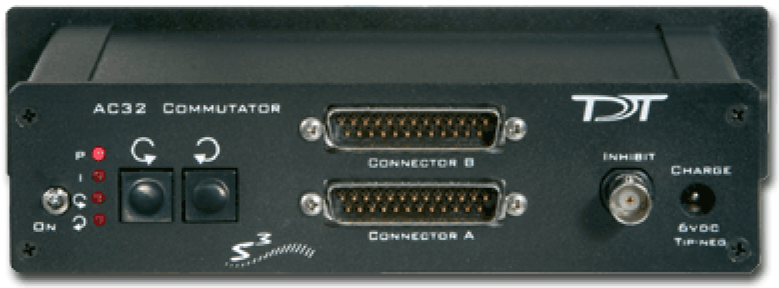

PA5 Total Harmonic Distortion
% THD
0.1
0.08
0.06
0.04
0.02
0
1
2
3
4
5
6
7
8
9
10
V-Peak

~

Part 12 Commutators

~

ACx Motorized Commutators



Overview

As part of a complete solution for research with awake, behaving subjects, TDT has developed a series of 16, 32, and 64-channel ultra quiet motorized commutators. Lightweight cables and connectors minimize the torque caused by subject motion relative to a fixed cable. Sensors on the commutator continuously measure the rotational angle applied to the headstage cable, and spin the motor to compensate, eliminating the turn-induced torque at the subject's end of the cable. Pushbuttons allow for optional manual control, and an input BNC can be used to inhibit the commutator motor during critical recording periods. A rechargeable Li-Ion battery powers the motorized commutators.

Part numbers:

AC16 - 16 Channel Commutator

AC32 - 32 Channel Commutator

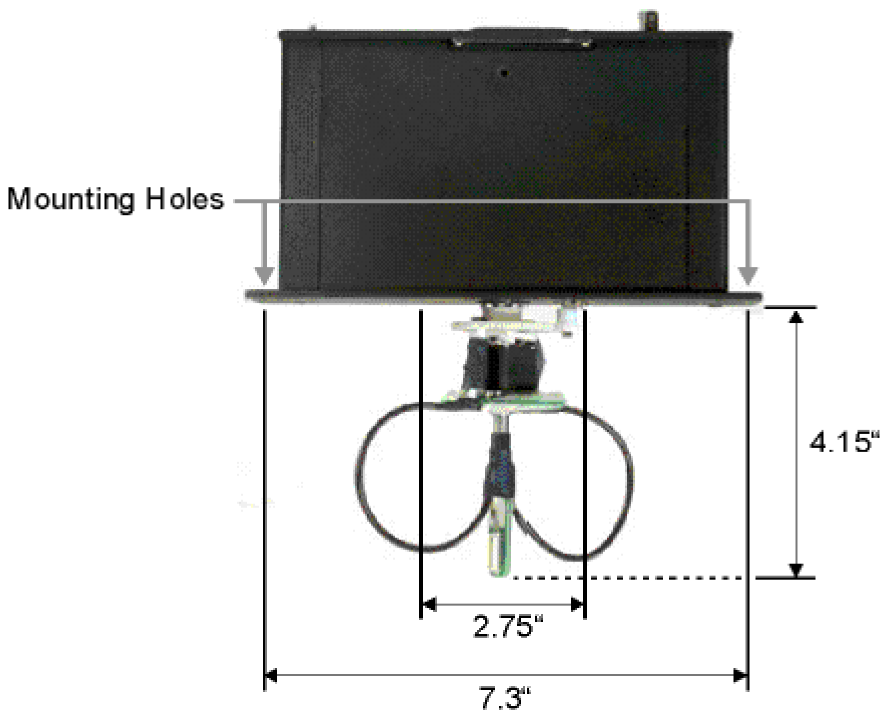
AC64 - 64 Channel Commutator

Power and Interface

The commutators are powered by a 1500 mAh Li-ion Battery. A 6-9 V DC, 500 mA, center negative adaptor (one provided) charges the unit. Low battery status is reported only by a decrease in rotational speed. No PC interface is required for operation.

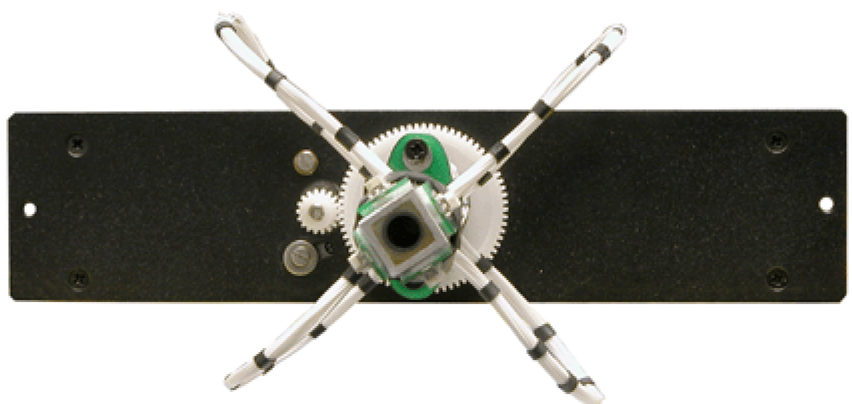
Mounting

The commutator assembly can be mounted above the subject by utilizing the two mounting holes provided. Depending on the mounting configuration, a 2.75" diameter access hole may have to be drilled into the support to which the commutator is mounted. Dimensions are provided below to determine clearance requirements.



Connection and Setup

Before using the AC32 and AC64 commutators, it is important to adjust the wire harness on the back so it is balanced. The AC32 harness should be in two loops 180 degrees apart and the AC64 harness should be in four loops 90 degrees apart (as shown below). Typically, preamps are connected to the DB25 connectors on the front of the commutator and headstages (with special splice connectors) are connected to the interface receptacles on the back of the commutator.



Amplifier Connections

The commutators interface with one or more preamplifers via connections on the user interface panel. All connections are designed for direct connections to TDT preamplifers. By default, the 16 and 32 channel versions feature DB25 connectors that match the pin configuration of the Medusa PreAmps. 64 channel versions feature flying leads with connectors that mate with the Z-Series PreAmp. Custom pinout configurations are available.

Default Device Configurations

Commutator	Use with
AC16	Medusa RA16PA
AC32	Medusa RA16PA
AC64	Z-Series PZ2 or PZ3

Channel Mapping Diagram

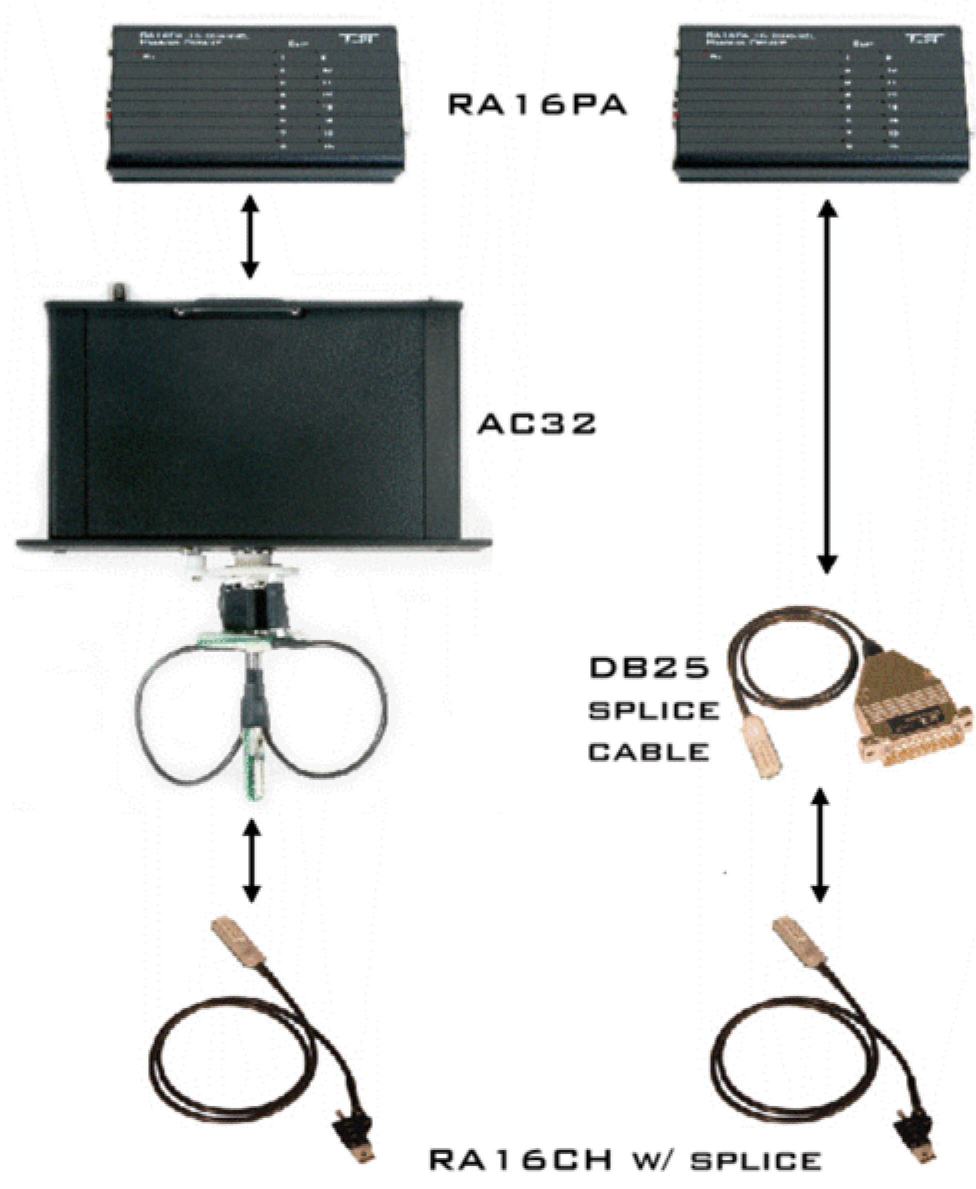
Device	Connector	Channels
AC16	Connector	1-16

Device	Connector	Channels
AC32	Connector A	1-16
	Connector B	17-32

Device	Connector	Channels
AC64	Connector #1	1-16
	Connector #2	17-32
	Connector #3	33-48
	Connector #4	49-64

Headstage Connections

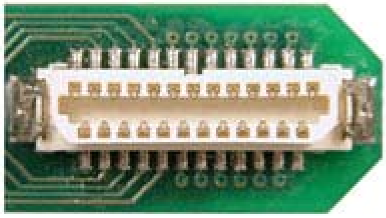
TDT offers a headstage with splice suitable for use with the commutator. A DB25 splice cable can also be provided to allow you to easily switch to a configuration that does not use the commutator. See the following illustration.



Important!: When using TDT's SH16 Switching Headstage with the AC64, the “control” connector of the headstage **MUST be connected using the #1 (ch1-16) connector.** The switching headstage CANNOT be connected to any other connector.

When using non-TDT switching headstages, contact TDT customer support for assistance.

Interface Receptacles



Interface receptacles (AC64 (4), AC32 (2), AC16 (1)) on the back of the commutator provide connections to headstages via standard interface headers. See technical specifications for pin mapping and see *Headstage Connections* below for direct connection solutions from TDT.

Features

LEDs

The four indicator LEDs on the front panel indicate power, the status of the Inhibit BNC input, clockwise rotation and counterclockwise rotation.

P	Power (~2 Hz flash when on, ~4 Hz flash when rotating)
I	Inhibit
	Counterclockwise rotation
	Clockwise rotation

Note: When the sensors on the commutator cause the motor to continuously rotate more than five revolutions in one direction, the unit will enter a hold state to prevent the wires from tangling. The commutator will not respond to commands and both the clockwise and counterclockwise LEDs will flash. Cycle power to reset the unit.

Manual Rotational Buttons

The commutators feature both clockwise and counterclockwise manual rotational buttons. When pressed, these buttons will rotate the commutator at approximately 12 RPM. Pressing either of these buttons also overrides the current rotational state of the commutator.

Inhibit BNC

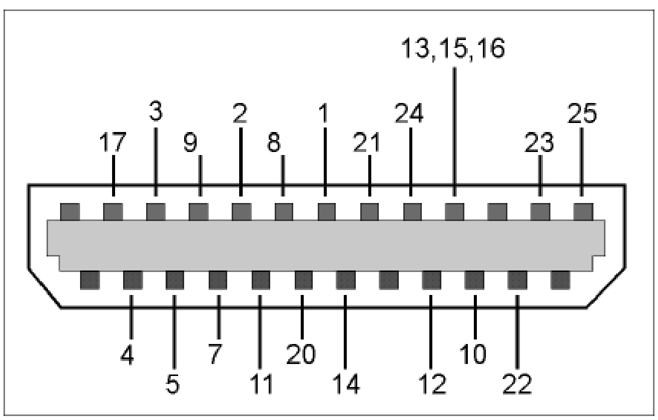
During critical recording periods it may be necessary to prevent rotation to ensure signal integrity. A logical low (0) on the Inhibit BNC will prohibit any rotation initiated by either the sensors on the commutator or the manual rotational button.

AC16, AC32, AC64 Technical Specifications

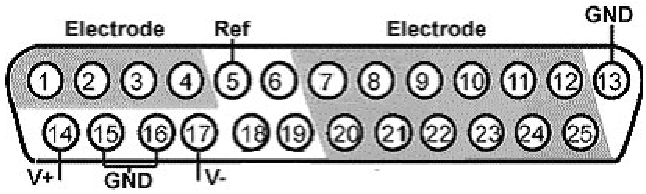
Channels:	16, 32, or 64
Signal/Noise:	120 dB (20 Hz to 25 kHz)
RPM (approx):	12
Digital Inputs:	1 Inhibit
Power Consumption:	35 mAh, quiescent 65 mAh, rotating
Power Supply:	Battery 1500 mAh Li-ion Battery. 1000 cycles of charging, not removable by user. Charger 6-9 V DC, 500 mA, center negative
Dimensions (in):	Backplate to end of connector 4.15 Minimum diameter for access hole 2.75 Distance between mounting holes 7.3
Weight (g):	~ 665 (AC16 and AC32) ~ 945 (AC64)

Interface Receptacles

The interface receptacle diagram shows how the pins on each receptacle map to the pins on the associated DB25 connector on the front of the commutator.



AC16 and AC32 Headstage Connector(s) Pinout

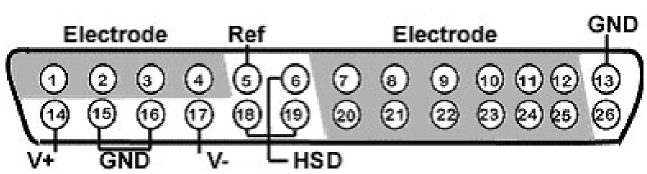


Pin	Name	Description
1	E1	Electrode Channels
2	E2	
3	E3	
4	E4	
5	Ref	Reference
6	N/A	Not Used
7	E5	Electrode Channels
8	E7	
9	E9	
10	E11	
11	E13	
12	E15	
13	GND	Ground

Pin	Name	Description
14	V+	Positive Voltage
15	GND	Ground
16	GND	Ground
17	V-	Negative Voltage
18	N/A	Not Used
19	N/A	Not Used
20	E6	Electrode Channels
21	E8	
22	E10	
23	E12	
24	E14	
25	E16	

AC64 Headstage Connectors Pinout

Important!: Connectors 2, 3, and 4 share common GND, V+, and V-.



Pin	Name	Description
1	E1	Electrode Channels
2	E2	
3	E3	
4	E4	
5	Ref	Reference
6	HSD	Headstage Detect
7	E5	Electrode Channels
8	E7	
9	E9	
10	E11	
11	E13	
12	E15	
13	GND	Ground

Pin	Name	Description
14	V+	Positive Voltage
15	GND	Ground
16	GND	Ground
17	V-	Negative Voltage
18	HSD	Headstage Detect
19	HSD	Headstage Detect
20	E6	Electrode Channels
21	E8	
22	E10	
23	E12	
24	E14	
25	E16	
26	N/A	Not Used

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Part 13 Transducers and Amplifiers

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CF1/FF1 Magnetic Speakers

Overview

TDT Magnetic Speakers offer high output and fidelity over a bandwidth from 1 – 50 kHz. These broadband speakers have more power at lower frequencies than our electrostatic speakers, making them well suited for laboratory species with lower frequency hearing. Their high output levels and broad bandwidth also make them excellent for noise exposure studies.

These 4-Ohm magnetic speakers are available in either free-field or closed-field models. The free-field model delivers signals of over 100 dB SPL with < 1% distortion over its entire bandwidth (+/- 4 V, 10 cm). The closed-field model has an internal parabolic cone designed to maximize output and minimize distortion. Its tapered tip can be inserted directly to the subject's ear or fitted with the provided tubing and used with most standard ear tips.

The FF1 and CF1 magnetic speakers can be driven using either TDT's SA1 or SA8 stereo amplifiers. The speaker input is connected via a BNC connector, which carries both bias and signal voltages from the stereo amplifier. Both models feature a rugged polymer enclosure with a stable base as well as a built-in, ¼"-20 threaded post for positioning with laboratory mounting hardware.

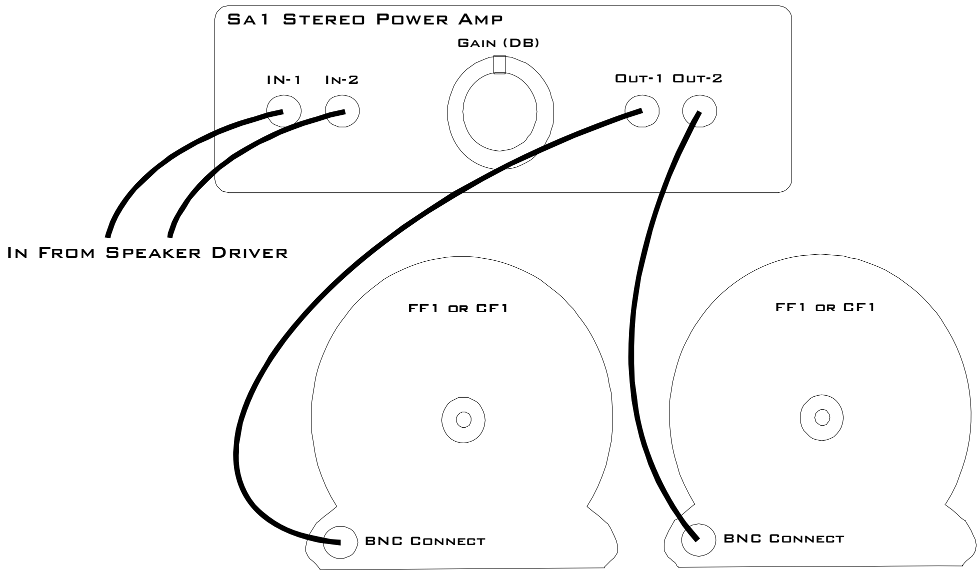
Part Numbers:

FF1 - Free-Field Magnetic Speaker

CF1 - Closed-field Magnetic Speaker (Provided with 6" of 1/8" O.D. PVC tubing)

Cable Connection

Connections to the speakers are made through a BNC connector located on the back of the FF1 and CF1 housing. If using the SA1 stereo amplifier, simply connect a BNC cable from the FF1 or CF1 to one of the output BNC connectors on the SA1 as shown in the following figure.



If you are using the SA8 See *the SA8 Eight Channel Power Amplifier*, page 13-27 for more information.

Routine Care and Maintenance

Inspect speakers for visual damage prior to use. Exposure to high temperatures will damage the speaker. The polymer used to construct the speaker's housing is very durable, however prolonged pressure, such as supporting the weight of the CF1 with the speaker's parabolic cone, may alter the original structure of the cone causing possible distortion and undesirable effects.

Unlike the closed-field model the free-field model's speaker is exposed and should be carefully handled. Sharp objects could puncture the speaker membrane causing damage to the unit.

If there is damage to the BNC connector or the speaker housing, contact TDT for an RMA for repair.

Closed Field Speaker Design Considerations

All speaker configurations should be calibrated to your specific configuration. If you are planning to deliver tone stimuli, SigCalRP can be used to normalize the desired stimulus signals. For questions about normalizing other types of stimuli, contact TDT.

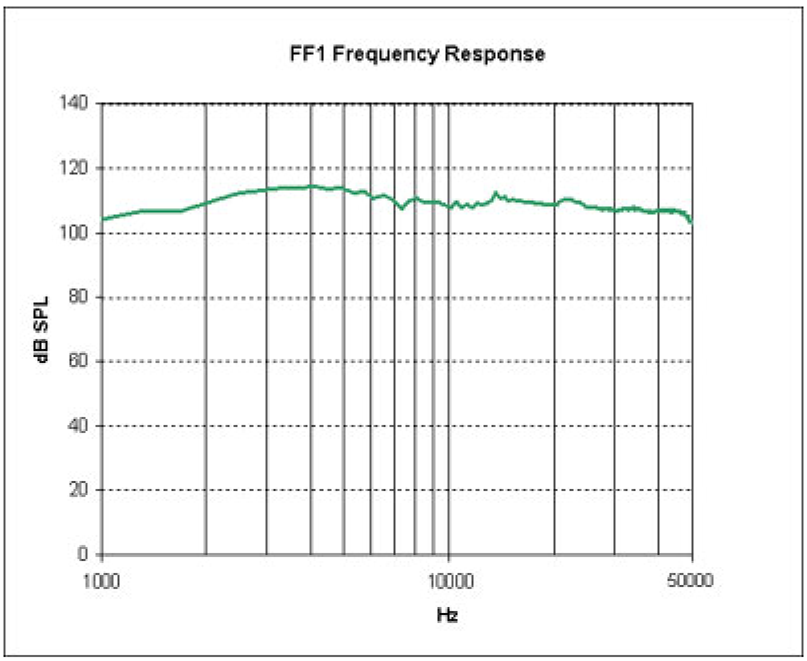
When using the CF1 speaker for experiments the provided PVC tubing will transfer the signal best when it is kept straight. Note that the speaker performance is dependent on the coupling system used and the ear of the subject. Users should test the device under experimental conditions to ensure it meets their requirements. Technical Specifications measured under specific controlled conditions are provided for comparison purposes.

Technical Specifications

FF1 Technical Specifications

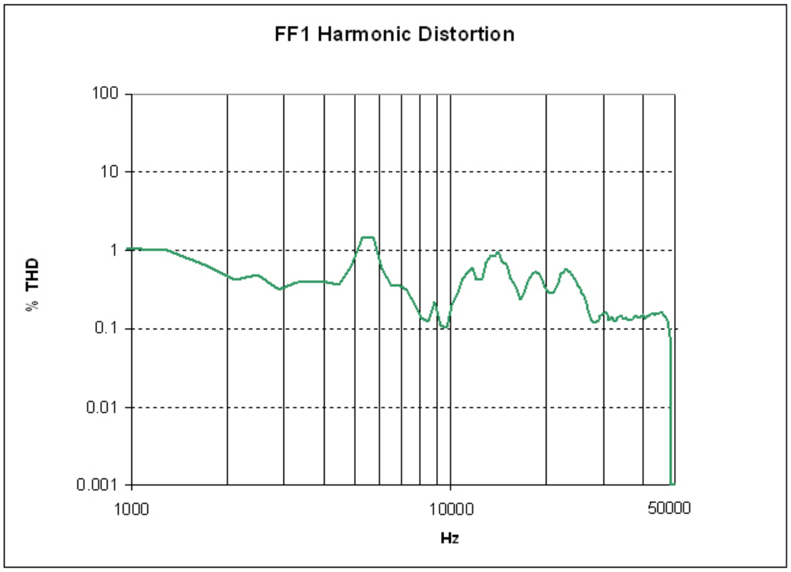
Frequency Response	+/- 8dB from 1 kHz to 50 kHz
Crossover Frequency	500 Hz High Pass
Weight	~550 Grams
Dimensions	7.62 cm outside diameter x 3.81 cm deep
Typical Output (+/- 1 V peak input)	108 dB SPL at 10 cm from 1 kHz to 50 kHz
THD	<= 1% from 1kHz to 50 kHz
Impedance	4 Ohms

Free-field Frequency Response at 10 cm



FF1 measurements typical at 10 cm using +/- 4V input.

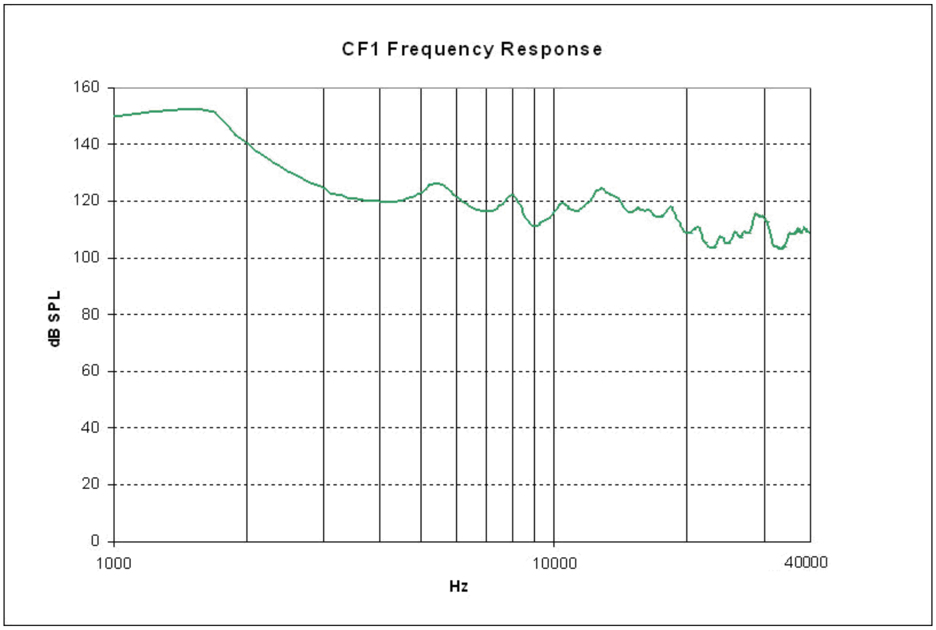
Harmonic Distortion at 4V Peak



CF1 Technical Specifications

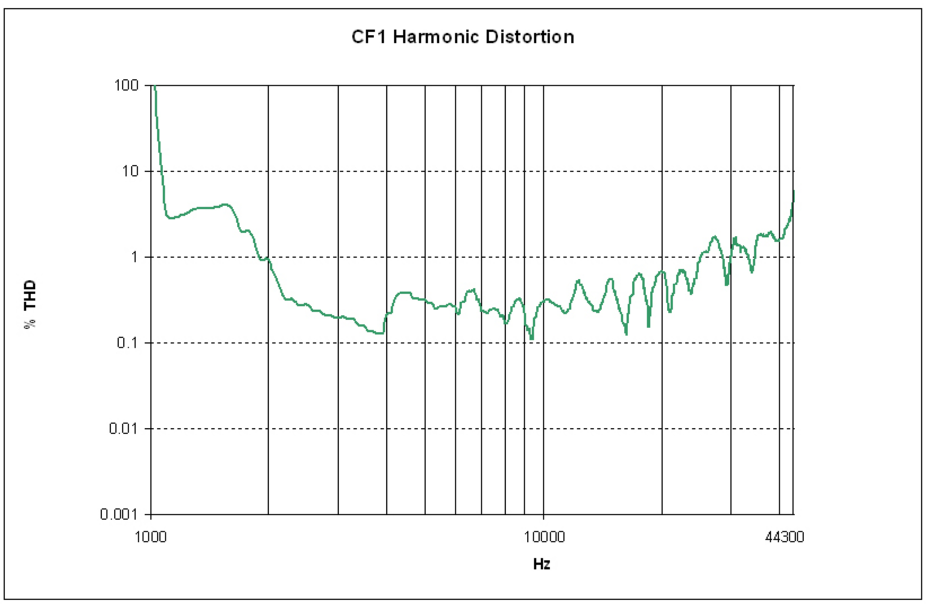
Frequency Response	+/- 24dB from 1 kHz to 40 kHz
Crossover Frequency	500 Hz High Pass
Weight	~590 Grams
Dimensions	7.62 cm outside diameter x 8.89 cm deep
Typical Output (+/- 1 V peak input)	120 dB SPL from 1 kHz to 40 kHz
THD	<= 1% from 1kHz to 40 kHz

Closed-field Frequency Response



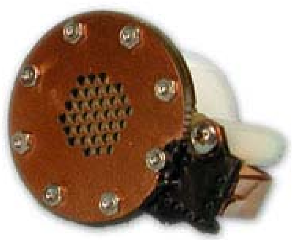
CF1 measurements typical closed field, approx 0.1cc pvc tube coupler using +/- 1V input.

Closed-field Harmonic Distortion



Harmonic Distortion at 1V Peak

EC1/ES1 Electrostatic Speaker



Overview

TDT Electrostatic Speakers (Patent No. US 6,842,964 B1) are designed specifically for ultrasonic signal production. The electrostatic design offers a thin, flexible membrane with an extremely low moving mass. Unlike conventional speakers, these speakers distribute the driving signal homogeneously over the surface of the membrane. These factors produce a small, lightweight speaker with an excellent ultrasonic response and very low distortion. Available with or without a coupler, both models are easy to position and are particularly well suited for studies with small animals that have hearing in the ultrasonic range.

Part Numbers (Patent No. US 6,842,964 B1):

ES1 - Free Field Electrostatic Speaker

EC1 - Electrostatic Speaker—Coupler Model

Cable Connection

The ES1 and EC1 electrostatic speakers work exclusively with the ED1 Electrostatic Speaker Driver. Input is via a 4-pin, mini-DIN connector, which carries both bias and signal voltages from the speaker driver. Connection to the speaker driver is through a standard 20' long cable. Other cable lengths can be special ordered, but will affect the speaker's frequency response. The speakers come fully enclosed to eliminate access to the high-voltage bias and driving signals. A 1/8" mounting hole at the base of the speaker accepts a standard 4-40 standoff. See the *ED1 Electrostatic Speaker Driver*, page 13-13 for information about gain settings.

The orientation of the cable connection is indicated with dots on the cable connector and on the speaker. The cable should be connected so that the dot on the cable faces towards the speaker.

When connecting the cable, ensure that the four pin connectors are fully seated on the speaker and the speaker driver. When the cable is repeatedly moved during the experiment, periodically check that the connectors are fully seated.

EC1 Coupled Electrostatic Speaker

The EC1 includes a small piece of Tygon® tubing coupled to the output. The tubing will transfer the signal best when it is kept straight. Note that the speaker performance is dependent on the coupling system used and the ear of the animal. Users should test the device under experimental conditions to ensure it meets their requirements. Technical Specifications measured under specific controlled conditions are provided for comparison purposes.

Maximizing the Life of the Speakers

The TDT electrostatic speakers are designed to operate with input signals between 4 and 110 kHz. Playing signals below 4 kHz causes a large amount of harmonic distortion that degrades the operation of the speakers over time, causing a decreased power output across all frequencies.

Broadband Signals

When using broadband signals, limit the amount of energy in the low frequency ranges whenever possible. For example, band limiting noise stimuli with a high pass filter at 4 kHz or above (the higher the better for the life of the speakers) and limiting complex harmonic signals, such as frequency sweeps, to frequencies above 4 kHz can increase the effective life of the speakers.

Click Stimuli

ABR experiments in both human and mouse studies typically use a 100 microsecond click stimuli, which has most of its energy in the 2 kHz to 8 kHz range. Because click stimuli are short impulses that generate signals across a broad frequency range, band limiting the frequencies is not feasible. TDT recommends that users attenuate the click stimuli so as to minimize the potential effects on the speaker. Also note that the shorter the stimuli the flatter the frequency response and the greater the energy in the higher frequencies. Moreover, the shorter the duration of the click the less total energy it has (for a given voltage).

Routine Care and Maintenance

Inspect speakers for visual damage or obstruction of the speaker holes prior to use. If there is damage to the copper shield around the components next to the connector or debris clogging the speaker holes, contact TDT for an RMA for repair.

Caution!: NEVER attempt to clean the holes in the baseplate of the speaker. Doing so can puncture the speaker membrane.

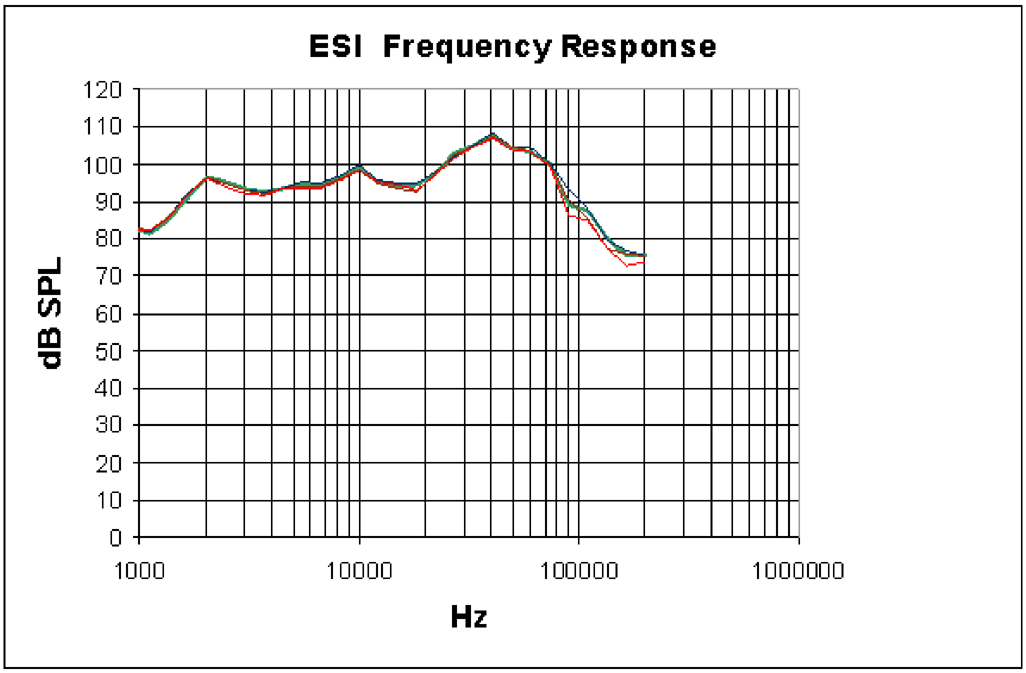
When using the EC1, check the end of the Tygon® tubing for cerumen and other debris and clean as necessary.

Technical Specifications

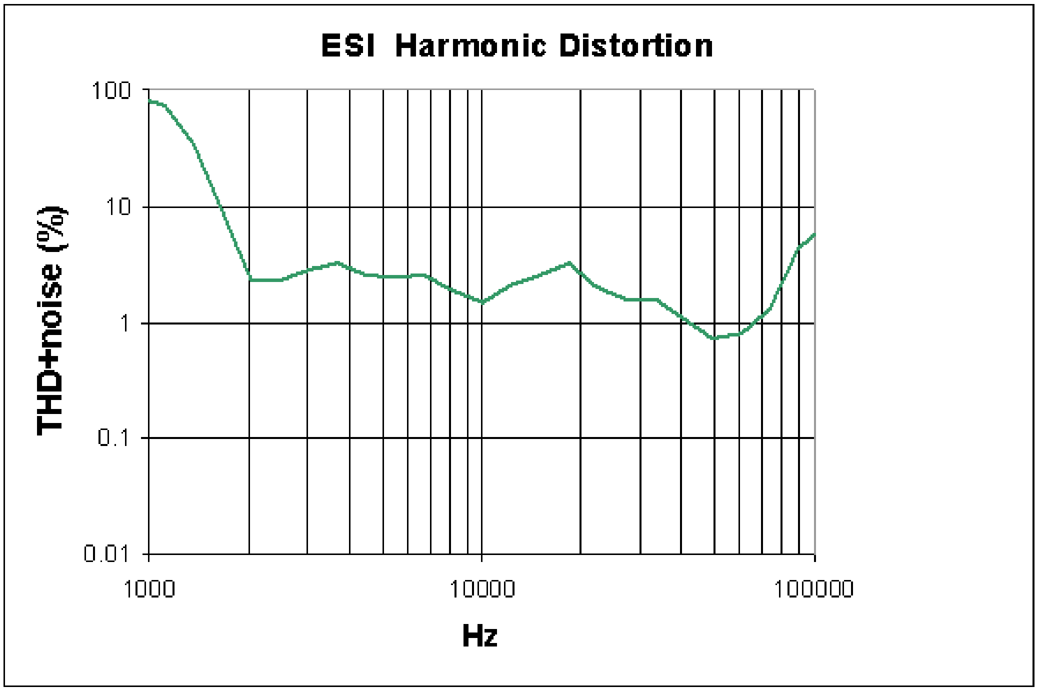
ES1 Technical Specifications

Frequency Response	+/- 11 dB from 4 kHz to 110 kHz
Weight	22 Grams
Dimensions	3.8 cm outside diameter x 2.6 cm deep
Typical Output (10V peak input)	95 dB SPL at 10 cm, 5kHz signal
THD	<3%, 2 kHz - 110 kHz, 4 Vp input

Free-field Frequency Response of Four Speakers at 10 cm



Harmonic Distortion at 4 V Peak



Noise as well as harmonic distortion is measured. Lower signal levels (e.g. above 75 kHz shown above) have higher THD+noise because of lower signal to noise ratios. When measured at higher signal levels, the THD above 75 kHz is actually <3% up to 110 kHz.

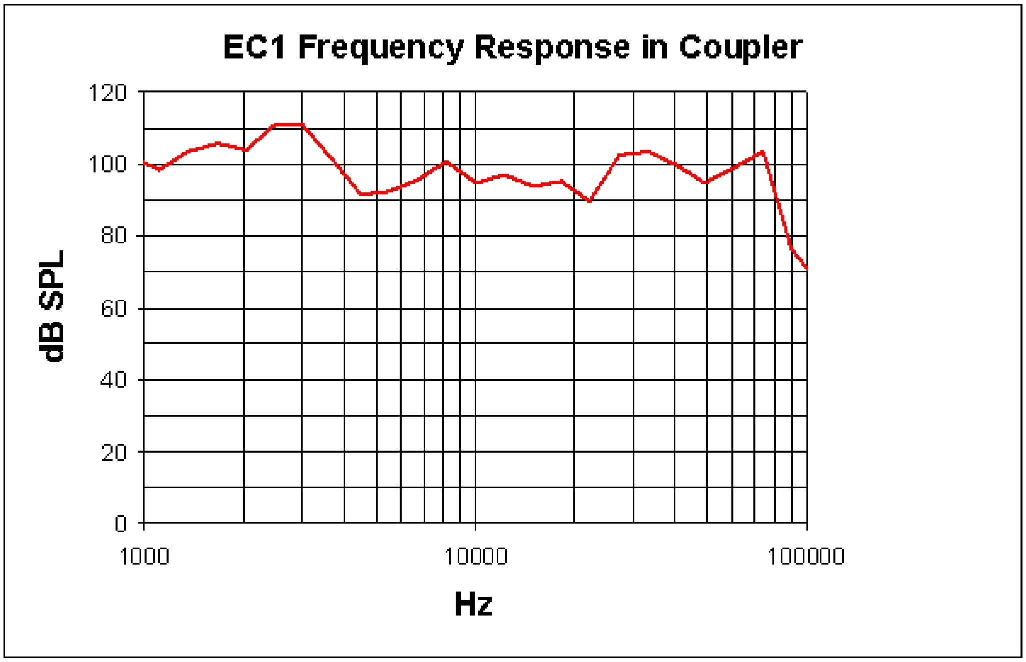
EC1 Technical Specifications

Frequency Response	+/- 9 dB from 4 kHz to 110 kHz
Weight	22 Grams
Dimensions	3.8 cm outside diameter x 2.6 cm deep
Typical Output	90 dB SPL, 5kHz signal* *Every experimental setup is unique. It is important to calibrate the response of the speaker in each experimental setup.*
THD	*Every experimental setup is unique. It is important to calibrate the response of the speaker in each experimental setup.*

Frequency Response in Plexiglas Coupler

*Measurements were made in a 1 cm x 0.5 cm coupler with a 20 cm length of 3/32" i.e. tubing attached to the fitting of the EC1. 4 V peak input tones were tested and frequency response was measured with a calibrated pressure microphone.

The results of the calibration will vary depending on the type of ear to which the speaker is coupled and the length of the tube that is coupled to the ear. This curve is provided as representative of the type of response that may be obtained in a closed field.



In this case, the low end of the response (< 5 kHz) is enhanced over the free-field response while the high end of the response (> 80 kHz) is attenuated.

Every experimental setup is unique. It is important to calibrate the response of the speaker in each experimental setup.

Important Note!: Modifying the EC1 or ES1 can result in unexpected changes in the transfer function. All modifications to the EC1 or ES1 should be performed by TDT. If you need to be 30-60 dB lower than specifications, or if you have one of these devices, contact TDT for assistance.

ED1 Electrostatic Speaker Driver

Overview

The ED1 is a broadband electrostatic driver that produces incredibly flat frequency responses reaching far into the ultrasonic range. The ED1 is designed especially for TDT's ES series electrostatic speakers. The ED1 Electrostatic speaker driver can drive two ES series speakers and is powered off the zBUS.

The ED1 is a TDT System 3 device, and receives power from the zBUS. It's two input BNCs accept input signals up to 10 Vpeak. The front panel gain control can be used to the control overall signal level of both channels from 0 to -27 dB in 3 dB steps. ED1 output is via two 4-pin, mini-DIN connectors, which carry both bias and signal voltages. The ED1 is designed to work exclusively with TDT ES series electrostatic speakers.

While the ED1 will accept a 10V input, it is possible to overdrive and ES1 when the ED1 is on the maximum gain setting. Always check that the output signal is not distorted. If the signal is distorted, turn down the gain on the ED1 until the distortion disappears. The SigCalRP software that is distributed with SigGenRP is useful for measuring the frequency response of the ES1 and to measure the Total Harmonic Distortion (THD) of the speaker. SigCalRP also generates normalization curves that can be used to flatten the frequency response of the ES1.

Power

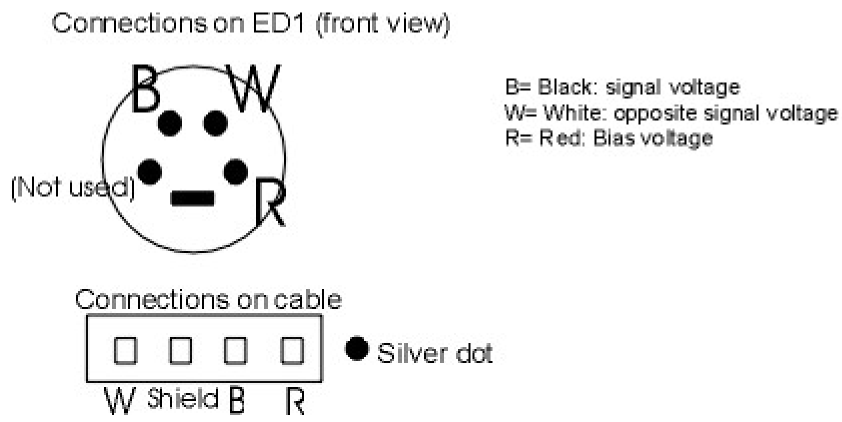
The ED1 Electrostatic Speaker Driver is powered via the System 3 zBus (ZB1PS). No PC interface is required.

ED1 Technical Specifications

Input Signal Range	+/- 10 V peak into ED1
Gain	0 dB to -27 dB on both channels, in 3 dB steps
Input Impedance	10 kOhm
Output Impedance	1 kOhm

Note: For further information, see ES series speaker specifications, page 13-10.

ED1 Pinouts


Connections on ED1 (front view)
B
W
(Not used)
R
B= Black: signal voltage
W= White: opposite signal voltage
R= Red: Bias voltage
Connections on cable
Silver dot
W Shield B R

FLYSYS FlashLamp System

Overview

The Flashlamp System includes a high intensity photic stimulator, lamp driver, and liquid light guide optic. Ideal for standard ERG, Visual Evoked Potential, and Visual Neurophysiology applications, the system features rapid flash rates, variable intensity control, high output, and a spectral range from UV to near infrared.

The modular design and supplied 9' cable allows for precise positioning of the Flashlamp (LS1130) and the 1 meter liquid light guide optic (FO1) offers additional positioning and focusing abilities.

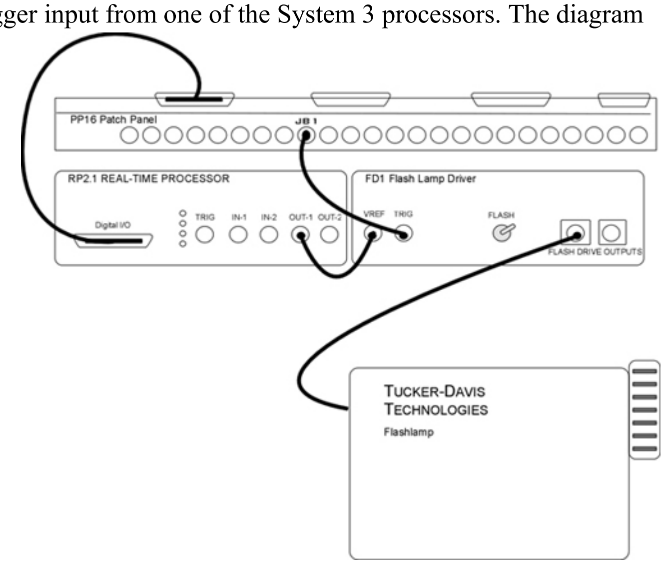
Power

The Flash Lamp Driver (FD1) provides power for the flashlamp and can control flashlamps that use their own power supply.

The FD1 Flashlamp Driver is powered via the System 3 zBus (ZB1PS). No PC interface is required for FD1 operation.

System Set-Up

The LS1130 output intensity and rate of stimulation are controlled via the FD1, which receives a variable voltage reference and trigger input from one of the System 3 processors. The diagram below shows how the system would be connected when using an RP2.1 module for control.



System Features

Vref Input Signal

The variable reference voltage controls flashlamp output intensity and can be supplied by any System 3 device with a DC level positive, such as the RP2.1 or RX processors (the RA16BA cannot be used), and must be set high for 10 mSec before the stimulus trigger.

Trig Input Signal

A TTL trigger controls stimulation rate and is typically supplied by a digital output line from one of the System 3 processors, such as the RP2.1 or RX6. Alternatively, the trigger line can be provided by an external source TTL source with a maxium voltage of 5 V and 1 mSec duration.

Flash Switch

This manual switch can be used to trigger the flashlamp. To trigger the lamp, push the switch up and then press down.

Flash Driver Output (LS1130 or MVS7000)

The Flashdrive LS1130 output will drive the standard LS1130 flashlamp that ships with the FLSYS. The MVS7000 output can be used to control other flashlamps. **Important note:** contact TDT for assistance before using any other flashlamps with the FD1.

Flash Intensity

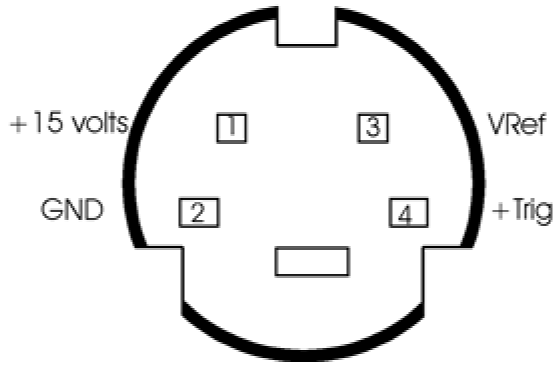
To calculate the flash intensity, use the following equation: J=1/2(0.50 µF) (Vref*100)^2

FLYSYS Technical Specifications

Includes FD1 Flash Lamp Driver, LS1130 Flashlamp, and FO1 Liquid Light Guide.

Flash Rate	0.1 - 200 Hz
Flash Duration	10 µsec
Trigger	TTL (5V max)
Flash Intensity (max)	0.235 Joules
Charge Time	30 msec
Spectrum	350 – 800 nm
Input Signal (Vref)	4 – 10 V
Life	10^9 flashes
Power and Communication	zBus required for FD1

LS1130 and MVS7000 Connector Pinout



Note: connectors are wired the same.

HB7 Headphone Buffer

Overview

The HB7 headphone buffer is used to amplify signals for headphones. The HB7 is a two channel device. The outputs include both a stereo headphone jack and Left and Right BNC connectors. The output level can be controlled with a Gain knob, and there is a Differential switch that allows the LEFT input to be output to the Left and Right outputs resulting in an additional 6 dB of gain.

Power

The HB7 Headphone Buffer is powered via the System 3 zBus (ZB1PS). No PC interface is required.

Features

Inputs

The HB7 has two inputs for signals up to ±10 V, accessed through front panel BNC connectors labeled LEFT and RIGHT.

Outputs

The outputs include both a stereo headphone jack labeled PHONES and Left and Right BNC connectors.

Note: When monitoring both output channels with only one input connected, users should short the unused input channel to ensure maximum channel separation.

Gain

A single GAIN knob provides control over the signal output level in 3 dB steps from 0 to -27 dB.

DC/AC Switch

The DC/AC switch can be used to switch from DC coupling to AC coupling mode. In AC coupling mode, DC shifts in the signals are removed.

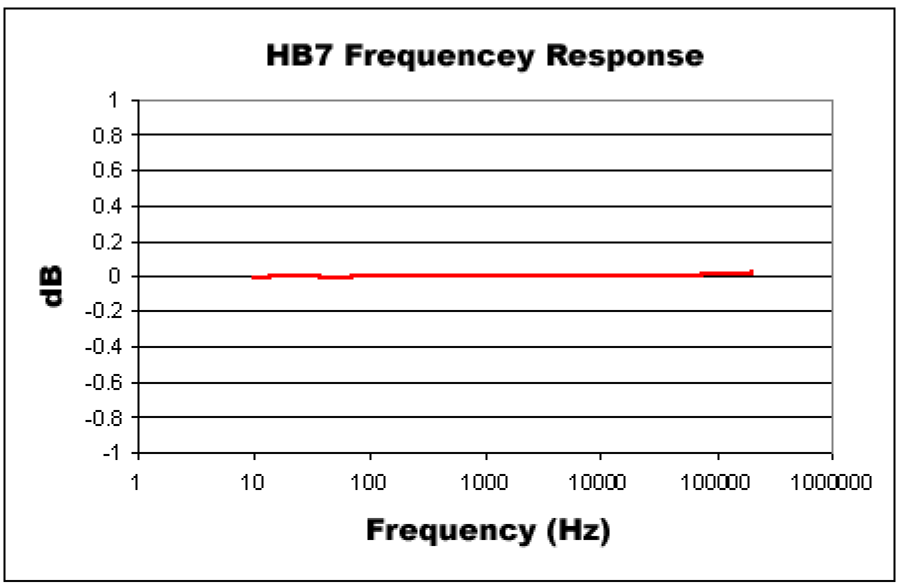
DIFF Switch

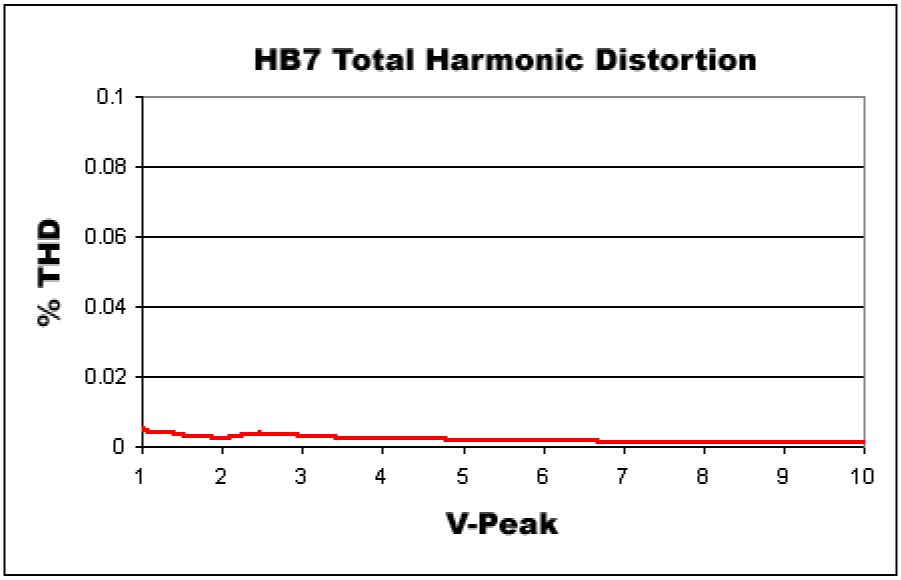
The DIFF switch will switch to a differential output mode that gives 6 dB of additional gain when connected to a speaker. When DIFF is switched on (the switch in the up position), the left channel input goes to both the left and right channels and is inverted on the right channel (the right input BNC is not used). The differential output will usually only be used with speakers, not headphones.

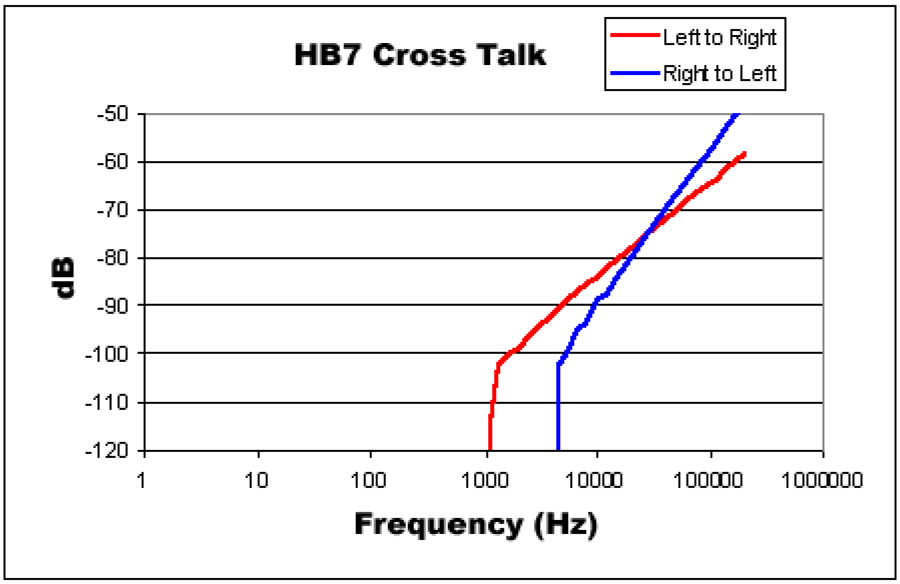
To connect the speaker, connect the left output to one pole of the speaker and the right output to the other pole of the speaker (neither ground of the left nor right output will be connected).

HB7 Technical Specifications

Input Signal Range	±10 V peak
Power Output	0.12 W into 4 Ohms, 0.25 W into 8 Ohms, 1.0 W into 32 Ohms
Spectral Variation	<0.1 dB from 10 Hz to 200 kHz
Signal/Noise	117 dB (20 Hz to 80 kHz)
Noise Floor	9.2 µV rms
THD	<0.0002% (1 kHz tone, +/- 7V peak)
Input Impedance	10 kOhm
Output Impedance	5 Ohm




HB7 Total Harmonic Distortion
% THD
0.1
0.08
0.06
0.04
0.02
0
1
2
3
4
5
6
7
8
9
10
V-Peak


HB7 Cross Talk
Left to Right
Right to Left
dB
-50
-60
-70
-80
-90
-100
-110
-120
1
10
100
1000
10000
100000
1000000
Frequency (Hz)

MA3: Microphone Amplifier

Overview

The MA3 is a two-channel microphone amplifier for auditory scientists. This high-quality low-cost system is designed for use with both ¼” audio jack microphones and balanced XLR inputs for optimum impedance and noise characteristics. The MA3 is able provide a bias voltage for those microphones that require it. Two BNC connectors provide output to the Analog input on the RP2. A variable gain knob provides amplification from 10 dB to 55 dB in 5 dB steps. A toggle switch provides 20 dB of additional gain for over five thousand fold amplification.

Power

The MA3 Microphone Amplifier is powered via the System 3 zBus (ZB1PS). No PC interface is required.

Features

Inputs

The MA3 comes with three inputs: an XLR microphone input and two ¼” TRS connector inputs. Signals from two microphones can be amplified simultaneously.

Bias

The Bias switch produces a bias voltage for microphones that require it.

Gain Control

The gain control amplifies the microphone input in 5 dB steps from 10-55 dB (3x-560x). The Gain Switch adds an additional 20 dB (10x) of gain for a maximum amplification of 5600.

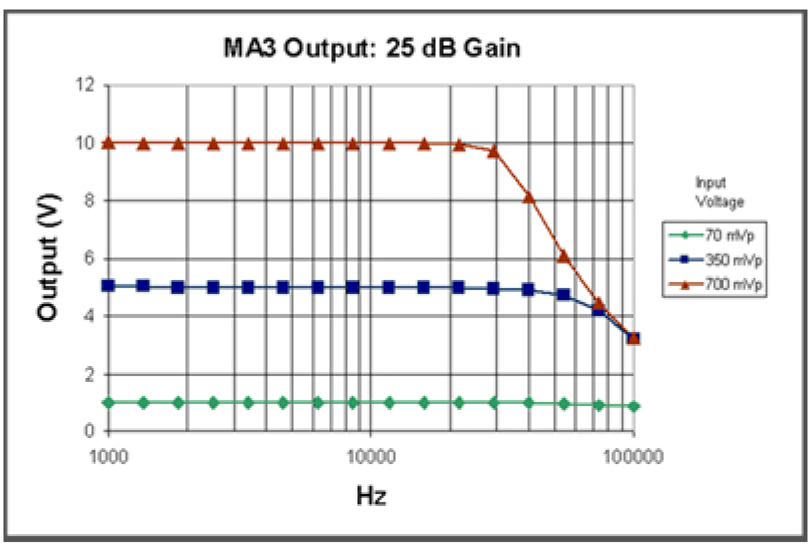
Outputs

Two BNC outputs give easy connection to any TDT System 3 device. The maximum voltage output is +/-10 volts. Clip lights indicate and overvoltage on the signal output.

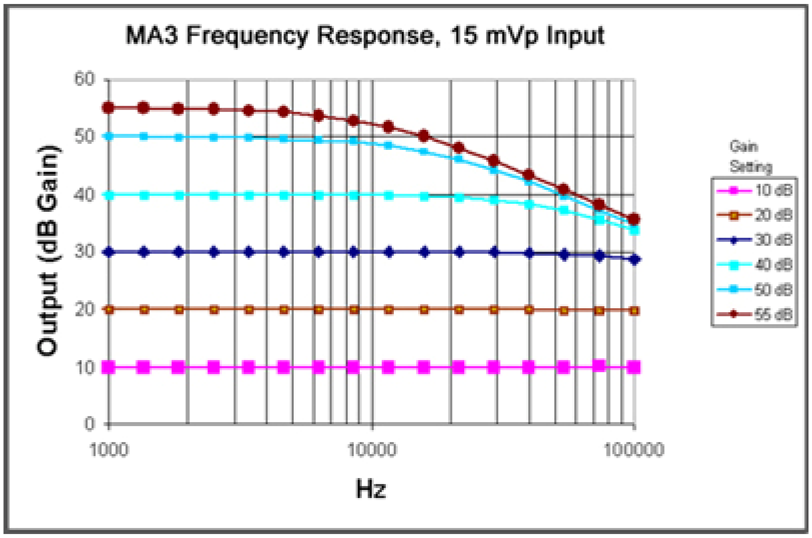
MA3 Technical Specifications

Input Signal Range	±10 V peak
-3dB Bandwidth	200 kHz @ 40 dB gain
Gain Accuracy	+/- 1 dB
Spectral Variation	1 dB from 20 Hz to 20 kHz
Signal/Noise	110 dB (20 Hz to 30 kHz at 9.9 V)
Noise Floor	9.2 µV rms
THD	< 0.002% (1 kHz tone, +/- 7 V peak)
Input Impedance	600 Ohm
Output Signal Range	+/- 10 V peak
Bias Voltage	10 V, 150 mA max, superimposed onto microphone
Output Impedance	5 Ohm

Output Diagram



Frequency Response Diagram



MS2 Monitor Speaker

Overview

The MS2 Monitor Speaker is used as an audio monitor for signals up to ± 10 V. The MS2 output level is controlled manually using a 1-turn potentiometer on the front panel interface. Maximum output is greater than 90 dB SPL at 10 cm. The frequency response ranges from 300Hz to 20 kHz. A typical use of the MS2 is for audio monitoring of electrophysiological potentials.

Power

The MS2 Monitor Speaker is powered via the System 3 zBus (ZB1PS). No PC interface is required.

Features

Manual control is via a single LEVEL knob, which provides control over the signal output level. The MS2 has one input channel for signals up to ±10 V, accessed through a front panel BNC connector

The MS2 is useful for monitoring the output signal that may be going to headphones in a soundproof room and for monitoring physiological signals that are being acquired, such as neurophysiology recordings.

MS2 Technical Specifications

Input Signal Range	±10V peak
Max Output	> 90 dB SPL at 10 cm
Input Impedance	10 kOhms

SA1 Stereo Amplifier

Overview

The SA1 is a power amplifier for the zBus that delivers up to 3 watts of power to speakers. It has excellent channel separation combined with low noise and distortion. The frequency response is flat from 50 hertz to 200 kilohertz. Gain can be varied over a 27 dB range in 3 dB increments.

Power

The SA1 Stereo Amplifier is powered via the System 3 zBus (ZB1PS). No PC interface is required.

Features

Inputs

There are two inputs (±10 V maximum) that connect through BNC's labeled IN-1 and IN-2.

Outputs

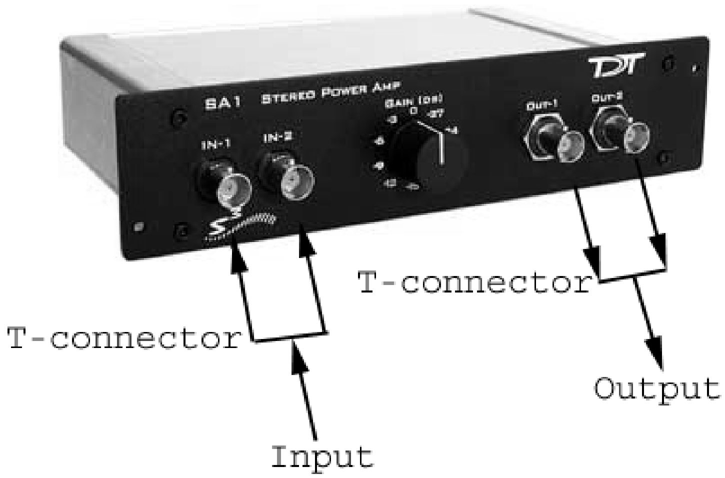
The outputs are two (OUT-1 and OUT-2) BNC connectors.

Gain

A single GAIN knob provides control over the signal output level in 3 dB steps from 0 to -27 dB.

Ganged Output Mode

A ganged output mode gives 6 dB of additional gain when connected to a speaker. Split the signal to the input; send one to the IN-1 and the other to IN-2. Take the outputs from OUT-1 and OUT-2 and combine them to boost the gain.



SA1 Technical Specifications

Input Signal Range	±10V peak
Power Output	1.5 W/channel into 8 ohms, 3.0 W with Ganged output.
Spectral Variation	< 0.1 dB from 50 Hz to 200 kHz
Signal/Noise	116 dB (20 Hz to 80 kHz)
THD	< 0.02% at 1 Watt from 50 Hz to 100kHz
Noise Floor	10.5 µV rms
Input Impedance	10 kOhm
Output Impedance	2 ohms, 1 ohm Ganged

SA8 Eight Channel Power Amplifier

Overview

The SA8 is an eight-channel power amplifier that delivers up to 1.5 watts of power per speaker to up to eight speakers. The unit features high channel separation with low cross talk combined with low noise and distortion. The gain for all eight channels can be set to 0, -6, -10 or –13 dB.

Power

The SA8 Power Amp is powered via the System 3 zBus (ZB1PS). No PC interface is required.

Features

Inputs

There are eight available inputs located on the DB9 connector on the front panel of the SA8.

Outputs

The eight output channels are accessible via the DB25 connector and are arranged for optional direct connection to a PP16 Patch Panel. For easy wiring and connection to a wide variety of transducers, the eight outputs are duplicated on the DB25 and sufficient ground pins are provided to allow for connections requiring a single ground for all channels or paired grounds for each channel. See *Mapping SA8 Output to PP16 Connectors*, page 13-28 for more information on easy access to SA8 output channels via the patch panel.

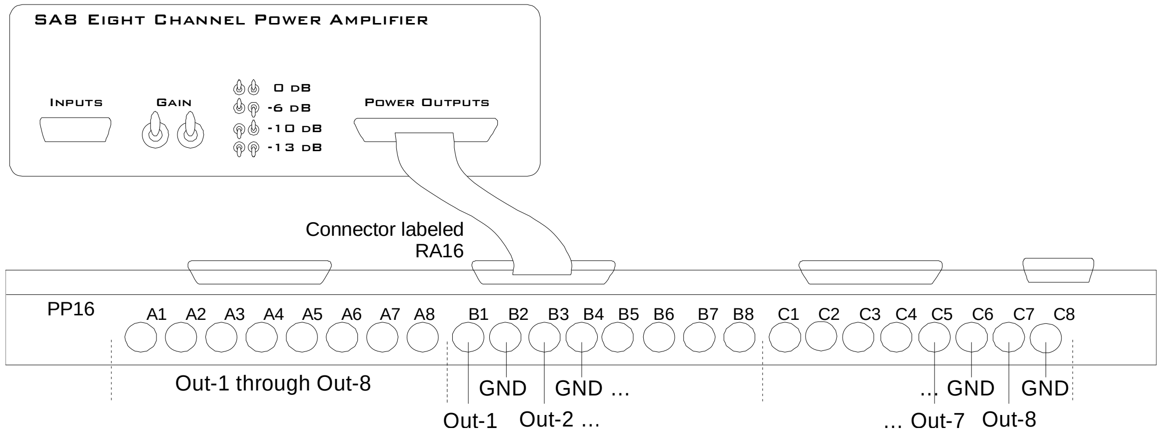
Gain

The gain is controlled by two toggle switches on the front panel of the SA8. The following table describes the selectable gain values.

Front Panel Diagram	Left Toggle	Right Toggle	dB Gain
	Up	Up	0
	Up	Down	-6
	Down	Up	-10
	Down	Down	-13

Mapping SA8 Output to PP16 Connectors

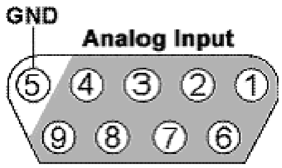
The picture below maps the SA8 signal out connection to the PP16.



SA8 Technical Specifications

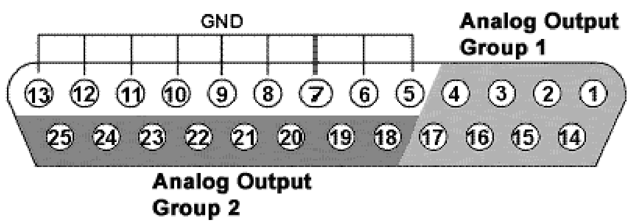
Input Signal Range	±10V peak
Power Output	1.5 W/channel into 8 ohms
Spectral Variation	< 0.1 dB from 50 Hz to 200 kHz
Signal/Noise	116 dB (20 Hz to 80 kHz)
THD	< 0.02% at 1 Watt from 50 Hz to 100kHz
Noise Floor	10.5 µV rms
Input Impedance	10 kOhm
Output Impedance	2 ohms
Cross Talk	< -60 dB

Analog Input Pinout Diagram



Pin	Name	Description
1	A1	Analog Input Channels
2	A3	
3	A5	
4	A7	
5	GND	Ground
6	A2	Analog Input Channels
7	A4	
8	A6	
9	A8	

Analog Output Pinout Diagram



Pin	Name	Description
1	A1	Analog Output Channels Group 1
2	A3	
3	A5	
4	A7	
5	GND	GND
6		
7		
8		
9		
10		
11		
12		
13		

Pin	Name	Description
14	A2	Analog Output Channels Group 1
15	A4	
16	A6	
17	A8	
18	A1	Analog Output Channels Group 2
19	A2	
20	A3	
21	A4	
22	A5	
23	A6	
24	A7	
25	A8	

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Part 14 Subject Interfaces

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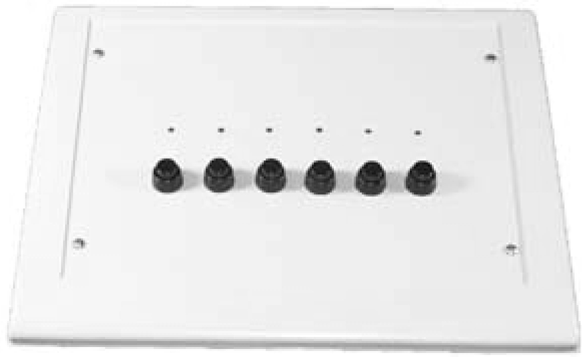
BBOX Button Box



Overview

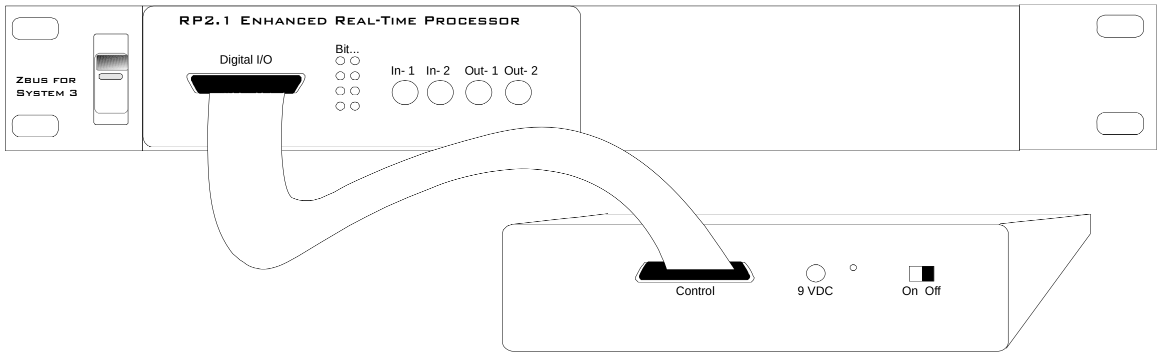
The button box is a complete subject response interface. It is an excellent system for psychoacoustics, including n-alternative forced choice, GO NO GO, Bekesy style presentation, and modified method of limits experiments. The button box provides accurate reliable performance. All inputs are debounced in the button box and a built-in rechargeable lithium-ion battery provides power for up to 24 hours of continuous use per charge.

The standard button box configuration includes six buttons and six high intensity LEDs. However, the button and LED organization can be configured to user specification. The button box can have up to eight buttons and 32 LEDs. The button box design allows experimenters a great deal of flexibility to control feedback based on subject response, reinforcing behavior for correct and incorrect choices.

The button box can be controlled from an RP2.1 or RV8 processor with button response acquisition and LED control through the digital input/output port of these modules. Data can be latched and then read from specialized RPvdsEx circuits using ActiveX and Matlab, or other programming languages. RPvdsEx circuits designed for button box control can be used with all TDT software.

Connecting the Button Box to the RP2.1 or RV8

The button box is controlled using the RP2.1 or RV8 processor. The button box connects from the DB25 connector (Control) directly to the digital input/output port on the RP2.1 or RV8 with the supplied ribbon cable. The button box is configured at the factory for the RP2.1. It can be configured for the RV8 by installing a jumper pin (Jumper for RV8) on the back of the button box.



RP2.1 to BBOX

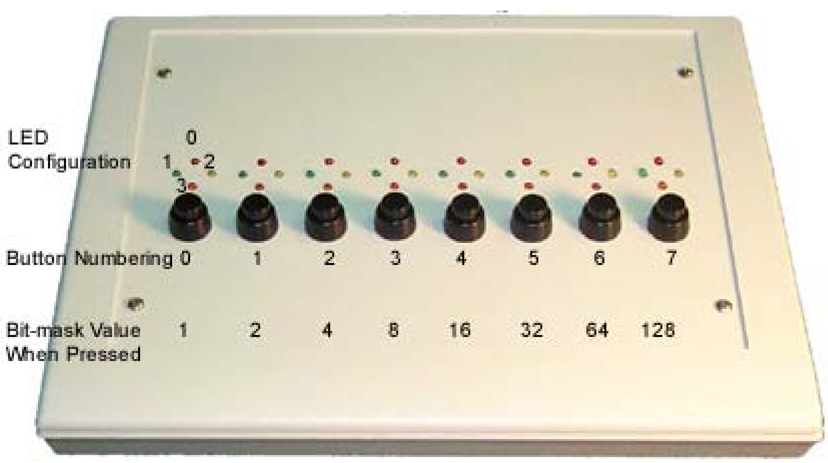
Power Requirements

The button box is supplied with a 3.3 Volt lithium-ion battery pack. This high current battery should provide up to 24 hours of continuous use per charge. The lithium-ion battery charges in under three hours with the supplied 9 Volt battery charger. The ON/OFF switch, the power connection for the battery charger, and a power indicator light are found on the back of the button box. The Power/(Low Bat) LED lights when the button box is on and flashes if the battery is low.

Important: To operate any features of the button box the power must be turned on and the device must be connected to an RP2.1 or RV8 that is powered on and connected to a PC.

Caution!: A low battery may give erroneous results. If the battery is low, the battery charger can be connected to the device. This will charge the battery and power the box at the same time.

Organization of Buttons and LED's



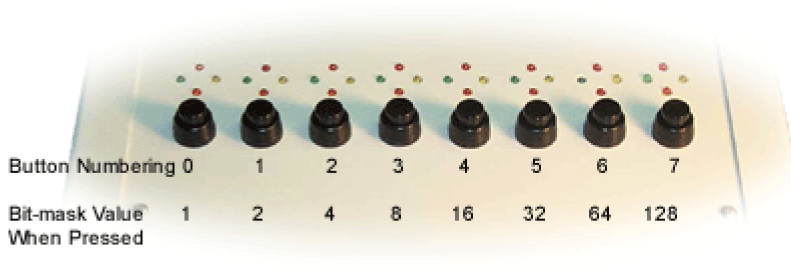
BBox Control

LEDs can be controlled and button presses can be acquired by including the necessary circuit segments in the RPvdsEx circuit that will be run on the controlling device. The button box can also be controlled using ActiveX and Matlab, or any programming language that supports ActiveX. Before designing or debugging circuits for the button box, ensure that the button box is connected to the RP2.1 or RV8 that will be used for control and that the button box power is turned on. The buttons will only operate when the button box is powered.

The remaining button box help topics provide the necessary information for basic button box control, including circuits that acquire button responses and test for correct or incorrect responses to button presses. The information provided assumes some knowledge of RPvdsEx and possibly ActiveX. Users with custom built button boxes should modify circuits based on the configuration of the buttons.

Acquiring BBox Button Presses

The most efficient way to acquire button presses is with the WordIn component in RPvdsEx. The WordIn checks all the digital input lines and returns a 16-bit value from the digital line addressed. Input values are generated as a bit-mask that determines which buttons were pressed. Users can also record the inputs from the individual digital I/O lines. The RPvdsEx examples in this topic use the WordIn method.

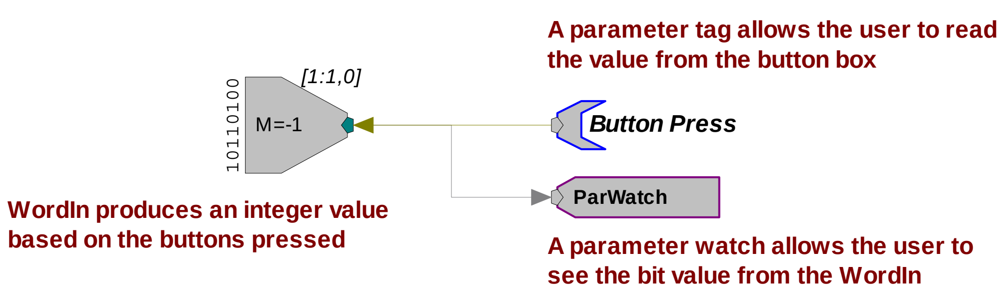


BBox Organization of Buttons

Note: In order for the buttons to operate the button box power supply must be turned on.

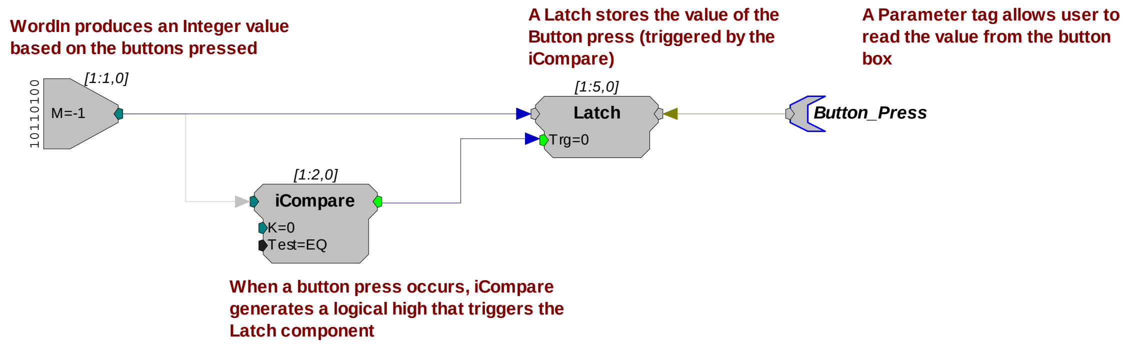
Many of the circuits shown below, as well as some MATLAB examples for use with ActiveX controls, are included with RPvdsEx (RPvdsEX|Examples|ButtonBox).

A simple circuit for acquiring button presses...



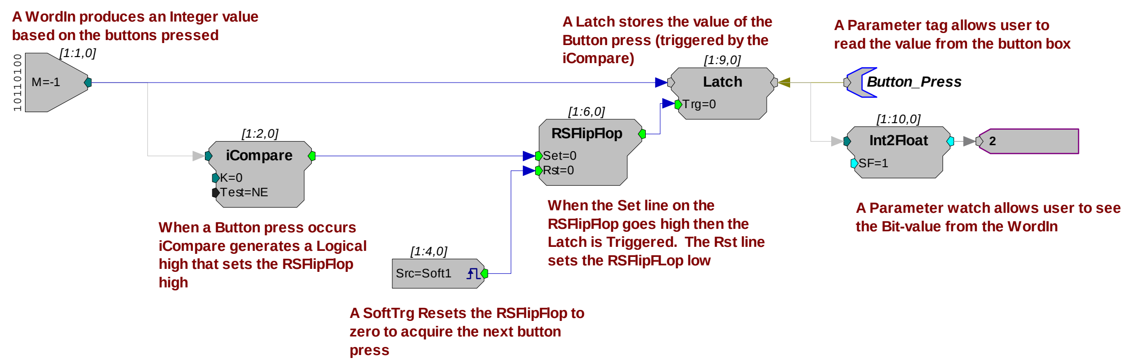
In this example, the user would continuously poll the component, from a program that acquired the value from the ButtonPress parameter, to determine which buttons are pressed. A simple circuit like this may be required if the RP2 that controls the button box is also used for stimulus presentation.

A more likely circuit design for button acquisition...



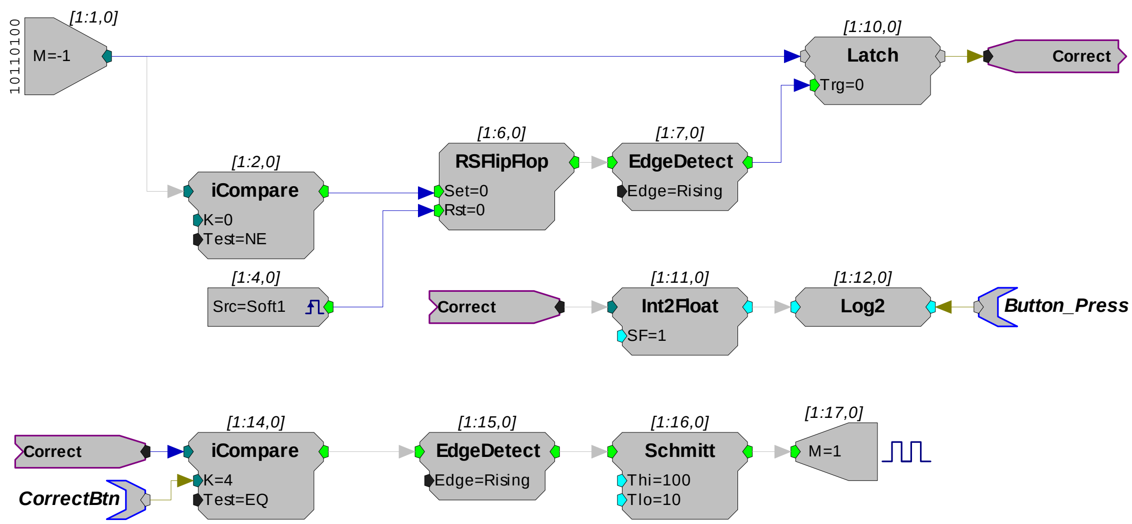
In this example, the WordIn produces an integer value based on the buttons pressed. When a button press occurs, an iCompare generates a logical high that triggers the Latch component. The Latch stores the value of the button press until the next button press occurs. The Button_Press parameter tag allows the user to read the value from the button box. If only the first button press is important then a reset line should be included in the circuit to rest the Latch.

Resetting the Latch...



In the previous examples all button presses are acquired, that is, if a person presses buttons simultaneously there is the chance that both responses will be obtained. This will happen infrequently with circuits that use an iCompare and Latch, but it is still possible. In some cases the user will want to determine if the proper button press was acquired or wait until a particular button press has happened. Additional circuitry can be added that checks for this.

Identifying the correct button press...



iCompare is only triggered when ther correct button is pressed. EdgeDetect then sets the Schmitt trigger, which turns on an LED for 100 milliseconds

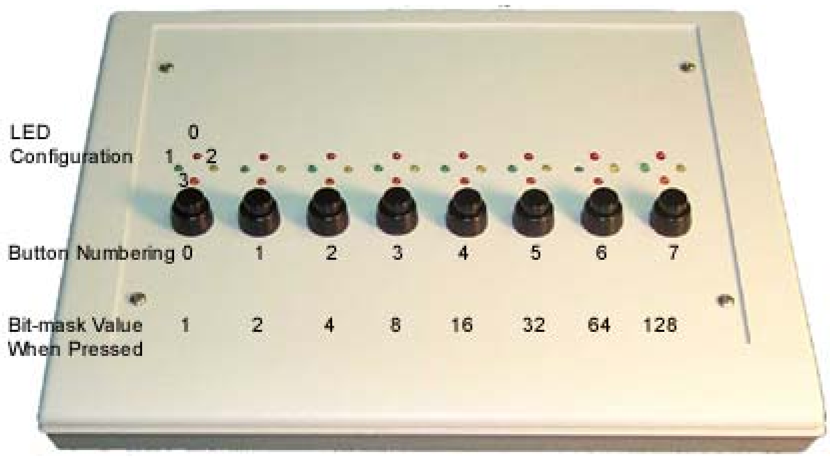
In this example, the top part of the circuit detects if a button is pressed. The button press value is also translated into a value representing which bit was read. For example, if the bit in bitmask value is 16, then Log2 converts the value to 4. This lets the user determine, via the Button_Press parameter tag, that bit 4 was high.

The lower part of the circuit tests to determine if the correct button was pressed. If so, an LED is flashed. A parameter tag is used to identify the correct button press. The iCompare is only triggered when the correct button is pressed. The EdgeDetect component then sets the Schmitt that turns on the first LED for 100 milliseconds.

Button box circuits can be incorporated in to all TDT System 3 software. For information on using the button box with other applications please see that application's documentation. If you have questions about how to design your own applications for the button box call 386-462-9622 for technical assistance.

Controlling the LEDs

This topic demonstrates several methods to control LEDs. The button box may have up to four LEDs for each button and each LED can be turned on and off independently of any other. Using the LEDs involves two steps: 1) designating the LED to turn on or off and 2) turning the LED on and off. LEDs are designated by specifying the column (button number) and position (LED number).



BBox Organization of LEDs and Buttons

Selecting an LED	
Bits 0, 1:	Control the position within a column
Bits 2, 3, 4:	Control which column is selected
Turning on/off LEDs	
Bit 5:	Turns on selected LED
Bit 6:	Turns off selected LED
Bit 7:	Turns off all LEDs

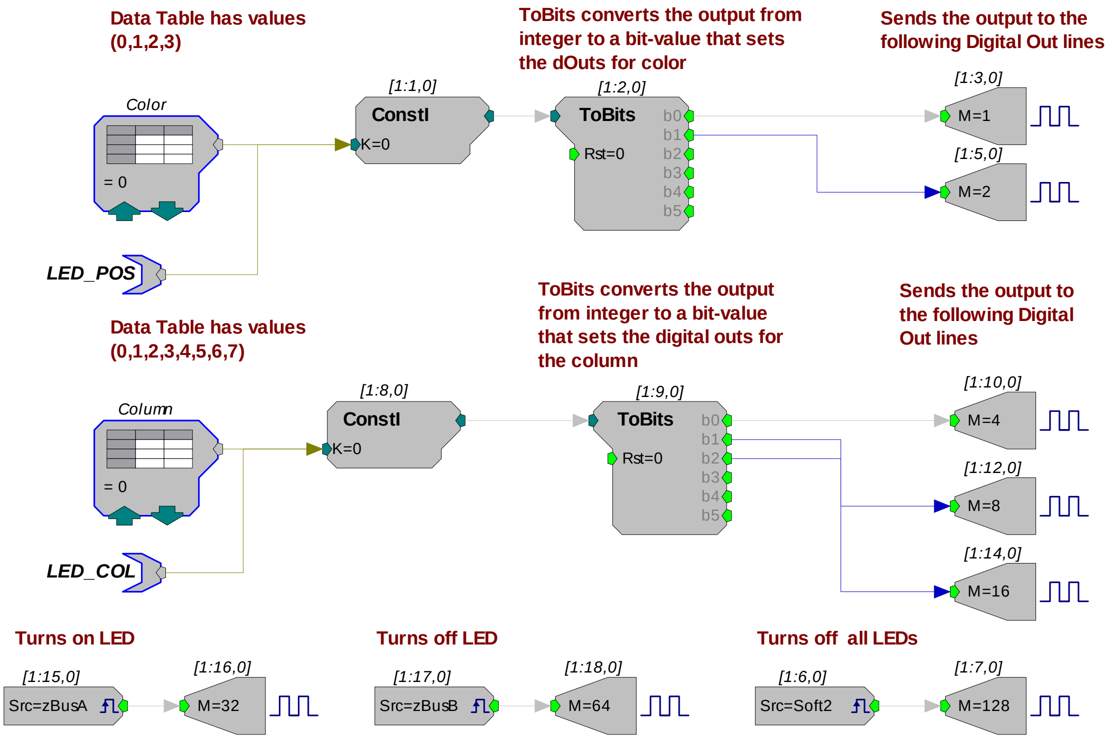
Bit Patterns Table

Note: Because the button box has its own power supply, the LED's will remain on until they are turned off via the RP2 or RV8 or until the power is turned off.

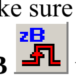
The circuits shown below, as well as some MATLAB examples for use with ActiveX controls, are included with RPvdsEx (RPvdsEX|Examples|ButtonBox). In the first design the user designates the LED and the button number or column position in two separate steps. In the second the steps are combined. In the final design LED designation and on/off information are combined in a single word.

Designating the LED and button number or column position in two separate steps...

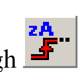
In the example above there are two sets of inputs used to specify the LED. The first controls which LED (LED position within a grouping) is lit while the second controls the column (button location) in which the LED is located. DataTables are used to test and run the circuit within RPvdsEx and parameter tags (LED_POS and LED_COL) are included to allow users to control the position and column values from another application.



To follow along with this example, open the *LED1* RPvdsEx file in the ButtonBox example folder (TDT|RPvdsEX|Examples|ButtonBox).

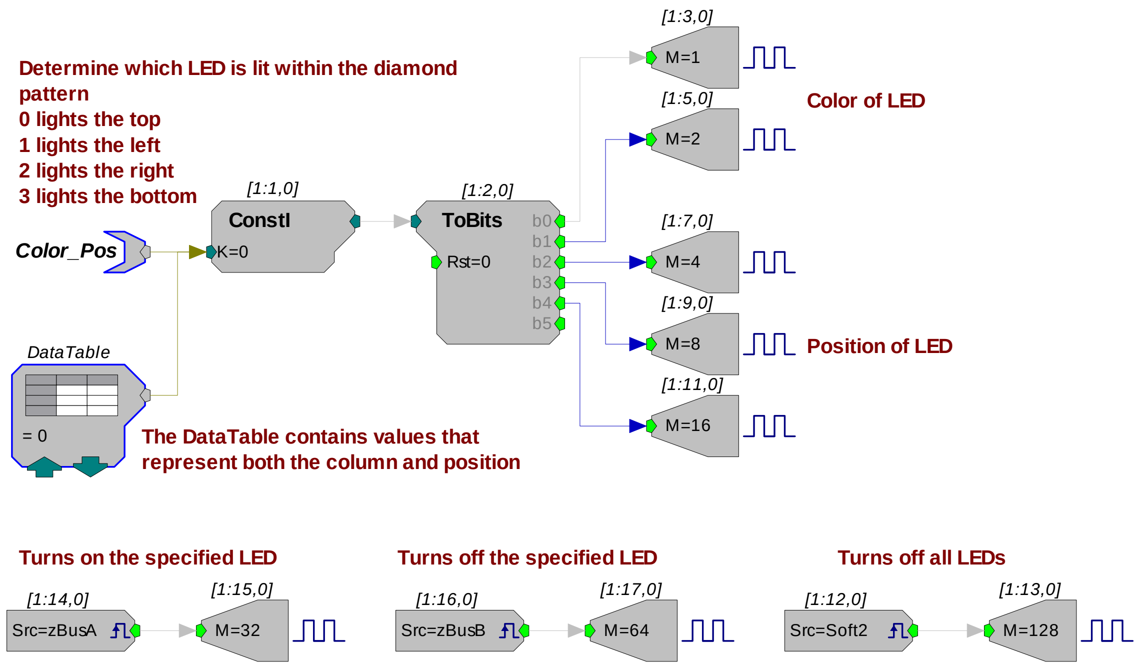
- To set the color or position of the LED (0 = Top, 1 = Left, 2 = Right, 3 = Bottom), click the **green up and down arrows** on the DataTable labeled **Color**.
- To determine which column the LED is in (0 = Far Left ... 7 = Far Right), click the **green up and down arrows** on the DataTable marked **Column**.
- To turn on the LED, press the **zBusA** trigger button in RPvdsEx. Make sure to click the **pulse**  button for the zTrig. To turn off the LED press the **zBusB**  trigger.

You can select (one at a time) several lights to turn on and off. For example, to light the top LED in the first column and the bottom LED in the last column perform the following steps:

1. Set the **Color** DataTable to **0** and the **Column** DataTable to **0**.
2. Turn on the LED by clicking the **zBusA** trigger button in RPvdsEx. This will turn on the top LED in the first column.
3. Set the **Color** DataTable to **3** and the **Column** DataTable to **7**.
4. Click the **zBusB** trigger button in RPvdsEx. Both LED's should now be on.
5. To turn off the latter LED, click the **zBusB** trigger button.
6. To turn off all LEDs, click the **Soft2**  button in RPvdsEx.
7. To turn on all LED's in succession, set the **zBusA** trigger line high  and then cycle through the DataTable values.
8. To reverse the operation set the **zBusA** trigger low , set the **zBusB** trigger high , then cycle through the DataTable values.

Combining the position and column setup...

The following example combines the two data tables and uses one ToBits component to control the button box's LEDs.



The single data table used in this example contains values that combine the column and position. For example:

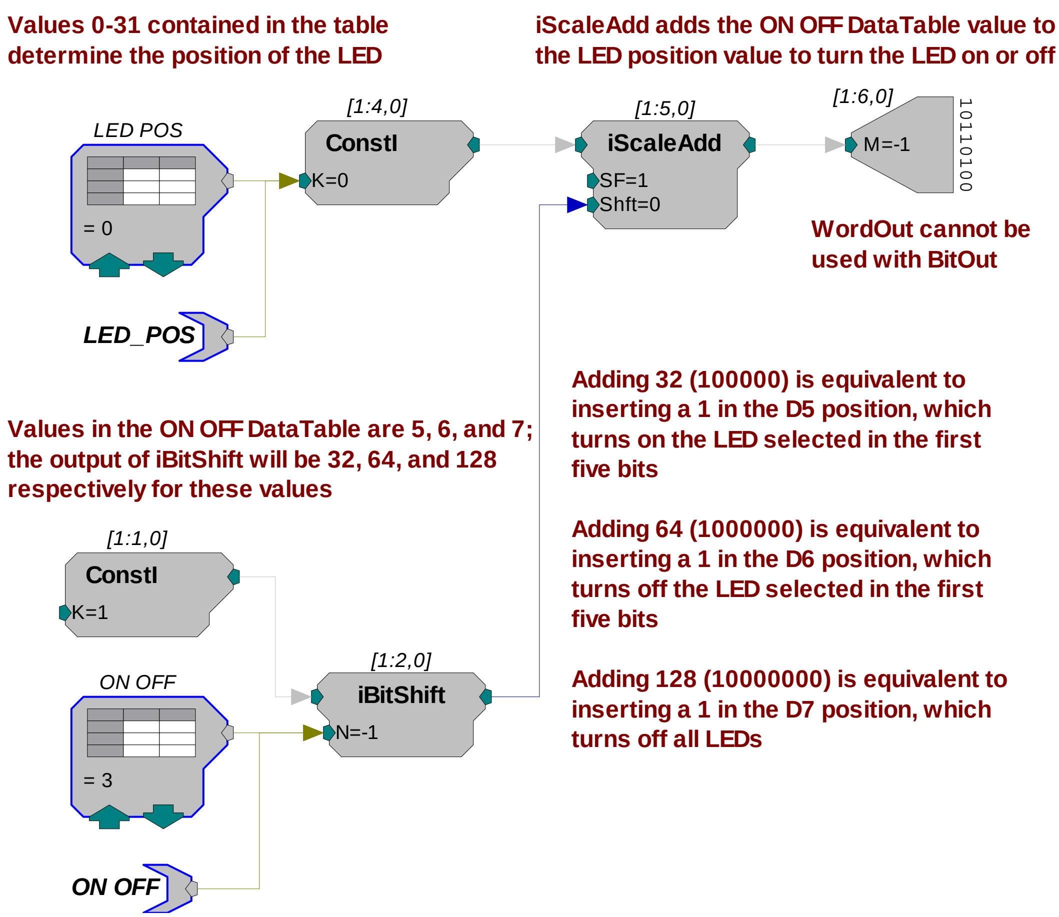
If 28 is used in the data table, the circuit selects the top LED in the seventh column. That's because the top position in the seventh column is represented by the digital number 11100 (as shown below), which equals 28.

Column Select Lines			**LED Position Select Lines**	
D_4	D_3	D_2	D_1	D_0
1	1	1	0	0

To learn more about this example, open the *LED2* RPvdsEx file in the ButtonBox example folder (TDT|RPvdsEX|Examples|ButtonBox).

Using a WordOut with a DataTable/ParTag for on/off actions...

The following example uses the WordOut component similarly to the way the WordIn is used in the button press example. As before, a DataTable is used to determine which LED to light. In the LED POS DataTable, values 0 - 31 are used to determine the position of the LED. In addition, another DataTable is used to set whether the LED is turned ON or OFF, all LED's are turned OFF, or if nothing is done when the LED is selected. This value gets added to the LED position value and is sent out via the WordOut component. The values for the second DataTable are 0 = 0 (nothing done), 1 = 32 (LED ON), 2 = 64 (LED OFF), and 3 = 128 (all LEDs OFF). The cycle usage for this example is half the cycle usage for the one above it. Notice that there are no BitOut components used. The WordOut and BitOut components cannot be used in the same circuit.



Note: See the *Bit Pattern Table* for a review of how each bit position is used.

This example is found in the LED3 RPvdsEx file in the ButtonBox example folder (TDT|RPvdsEX|Examples|ButtonBox).

RBOX Response Box

The RBOX has four buttons for user response, and four LEDs that can be used to provide subjects with feedback. This small and lightweight response box is an affordable solution to collecting simple subject response data. The RBOX has three models: RBOX is used with the RP2.1 processor, RBOX4 with the RM-series processors, and RBOX_RX6 with the RX-series processors.

Part numbers:

RBOX – Response Box for RP2.1

RBOX4 – Response Box for PI2, RM1, or RM2

RBOX_RX6 – Custom Response Boxes

Connecting the RBOX to the RP2.1

The standard RBOX connects via the DB25 connector directly to the digital input/output port on the RP2.1 with the supplied cable.

Connecting the RBOX4 to the RM1 or RM2

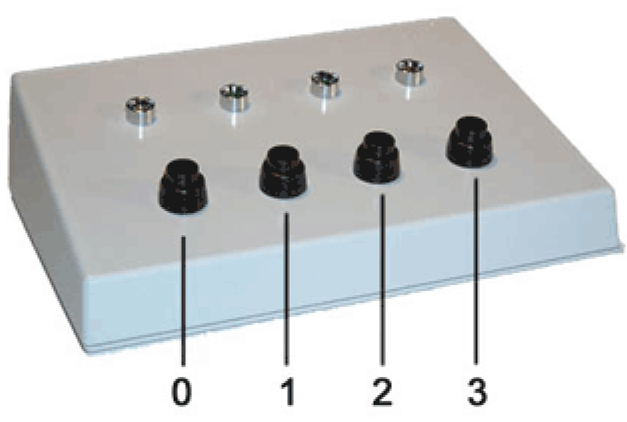
The RBOX4 connects via the DB9 connector directly to the digital input/output port on the back panel of the RM1 or RM2 with the supplied cable. See *Configuring an RM Processor for the RBOX4*, page 14-13 for set-up information.

Connecting the RBOX_MISC to an RX-series processor

An RBOX can be requested for use with RX devices and will connect via the DB25 connector directly to the digital input/output port on an RX-series processor with the supplied cable. See *Configuring an RX Processor for the RBOX*, below, for set-up information.

Buttons and LEDs

The buttons and LEDs are numbered as follows. Contact TDT for assistance with custom button or LED configurations.



Note that the logic on the inputs to the RP/RM/RX processors is reversed logic. Therefore, when polling the lines to determine if a button has been pressed, a logic high or ‘1’ means that no button is pressed and a logic low or ‘0’ indicates a button press.

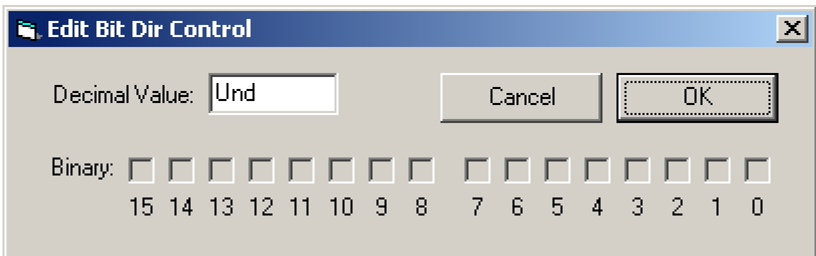
Software Support

The response box can be used directly with PsychRP, SykoFizX or custom designed software. More information on RBOX operation can be found in PsychRP Help.

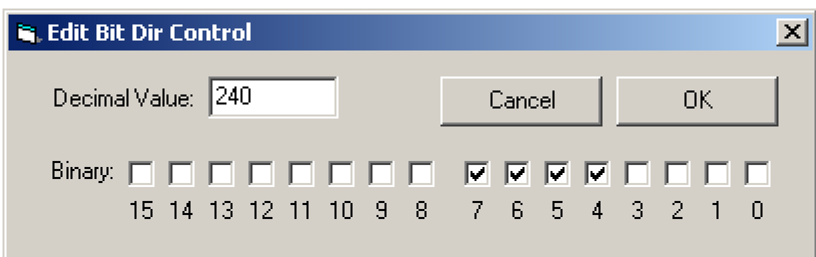
Configuring an RM Processor for the RBOX4

The RBOX4 uses the ground connection (pin 1) and the 8 bits of digital I/O on an RM-series processor Digital I/O port. Bits 0 through 3 are used as button inputs and Bits 4 through 7 are used as LED outputs. To use the response box with an RM processor, configure the bits in the RPvdsEx configuration register as follows:

1. Click the **Device Setup** command on the **Implement** menu.
2. In the **Set Hardware Parameters** dialog box, click the **Type** drop-down box and select either the **RM1** or **RM2** from the list.
3. The dialog expands to display the **Edit Bit Dir Control** dialog box.
4. Click **Modify** to display the **Edit Bit Dir Control** dialog box. In this dialog box, a series of check boxes are used to create a bitmask that is used to program all bits.



5. To enable the check boxes, delete **Und** from the **Decimal Value** box and enter **240**. This configures Bits 4 through 7 as outputs.



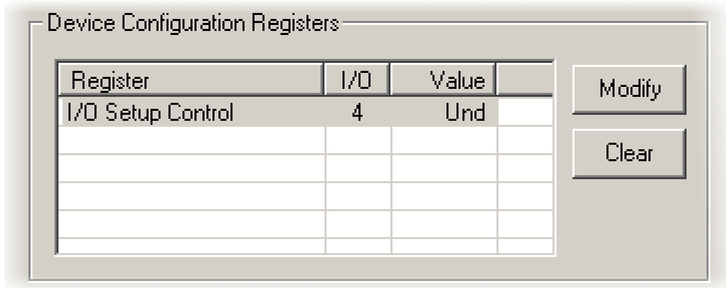
6. When the configuration is complete, click **OK** to return to the **Set Hardware Parameters** dialog box.

Configuring an RX Processor for the RBOX_RX6

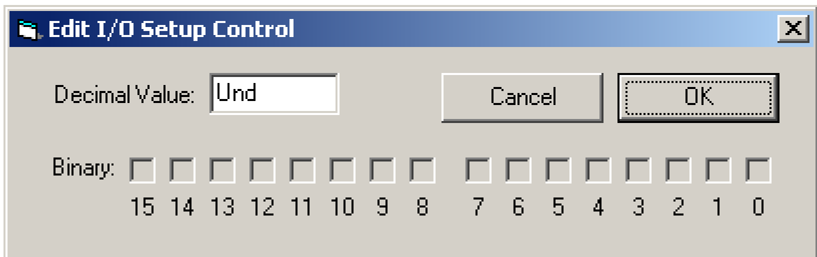
The RBOX_RX6 uses the ground connection (pin 5) and the 8-bits of bit-addressable digital I/O on an RX-series processor Digital I/O port. Bits 0 through 3 are used as button inputs and Bits 4 through 7 are used as LED outputs. To use the response box with an RX processor, configure the bits in the RPvdsEx configuration register as follows:

1. Click the **Device Setup** command on the **Implement** menu.
2. In the **Set Hardware Parameters** dialog box, click the **Device Type** box and select any **RX** device from the list.

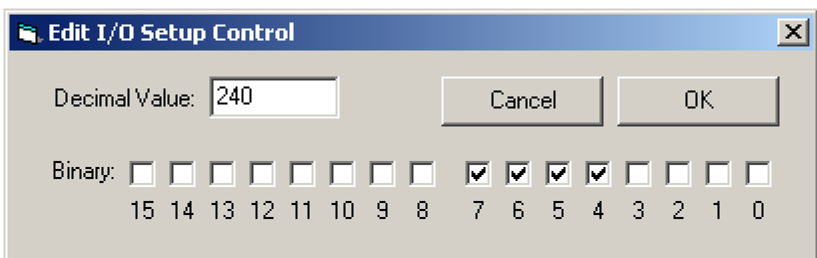
3. The dialog expands to display the **Device Configuration Register**.



4. Click **Modify** to display the **Edit I/O Setup Control** dialog box. In this dialog box, a series of check boxes are used to create a bitmask that is used to program all bits.



5. To enable the check boxes, delete **Und** from the **Decimal Value** box and enter **240**. This configures Bits 4 through 7 as outputs.



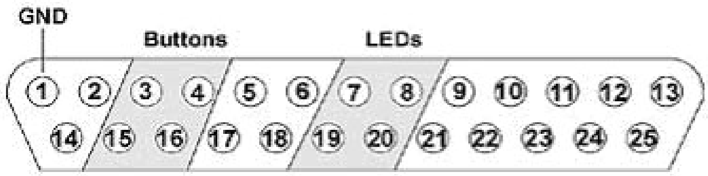
6. When the configuration is complete, click **OK** to return to the **Set Hardware Parameters** dialog box.

RBOX Technical Specifications

Response Box for RP2.1

Buttons	4
LEDs	4
Connection	25-pin
Cable Length	6'

RBOX DB25 Pin Out



Pins	Name	Description
1	GND	Ground
2	NA	Not Used
3	B1	Button Bit 1
4	B3	Button Bit 3
5	NA	Not Used
6		
7	L1	LED Bit 1
8	L3	LED Bit 3
9	NA	Not Used
10		
11		
12		
13		
14		

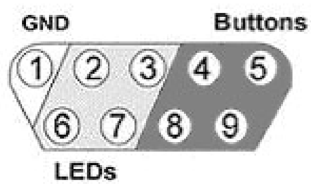
Pins	Name	Description
15	B0	Button Bit 0
16	B2	Button Bit 2
17	NA	Not Used
18		
19	L0	LED Bit 0
20	L2	LED Bit 2
21	NA	Not Used
22		
23		
24		
25		

RBOX4 Technical Specifications

Response Box for RM1, RM2, or PI2

Buttons	4
LEDs	4
Connection	9-pin
Cable Length	6'

RBOX4 DB9 Connector Pin Out



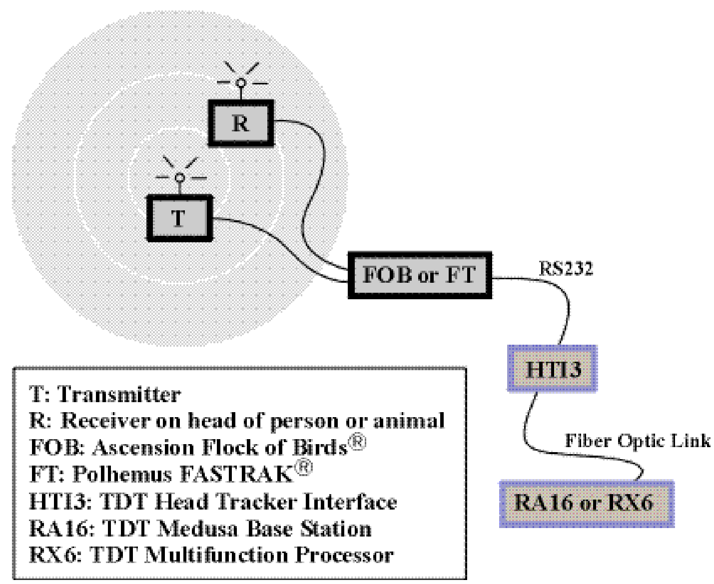
Pin	Name	Description
1	GND	Ground
2	L2	LED Bit 2
3	L0	LED Bit 0
4	B2	Button Bit 2
5	B0	Button Bit 2
6	L3	LED Bit 3
7	L1	LED Bit1
8	B3	Button Bit 3
9	B1	Button Bit 1

HTI3 Head Tracker Interface

Overview

The HTI3 is an interface between your System 3 processor and either the Polhemus FASTRAK® or Ascension Flock of Birds® or miniBIRD® motion trackers and can acquire X, Y, and Z coordinates as well as azimuth, elevation, and roll (AER) data from two receivers/sensors. A boresight signal can be used to zero the AER values to a relative position. This can be accomplished by a manual button press on the front panel of the HTI3 or from an external 3V digital source via the boresight input BNC.

Data can be transferred directly to any System 3 processor with a fiber optic input, bypassing the host computer and enabling movement and positional information to be integrated into experiments in real-time without any increase in latency. Positional information from motion trackers can be efficiently stored and synchronized with biological signals such as EMG, EEG and extracellular neurophysiology or used to update a 3D audio signal presentation in real-time.



The HTI3 parses the incoming signals from the motion tracker into the following data components:

Receiver #: Each HTI3 can handle up to 2 channels of motion tracker receivers.

Error code: The HTI3 will generate four channels that encode the decimal error codes from the Fastrack motion tracker.

XYZ coordinate space: The HTI3 will generate three channels of coordinate space distance from each receiver in either inches or centimeters based on information from the motion tracker.

Azimuth, Elevation and Roll (AER): The HTI generates three channels of AER information for each receiver based on signal information from the motion tracker.

NOTE: The XYZ space is absolute distance from the transmitter while the AER information is relative to the boresight point.

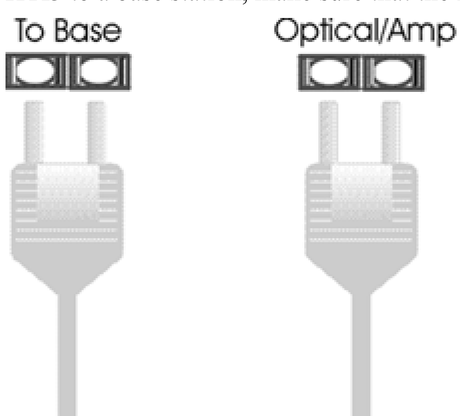
The raw HTI3 output signals must be scaled to achieve the appropriate signal range before the data can be used. Special processing must be implemented in RPvdsEx to perform the necessary scaling and to reduce redundancy in the data. See *HTI3 Circuit Design* for more information about this processing and techniques for using the data with HRTF filter components.

Power and Interface

The device is powered via the System 3 zBus (ZB1) and requires an interface to the PC. If the HTI3 is housed in one of several ZB1 caddies in your system, ensure that it is connected in the interface loop according to the installation instructions: Gigabit, Optibit or USB Interface.

To Base

The HTI3 sends information to the base station over a fiber optic cable. When connecting the HTI3 to a base station, make sure that the fiber optic cable is connected as shown below.



Features

Reset/Boresight

Pressing the Reset/Boresight button momentarily will issue a boresight command to the tracker unit. This signal will zero the AER values respective to the boresight position. Holding the button down for one second will issue a reset command to the tracker unit and undo the boresight command. The AER values will now be returned with respect to the default initial positioning.

To Tracker

The **To Tracker** DB9 input connects the motion tracker to the HTI3.

Note: When using the FOB or miniBIRD® motion tracker, data will be properly transferred to the interface if only pins 2, 3 and 5 are connected. A special connector is shipped with the HTI3 to make this transition from the RS232 cable to the tracker. This connector also performs the required RS232 gender change.

Polhemus/FOB

The toggle switch is provided to select between the FT or FOB motion tracker. This switch must be in the correct position on power up of the HTI3 for correct operation.

To use the miniBIRD® set to FOB.

The miniBIRD® tracker must be set to Normal Addressing Mode and the DIP settings should be configured as below:

1	2	3	4	5	6	7	8
ON	ON	ON	OFF	OFF	OFF	ON	OFF

Boresight

A boresight command can be issued from an external 3V digital source via the Boresight BNC input. This signal needs to be a logical high (‘1’) pulse of at least 200 ns in length. The signal then needs to be set logic low (‘0’) for at least 200 ns before another boresight command can be issued.

Activity Lights

Active

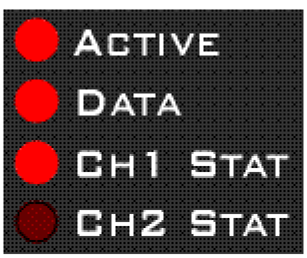
The Active LED indicates if the HTI3 is connected to a base station via a fiber optic cable. This LED will flash slowly (~1 Hz) if this connection is not properly made.

Data

The Data LED indicates if the HTI3 is receiving data from the motion tracker unit. This LED will also flash slowly (~1 Hz) if the tracker is not properly connected to the HTI3.

CH1 Stat/Ch2 Stat

The Ch1 Stat and Ch2 Stat LEDs indicate if the interface is receiving data from receiver 1, receiver 2 or both. The figure below shows the LED pattern for the HTI3 properly connected to a base station and a motion tracker while acquiring data from receiver 1.



HTI3 Circuit Design

The HTI3 parses incoming signals from a motion tracker into 16 channels of data and sends it to a base station (such as RX5, RX6, or RA16BA) at rates up to 25 kHz. Most motion trackers send data at a slow rate (~120 Hz). This means that there is a large amount of redundancy in the data acquired by the base station. The circuit designs described below will reduce the resulting redundancy and convert the raw HTI3 output signals into useful information such as error codes, distance measures and relative positional information such as Azimuth, Elevation, and Roll.

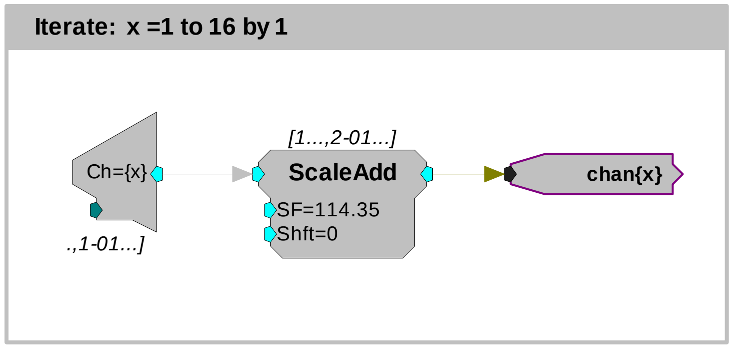
Acquiring and Scaling Motion Tracker Signals

Motion tracker signals are acquired via a fiber optic cable connecting the HTI3 to a base station. The most common signals input via the fiber optic port are biological signals amplified using one of the TDT preamplifiers; so all signals input through one of these ports are automatically scaled accordingly. When the fiber optic inputs are used to acquire signals from other devices, such as the HTI3, the signals must be scaled according to the signal characteristics of the specific device. In the case of the HTI interface, the signal from each channel must be scaled by 114.35. This adjusts the signal to a range of +/- 1.0V. Additional scaling is required to convert signals on some input channels to the appropriate units or values. The table below describes the scale factor(s) for each signal input from the HTI3 and for each device.

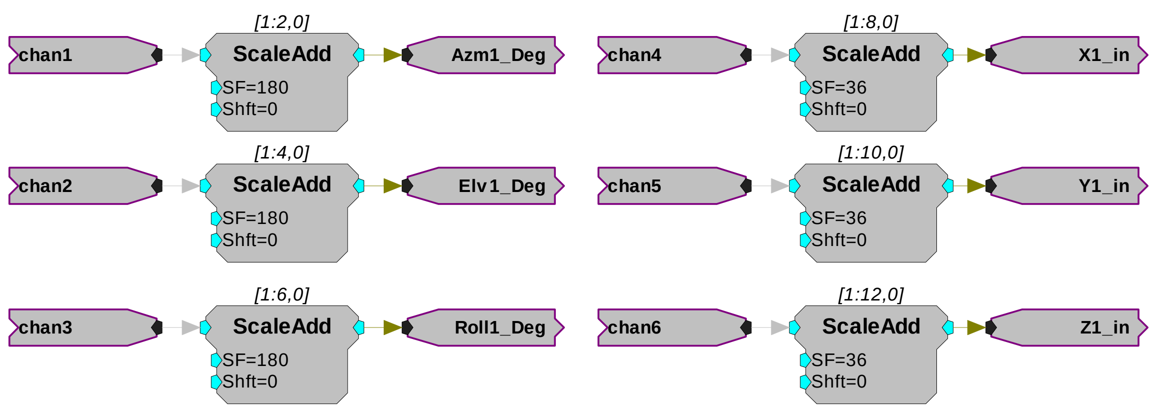
Device	Receiver	Channel	Data	SF (base)	SF (cm) or SF(ASCII) for err	SF (in)	SF (rad)	SF(deg)
FT	1	1	Azm	114.35	NA	NA	3.14159	180
		2	Ele		NA	NA	3.14159	180
		3	Roll		NA	NA	3.14159	180
		4	X		300	118.11	NA	NA
		5	Y		300	118.11	NA	NA
		6	Z		300	118.11	NA	NA
	2	7	Azm		NA	NA	3.14159	180
		8	Ele		NA	NA	3.14159	180
		9	Roll		NA	NA	3.14159	180
		10	X		300	118.11	NA	NA
		11	Y		300	118.11	NA	NA
		12	Z		300	118.11	NA	NA
	1	13	err		16384.2			
		14	err		16384.2			
		15	err		16384.2			
		16	err		16384.2			
FOB	1	1	Azm	114.35	NA	NA	3.14159	180
		2	Ele		NA	NA	3.14159	180
		3	Roll		NA	NA	3.14159	180
		4	X		91.44	36	NA	NA
		5	Y		91.44	36	NA	NA
		6	Z		91.44	36	NA	NA
	2	7	Azm		NA	NA	3.14159	180
		8	Ele		NA	NA	3.14159	180
		9	Roll		NA	NA	3.14159	180
		10	X		91.44	36	NA	NA
		11	Y		91.44	36	NA	NA
		12	Z		91.44	36	NA	NA
	1	13	NA					
		14	NA					
		15	NA					
		16	NA					

Note: The scale factor for the FT error codes converts the values to ASCII codes.

These scale factors must be incorporated into any circuit design. The circuit below performs the initial scale factor. The circuit uses the iterate function to efficiently scale all 16 channels. The circuit uses only single processor components and works on all devices. The iterate function duplicates the construct 16 times, with an input signal from channel 'x' scaled by 114.35 and then sent to a hop out.



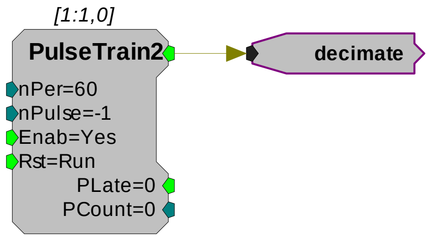
The next circuit segment scales each channel based on the table above for the FOB motion tracker. The first three channels in this example scale Azimuth, Elevation, and Roll. If the input to the HTI3 includes two motion tracker channels, then channels 7, 8 and 9 will contain the Azimuth, Elevation, and Roll information for the second motion tracker. To return this information in radians, the scale factor should be changed to 3.14159. Channels 4-6 are scaled to inches. To scale the XYZ coordinate space to centimeters the scale factor would be 91.44. This circuit can be easily modified to use with the FT motion tracker by inserting the appropriate scale factors from the table above.



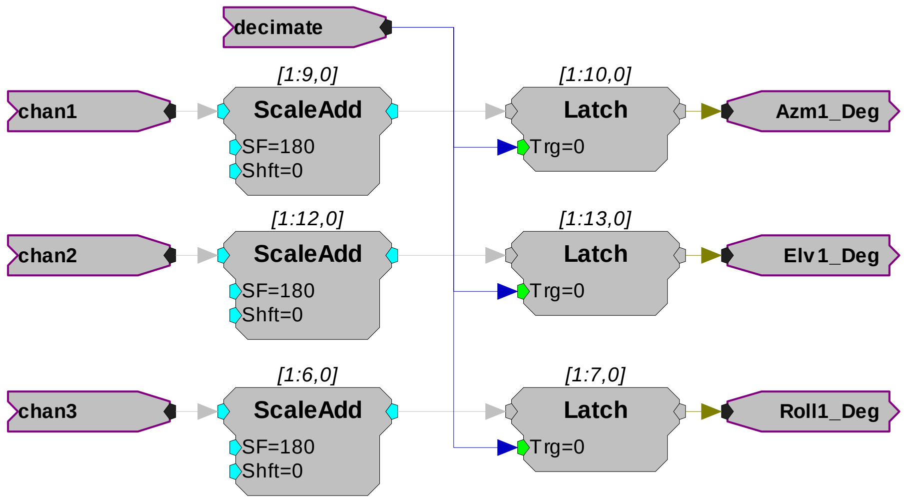
Data Storage and Visualization of Signal Input

Motion tracker signals are updated/transferred to the HTI3 at rates up to 120Hz. The HTI3 sends signals to the RX/RP device at sample rates up to 25 kHz. This means that each value from the motion tracker may be repeated on the DSP up to 200 times. To minimize the redundancy of the signal, the channel outputs can be decimated by a fixed value. This will decrease the amount of data stored on either the DSP or transferred to a computer. The construct below shows two ways to decimate the signal. One way shows real-time visualization of the signal and the other illustrates storage of the signal to disk.

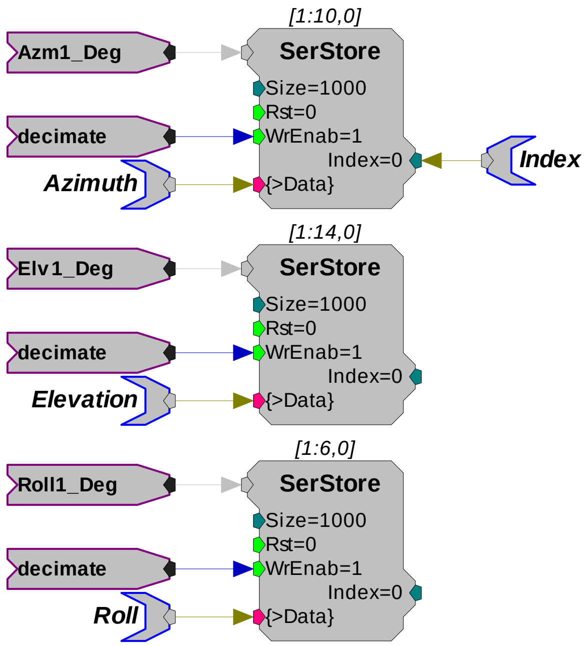
Since the following circuit segments are based on the data transfer rate of the motion tracker itself, users should review the documentation provided with their device before using the parameter values shown.



The PulseTrain2 component sends out a pulse every 60 samples. The output from the PulseTrain2 is sent to the Trigger line on a latch. Therefore the output from the latch is updated once every 60 samples. This generates an updated output that more closely matches the data transfer rate of the motion tracker. The output can then be sent to a head related transfer function (HRTF) coefficient generator (see *Using the HTI3 with HRTF Filters*).

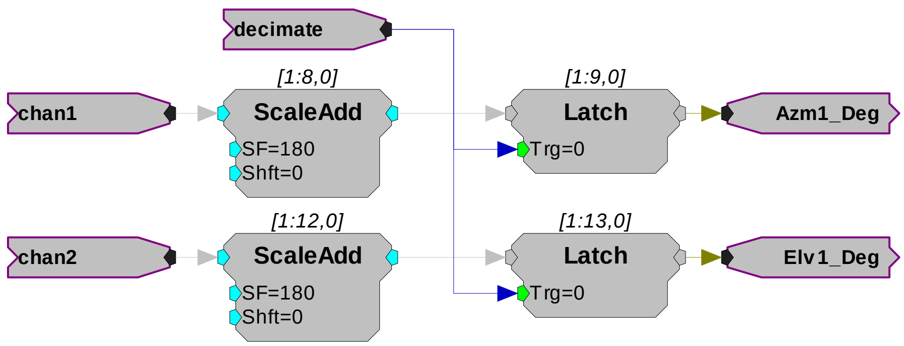


Another way to use the decimated signal would be to send it to a Serial Buffer input. In this case the values are stored once every 60 samples. If you were using this with OpenEx this would be the primary way to save the data set.

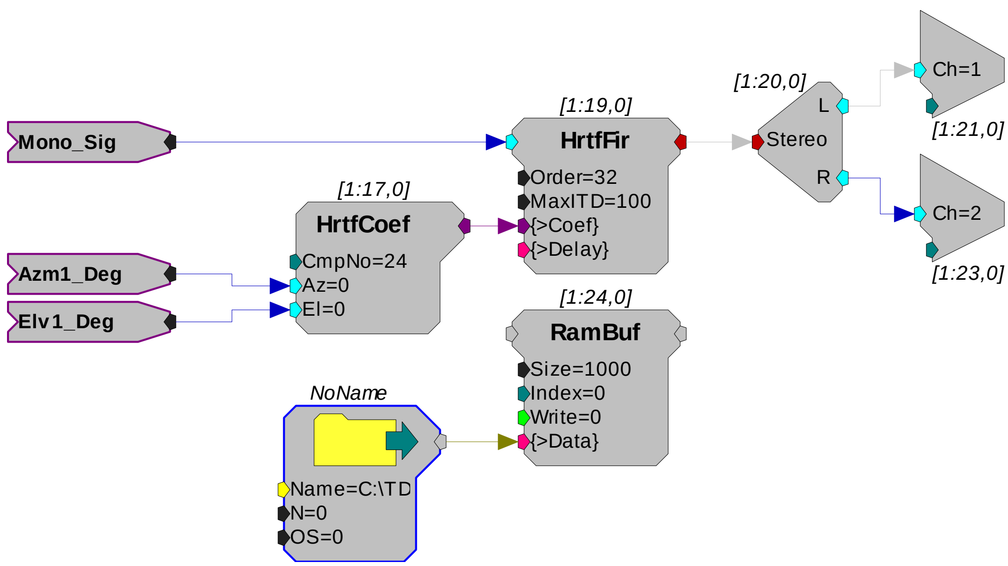


Using the HTI3 with HRTF Filters

One great advantage of the HTI3 setup is that users can connect the device to an RX6 Multifunction Processor. With the RX6 system, a virtual 3D audio environment can be generated. The following circuit uses the Azimuth and Elevation information to change the perception of a signal input. Channels 1 and 2 are latched via the PulseTrain2 decimation construct discussed earlier.



The output of the HTI3 is sent to an HRTF filter that converts the mono input into a stereo output that can be sent to Headphone buffers etc. A random access buffer stores the HRTF filter values.



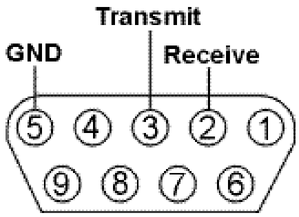
About the Sample Circuit

The sample circuit HTIFLOCKOFBIRDS.rpx illustrates the scale factors for all incoming channels from the FOB motion tracker. Page 0 shows the initial scaling and the secondary scaling for channels 1-3 (deg) and 4-6 (in). Page 1 shows the scaling of the channels relating to the optional 2nd motion tracker input (channels 7-12).

HTI3 Technical Specifications

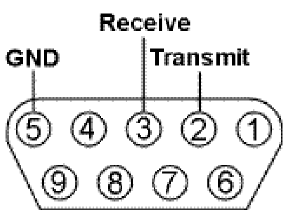
Max update rate	120 Hz
Boresight trigger	External
RS232 acquisition rate	115 kbaud

To Tracker - DB9 Pinout for Ascension Flock of Birds®



Pin	Name	Description
1	NA	Not Used
2	Receive	Serial Receive Line
3	Transmit	Serial Transmit Line
4	NA	Not Used
5	GND	Ground
6	NA	Not Used
7		
8		
9		

To Tracker - DB9 Pinout for Polhemus FASTRAK®



Pin	Name	Description
1	NA	Not Used
2	Transmit	Serial Transmit Line
3	Receive	Serial Receive Line
4	NA	Not Used
5	GND	Ground
6	NA	Not Used
7		
8		
9		

~

Part 15 Signal Handling

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PM2Relay

Overview

The PM2Relay (PM2R) is a 16 channel multiplexer for delivering powered and unpowered signals to a device. When coupled to a power amplifier such as the SA1, the PM2R can transfer several watts of power to standard four ohm and eight ohm speakers.

The PM2R is designed to be used as a "de-multiplexer", that is, one input switched to 16 possible outputs. However, it can also be used as a straight multiplexer (16 inputs to one output). This is accomplished by sending signals in to the 16 "signal out" channels. The selected channel will be output on the "signal in" channel. Users that are doing this should be very careful, as it is easy to exceed the maximum input values when sending in 16 input signals. **The aggregate input of all signals should never exceed two amps, or 15 Volts, because severe damage can be caused to the module.**

Each RP2 can control up to four PM2R devices and each PM2R can have one active channel. Therefore, a maximum of four signals can be played out simultaneousely when using four PM2Rs.

To connect to a System 3 module, attach the 25-pin, blue ribbon cable from the RP2 device to the PM2R. Connect your powered signal source to the Signal In and connect the signal out to the RP2 connection on the PP16, or your own connectors. The channel outs on the PP16, from the left to right, correspond to the 16 channels (0-15) on the device.

Power

The PM2Relay is powered via the System 3 zBus (ZB1PS). No PC interface is required.

Features

The PM2R uses a bit pattern code to control the output of a powered signal to one of sixteen output channels. The powered signal can come from any power amplifier such as the SA1 (Stereo Amplifier) or the HB7 (Headphone Buffer). The PM2R is designed to use a bit-code pattern from an RP2 Real-Time Processor or RV8 Barracuda Processor.

RP Control Input

The male DB25 connector on the left is the interface to the RP2. A blue ribbon connector is used to directly connect the RP2 and the PM2R. The PM2R uses all the bit outputs from the RP2. If you require additional bit outs, TDT recommends purchasing an RV8.

In addition, any System 3 processor that has at least eight digital outputs, including the RX family of devices, can be used to control the PM2R (a special connector may be required).

Signal In

The BNC connector is the powered signal input. The maximum power input is a two amp, 15 Volt continuous signal or approximately 30 watts of continuous power.

Signal Out

The female DB25 connector on the right is the interface for the powered signal output. Users can also connect the PM2R output to the patch panel (PP16) connector labeled for the RP2 for easy BNC access to the powered signal.

Channel...

Sixteen LEDs indicate which channel is active. One channel can be active at a time. It is also possible to inactivate all channels.

PM2R Bitcode Pattern

The bitcode pattern from the RP2 consists of an 8-bit word that contains the following information; the device ID, the channel ID, and a set-bit. A final bit shuts off all channels. To control the PM2R, generate the bitcode pattern associated with the device and channel then send out the set-bit to change the channels. Be aware that the relays on the PM2R have a transition time of around one millisecond.

Bits 0 - 3 identify the channel number. Integer 0, or bitpattern (xxxx 0000), is channel 0 and integer 15, or bitpattern (xxxx 1111), is channel 15.

Bits 4 and 5 identify the device number. Integer value 0, or bit pattern (xx00xxxx), is device number 0 and integer value 48, or bit pattern (xx11xxxx), is device number 3. The device number is set internally for each PM2R and allows for an RP2 to control up to four PM2R modules. If only one PM2R is being used, it should have device number 0.

Bit 6 is the set-bit. When this bit is set high, the channel and device from the previous six bits is activated.

Bit 7 deactivates all channels across only the specified device.

The chart below shows the bit ID, its integer value, and its function.

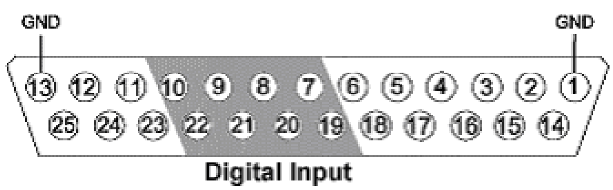
Bit Number	Integer Value	Function
0	1	Least significant bit of channel number
1	2	Bit 2 of channel number
2	4	Bit 3 of channel number
3	8	Most significant bit of channel number
4	16	Least significant bit of device number
5	32	Most significant bit of device number
6	64	Turns on the channel of the specified device
7	128	Turns off all channels on specified device only

Note: Make sure to put a delay of one sample between setting the channel number and turning the channel on. Trying to do both at the same time will not work correctly. For example, send "00000111" to select channel 7, and then send "01000000" one sample later to turn the channel on.

PM2R Technical Specifications

Switching Mode	Single 1-to-16/16-to-1
Switching Time	2 mSec
Input/output Level	+/- 15 Volts
Channel Cross-Talk	< -80 dB
S/N (typical)	90 dB
Maximum Allowable Current	2 Amps continuous

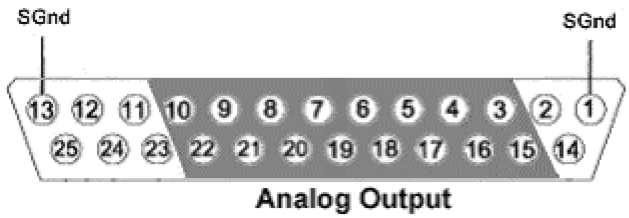
RP Control Input - DB25 Pinout



Pin	Name	Description
1	GND	Ground
2	NA	Not Used
3	NA	
4	NA	
5	NA	
6	NA	
7	D1	Digital Input Channels
8	D3	
9	D5	
10	D7	
11	NA	Not Used
12	NA	
13	GND	Ground

Pin	Name	Description
14	NA	Not Used
15	NA	
16	NA	
17	NA	
18	NA	
19	D0	Digital Input Channels
20	D2	
21	D4	
22	D6	
23	NA	Not Used
24	NA	
25	NA	

Signal Output - DB25 Pinout



Pin	Name	Description
1	SGND	Signal Ground
2	NA	Not Used
3	A1	Analog Output Channels
4	A3	
5	A5	
6	A7	
7	A9	
8	A11	
9	A13	
10	A15	
11	NA	Not Used
12	NA	
13	SGND	Signal Ground

Pin	Name	Description
14	NA	Not Used
15	A0	Analog Output Channels
16	A2	
17	A4	
18	A6	
19	A8	
20	A10	
21	A12	
22	A14	
23	NA	Not Used
24	NA	
25	NA	

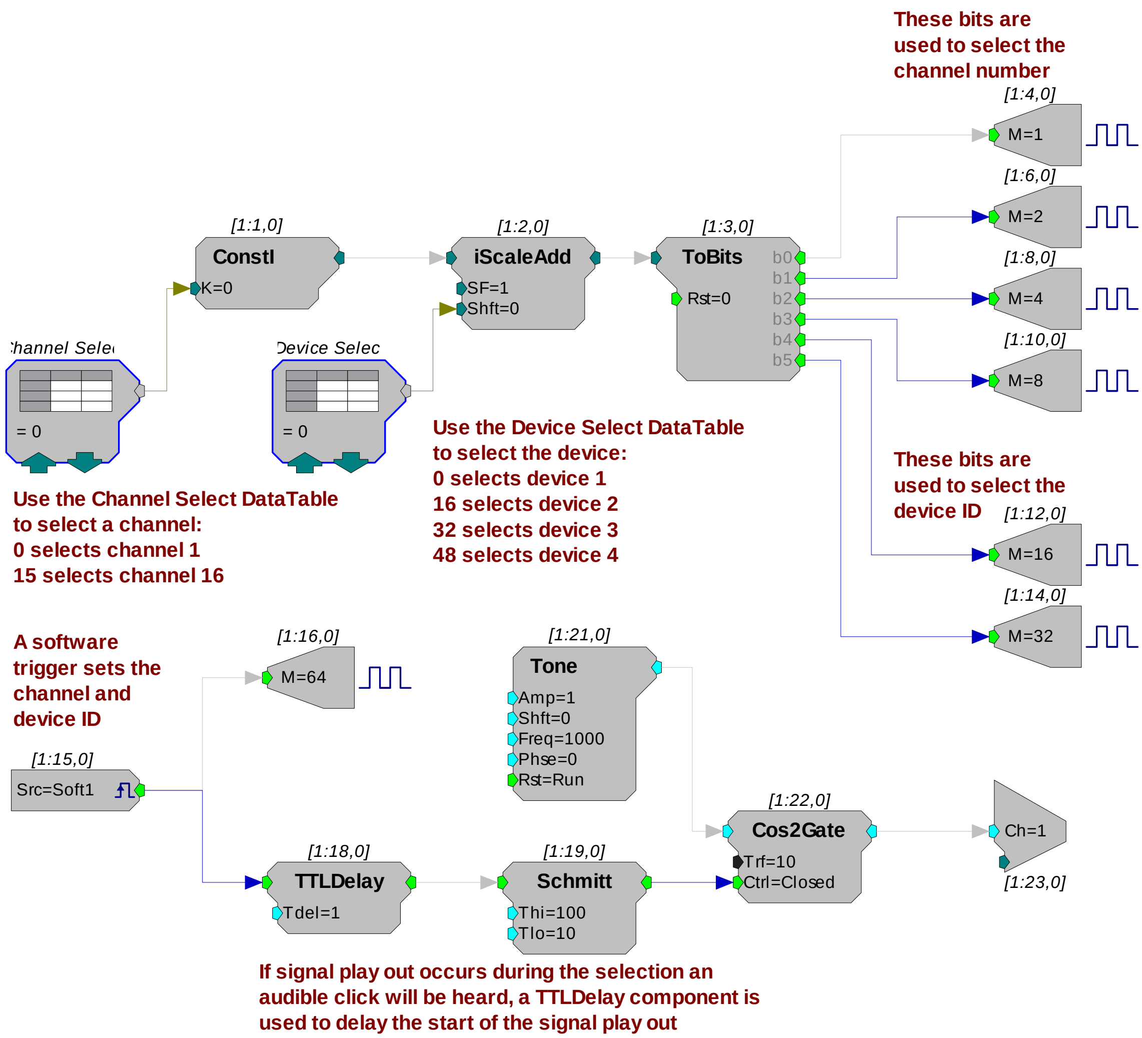
PM2R - Controlling Signal Presentation

The circuits described here use typical techniques for controlling the signal presentation when using a PM2R. These circuits have been designed as tutorials and will need to be modified to meet the needs of the individual researcher.

Controlling the PM2R with BitOuts:

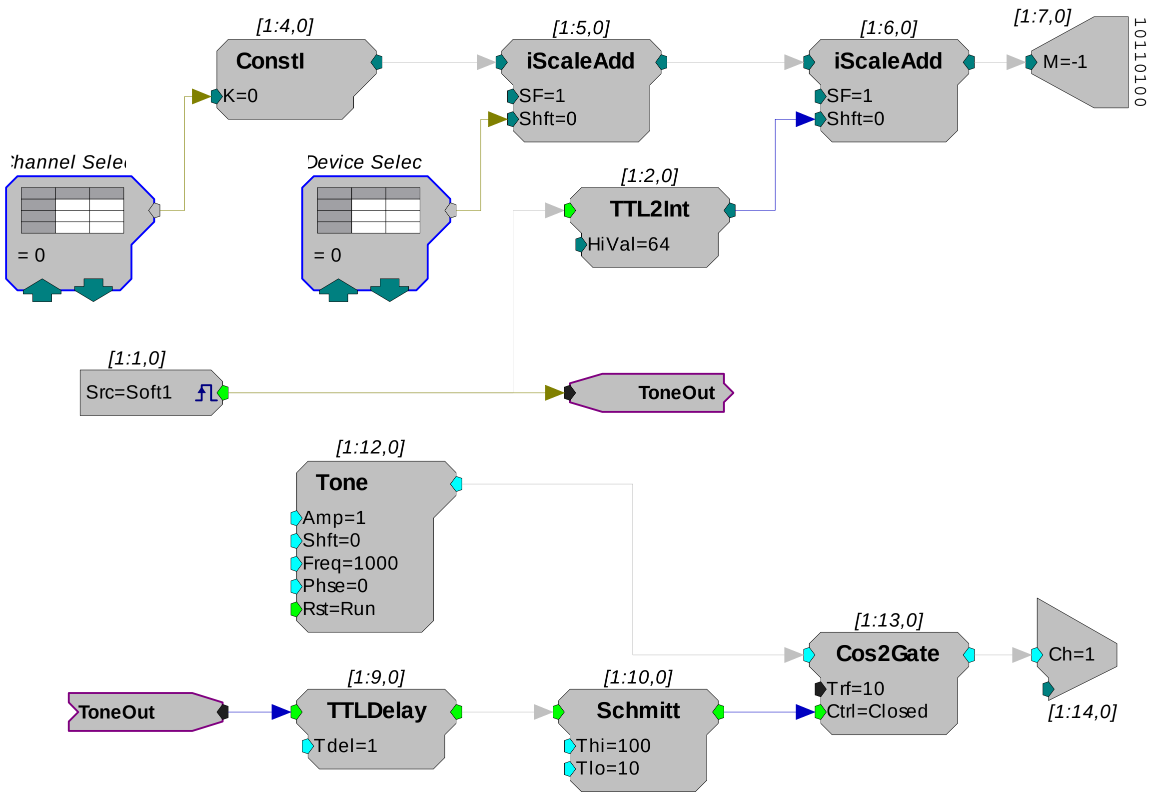
In this example several BitOuts are used to control the PM2R (via an RP2.1) from within RPvdsEx. The bit pattern is generated by two DataTable components. DataTables are commonly used to send values from the PC to the RP devices. While working in RPvdsEx, the selection can be changed by clicking the green up and down arrows near the bottom edge of the components. The first DataTable (Channel Select) stores the values for the channel number. Channel numbers start at zero and go to fifteen. Each RP2.1 is capable of controlling up to four PM2R devices. The second DataTable (DeviceSelect) stores the values for the device ID. The values in the table are 0 (device 0), 16 (device 1), 32 (device 2), and 48 (device 3). The iScaleAdd is used to add the integer values from both tables and the ToBits component changes the resulting integer to the bitcode pattern. The first four bits are used to select the channel number and the last two bits are used to select the device ID.

A software trigger is used to change devices and initiate a tone burst of 100 milliseconds duration. The software trigger causes the Schmitt trigger to open a gate for 100 milliseconds. The Schmitt trigger is delayed by one millisecond relative to the channel select. This removes the transient associated with the relays.



Controlling the PM2R with WordOut:

In this example a WordOut is used to control the PM2R (via an RP2.1) from within RPvdsEx. This simplified format decreases cycle usage. An additional iScaleAdd is required because the BitOut and WordOut components function differently and should not be used in the same circuit. As before, a software trigger initiates the start of the stimulus presentation. The triggered signal adds 64 to the output to change the channel.



Controlling the PM2R from a run-time application:

The examples described here could easily be modified to allow control from run-time applications. Parameter tags can be included and used in other applications such as BioSigRP or OpenEx.

SM5 Signal Mixer

Overview

The SM5 is a three-channel signal mixer. The relative contribution of the three inputs to the final output can be adjusted using a variable gain for two of the inputs. In addition, the signal on the two adjustable channels can be inverted before addition. The input signal range is ±10V for each channel, with the additional caveat that the amplified signal for each channel may not exceed ±10V without clipping. The range for the summed output is ±10V.

Power

The SM5 Signal Mixer is powered via the System 3 zBus (ZB1PS). No PC interface is required.

Features

The SM5 Signal Mixer is a three-channel weighted summer with variable input weighting and channel inverting. The SM5 is a zBus rack mounted device, through which it receives power.

Inputs

Three signals input channels (A, B, and C), with a range up to ±10 V peak, are accessed through front panel BNC connectors. Input channels A and B are multiplied by a weighted, signed constant, K, before being added to the final output. The weighting range for these two channels is adjustable from -20 dB to +20 dB (i.e. |K| = 0.1 to 10) using a GAIN knob on the front panel. The sign of K for channels A and B can also be selected using front panel toggle switches, labeled INV-A and INV-B.

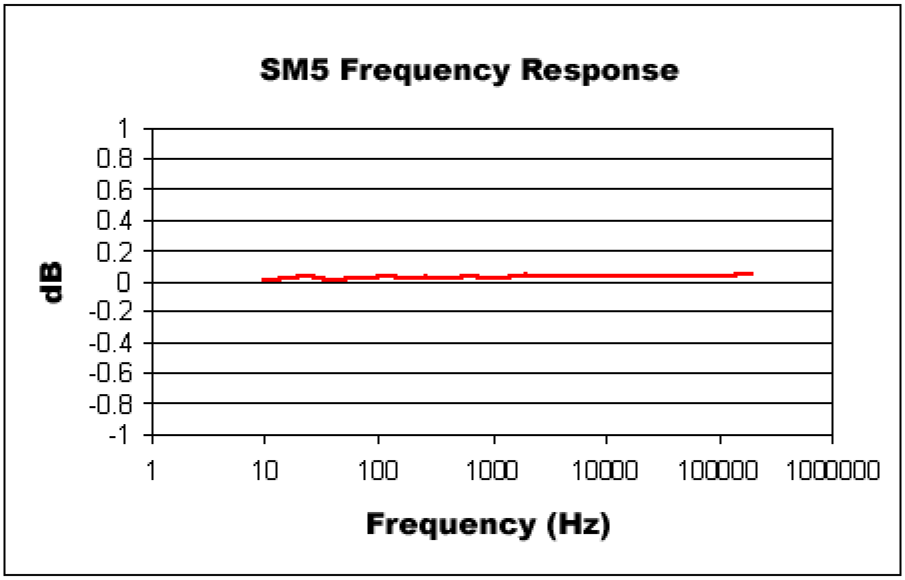
If an input is not being used, it should be grounded by attaching a shorted BNC cable. This will prevent unwanted noise from being added to the output.

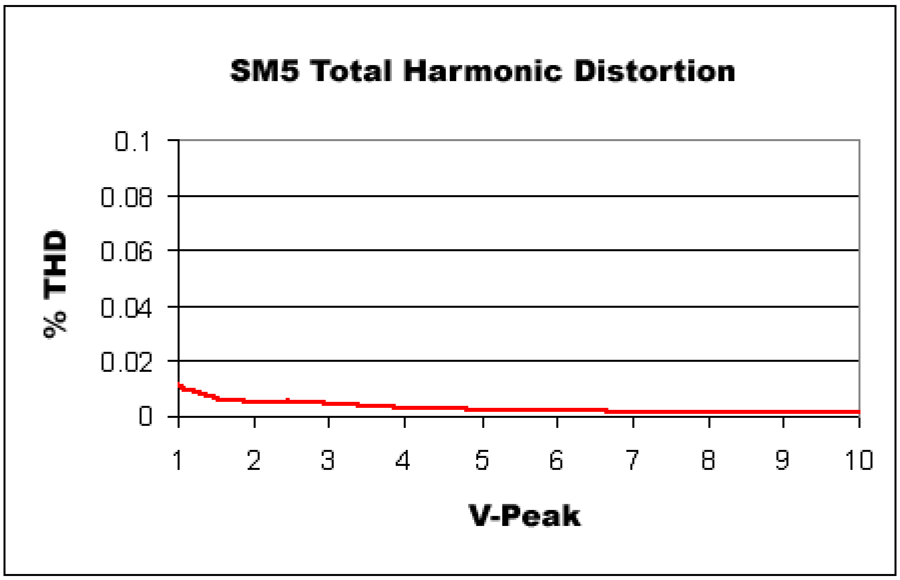
Clipping

The variable weighting provides a great deal of flexibility in input and output signals. However, care should be taken to avoid clipping any signal component. The SM5 output signal = (Ka*A) + (Kb*B) + C is limited to ±10V peak. In addition, the raw inputs, A, B, and C, as well as the weighted inputs, Ka*A, and Kb*B, are limited to ±10V peak.

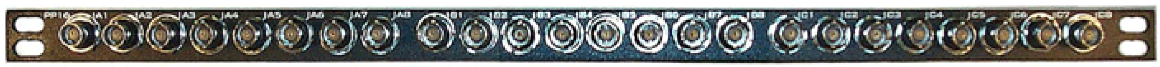
SM5 Technical Specifications

Input Signal Range	±10V peak
Weighting Range	-20.0 to +20.0 dB
Max Output	±10V
Spectral Variation	< 0.1 dB from 10 Hz to 200 kHz
S/N (typical)	111 dB (20 Hz to 80 kHz)
THD	< 0.002% (1kHz tone +/- 7V peak)
Noise Floor	19 µV rms
Output Impedance	20 Ohm
Input Impedance	10 kOhm
Inversion	Channels A & B


SM5 Frequency Response
dB
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
1
10
100
1000
10000
100000
1000000
Frequency (Hz)


SM5 Total Harmonic Distortion
% THD
0.1
0.08
0.06
0.04
0.02
0
1
2
3
4
5
6
7
8
9
10
V-Peak

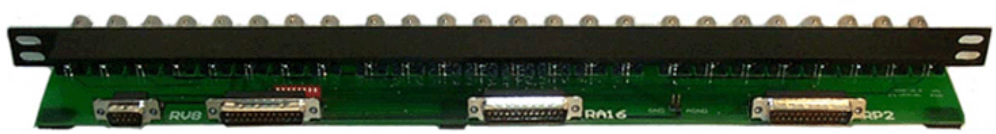
PP16 Patch Panel



The PP16 Patch Panel provides convenient BNC connections for easy access to the digital and analog inputs and outputs of a variety of System 3 devices. Originally designed for use with the RP2 Real-time Processor, RA16 Medusa Base Station, and RV8 Barracuda; the PP16 back edge is equipped with a nine pin and three 25-pin connectors, which have been marked with the corresponding device label to minimize the possibility of miswiring.

To connect the PP16 to a device:

Connect the male end of the 9- or 25-pin ribbon cable to the desired module and connect the female end to the correct PP16 input according to the following table.



RV8 (Barracuda) RA16 (Medusa) or RX5 (Pentusa) RP2

Connector:	RV8 9-Pin	RV8 25-Pin	RA16 25 Pin	RP2 25 Pin
Devices:	RV8 Optional I/O	RV8 Digital I/O	RA16BA RA8GA SA8 RX5 RX6 RX7 RX8	RP2 RP2.1 PM2R

Mapping the Inputs and Outputs for Each Device

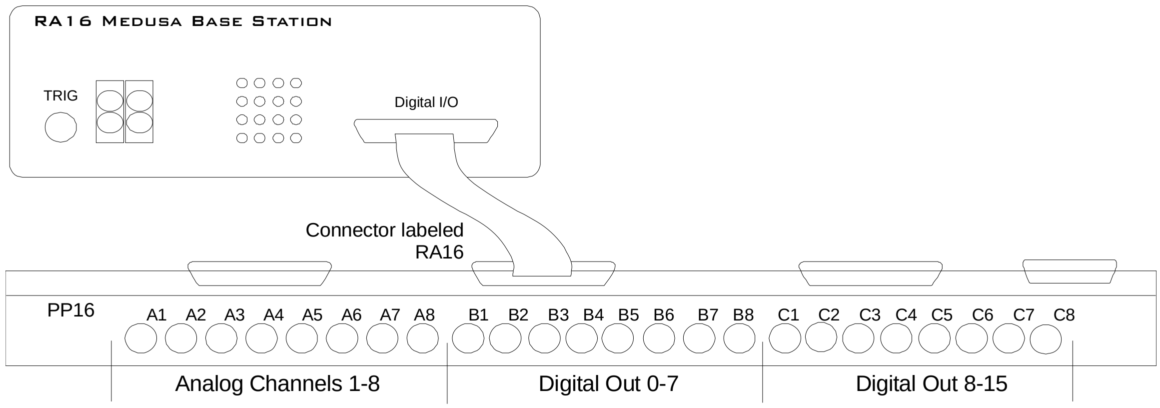
Each device has a unique input and output configuration. The table below shows the configuration of the BNC connectors.

Device & Connector	A1-A8	B1-B8	C1-C8
RP2, RP2.1 Digital I/O Connector	Digital Inputs Channels 1-8	Digital Outputs Channels 1-8	C1=Trigger C2=Volt out (3.3v)
RA16BA Analog/Digital I/O Connector	Analog Outputs Channels 1-8	Digital Outputs Channels 0-7	Digital Outputs Channels 8-15
RV8, RV8D Digital I/O Connector	Digital Inputs Channels 8-15	Digital Inputs Channels 0-7	Digital Outputs Channels 0-7
RV8D* Optional I/O Connector	Analog Outputs Channels 1-8	Not Used	Not Used
RA8GA Analog I/O Connector	Analog Input Channels 1-8	Not Used	Not Used
PM2R Signal Out Connector	Analog Output Channels 0-7	Analog Output Channels 8-15	Not Used
SA8 Power Outputs Connector	Analog Output Channels 1-8	Analog Output Signal and Ground: Channels 1-4	Analog Output Signal and Ground: Channels 5-8
Note: The PP16 can also be used with the RX devices, however, the PP24 is recommended for these devices.			
RX5, RX6, RX7, RX8 Digital I/O Connector	Bit Addressable Digital I/O Channels 0-7	Digital I/O, Byte A Channels 0-7	Digital I/O, Byte B Channels 8-15
RX5, RX7 Multi I/O Connector	Analog Outputs A2, A4, A6, A8 = Channels 1-4 A1, A3, A5, A7 = Not Used	Digital I/O, Byte C Channels 16-23	Digital I/O, Byte D Channels 24-31
RX8 Analog I/O Connector	Analog I/O Block A Channels 1-8	Analog I/O Block B Channels 9-16	Analog Output Block C Channels 17-24

*When the RV8D uses the Optional I/O analog output connector, digital inputs 8-15 are not accessible.

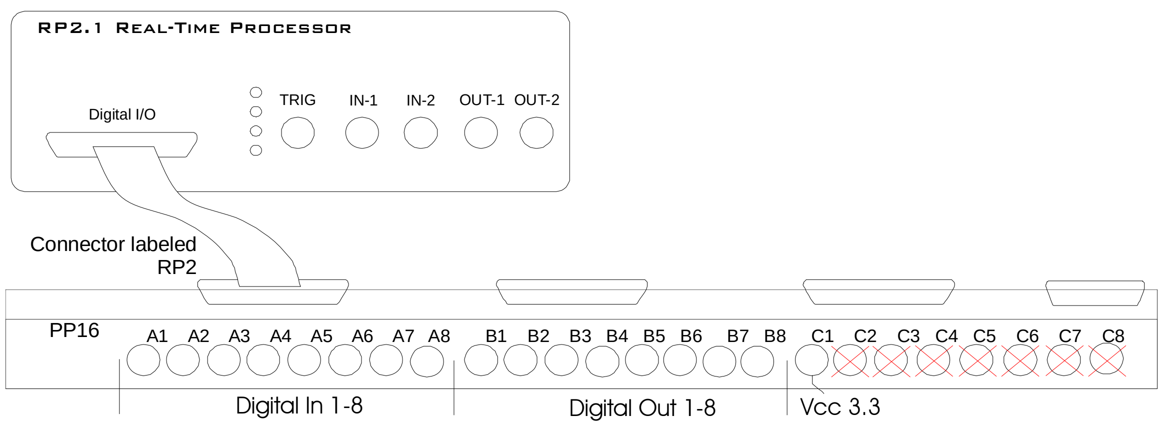
Mapping RA16BA I/O

The diagram below maps the RA16BA Digital I/O connection to the PP16.



Mapping RP2/RP2.1 I/O

The diagram below maps the RP2 Digital I/O connection to the PP16. The last seven BNC connectors are not used. BNC C1 maps to V_{CC} 3.3.



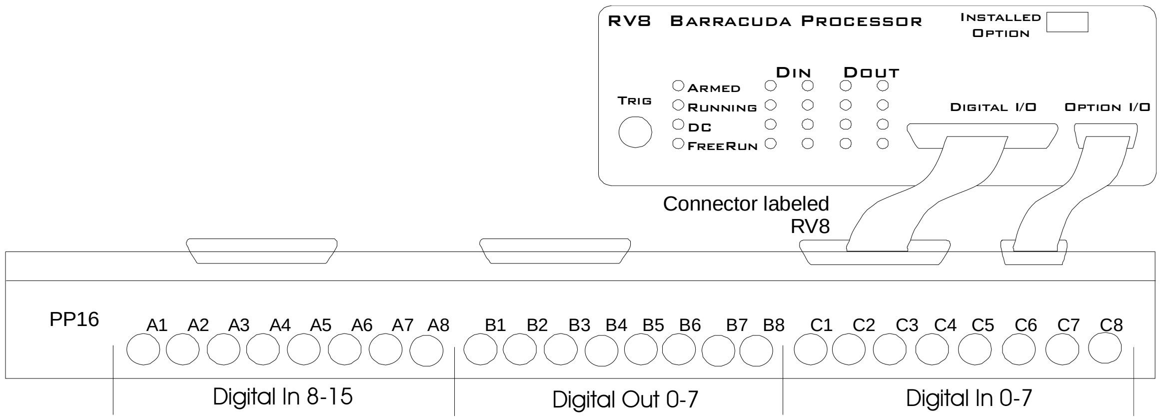
Mapping RV8 I/O

There are two connectors for the Barracuda on the rear edge of the PP16. The optional analog channels are on the DB9 connector and the digital I/O are on the DB25 connector. The PP16 is configured to accommodate 24 of the 32 inputs, outputs, and channels on the Barracuda, at any given time.

TDT ships a special cable that connects between the DB9 connector and the RV8.

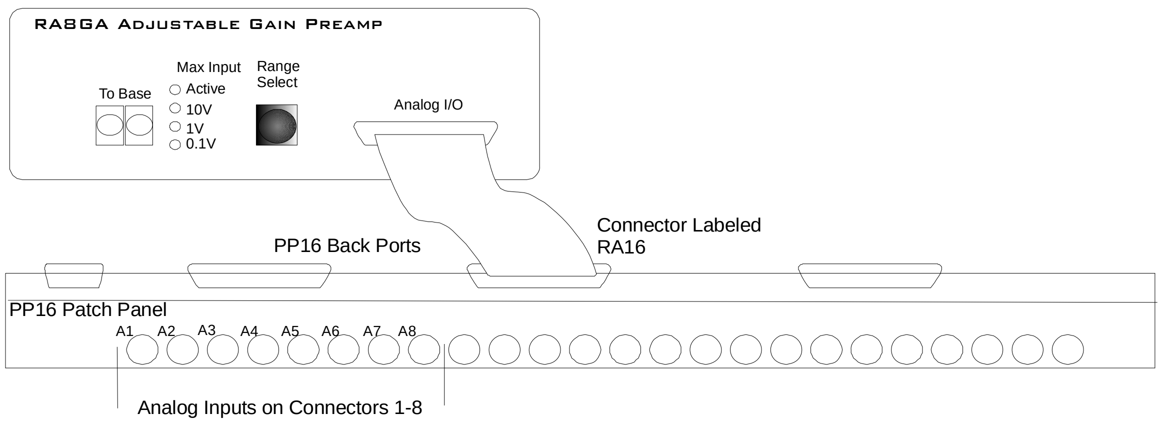
Connect the analog ground on the back of the PP16 to produce adequate signal quality.

The default connection for the Barracuda is shown below. In this format, eight digital inputs, eight digital outputs, and the eight optional analog channels are configured as follows:



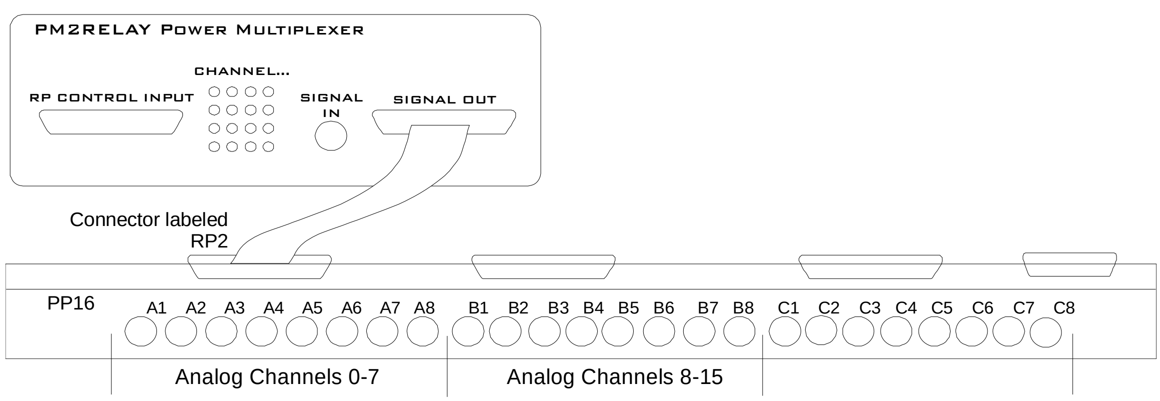
Mapping RA8GA

A PP16 patch panel can be used to simplify connection to the preamplifier's analog inputs. A ribbon cable can be connected from the RA8GA Analog I/O connector to the RA16 connector on the back of the PP16 allowing acquisition of signals via the first eight BNC connectors on the front of the PP16.

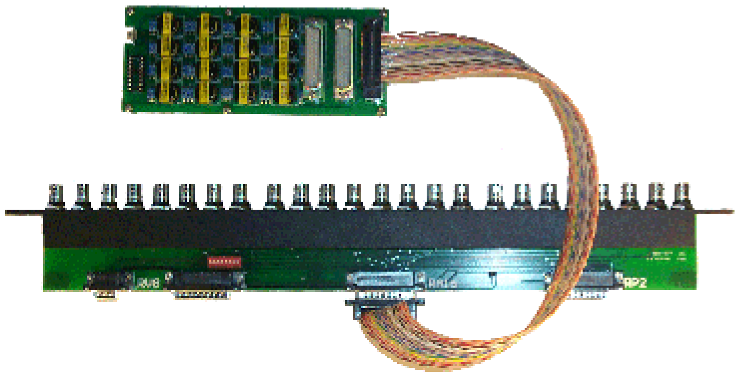


Mapping PM2R I/O

The picture below maps the PM2R signal out connection to the PP16.



Connect to the ETM1



The connector labeled J1 is used to connect the ETM1 to a PP16. Plug one end of a serial DB25 male-female cable into the J1 connector and plug the other end into the RA16 port of the PP16. Channels 1 - 8 and 9 - 16 of the headstages can be accessed through the patch panel BNCs labeled A1-A8 and B1 - B8, respectively. Also, a custom cable can be fabricated to connect the ETM1 (connector J1) to virtually any signal source.

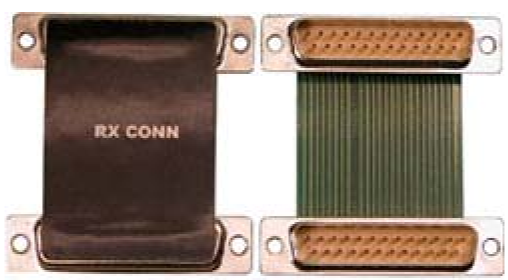
PP24 Patch Panel

Overview

The PP24 Patch Panel provides front panel, BNC connections for easy access to the digital and analog inputs and outputs of the RX family of processors. The PP16 Patch Panel is recommended for use with devices such as the RP2.1 and RA16BA processors, Power Multiplexer (PM2R), and Power Amplifier (SA8). The PP24 can also be used with the RZ5.

The PCB Adapter Advantage

The PP24 is supplied with a single device specific PCB adaptor that can be used with either RX or RZ processors. The PCB provides better performance than ribbon cables, facilitating faster data transfer rates and improved signal to noise ratios.



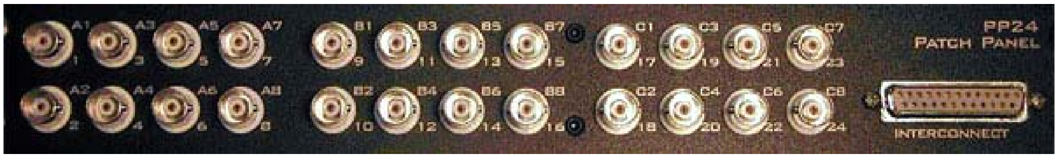
Adjustable Positioning

The PP24 is equipped with a 25-pin connector on the front panel. The PCB Adapter can be used to connect the PP24 to an RX device positioned either directly above or directly below the PP24 or an RZ processor positioned above the PP24. Four thumbscrews located on each corner of the PP24 front panel allow the user to slide the BNC array into the correct position to align the connector with the target device.

Caution: The thumbscrews should never be completely removed. Avoid loosening the thumbscrews too far.

Mapping the Inputs and Outputs for Each Device

The PP24 consists of 3 banks of BNC connectors, Bank A, B, and C. Each of the banks is labeled 1-8 within the set and each BNC is also numbered as part of the entire group from 1 – 24.



The following table shows the configuration of the BNC connectors for each I/O connector of the RX and RZ devices.

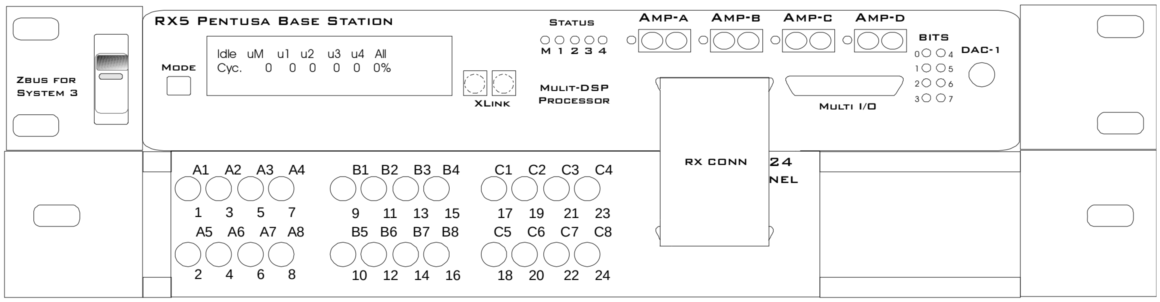
Device & Connector	A1-A8	B1-B8	C1-C8
RX5, RX6, RX7, RX8 Digital I/O Connector	Bit Addressable Digital I/O Channels 0-7	Digital I/O, Byte A Channels 0-7	Digital I/O, Byte B Channels 8-15
RX5, RX7 Multi I/O Connector	Analog Outputs A2, A4, A6, A8 = Channels 1-4 A1, A3, A5, A7 = Not Used	Digital I/O, Byte C Channels 16-23	Digital I/O, Byte D Channels 24-31
RX8 Analog I/O Connector	Analog I/O Block A Channels 1-8	Analog I/O Block B Channels 9-16	Analog Output Block C Channels 17-24
RZ5 Digital I/O Connector	Bit Addressable Digital I/O Channels 0-7	Digital I/O, Byte A Channels 0-7	Digital I/O, Byte B Channels 0-7
Analog I/O Connector	Not Used	Analog Inputs Channels 1-4	Analog Outputs Channels 9-12

For more information, see the diagrams for the desired device below. Note that the RX5 and RX7 use the same Digital and Multi I/O mappings.

Mapping RX5 or RX7 I/O

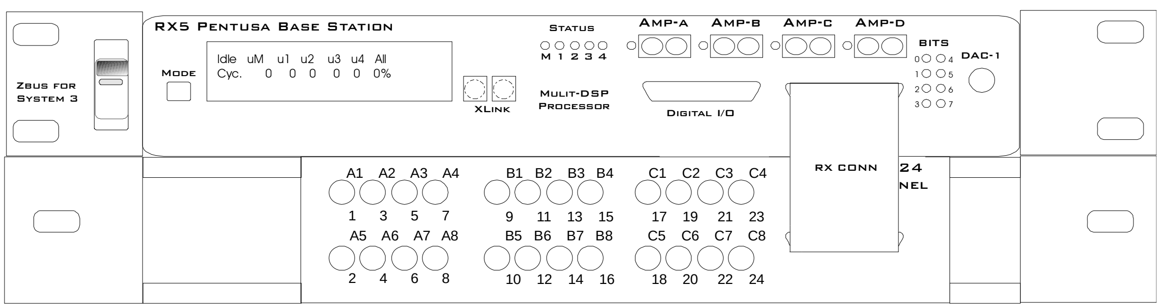
Note: The PP24 can be mounted above or below the RX5.

The diagram below maps the **RX5 or RX7 Digital I/O connections** to the PP24. All digital bits are programmable as input or output.



A1-A8	B1-B8	C1-C8
Bit Addressable Digital I/O Channels 0-7	Digital I/O, Byte A Channels 0-7	Digital I/O, Byte B Channels 8-15

The diagram below maps the **RX5 or RX7 Multi I/O connections** to the PP24. All digital bits are programmable as input or output.

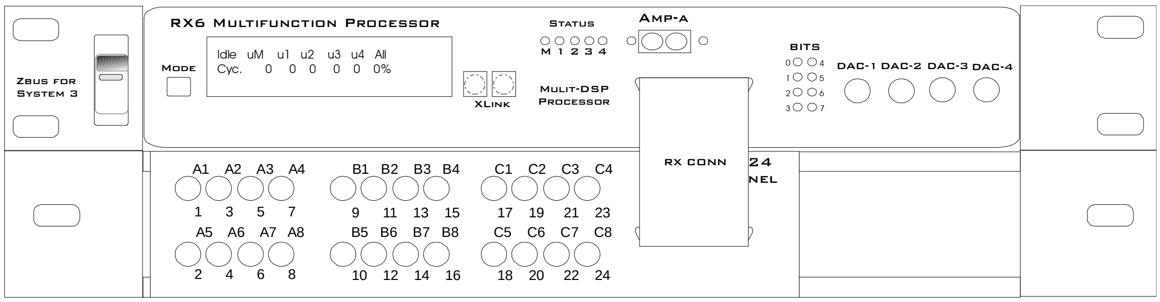


A1-A8	B1-B8	C1-C8
Analog Outputs A2, A4, A6, A8 = Channels 1-4 A1, A3, A5, A7 = Not Used	Digital I/O, Byte C Channels 16-23	Digital I/O, Byte D Channels 24-31

Mapping RX6 I/O

Note: The PP24 can be mounted above or below the RX6.

The diagram below maps the RX6 Digital I/O connection to the PP24. All digital bits are programmable as input or output.

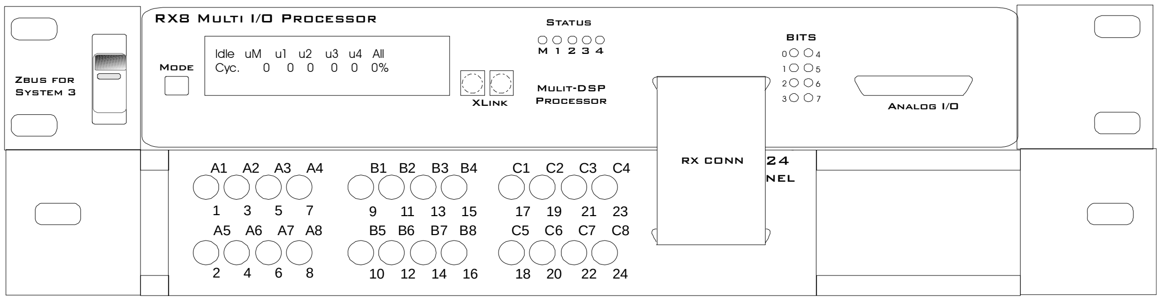


A1-A8	**B1-B8**	**C1-C8**
Bit Addressable Digital I/O Channels 0-7	Digital I/O, Byte A Channels 0-7	Digital I/O, Byte B Channels 8-15

Mapping RX8 I/O

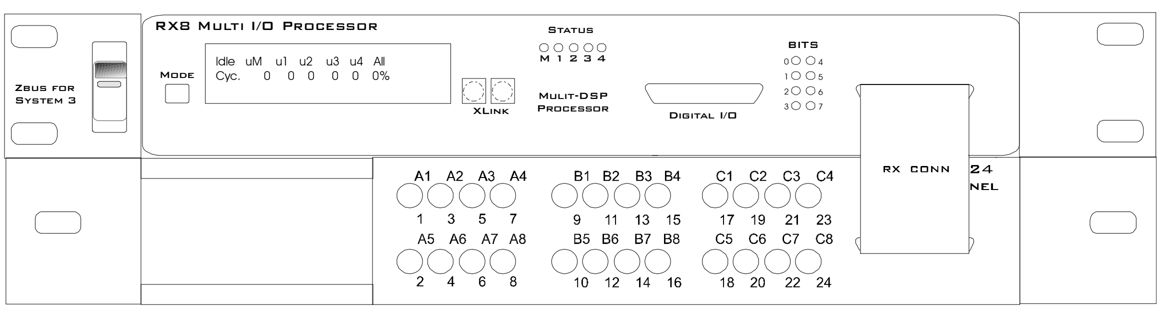
Note: The PP24 can be mounted above or below the RX8.

The diagram below maps the RX8 Digital I/O connection to the PP24. All digital bits are programmable as input or output.



A1-A8	B1-B8	C1-C8
Bit Addressable Digital I/O Channels 0-7	Digital I/O, Byte A Channels 0-7	Digital I/O, Byte B Channels 8-15

The diagram below maps the RX8 Analog I/O connection to the PP24. All digital bits are programmable as input or output.

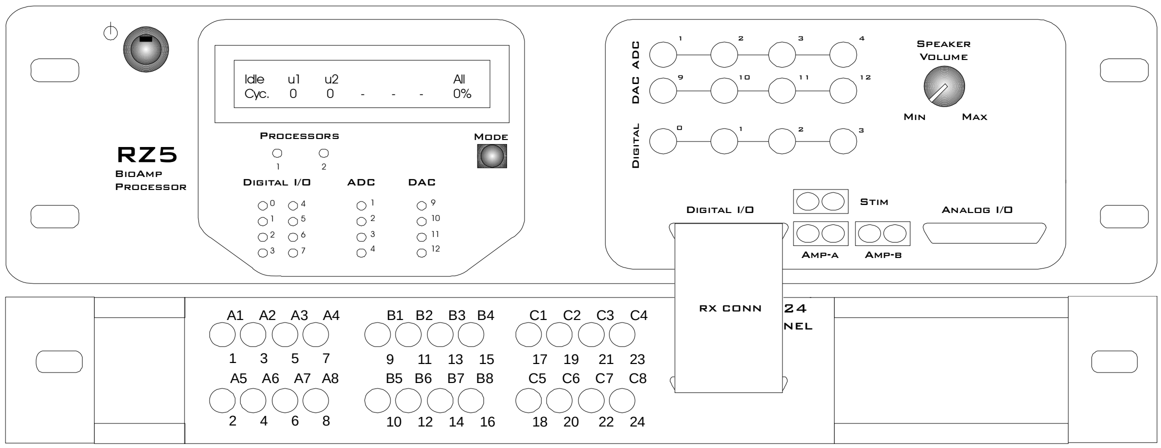


A1-A8	B1-B8	C1-C8
Analog I/O Block A Channels 1-8	Analog I/O Block B Channels 9-16	Analog Output Block C Channels 17-24

Mapping RZ5 I/O

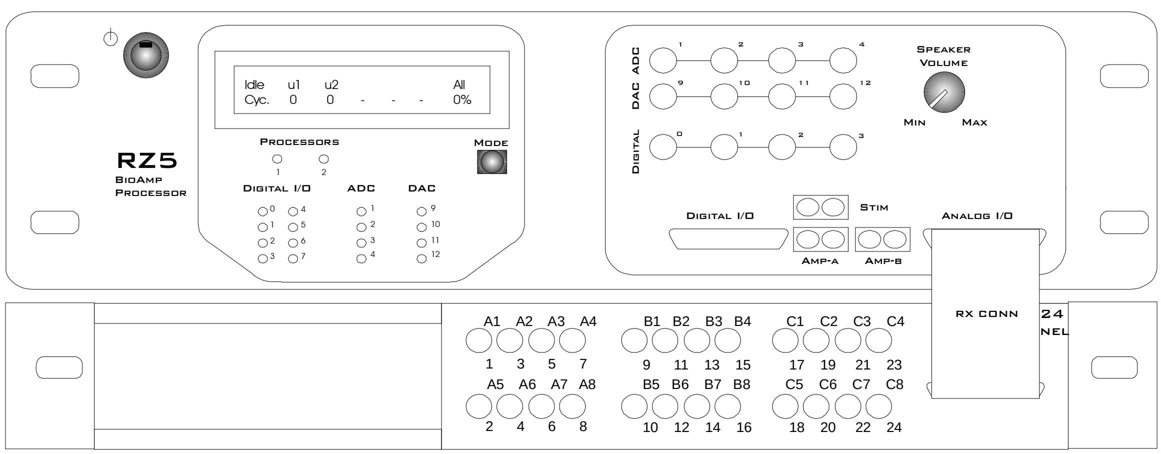
Note: The PP24 is mounted below the RZ5.

The diagram below maps the RZ5 Digital I/O connection to the PP24. All digital bits are programmable as input or output.



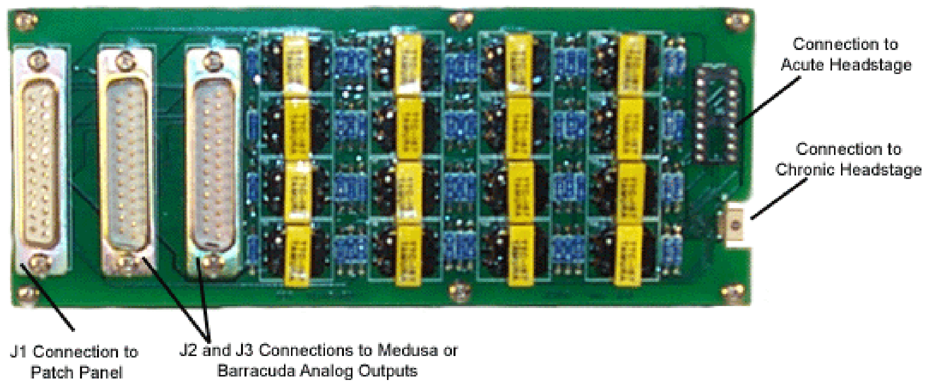
A1-A8	B1-B8	C1-C8
Bit Addressable Digital I/O Channels 0-7	Digital I/O, Byte A Channels 0-7	Digital I/O, Byte B Channels 0-7

The diagram below maps the RZ5 Analog I/O connection to the PP24.



A1-A8, B5-B8, C5-C8	B1-B4	C1-C4
Not Used	Analog Input Channels 1-4	Analog Output Channels 9-12

ETM1 Experiment Test Module



Overview

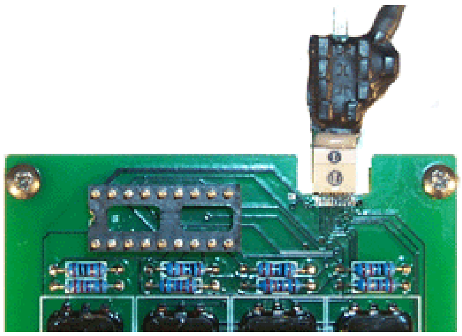
The Experiment Test Module (ETM1) allows you to design and test experimental protocols before running critical experiments and can be used to input signals into either the chronic (RA16CH) or acute (RA16AC) headstages from the analog outputs of the Medusa (RA16BA) or Barracuda Processor (RV8). The ETM1 also accepts signals via the Patch Panel (PP16). A processor can be used to generate signal spikes that simulate a physiological recording. The simulated spike signals can then be passed through the ETM1 and acquired by the connected headstage. The ETM1 also includes a connection to receive signals via the Patch Panel (PP16). Using the PP16, virtually any signal source can be used. The ETM1 allows the experimental setup to be tested without using a subject.

There is 1000 to 1 signal attenuation in the ETM1. Therefore, 1V on the input is equivalent to 1mV on the output to the headstage. The ETM1 uses transformer isolation of the incoming signal to the resulting output to the headstages.

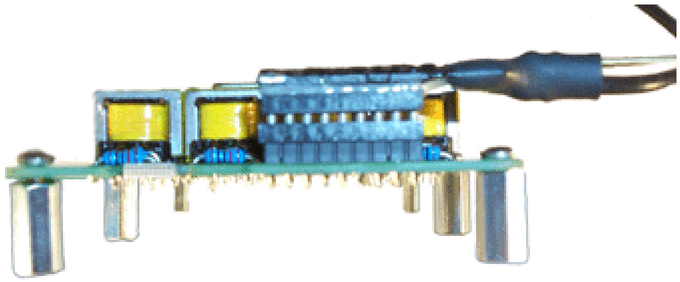
Inputs, or processor and patch panel connections, are located on one end of the device and output, or headstage connections, are located on the other end of the device.

Connecting the Headstage

Connect the headstage to the corresponding connector on the ETM1.



Chronic Headstage connected to ETM1



Acute Headstage connected to ETM1

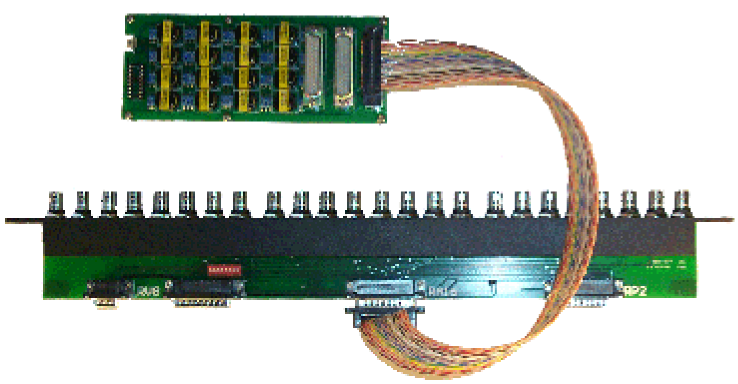
Connecting the Signal Source

The connectors labeled J1, J2 and J3 are used to connect the ETM1 to signal sources. The first eight-headstage channels (1-8) are wired to connector J2. The other eight-headstage channels (9-16) are wired to connector J3. All 16 channels are wired to connector J1. See technical specifications, page 15-25 for pinouts.

Connecting to an RA16BA or RV8

For headstage channels 1-8, plug one end of a serial DB25 male-female cable into the J2 connector and plug the other end into the Analog/Digital I/O Port of an RA16BA or RV8. For headstage channels 9-16 plug one end of a serial DB25 male-female cable into the J3 connector and the other end into the Analog/Digital I/O port of a second RA16BA or RV8.

Connect to the PP16



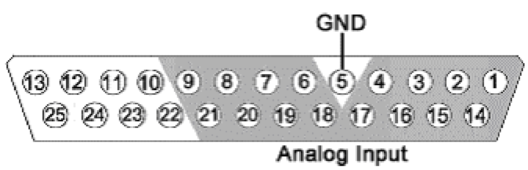
The connector labeled J1 is used to connect the ETM1 to a PP16. Plug one end of a serial DB25 male-female cable into the J1 connector and plug the other end into the RA16 port of the PP16. Channels 1 - 8 and 9 - 16 of the headstages can be accessed through the patch panel BNCs labeled A1-A8 and B1 - B8, respectively. Also, a custom cable can be fabricated to connect the ETM1 (connector J1) to virtually any signal source.

ETM1 Technical Specifications

Maximum Input	Should not exceed the maximum input for your amplifier (such as 4V for the RA16PA)
Frequency Response	Flat from 500 - 20,000 Hz
Highpass Filter (Fc)	20 Hz
S/N (typical)	70 dB
THD (Typical)	0.01% for 1 kHz input at 1 V peak-to-peak
Cross-Talk	< -70 dB
Attenuation	60 dB

J1 DB25 Pinout

Analog input channels 1-16. The J1 connector is typically used to input signals from the PP16 Patch Panel.

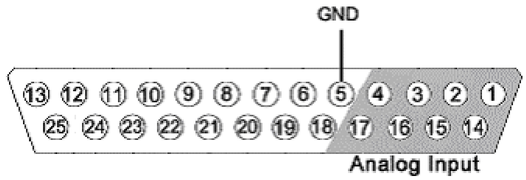


Pin	Name	Description
1	A1	Analog Input Channels
2	A3	
3	A5	
4	A7	
5	NA	Not Used
6	A10	Analog Input Channels
7	A12	
8	A14	
9	A16	
10	NA	Not Used
11		
12		
13		

Pin	Name	Description
14	A2	Analog Input Channels
15	A4	
16	A6	
17	A8	
18	A9	
19	A11	
20	A13	
21	A15	
22	NA	Not Used
23		
24		
25		

J2 DB25 Pinout

Analog input channels 1-8. Typically used to input signals from the RA16BA or the RV8.

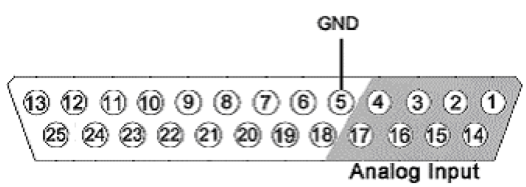


Pin	Name	Description
1	A1	Analog Input Channels
2	A3	
3	A5	
4	A7	
5	GND	Ground
6	NA	Not Used
7		
8		
9		
10		
11		
12		
13		

Pin	Name	Description
14	A2	Analog Input Channels
15	A4	
16	A6	
17	A8	
18	NA	Not Used
19		
20		
21		
22		
23		
24		
25		

J3 DB25 Pinout

Analog input channels 9-16. Typically used to input signals from the RA16BA.



Pin	Name	Description
1	A9	Analog Input Channels
2	A11	
3	A13	
4	A15	
5	GND	Ground
6	NA	Not Used
7		
8		
9		
10		
11		
12		
13		

Pin	Name	Description
14	A10	Analog Input Channels
15	A12	
16	A14	
17	A16	
18	NA	Not Used
19		
20		
21		
22		
23		
24		
25		

Part 16 PC Interfaces

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Interface Transfer Rates

Transfer rates depend on a number of factors, including the device accessed the type of transfer, and cycle usage.

The table below includes typical transfer rates for the Optibit, Gigabit, and USB interfaces at a 50% cycle usage with RP/RX and RZ devices. All values are MB/s.

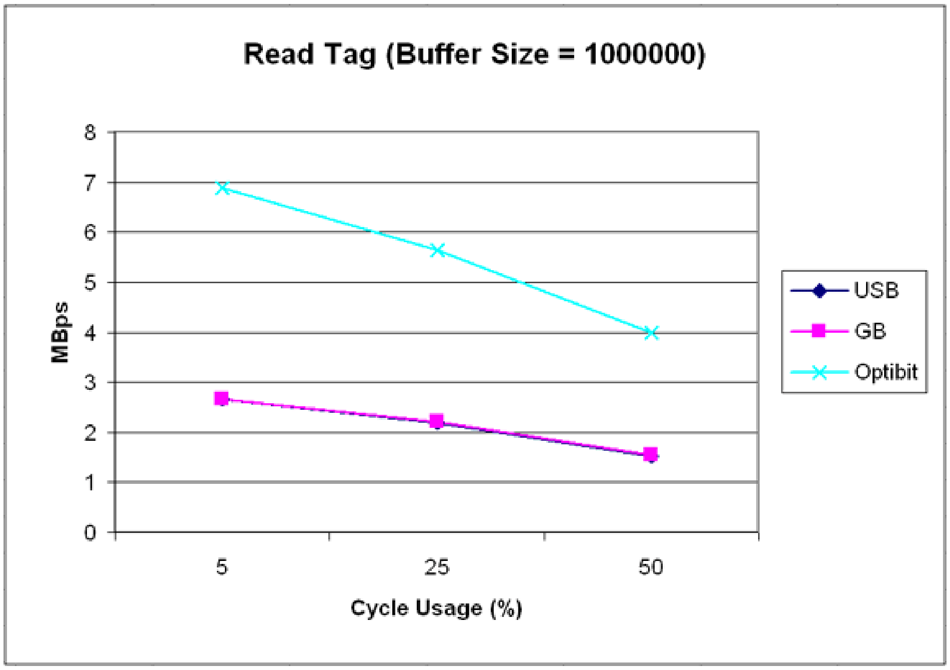
Interface	**Transfer Type**	**RP/RX**	**RZ**
PO5/FO5	Read	1.5/4.0	8.0
	Write	1.0	8.0
PI5/FI5	Read	1.5	NS
	Write	1.0	NS
UZ2	Read	1.5	NS
	Write	1.0	NS

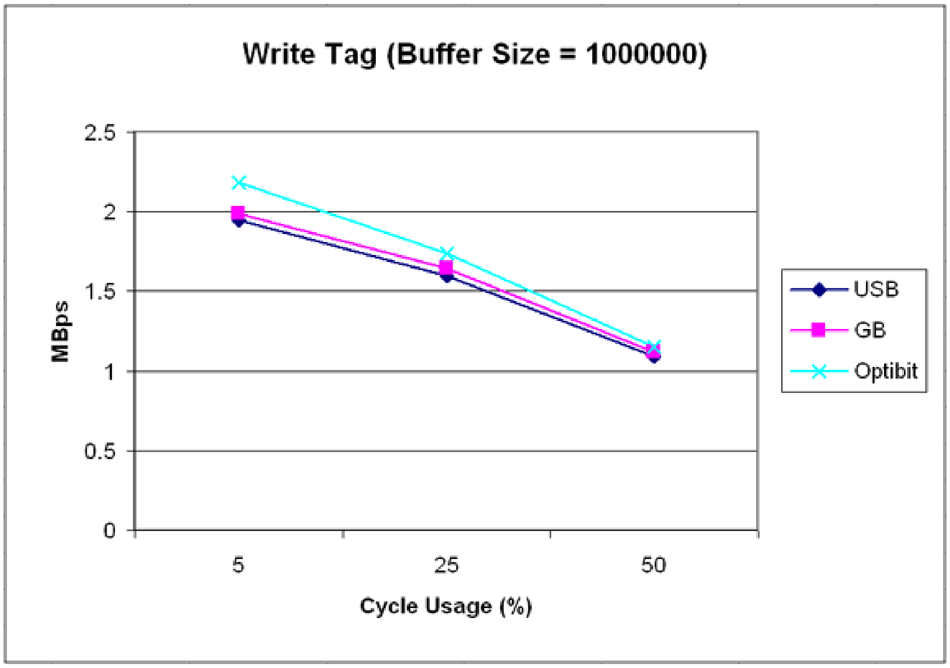
Because of the overhead required to poll the hardware or run single commands with the USB interface, users should be aware of the following relationships when performing small data transfers with the UZ2.

Interface	**Transfer Type**	**RP/RX**
UZ2	Snippet Transfers (~100)	0.3 MB/s
	Single Commands	1000 Commands/s

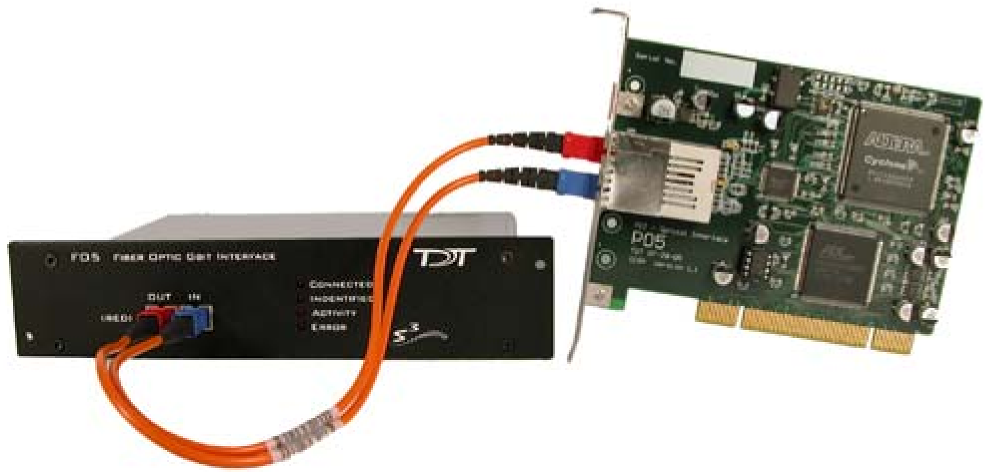
Cycle Usage and Large Transfers

The following graphs show how the cycle usage affects the transfer rate for large transfers with the Optibit, Gigabit, and USB 2.0 interfaces with an RX device. The data was collected using a buffer size of 1,000,000 for the Read Tag and Write Tag commands. The transfer rates were tested using both the RP2.1 (a single processor device) and only the main processor of an RX6 and using circuits generating cycle usages of 5, 25, and 50 percent.





Optibit Interface



Overview

The Optibit system (Optical Gigabit) is designed for users that require high-speed real-time control of System 3 devices or precise system-wide device synchronization. The Optibit interface consists of a PCI card (PO5) that must be installed in the computer and one or more Optibit-to-zBUS interface modules (FO5) that mount in the rear slot of a zBUS device chassis. It is up to 8x times faster than the original gigabit interface and also reduces the system's susceptibility to EMF. Devices are connected in a simple loop using provided high speed noise immune fiber optic cabling. Also, when using the Optibit interface, all devices (across all chassis) are automatically phase locked to a single clock.

Part Numbers:

PO5 – Optical PCI Card for Hardware/Software Control

FO5 – PO5 to zBus Interface

Status LEDs

Four status LEDs on the face of the FO5 indicate the connection status of the interface.

Connected – The Connected LED is lit when the interface is powered on and the fiber optic cable labeled IN is connected properly. Although the Connected LED will light if only the IN cable is connected, both cables have to be connected properly for communication to take place.

Identified – The Identified LED lights when a software signal sent from the PC is recognized by the interface. This takes place when launching TDT software such as zBusMon, RPvdsEx or loading an OpenEx project.

Activity – The Activity LED is lit when data is being sent to or from the TDT hardware.

Error – The Error LED lights when there is a connection or communication error. For example, this LED will light if the fiber optic cables are not connected properly.

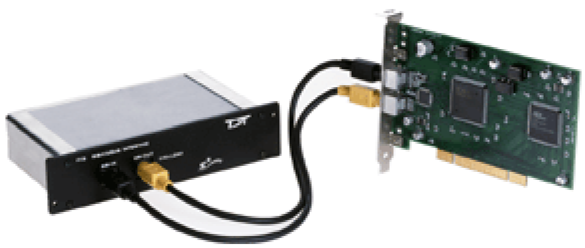
PO5 Technical Specifications

The PI5 and PO5 *zBus to PC interface cards* must be installed in a standard size, (PCI v 2.2 or greater) compliant 3.3 V slot.

Notes:

- Do not install in a PCI-X slot—the interface might fail.
- Do not attempt to install in low-profile PCI slots. While low profile and standard PCI cards maintain the same electricals, protocols, PC signals, and software drivers as standard PCI expansion cards, the low profile bracket is not compatible with standard cards.
- Maximum cable length: 30 meters
- Interface Transfer Rates vary by transfer type and device.

Gigabit Interface



Overview

The Gigabit system is designed for users that require high-speed real-time control of System 3 devices or precise system-wide device synchronization. The gigabit interface consists of a PCI card (PI5) that fits in the computer and one or more GBit-to-zBUS interface modules (FI5) that mounts in the rear slot of a zBUS device chassis. Devices are connected in a simple loop using provided cabling. When using the gigabit-interface all devices (across all chassis) are automatically phase locked to a single clock. Over 100 devices can be connected in a single Gigabit loop with automatic device identification and system initialization.

Part Numbers:

PI5 – PCI Card for Hardware/Software Control

FI5 – PI5 to zBus Interface

PI5 Technical Specifications

The PI5 and PO5 *zBus to PC interface cards* must be installed in a standard size, (PCI v 2.2 or greater) compliant 3.3 V slot.

Notes:

- Do not install in a PCI-X slot—the interface might fail.
- Do not attempt to install in low-profile PCI slots. While low profile and standard PCI cards maintain the same electricals, protocols, PC signals, and software drivers as standard PCI expansion cards, the low profile bracket is not compatible with standard cards.
- Maximum cable length: 30 meters
- Interface Transfer Rates vary by transfer type and device.

Gigabit Anomalies and Tech Notes

The PI5 must be installed in a computer that has a 3.3 V compliant PCI slot (v2.2 or greater).

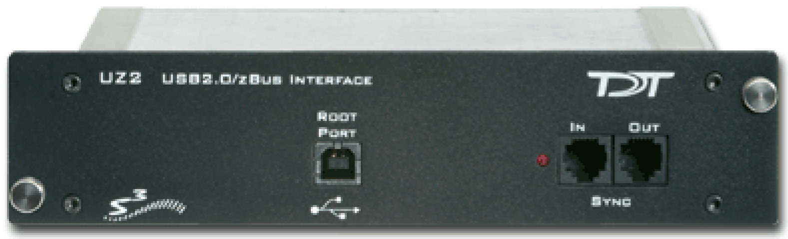
The PI5 is not compatible with the WindowsXP and 2000 Standby and Hibernate features. We recommend configuring PC Power Options to never use these modes for any PC used to run TDT applications.

Problems loading drivers may occur when the C:WINNT/inf folder is not visible. In Windows Explorer choose Tools|Folder Options, then choose View|Hidden Files and Folders, and select Make Visible.

When data is being transferred from the TDT hardware to the computer, CPU usage on the computer goes up to 100%. The computer is still usable (can ran other programs, etc.) despite the high CPU usage, however, other programs that are running on the computer may slow down.

After installing the Gigabit PCI card in your computer, there may be a conflict with how the PC communicates with the card and other devices in the system. This could lead to the following error message when performing a transfer test in zBUSmon: “System Test Error: Cycle power on system and test again.” If you experience system problems and find the IRQ number to be the same on another device, then you should move the PI5 card to another PCI slot in your machine.

UZ2 USB 2.0 Interface



Overview

The USB 2.0 zBus Interface mounts in the rear bay of a zBus device chassis and handles communication and data transfer between your computer and zBus mounted programmable devices, such real-time processors or programmable attenuators. Most nonprogrammable devices, such as speaker drivers or signal mixers, do not require an interface. You will need a USB2.0 port available on the host PC for each UZ2 in a multi-chassis system. We recommend upgrading to an Optibit interface if a system requires more than three chassis.

Note: The USB 2.0 interface requires Windows XP (with either Service Pack 1 or 2) or Windows 2000 (with Service Pack 5).

Connecting the UZ2

The UZ2 connects to your computer with standard USB 2.0 A to B cables (provided with each module). Interface drivers are bundled with the TDT Drivers and will be installed when the device is connected to the host computer for the first time. The UZ2 can be safely connected or unconnected while the computer is running.

Important Note: Wait ten seconds after devices have gone through the boot sequence or 30 seconds after turning on devices (with the computer already running) before running applications that use TDT devices. We also recommend using zBUSmon to verify the logical order of devices before beginning any experiment. See *Boot Up Sequence*, below, for more information.

Sync

The Sync allows users to synchronize several modules that are mounted in different device caddies. Each USB module has its own clock. Clocks on multiple USB devices will drift relative to each other. The Sync line uses the clock from one USB module, the master, to synchronize the clocks across all zBUS device caddies.

To connect several zBUS caddies, one module (the highest logical module) is designated as the master and the other clocks are slaved to the master clock. Connect the Sync Out of the master clock to the Sync In of the slave with a short patch cable. To connect several device caddies, daisy chain the connections between the slave caddies as shown below. When the Sync lines are connected correctly the LED to the left of the Sync connectors should be lit on each slave devices. The LED on the master will remain unlit. The LED should only flash when the Sync lines are not connected.

Sync LEDs	Indicates
Flashing (on slave)	Connected incorrectly
Master device not lit and slave devices lit	Connected correctly
No devices lit	Not synced to any device

Logical Order of Devices

The logical order of devices is determined each time the zBus caddies are powered on. You can verify the current logical order using the zBUSmon software.

Boot Up Sequence

The boot up sequence for the USB 2.0 interface is driven from the PC and follows the communication protocol described below.

1. The first time the hardware is turned on a device driver is loaded to the interface. Depending on your operating system, the PC might beep to indicate that the device driver has been loaded
2. A second set of drivers will be loaded and the devices will reboot.
3. The TDT hardware is queried to determine the logical order of the devices and zBus caddies.

Important!: If the zBUS is accessed during step three, the devices will fail to ID. To ensure that step three is completed, wait ten seconds after the devices have rebooted (step two) before loading any TDT application or viewing the devices in zBUSmon. If the hardware fails to ID shut down the TDT hardware and restart the device.

Part 17 The zBus and Power Supply

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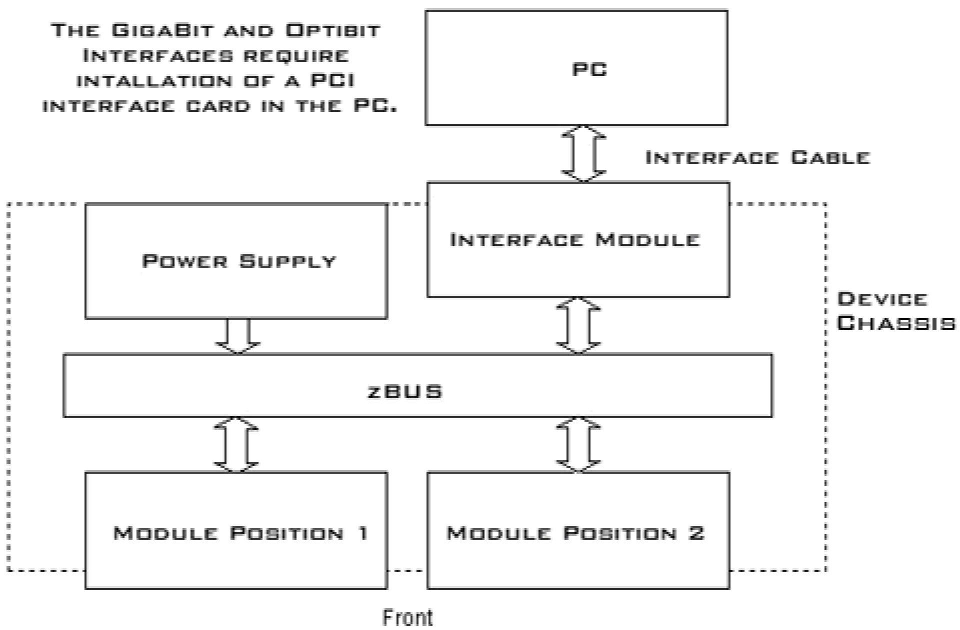
ZB1PS Chassis - Powered zBUS Device Chassis

Overview

zBUS is TDT's high-speed, low-noise bus for System 3 modules. The bus is integrated into a device chassis, which serves as a rack mountable housing for most modular devices in the System 3 line. As seen in the functional diagram below, the bus distributes communication and power throughout the system.



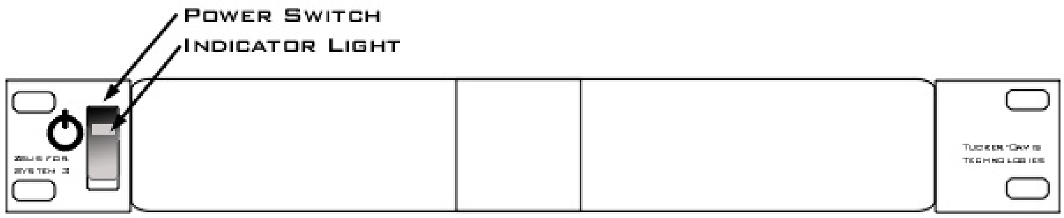
Functional Diagram

One or two modular devices can be mounted in the chassis' front bays, providing easy access to front panel connections. An interface module can be mounted in the second rear bay for chassis housing a programmable device. Multiple chassis can be interfaced for custom system configurations and individual modules can be added or removed as needed.

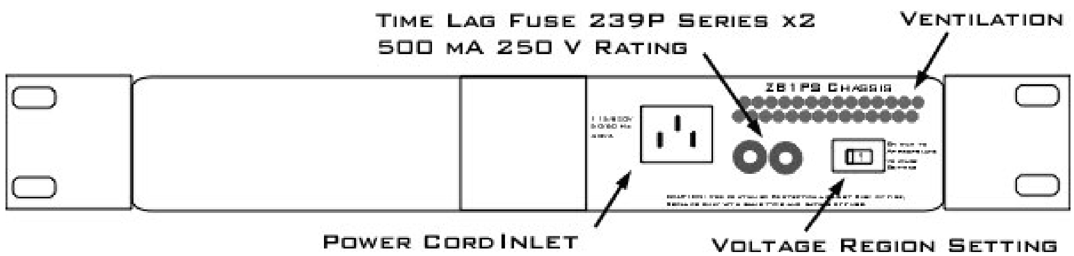
Power Supply

The ZB1PS chassis features an onboard, switchable (115V/220V) power source. The power supply is integrated into the chassis and cannot be removed. A small fan is located inside of the power supply and provides cooling while the power supply is active.

Using the ZB1PS



Front View



Back View

Applying Power to the Chassis

CAUTION! Allow at least 2 cm clearance from each side of the chassis for proper cooling. A ventilation fan is provided on the right side of the chassis. Ventilation holes are also provided on the power supply panel and another internal fan is provided inside the power supply housing. Installation of the chassis with the ventilation obstructed may cause a malfunction or fire.

Use only the supplied power cord.

To turn the ZB1PS on:

1. Position the chassis so that both the power switch and power cord may be accessed easily.
2. Ensure that the power switch is off and connect the power cord.
3. Ensure that the voltage region switch is set correctly. For standard outlets in the United States it should be switched to 115 V.
4. Turn the power switch on and check that the power switch's green LED is illuminated.

The Indicator Light

A front panel switch turns on the chassis power supply and includes an indicator light. The power switch's green LED will illuminate when the chassis is switched on. The light will flash rapidly when it receives a command from software and slowly to indicate a communications error (check all cable connections).

Disconnecting Power from the Chassis

CAUTION! When removing the power cord from either the power supply or socket outlet, grasp the plug, not the cord, in order to avoid damaging the cable.

To disconnect the ZB1PS:

1. Turn off the power switch.
2. Disconnect the power cord from the power supply.
3. Disconnect the power cord from the wall socket plug.

Adding and Removing Modules

Before adding or removing modules, make sure the zBus is powered off.

To remove a module from the chassis:

1. Unscrew the two thumb screws on the corner of the module faceplate.
2. Pull straight out on the front-panel BNC connectors. A BNC 'T' connector makes a great handle for removing zBus devices.

To add a module to a chassis:

1. Insert the module into an empty bay and push straight back until it seats onto the connector.
2. Hold the module in place with the thumb screws.

Maintaining the ZB1PS

Safety Notices

This device has passed rigorous testing by Underwriters Laboratories and is UL compliant for CAT II installation in laboratories and other indoor environments. Before applying power to the zBUS caddie, verify that the correct safety precautions are taken.

WARNINGS! Read the following warnings prior to operation.

- If the device is damaged, or fails to operate according to the specifications described in this manual, disconnect the power cord and contact TDT support immediately.
- Before applying power to the device, you must correctly connect the power cord to a standard socket outlet provided with a protective earth contact.
- In the event of impaired ground protection, avoid using the device to prevent possible damage.
- When removing the power cord from either the power supply or socket outlet, grasp the plug, not the cord, in order to avoid damaging the cable.
- Do not attempt to disassemble the power supply or caddie. If you experience any issues, contact TDT support immediately.
- Only fuses with the required rated voltage, current, and specified type should be used with this device. Do not attempt to alter or disassemble the power supply fuses.
- Do not attempt to alter this device in any way that deviates from its intended operation as described in this user manual.

- Capacitors contained inside the device may retain their charge even after power has been disconnected from its supply source.
- Operation of this device in the presence of flammable gases or fumes is strictly prohibited to avoid definite safety hazards.
- Do not subject this device to excessive amounts of vibration or shocks during handling or shipping, and avoid dropping the device.
- Although there is a protective screen over the ventilation fan, do not attempt to stick any objects into the fan. This may result in injury or damage the device.
- Do not attempt to store this device where it may be exposed to prolonged periods of excessive sunlight, high temperatures, high humidity, or condensation. If exposed to such conditions, the device may no longer work properly and its specifications may no longer be satisfied.
- The device is designed for indoor use only and is not waterproof; do not get the device wet.
- Do not attempt to use this device in a manner unspecified by TDT.

Changing the Power Supply Fuses

CAUTION! Only fuses with the required rated current, voltage, and specified type should be used with this device. Use only 500 mA, 250 V rated Time Lag fuses.

To change the power supply fuses:

1. Turn off the power switch.
2. Disconnect the power cord from the power supply.
3. Using a flathead screwdriver gently push the fuse plate inward.
4. Once the fuse plate is pressed inward gently turn the screwdriver counterclockwise until the fuse plate tab is visible.
5. Depress the fuse plate and it will pop up.
6. Grab both ends of the fuse plate and slide the fuse housing out of the power supply.
7. Replace the defective/broken fuse with a new 500 mA 250 V rating Time Lag fuse by gently pushing the end of the fuse into the fuse housing.
8. Push the fuse housing back into the power supply again by pressing the screwdriver inward.
9. Rotate the screwdriver clockwise until the fuse tab is correctly locked back into its original position.
10. Repeat for the other fuse if necessary.

Cleaning the ZB1PS Chassis

To clean the device:

1. Remove power from the ZB1PS chassis.
2. Clean the external surfaces of the device with a soft, dry cloth.
3. Do not attempt to disassemble and clean the inside of the device.

ZB1PS Technical Specifications

Chassis	
Height	1U
Width	Standard 19'' rack mount
Power Supply (Integrated)	
Maximum Working Voltage	HI to earth ground 230V max LO to earth ground 230V max
Main Voltage Rating	115/230 V, 50/60 Hz, 40 VA AC
Installation Category	CAT II
Environmental	
Operating Temperature	0 to 45°C
Storage Temperature	5 to 40°C
Humidity	80% for temperatures up to 31°C, decreasing linearly to 50% RH at 40°C
Maximum Altitude	2,000 m
Pollution Degree	2 (Indoor use only)
Power Supply Fuses	
Time Lag Fuse 239P Series	2 fuses
Operating Temperature	-55˚C to 125˚C
Ampere Rating	0.500 A
Voltage Rating	250 V
Interrupting Rating	10,000 amperes at 125 VAC, 0.7-0.8 power factor 35 amperes at 250 VAC, 0.7-0.8 power factor

ZB1 Device Caddie and PS25F Power Supply

The ZB1 and PS25F are TDT’s legacy zBUS caddie and power supply. The ZB1 device caddie is similar to the newer ZB1PS; however, it does not have onboard power and must be used in conjunction with the PS25F.

WARNINGS! The PS25F power supply **must** be placed in the right hand bay of a ZB1 Device Caddie as you look at the back of the caddie. It can damage the system if it is placed in any other bay.

No other power supply can be used to power the zBUS.

The two voltage switches should be switched to the mains voltage for your country. For example, in the United States these should both be switched to 115 V.

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Part 18 System 3 Utilities

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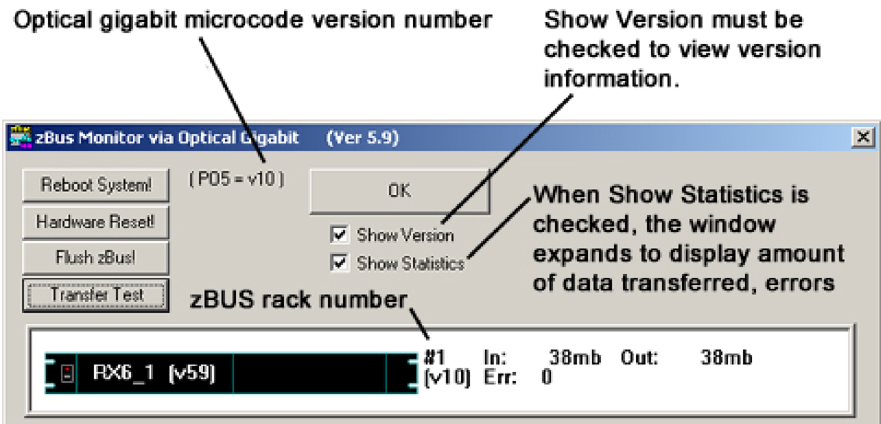
zBUSmon – Bus/Interface Test Utility

The zBUS Monitor program is a tool used to test the USB, Gigabit, or Optibit connection to System 3. This program is installed in the C:\TDT\zDrv3 directory by default and a shortcut is added to the TDT Sys3 Directory in the Start menu.

The zBUSmon Window

When the utility is run a small monitor window is opened. All correctly connected zBUS or built-in device chassis housing a programmable device, such as the RP2 and PA5, are represented in the system diagram. Chassis housing non-programmable devices, such as the SM5 or HB7, are not displayed.



Reboot System!

The **Reboot System!** button resets hardware and reloads device drivers.

Hardware Reset!

The **Hardware Reset!** button resets connected hardware.

Flush zBus!

The **Flush zBus!** button flushes interface line of commands or data.

Transfer Test

The **Transfer Test** button tests communication between the TDT modules and the PC. This will test data transfer both to and from the PC. A status bar is displayed indicating how much time is remaining in the test. Click anywhere in the zBUSmon window to end the test early.

Show Version Check Box

When the **Show Version** box is checked, the version number of each programmable device's firmware (TDT Microcode) are displayed in the hardware diagram. The microcode version number is shown within parentheses next to each device. For processor devices, the version number shown should be the same as the version number of the TDT Drivers installed on the PC

(**Note:** this does not occur in the PA5). The RP2.1 and the RL2 have a 1 in front of the microcode version number.

Microcode and driver version numbers should always be the same. Microcode versions displayed with red text are significantly outdated (such as versions older than v62) and should be updated immediately.

Show Statistics

The zBUSmon program, when used with the Optical Gigabit interface, provides an additional option to view system statistics. When **Show Statistics** is checked, the window expands to display the amount of data transferred and error codes if necessary. Rebooting the system, resetting the hardware, or cycling power on the zBUS racks will reset the data in the expanded window.

RPProg - Microcode Update Utility

About the Microcode

The microcode is low-level software that resides in flash memory on the System 3 processor devices. The microcode contains the DSP instructions for the RPvdsEx processing components. Because the System 3 design allows users to update this software quickly and efficiently, users can take advantage of the latest software tools available without purchasing new equipment or sending devices to our manufacturing facility for upgrades.

Updating the Microcode

When should the microcode be updated?

Every time a new version of RPvdsEx is installed on the host PC, the microcode should be updated on all processors in the system. This includes programmable devices that may have been purchased prior to your new system. New versions of the files need to update the microcode are always included in the TDT Drivers installation.

How is the microcode updated?

Users must update the microcode using the System 3 Device Programmer (**PrgG21K.exe**). This program is copied to the host PC during TDT Drivers installation and is stored in the following directory: **C:\TDT\RPvdsEx\RPProg**.

Important Notes: You should not use your PC for other tasks while devices are being reprogrammed. Most processors can be programmed in four minutes; however, the RZ processors may take up to 40 minutes (five minutes per DSP).

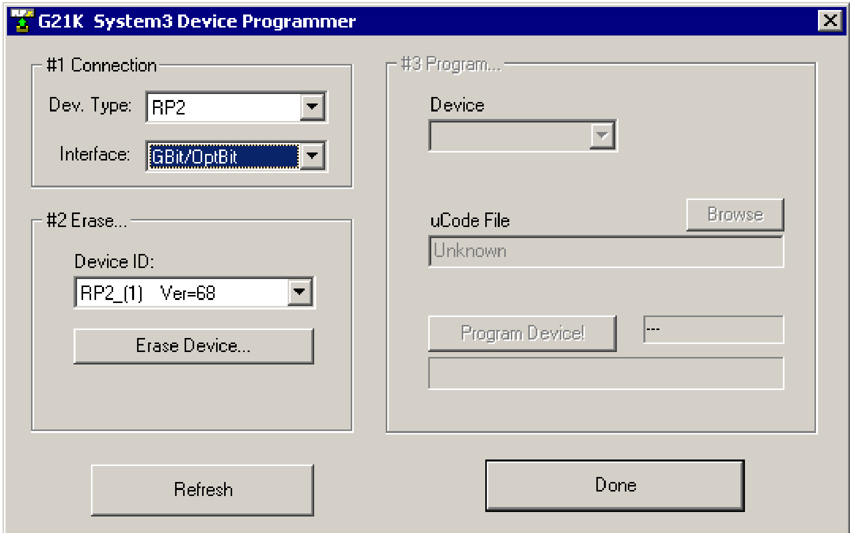
For instructions on updating an RL2 contact TDT Support.

To update the microcode:

1. Run the System 3 Device Programmer.

   To run the System 3 Device Programmer, click the **Start/Programs** menu, point to **TDT Sys3** and click **RPProg**.

2. Select the Device and System Interface Type.



   a. Under **#1 Connection**, select the device type to be programmed from the **Dev. Type** drop-down list.
   b. Select your system's interface type from the **Interface** drop-down list.

      Connected devices (of the type selected) will appear in the Device ID drop-down list.

3. Erase (Prepare) the Device.

Important Note: High performance processors, such as the RX5, are erased using a different method from other real-time processors. Please note your device type and follow the appropriate procedure for erasing the device.

Classic, Single-DSP Processors and Z-Series Processors

   a. To erase the first device in the list, click the button below the Device ID list under **#2 Erase**. (**Erase Device!** or **Prepare Device...**)

      A warning message will be displayed.
   b. Click **Yes** to continue.

      When the device has been erased, a message is displayed.
   c. Click **OK**.

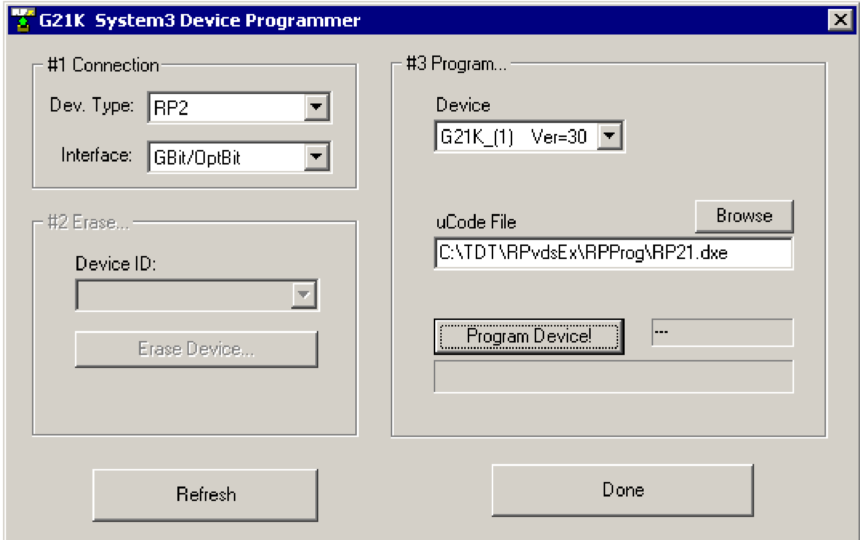
RX-Series Processors

   a. To erase the device, press and hold the **Mode** button on the front panel of the device and click **Refresh** in the programmer window. Release the **Mode** button.

      After the device is erased, the display on the device should read: **FirmWare: BLANK**.

      After a device is erased it appears in the **# 3 Program** area. In that list it appears with a generic name such as **G21K_1** the remaining programmed devices are renumbered. This can sometimes make it difficult to identify devices if more than one device is erased at a time.

Be sure to program this device before erasing others.



4. Program the Device

   a. Click **Browse** next to the **uCode File** box, then select the appropriate microcode file for the selected device.

File	Device
RP2.dxe	RP2 Real-Time Processor
RP21.dxe	RP2.1 Enhanced Real-Time Processor
RA16.dxe	RA16BA Medusa Base Station
RV8.dxe	RV8 Barracuda Processor
RMX.dxe	RM1/RM2 Mobile Processors
RXn.dxe	RX5 Pentusa Base Station
	RX6 MultiFunction Processor
	RX7 Micro Stimulator Base Station
	RX8 Multi I/O Processor
RZn.dxe	Z-Series Processors

   b. Click **Program Device!**.

   A warning message will be displayed.

   c. Click **Yes** to continue.

   Important! Wait until the device is programmed before doing anything else with your PC. Most processors can be programmed in four minutes; however, the RZ processors may take up to 40 minutes (five minutes per DSP).

   d. Click **OK**.
   The selected Real-Time Processor has now been reprogrammed.

5. Programming Additional Devices

   If you have additional devices to program, click **Refresh**, then **repeat** beginning with Step 2, Select the Device and System Interface Type.